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Stochastic Impulse Control Problems: Dynamic Programming, Viscosity Solutions, and Deep Reinforcement Learning

With Applications to Optimal Foreign Exchange Intervention and Dynamic Portfolio Allocation

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ABSTRACT

This thesis studies finite-horizon stochastic impulse control problems for jump-diffusion processes and considers two approaches for the problem's solution. First, a review of the dynamic programming principle is provided. Second, the problem is analysed through its parabolic quasi-variational inequality (QVI) formulation. Because solutions to QVIs are generally non-smooth, a viscosity solution approach is used to show that the QVI characterises the impulse control problem. The QVI is then solved numerically using a finite-difference scheme.

The impulse control problem and deep reinforcement learning (DRL) are then connected by means of dynamic programming. Two simulation-based DRL algorithms are of particular interest in this thesis, namely the value-based Double Deep Q-Network (DDQN) and the policy-based Proximal Policy Optimisation algorithm augmented with self-imitation learning (PPO-SIL).

These algorithms are applied to two financial problems. The first considers optimal control of a foreign exchange (FEX) rate with transaction costs, where the DDQN and PPO-SIL algorithms are compared against the finite-difference scheme approximation. The second addresses the dynamic portfolio allocation problem with transation costs, where PPO-SIL is used to trade a 20-asset equity portfolio over a one-year horizon. Asset prices are simulated using correlated multidimensional Variance–Gamma processes with common subordinators and calibrated to European call option prices. In this setting, the learned trading strategy achieves average cumulative returns of approximately 10% and an average annualised Sharpe ratio of 3.26.

SAMMENDRAG

Denne oppgaven handler om stokastiske impulskontrollproblemer hor hoppeprossesser over en endelig tidshorisont. Teoretiske løsningsmåter som dynamisk programmering og den kvasi-variasjonsulikheten (QVI) er gjennomgått. Fordi løsninger av QVI-er generelt sett ikke er glatte, brukes viskositetsløsninger for å bevise at løsningen på QVI-en er løsningen på impulskontrollproblemet. QVI-en løses numerisk ved bruk av differansemetoder.

Impulsekontrollproblemet kobles sammen med dyp forsterkningslæring (DRL) ved hjelp av dynamic programmering. To simulasjonsbaserte DRL algoritmer er tatt i betraktning: den verdibaserte Double Deep Q-Network-metoden (DDQN) og den polisebaserte Proximal Policy Optimisation-algoritmen som er utvidet med selvimitasjonslæring (PPO-SIL).

Til slutt anvendes metodene på to problemer innen finans. Det første problemet handler om den optimale kontrollem av valutakurset med kurtasje, , hvor både DDQN- og PPO-SIL-algoritmene sammenlignes med løsningen fra endelig-differansemetoden. I det andre problemet, som er en dynamisk porteføljeallokeringsproblem med kurtasje, brukes PPO-SIL til å forvalte en portefølje med 20 aksjer over en tidsperiode av ett år. Aksjepriser antas til å være korrelerte, multidimensjonale varianse-gamma-prosesser med en felles subordinator, og Europeiske kjøpsopsjoner brukes til å estimere parameterne inni aksjeprismodellen. PPO-modellen oppnår en gjennomsnittlig kumulativ avkastning på rundt 10% og en gjennomsnittlig årlig Sharpe-ratio på 3.26.

PREFACE

This thesis marks the conclusion of five years in the M.Sc. programme in Applied Physics and Mathematics at the Norwegian University of Science and Technology (NTNU), with specialisation in Industrial Mathematics and one year of exchange at Ecole Polytechnique Fédérale de Lausanne (EPFL). The work presented in this thesis was carried out in the spring of 2026.

The motivation for this thesis comes from my interest in quantitative finance and, more specifically, from the curiosity of exploring mathematical ways of supporting trading decisions for retail investors. Funnily enough, the initial idea of combining mathematics and reinforcement learning for the purpose of trading originated during a dinner with friends and was gradually developed into the study presented here.

I wish to express my sincere gratitude to my supervisor, Professor Espen Robstad Jakobsen, for giving me the freedom to choose my own thesis topic, for our weekly meetings, as well as for his guidance in finding relevant literature and refining the direction of the thesis.

Finally, I would like to thank my friends, family, fellow students, and my partner for their support, motivation, and companionship during the past five memorable years.

Français

Et voilà ! Cinq années d'études en physique et mathématiques appliquées à l'Université norvégienne de sciences et de technologie (NTNU) s'achèvent avec cette thèse. Je tiens à remercier mon directeur de mémoire, le Professeur Espen Robstad Jakobsen, pour m'avoir laissé choisir mon propre sujet, pour nos réunions hebdomadaires, ainsi que pour ses précieux conseils. Je remercie également mes amis, ma famille et ma partenaire pour leur soutien tout au long de ces cinq dernières années.

Deutsch (CH)

Diese Masterarbeit schliesst meine Zeit als Student an der Norwegischen Technisch-Naturwissenschaftlichen Universität (NTNU) ab. Ich möchte mich bei meinem Betreuer, Professor Espen Robstad Jakobsen, bedanken, insbesondere dafür, dass er mir die Freiheit gegeben hat, mein eigenes Thema zu wählen. Ebenso danke ich ihm für die wöchentlichen Besprechungen und seine wertvollen Ratschläge. Ein grosser Dank gilt auch meinen Mitstudierenden, meinen Freunden, meiner Familie und meiner Partnerin für ihre Hilfe und Unterstützung während der letzten fünf Jahre. Ohne euch hätte es nicht besser gehen können!

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Gonchigsuren (Gosha) Bor

CONTENTS

LIST OF FIGURES

LIST OF TABLES

CHAPTER

ONE

INTRODUCTION

Many problems in economics and finance involve decision-making that cannot be taken continuously in time, but rather at discrete and irregular intervals. For example, a central bank does not continuously intervene in the foreign exchange market, a (retail) investor facing transaction costs does not rebalance her portfolio at every instant, and a firm does not continuously inject capital into a project for every small fluctuation in profitability. In all these cases, the controller, whether a central bank, investor, or firm, typically waits until an intervention is sufficiently valuable. The controller must therefore decide both *when* to intervene and *what* intervention to apply, resulting in a stochastic impulse control problem.

Between interventions, the state process is assumed to evolve according to a stochastic differential equation (SDE). In this thesis, the uncontrolled state is modelled as a jump-diffusion of the form

$$\mathrm{d}X(t) = \mu(t, X(t^-))\,\mathrm{d}t + \sigma(t, X(t^-))\,\mathrm{d}W(t) + \int_{\mathbb{R}^\ell} \gamma(t, X(t^-), z)\tilde{N}(\,\mathrm{d}t,\,\mathrm{d}z),$$

where the Brownian motion W models continuous uncertainty and the compensated Poisson random measure $\tilde{N}$ models jumps.

An impulse control, modeled through a double-sequence $\alpha = \{(\tau_j, \zeta_j)\}_{j\geq 1}$, controls the jump diffusion process at intervention times τ_j through an impulse ζ_j. If the performance of an impulse control is quantified with the performance criterion J, the impulse control problem translates into the maximization problem

$$v(t, x) = \sup_{\alpha\in\mathcal{V}} J^{(\alpha)}(t, x).$$

There exist two main approaches to impulse control problems. The first is the dynamic programming approach, based on Bellman's principle of optimality [13]. In this formulation, the value function is characterised recursively. That is, after the next decision time, the problem restarts from the new state. For impulse control, this leads to an approximation by a sequence of optimal stopping problems, as described by Øksendal and Sulem [84]. The second way to solve the impulse-control problem is through the parabolic Hamilton-Jacobi-Bellman Quasi-Variational Inequality

(HJBQVI), or simply QVI since continuous controls are not considered in this thesis. The QVI approach was first introduced by Bensoussan and Lions [14], where the authors characterise the value function through the free-boundary problem

$$\min\{\underbrace{-(\partial_t v + \mathcal{L}v + f)}_{\text{No intervention}}, \underbrace{v - \mathcal{M}v}_{\text{Intervene now}}\} = 0, \tag{QVI}$$

Here, $\mathcal{L}$ is the infinitesimal generator of the uncontrolled state process, and $\mathcal{M}$ is the nonlocal intervention operator. The first term in (QVI) corresponds to continuation without intervention, while the second term corresponds to immediate intervention.

In general, the value function v is not expected to be smooth as fixed intervention costs may create (non-differentiable) kinks at the intervention boundary. Moreover, as the underlying stochastic process is a jump-diffusion, the generator $\mathcal{L}$ contains a non-local and, possibly, singular integral, which further increases the complexity of the problem. As a consequence, classical solutions are either too restrictive or may not even exist in the specified domain, and one instead opts for a weaker solution in the form of *viscosity solutions*. The theory of viscosity solutions was first developed by Crandall and Lions [32] for local first-order partial differential equations (PDEs) and later extended to local second-order PDEs by Crandall, Ishii and Lions [31]. The nonlocal viscosity theory for partial integro-differential equations (PIDEs) is treated in Barles and Imbert [7], as well as Jakobsen and Karlsen [60]. For impulse control of jump-diffusions, Seydel [92] gives the central existence and uniqueness result for the elliptic and parabolic HJBQVIs used in this thesis.

Closed-form solutions are rarely available for impulse control problems, so numerical methods are typically required. However, (QVI) is a nonlinear equation with a nonlocal intervention operator, making numerical approximation difficult. While classical convergence arguments of Barles and Souganidis [8] prove that monotone, stable, and consistent schemes converge to the correct solution, the presence of a nonlocal intervention operator requires additional analysis. An extension of the Barles–Souganidis theorem to HJBQVIs is provided in [3, 5, 6]. Nevertheless, finite-difference schemes scale poorly with dimension, which limits their applicability to high-dimensional portfolio problems.

This motivates the second computational direction covered in this thesis. Deep reinforcement learning (DRL) methods have been widely applied in finance, including portfolio optimisation by Jiang, Xu and Liang [62], derivative hedging by Buehler et al. [22] and Buehler, Murray and Wood [23], as well as trading applications by Jain et al. [59] and Anchuri [1]. In this thesis, two DRL approaches are used to approximate impulse control problems. The value-based approach builds on Double Deep Q-Networks [49, 50], while the policy-based approach uses Proximal Policy Optimisation [91] combined with self-imitation learning [83].

This thesis focuses on two particular applications of impulse controls in finance: Optimal foreign exchange (FEX) intervention and dynamic portfolio allocation with transaction costs. In the FEX intervention problem, a central bank uses costly market interventions to keep an exchange rate close to a target level. This problem is one of the oldest examples of impulse control in finance, first studied in [81, 25, 26]. In the portfolio allocation problem, continuous rebalancing is no longer optimal when trading is costly. Instead, the portfolio manager waits until the portfolio has moved sufficiently far away from the desired allocation before rebalancing. This problem is studied in [39, 80, 65].

Thesis contribution

This thesis studies known theory on finite-horizon impulse control problems for jump-diffusion processes in a risk-neutral objective setting, in the sense that the controller maximises an expected performance criterion rather than an expected utility. The theoretical part formulates the impulse-control problem, relates it to the dynamic programming principle, and studies the corresponding QVI using viscosity solutions. Existence and uniqueness results are mainly based on Seydel [92]. However, while Seydel states the parabolic comparison theorem, the proof is concise and largely refers to the elliptic argument. This thesis provides a detailed proof of the comparison result for parabolic QVIs in the finite-horizon and jump-diffusion settings.

In addition, Appendix C derives a local dynamic-programming approximation in the continuation region. Under suitable smoothness and growth conditions, Proposition C.2 establishes a residual bound of order $O(h^{3/2})$. To the best of our knowledge, this specific result is not explicitly proven in the references used in this thesis. This result is then used to bridge the impulse-control QVI to the Bellman equations, the latter of which is approximated using value-based DRL methods.

From a numerical perspective, this thesis provides an empirical study of how value-based and policy-based DRL algorithms approximate finite-horizon impulse control problems and intervention regions. In particular, the DRL models are compared against a finite-difference scheme in the foreign exchange (FEX) intervention problem. In the second numerical experiment, the Proximal Policy Optimisation algorithm is applied to a dynamic portfolio allocation problem with transaction costs, where trading decisions are modelled as impulse controls. As such, this thesis aims to provide retail investors with a practical framework to support informed portfolio rebalancing decisions.

Contribution to sustainability

Although the main focus of this thesis is the application of impulse control in finance, the theory is applicable in several other fields, including optimal harvesting in forest management [53], river restoration [101], animal population management [100], and life insurance [3, 70]. For instance in forest management, impulse control can be used to model optimal harvesting, replanting, or fire-prevention interventions, while applications in life insurance can improve retirement plans. These examples illustrate how impulse controls can help in cases involving responsible resource use, ecosystem protection, and long-term economic planning. As such, they are relevant to several Sustainable Development Goals, including Decent Work and Economic Growth (SDG 8), Industry, Innovation and Infrastructure (SDG 9), Responsible Consumption and Production (SDG 12), Climate Action (SDG 13), and Life on Land (SDG 15). More generally, impulse-control models can support transparent and quantitatively disciplined decision-making in systems where interventions are costly, uncertain, and socially consequential.

Thesis structure

The thesis is structured in the following way.

- Chapter 2 introduces the stochastic preliminaries used throughout the thesis, such as filtered probability space, jump-diffusion SDEs, infinitesimal generators, and stopping times.
- Chapter 3 formulates the impulse-control problem for jump-diffusion processes together with a verification theorem and the dynamic programming principle.
- Chapter 4 studies the existence of viscosity solutions for QVIs. The chapter introduces additional assumptions and proves that the value function is a viscosity solution.
- Chapter 5 proves the uniqueness of the viscosity solution by means of the comparison-principle for the parabolic QVI.
- Chapter 6 presents a finite-difference scheme used to approximate the QVI in low-dimensional problems.
- Chapter 7 gives a brief overview of reinforcement learning and applies DRL two algorithms on the impulse control problem.
- Chapter 8 approximates the FEX rate problem using a finite-difference scheme and compares the scheme against DRL algorithms. As such, this chapter can be viewed as numerical benchmark for the DRL methods.
- Chapter 9 studies the use of impulse controls in dynamic portfolio allocation and uses the Proximal Policy Optimisation algorithm to approximate a trading policy in a portfolio consisting of 20 risky assets.
- Chapter 10 discusses the theoretical and numerical findings, outlines possible directions for future work, and concludes the thesis.

CHAPTER

TWO

PRELIMINARIES

This chapter introduces the preliminary concepts from stochastic analysis that are required to formulate and analyse finite-horizon impulse control problems, including filtereed probability spaces, jump-diffusions, and stopping times. Much of the theory presented in this chapter can be found in [2, 61, 84].

2.1 Stochastic setup

Throughout this thesis, we denote by $0 \leq T < \infty$ a finite time horizon and work with a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}\}_{0\leq t\leq T}, \mathbb{P})$. Here, Ω denotes the sample space, $\mathcal{F}$ the σ-algebra of events, $\mathbb{P}$ the real-world probability measure, and $\{\mathcal{F}_t\}_{0\leq t\leq T}$ the filtration. The interpretation of the filtration is that $\mathcal{F}_t$ contains the information available up to time $t \geq 0$. In many financial problems, decisions should depend only on past and present information, but not on future information. Thus, filtrations provide a way to capture this restriction on the flow of information.

The filtration is assumed to satisfy the usual hypotheses. That is, $\mathcal{F}_0$ contains all $\mathbb{P}$-null sets of $\mathcal{F}$, and the filtration is right-continuous, i.e.

$$\mathcal{F}_t = \bigcap_{u>t} \mathcal{F}_u.$$

In addition to the filtered probability space, we commonly use terminologies such as random variables, stochastic processes, and sample paths on $(\Omega, \mathcal{F}, \{\mathcal{F}\}_{0\leq t\leq T}, \mathbb{P})$. These are defined as follows:

Definition 2.1 (Random variable)**.** For Ω a sample space, a random variable is a measurable map $X : \Omega \to \mathbb{R}^d$.

Definition 2.2 (Stochastic process)**.** A stochastic process is a family of random variables $X = \{X(t)\}_{0\leq t\leq T}$, where each $X(t)$ is a random variable. Equivalently, a stochastic process is a map $X : [0, T] \times \Omega \to \mathbb{R}^d$, where Ω is the sample space.

Definition 2.3 (Sample paths)**.** Let X be a stochastic process. For a fixed atom $\omega \in \Omega$, the function $t \mapsto X(t, \omega)$ is called a sample path of X.

In this thesis, we adopt the short-hand notation $X(t) := X(t, \omega)$, for $\omega \in \Omega$, to denote a time-t sample path of X. Furthermore, we will assume that every stochastic process mentioned in this paper is adapted to the filtration $\{\mathcal{F}_t\}_{0\le t\le T}$ and is càdlàg[1]. In the following, if X is a jump process, we denote by $X(t^-)$ a sample of X before a potential jump occurs at time t. We use the short-hand notation $\mathbb{E}$ to denote the expectation taken under the probability measure $\mathbb{P}$, i.e. $\mathbb{E}[\cdot] := \mathbb{E}_{\mathbb{P}}[\cdot]$.

2.2 Brownian motions, Poisson random measures, and Lévy processes

We assume that stochastic processes are driven by continuous and discontinuous jump noises. Specifically, the continuous noise is modelled by Brownian motions, while the discontinuous jump part is a Lévy process.

Definition 2.4 (Brownian motion)**.** An m-dimensional Brownian motion is an adapted, continuous stochastic process $W = \{W\}_{0\le t\le T}$ with values in $\mathbb{R}^m$, satisfying

(i) $W(0) = 0$ a.s.[2],

(ii) $W(t) - W(s)$ is $\mathcal{F}_s$-independent for $0 \le s < t \le T$, and

(iii) $W(t) - W(s) \sim \mathcal{N}(0, (t-s)I_m)$, where $\mathcal{N}$ denotes the normal cumulative distribution function and I_m is an m-dimensional identity matrix.

Definition 2.5 (Lévy process)**.** A càdlàg adapted process $L = \{L(t)\}_{0\le t\le T}$ is a Lévy process if

(i) $L(0^-) = 0$ a.s.,

(ii) L has independent and stationary increments,

(iii) X is stochastically continuous in the sense that for all $a > 0$ and for all $s \ge 0$,

$$\lim_{s\to t} \mathbb{P}\big(|X(t) - X(s)| > a\big) = 0.$$

Comparing the Brownian motion definition with Lévy processes, it can be shown that the Brownian motion is a Lévy process, see [2, Example 1.3.8].

The Lévy measure ν satisfies

$$\int_{\mathbb{R}^\ell\setminus\{0\}} \big(|z| \wedge 1\big)\nu(\mathrm{d}z) < \infty. \tag{2.2.1}$$

Furthermore, we assume that for a given ν, there exist real numbers $p^* > 0$ and $q^* \ge 0$ such that

$$\int_{|z|\ge 1} |z|^{p^*}\nu(\mathrm{d}z) < \infty \quad \text{and} \quad \int_{|z|<1} |z|^{q^*}\nu(\mathrm{d}z) < \infty \tag{2.2.2}$$

holds. This assumption is needed for the existence proof of viscosity solutions in Chapter 4.

[1] The abbreviation càdlàg stands for *continu à droite, limite à gauche*, i.e. *right-continuous with left limits*.

[2] The acronym *a.s.* stands for *almost surely*.

Example 2.1 (The Variance-Gamma Lévy measure)**.** A recurring Lévy measure used throughout this thesis is the Variance-Gamma Lévy measure, defined by

$$\nu(\,\mathrm{d}z) = \frac{C}{|z|}\left(e^{Gz}1_{\{z<0\}} + e^{-Mz}1_{\{z>0\}}\right)\,\mathrm{d}z, \tag{2.2.3}$$

where $C, G, M > 0$ are constant parameters and $z \in \mathbb{R}$.

Remark 2.6. Note that the Variance-Gamma Lévy measure has infinite activity, i.e. $\nu(\mathbb{R}) = \infty$, due to the singularity at $z = 0$. Moreover, it has finite variation, that is, $\int_{\mathbb{R}} z^2\nu(\,\mathrm{d}z) < \infty$.

The discontinuous jumps in X are caused by a Poisson random measure $N(\,\mathrm{d}t,\,\mathrm{d}z)$ on $[0,T]\times\mathbb{R}^\ell$, with intensity measure $\nu(\,\mathrm{d}z)\,\mathrm{d}t$. For a Borel set $A \subset \mathbb{R}^\ell\backslash\{0\}$ with $\nu(A) < \infty$, the process

$$N(t, A) := N((0,t]\times A)$$

counts the number of jumps with marks in A up to time t. It is a Poisson process with intensity $\nu(A)$. As is often the case in financial applications, we use the compensated Poisson random measure

$$\tilde{N}(\,\mathrm{d}t,\,\mathrm{d}z) = N(\,\mathrm{d}t,\,\mathrm{d}z) - 1_{\{|z|<1\}}\nu(\,\mathrm{d}z)\,\mathrm{d}t,$$

instead of N.

2.3 Jump-diffusion processes

Let X be a d-dimensional, stochastic process in the filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}\}_{0\le t\le T}, \mathbb{P})$ with $X(t^-) = x \in \mathbb{R}^d$. We call X a jump-diffusion process if, for $0 \le t \le T$, it satisfies

$$\begin{aligned} X(s) = x &+ \int_t^s \mu\left(u, X(u^-)\right)\,\mathrm{d}u + \int_t^s \sigma\left(u, X(u^-)\right)\,\mathrm{d}W(u) \\ &+ \int_t^s \int_{\mathbb{R}^\ell} \gamma\left(u, X(u^-), z\right)\tilde{N}(\,\mathrm{d}u,\,\mathrm{d}z), \end{aligned} \tag{2.3.1}$$

where $\mu : [0,T]\times\mathbb{R}^d \to \mathbb{R}^d$ is the drift function, $\sigma : [0,T]\times\mathbb{R}^d \to \mathbb{R}^{d\times m}$ the diffusion matrix, and $\gamma : [0,T]\times\mathbb{R}^d\times\mathbb{R}^\ell \to \mathbb{R}^d$ the jump-amplitude function. We note that the instantaneous covariance matrix $\sigma\sigma^\top$ is positive semidefinite.

Throughout the thesis, we refer to (2.3.1) in its differential form, i.e.

$$\begin{aligned} &\mathrm{d}X(s) = \mu\left(s, X(s^-)\right)\,\mathrm{d}t + \sigma\left(s, X(s^-)\right)\,\mathrm{d}W(s) + \int_{\mathbb{R}^\ell} \gamma\left(s, X(s^-), z\right)\tilde{N}(\,\mathrm{d}s,\,\mathrm{d}z), \\ &X(t^-) = x. \end{aligned} \tag{2.3.2}$$

To guarantee the well-posedness of the SDE (2.3.2), i.e the SDE has a unique and stable solution, we assume that there exist $C > 0$ such that

$$\int_{\mathbb{R}^\ell} |\gamma(t,x,z)|^2\nu(\,\mathrm{d}z) \le C < \infty, \tag{2.3.3}$$

and that the Lipschitz and linear growth bound-like conditions

$$|\mu(t,x)-\mu(t,y)|^2+|\sigma(t,x)-\sigma(t,y)|^2 \\ +\int_{\mathbb{R}^\ell}|\gamma(t,x,z)-\gamma(t,y,z)|^2\nu(\,\mathrm{d}z)\le C|x-y|^2. \tag{2.3.4}$$

$$|\mu(t+h,x)-\mu(t,x)|^2+|\sigma(t+h,x)-\sigma(t,x)|^2 \\ +\int_{\mathbb{R}^\ell}|\gamma(t+h,x,z)-\gamma(t,x,z)|^2\le C(1+|x|^2)b(h) \tag{2.3.5}$$

hold for $x,y\in\mathbb{R}^d$, $t\in[0,T]$, and a positive function $b(h)\downarrow 0$ for $h\to 0$. Note that conditions (2.3.3)–(2.3.5) need not hold globally for all $x,y\in\mathbb{R}^d$, and may be relaxed to hold locally on a compact set $U\subset\mathbb{R}^d$.

Let $u\in C^{1,2}([0,T]\times\mathbb{R}^d)$ be a function once differentiable in time and twice differentiable in space. The infinitesimal generator of the jump-diffusion process X is given by

$$\mathcal{L}u(t,x)=\langle\mu(t,x),\nabla_x u(t,x)\rangle+\frac{1}{2}\operatorname{Tr}(\sigma(t,x)\sigma^\top(t,x)H_x u(t,x)) \\ +\int_{\mathbb{R}^\ell}\{u(t,x+\gamma(t,x,z))-u(t,x)-\langle\nabla u(t,x),\gamma(t,x,z)\rangle 1_{\{|z|<1\}}\}\nu(\,\mathrm{d}z), \tag{2.3.6}$$

where H_x denotes the Hessian operator in x and is defined as

$$H_x u(t,x)=\left(\partial_{x_i x_j}u(t,x)\right)_{i,j=1}^d\in\mathbb{R}^{d\times d}.$$

The corresponding parabolic operator is $\partial_t+\mathcal{L}$. We note that the integral term in (2.3.6) is nonlocal as it depends on the value of u at the shifted point $x+\gamma(t,x,z)$. The subtraction of the first-order term inside the integral is the compensation term associated with small jumps.

Stochastic processes following common SDEs can be found in Examples 2.2 and 2.3. These will be used in the numerical experiments presented in Chapters 8 and 9.

Example 2.2 (Geometric Brownian motion)**.** The possibly most famous SDE in finance is the geometric Brownian motion (without jumps). Let S denote a one-dimensional stock price process with initial condition $S(0)=s$, and define, by abuse of notation, $\mu(t,s):=\mu s$ and $\sigma(t,s):=\sigma s$, where $\mu\in\mathbb{R}$ and $\sigma>0$ are one-dimensional constants.

Then, the geometric Brownian motion SDE is given by

$$\mathrm{d}S(t)=\mu S(t)+\sigma S(t)\,\mathrm{d}W(t),\qquad S(0)=s,$$

and the unique solution to the SDE is

$$S(t)=s\exp\left\{\left(\mu-\frac{1}{2}\sigma^2\right)\,\mathrm{d}t+\sigma W(t)\right\}.$$

If $u\in C^{1,2}([0,T]\times\mathbb{R})$ is a function of t and s, the infinitesimal generator of S is given by

$$\mathcal{L}u(t,s)=\mu s\partial_s u(t,s)+\frac{1}{2}\sigma^2 s^2\partial_{ss}u(t,s),$$

which is local due to the lack of the jump term.

Example 2.3 (Variance Gamma)**.** A common example of pure-jump processes is the Variance-Gamma process. Recall that the Lévy measure of a Variance-Gamma process is given by (2.2.3). If X denotes a one-dimensional Variance-Gamma process, its solution is given by

$$X(t) = \sigma W(G(t)) + \theta G(t), \qquad X(0^-) = x$$

where $\sigma, \nu > 0$ and $\theta \in \mathbb{R}$ relate to the parameters C, G, and M through

$$\begin{aligned} C &= \frac{1}{\nu} > 0, \\ G &= \frac{1}{\sqrt{\frac{1}{4}\theta^2\nu^2 + \frac{1}{2}\sigma^2\nu} - \frac{1}{2}\theta\nu} > 0, \\ M &= \frac{1}{\sqrt{\frac{1}{4}\theta^2\nu^2 + \frac{1}{2}\sigma^2\nu} + \frac{1}{2}\theta\nu} > 0, \end{aligned}$$

and the one-dimensional process G Gamma distributed with shape $1/\nu$ and scale $1/\nu$. G is often referred to as the subordinator.

2.4 Stopping times

Stopping times formalise random times at which decisions may be taken without looking into the future, and as such are important to impulse control problems. Stopping times are defined as follows:

Definition 2.7 (Stopping time)**.** A random time $\tau : \Omega \to [0, \infty]$ is an $\mathcal{F}_t$-stopping time if
$\{\tau \leq t\} \in \mathcal{F}_t$ for every $t \geq 0$.

For an adapted process X and a stopping time τ, $X(t \wedge \tau)$ is a stopped process. The σ-algebra at the stopping time τ is $\mathcal{F}_\tau = \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \ \forall t \geq 0\}$.

Example 2.4 (First exit time)**.** A recurring example of stopping times used throughout this thesis is the so-called first exit time. Let $G \subset \mathbb{R}^d$ be an open set and X be a càdlàg adapted process with initial condition $X(0^-) = x \in D$. Then, the first exit time from G is given by

$$\tau_G = \inf\{t \geq 0 : X(t) \notin G\}.$$

Under the usual assumption on the filtration, τ_G is a stopping time.

An important result involving the stopping time, infinitesimal generator, and jump-diffusion is the Dynkin formula.

Theorem 2.8 ([84, Theorem 1.24], Dynkin formula)**.** *Let $X(s) \in \mathbb{R}^d$ be a jump diffusion with initial condition $X(t^-) = x$, $G \subset \mathbb{R}^d$ be an open set and let $u \in C^{1,2}([0,T] \times \mathbb{R}^d)$. Let $\tau < \infty$ be a stopping time. Suppose that*

$$\tau \leq \tau_G := \inf\{t \geq 0 : X(t) \notin G\}$$

$$\mathbb{E}^{(t,x)}\left[|u(\tau, X(\tau))| + \int_0^{\tau} |\partial_t u(t, X(t)) + \mathcal{L}u(t, X(t))| \, \mathrm{d}t\right] < \infty. \tag{2.4.1}$$

Then we have

$$\mathbb{E}^{(t,x)}\left[u(\tau, X(\tau))\right] = u(t,x) + \mathbb{E}^{(t,x)}\left[\int_t^{\tau} \partial_s u(s, X(s)) + \mathcal{L}u(s, X(s)) \, \mathrm{d}s\right].$$

Remark 2.9. The notation $\mathbb{E}^{(t,x)}$ denotes expectation conditional on $X(t^-) = x$, that is, $\mathbb{E}^{(t,x)}[\cdot] = \mathbb{E}[\cdot|X(t^-) = x]$.

Throughout this thesis, we assume that the second Dynkin condition (2.4.1) holds.

2.5 Martingales

Finally, we consider martingales and supermartingales, which are used in dynamic programming and verification arguments in the chapters to come.

Definition 2.10 (Martingales)**.** Let $M = \{M(t)\}_{0 \leq t \leq T}$ be a real-valued càdlàg process with $\mathbb{E}[|M(t)|] < \infty$ for all $t \in [0, T]$. Then M is an $\mathcal{F}_t$-martingale if for all $0 \leq s \leq t \leq T$,

$$\mathbb{E}[M(t)|\mathcal{F}_s] = M(s).$$

Definition 2.11. The process M defined in Definition 2.10 is a supermartingale if $\mathbb{E}[|M(t)|] < \infty$ and $\mathbb{E}[M(t)|\mathcal{F}_t] \leq M(s)$ for all $0 \leq s \leq t \leq T$.

CHAPTER

THREE

IMPULSE CONTROL OF JUMP-DIFFUSION PROCESSES

This chapter formulates the finite-horizon impulse control problem for jump-diffusion processes, where the uncontrolled jump-diffusion X introduced in Chapter 2 is turned into a controlled process through impulse controls. The theory presented in this chapter is taken from [84, Chapters 9 and 10]. However, some proofs and intermediate arguments have been expanded in order to provide more clarity to the reader. In Chapter 7.1, we present the impulse control problem and state a verification theorem. Chapter 3.2 then reformulates the impulse control problem as an iterated optimal stopping time problem using the dynamic programming principle.

3.1 A primer on impulse control

Recall the jump-diffusion process X defined in Chapter 2.3, that is,

$$\begin{aligned}
&\mathrm{d}X(s) = \mu(s, X(s^-))\,\mathrm{d}s + \sigma(s, X(s^-))\,\mathrm{d}W(s) + \int_{\mathbb{R}^\ell} \gamma(s, X(s^-), z)\tilde{N}(\mathrm{d}s, \mathrm{d}z),\\
&X(t^-) = x \in \mathbb{R}^d.
\end{aligned} \tag{3.1.1}$$

The generator $\mathcal{L}$ of X is given by (2.3.6), which we rewrite for the reader's convenience:

$$\begin{aligned}
\mathcal{L}u(t,x) = {}& \langle \mu(t,x), \nabla_x u(t,x)\rangle + \frac{1}{2}\operatorname{Tr}(\sigma(t,x)\sigma^\top(t,x)H_x u(t,x))\\
&+ \int_{\mathbb{R}^\ell} \{u(t, x+\gamma(t,x,z)) - u(t,x) - \langle \nabla u(t,x), \gamma(t,x,z)\rangle 1_{\{|z|<1\}}\}\nu(\mathrm{d}z),
\end{aligned} \tag{3.1.2}$$

where $u \in C^{1,2}([0,T) \times \mathcal{S})$.

In a finite-horizon impulse control problem, the process X is controlled with the impulse control α on a finite time horizon $T < \infty$. The controlled process is therefore denoted by $X^{(\alpha)}$. The main goal of the controller is to maximise a given performance criterion, while keeping the controlled process within the *solvency region* $\mathcal{S} \subset \mathbb{R}^d$. The solvency region is assumed to be an open set in order to omit boundary cases. Furthermore, we denote by $\mathcal{Z}$ the set of admissible impulses and assume it is non-empty at any time t and state x. We define notions such as *impulse control*, *post-jump, pre-intervention state*, and *post-intervention map* in what follows.

Definition 3.1 (Impulse Control)**.** We say that an impulse control is a double (possibly finite) sequence

$$\alpha = \{(\tau_i, \zeta_j)\}_{j=1}^{M}, \quad M \leq \infty,$$

where the intervention times $t \leq \tau_1 \leq \tau_2 \leq \cdots$ are $\mathcal{F}_t$-stopping times and $\zeta_1, \zeta_2, \ldots$ are the corresponding impulses at these times. We assume that ζ_j is $\mathcal{F}_{\tau_j}$-measurable for all j and that $\zeta \in \mathcal{Z}(t,x) \subset \mathbb{R}^d$.

Definition 3.2 (Post-Jump, Pre-Intervention State)**.** Let $\alpha = \{(\tau_j, \zeta_j)\}_{j=1}^{M}$, where $M \leq \infty$, be as in Definition 3.1. We denote by $\check{X}^{(\alpha)}$ the state immediately after any exogenous Poisson jump at time τ_j, but before the possible impulse at τ_j, and define it as

$$\check{X}^{(\alpha)}(\tau_j^-) = X^{(\alpha)}(\tau_j^-) + \Delta_N X(\tau_j),$$

Here, $\Delta_N X(\tau_j)$ denotes the jump of X caused by the jump of N only and is defined by

$$\Delta_N X(\tau_j) = \int_{\mathbb{R}^d} \gamma(t, X(\tau_j^-), z) \tilde{N}(\{\tau_j\}, \mathrm{d}z),$$

where

$$\tilde{N}(\{\tau_j\}, \mathrm{d}z) = \tilde{N}(\tau_j, \mathrm{d}z) - \tilde{N}(\tau_j^-, \mathrm{d}z).$$

Remark 3.3. In essence, this definition imposes that the controller may only apply impulses after exogenous Poisson jumps. For example, in the context of dynamic portfolio allocation, we assume that the asset manager is only allowed to rebalance her portfolio by means of impulses only after an exogenous market jump occurs.

Definition 3.4 (Post-intervention Function)**.** Let $(t,x) \in [0,T] \times \mathcal{S}$ be fixed, $\mathcal{S} \subseteq \mathbb{R}^d$ be the state space and $\mathcal{Z}$ the set of admissible impulses. The post-intervention function

$$\Gamma : [0,T) \times \mathcal{S} \times \mathcal{Z}(t,x) \to \mathcal{S}$$

applies the impulse ζ to the state x. Hence, if an impulse $\zeta_j \in \mathcal{Z}$ is chosen at the intervention time τ_j, the controlled state process $X^{(\alpha)}$ restarts at

$$X^{(\alpha)}(\tau_j) = \Gamma(\tau_j, \check{X}^{(\alpha)}(\tau_j), \zeta_j).$$

Remark 3.5. We note that in [84], the function Γ is assumed to be a Borel-measurable function. In [92], however, a stronger assumption[1] of continuity is needed to prove the existence and uniqueness of a viscosity solution. We therefore adopt the stronger continuity assumption on Γ.

[1] Note that any continuous function is Borel-measurable, but the converse implication does not always hold.

With Definitions 3.1, 3.2, and 3.4 in mind, suppose now that at any time t and any state x, we are free to intervene and give the system an impulse $\zeta \in \mathcal{Z} \subset \mathbb{R}^d$. Here $\mathcal{Z}$ denotes the set of admissible impulse values, and is assumed to be non-empty and given at any time t and any state x. Let α denote an impulse control. Furthermore, suppose that sending an impulse of size ζ to the state $X(\tau^-)$ immediately moves the state from $X(\tau^-)$ to $\Gamma(\tau, \check{X}^{(\alpha)}(\tau^-), \zeta) \in \mathcal{S}$. Then, for some initial time $0 \le t \le T$, we define the corresponding controlled process $X^{(\alpha)}$ by

$$
\begin{aligned}
&X^{(\alpha)}(t^-) = x \quad \text{and} \quad X^{(\alpha)}(s) = X(s), \quad \text{for } t \le s \le \tau_1, && (3.1.3)\\
&X^{(\alpha)}(\tau_j) = \Gamma(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j), \quad j = 1, 2, \dots && (3.1.4)\\
&(\text{If } \tau_1 = t, \text{we put } X^{(\alpha)}(\tau_1) = \Gamma(t, X^{(\alpha)}(t^-), \zeta_1) = \Gamma(t, x, \zeta_1).)\\
&\mathrm{d}X^{(\alpha)}(s) = \mu(s, X^{(\alpha)}(s^-))\,\mathrm{d}s + \sigma(s, X^{(\alpha)}(s^-))\,\mathrm{d}W(s)\\
&\qquad\qquad + \int_{\mathbb{R}^l} \gamma(s, X^{(\alpha)}(s^-), z)\tilde{N}(\,\mathrm{d}s,\,\mathrm{d}z) \quad \text{for } \tau_j < s < \tau_{j+1}. && (3.1.5)
\end{aligned}
$$

We note that if $\tau_{j+1} = \tau_j$, that is when two intervention times coincide, the controlled process jumps from $\check{X}^{(\alpha)}(\tau_j^-)$ to $\Gamma\big(\tau_j, \Gamma(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j), \zeta_{j+1}\big)$. However, throughout this thesis it is assumed that executing two rebalancing steps at the same time does not add any economic value relative to a single equivalent intervention. Hence, assume from now on that $\tau_j < \tau_{j+1} < \cdots$ for $j = 1, 2, \dots$. Moreover, the drift, diffusion, and jump-amplitude functions in (3.1.5) satisfy the Lipschitz and linear-growth bound conditions (2.3.4) and (2.3.5) respectively.

Definition 3.6 (Time of Insolvency)**.** We define the time of insolvency as the smallest stopping time for which the controlled process $X^{(\alpha)}$ becomes insolvent, that is,

$$\tau_{\mathcal{S}} = \inf\{s \in (t, \infty) : X^{(\alpha)}(s) \notin \mathcal{S}\}.$$

We now proceed to specify the performance criterion and the resulting impulse control problem. Suppose that we are given a continuous *profit function* $f : [0, T] \times \mathcal{S} \to \mathbb{R}$ and a continuous *bequest* function $g : [0, T] \times \mathbb{R}^d \to \mathbb{R}$. Let $K : [0, T] \times \mathcal{S} \times \mathcal{Z} \to \mathbb{R}$ be the cost-of-intervention function (or simply cost function) and assume K is a continuous, non-positive function. Furthermore, denote by $\mathcal{V}$ the set of admissible controls α. In the case if α is an infinite sequence, we assume that

$$\lim_{j \to \infty} \tau_j = \tau_{\mathcal{S}}, \tag{3.1.6}$$

where $\tau_{\mathcal{S}}$ is the time of insolvency in Definition 3.6. Finally, assume the following holds

$$\mathbb{E}^{(t,x)}\left[\int_t^{\tau_{\mathcal{S}} \wedge T} \big|f(s, X^{(\alpha)}(s))\big|\,\mathrm{d}s\right] < \infty \qquad \text{for all } t \in [0, T],\ x \in \mathbb{R}^d,\ \alpha \in \mathcal{V}, \tag{3.1.7}$$

$$\mathbb{E}^{(t,x)}\left[g^-\big(\tau_{\mathcal{S}} \wedge T, X^{(\alpha)}(\tau_{\mathcal{S}} \wedge T)\big)\right] < \infty \qquad \text{for all } x \in \mathbb{R}^d,\ \alpha \in \mathcal{V}, \tag{3.1.8}$$

$$\mathbb{E}^{(t,x)}\left[\sum_{t \le \tau_j < \tau_{\mathcal{S}} \wedge T} K^-\big(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j\big)\right] < \infty, \qquad \text{for all } x \in \mathbb{R}^d,\ \alpha \in \mathcal{V}, \tag{3.1.9}$$

where $g^-(t,x) := \max\{-g(t,x),0\}$ and $K^-(t,x,\zeta) := \max\{-K(t,x,\zeta),0\}$. Then, with $\tau' := \tau_{\mathcal{S}} \wedge T$, the performance criterion is read as follows:

$$J^{(\alpha)}(t,x) = \mathbb{E}^{(t,x)}\left[\int_t^{\tau'} f\big(s, X^{(\alpha)}(s)\big)\,\mathrm{d}s + g\big(\tau', X^{(\alpha)}(\tau')\big) + \sum_{t\le\tau_j<\tau'} K\big(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j\big)\right]. \tag{3.1.10}$$

The impulse control problem over a finite time horizon reads as:

Find $v(t,x)$ and $\alpha^* \in \mathcal{V}$ such that

$$v(t,x) = \sup_{a\in\mathcal{V}} J^{(\alpha)}(t,x) = J^{(\alpha^*)}(t,x). \tag{3.1.11}$$

We refer to v as the *value function.*

With the definition of the impulse control problem, we first introduce the intervention operator in Definition 3.7 and then turn to the verification theorem (Theorem 3.9), the latter of which lets us verify if a potential solution to the impulse control problem indeed solves 3.1.11.

Definition 3.7 ([84, Definition 9.1] Intervention Operator)**.** Let $\mathcal{H}$ be the space of all measurable function $h : \mathcal{S} \to \mathbb{R}$. The intervention operator, $\mathcal{M}^{(t,x)} : \mathcal{H} \to \mathcal{H}$ is defined by

$$\mathcal{M}^{(t,x)}h(t,x) = \sup_{\zeta\in\mathcal{Z}(t,x)} \{h(\Gamma(t,x,\zeta)) + K(t,x,\zeta)\}.$$

In the case when $\mathcal{Z}(t,x) = \emptyset$, then we set $\mathcal{M}h(t,x) = -\infty$[2].

Remark 3.8. For brevity, we denote the intervention operator in Definition 3.7 by $\mathcal{M} = \mathcal{M}^{(t,x)}$.

Theorem 3.9 ([84, Theorem 9.2] Quasi-Variational Inequalities for Impulse Control)**.** *Let $T > 0$ be a fixed, finite time horizon, recall $\tau_{\mathcal{S}}$ from Definition 3.6, and let $v(t,x)$ denote the value function for the impulse problem (3.1.11) on $[0,T]\times\mathcal{S}$. Furthermore, let the infinitesimal generator for the random process X be given by (3.1.2), recall the intervention operator from Definition 3.7, and define*

$$\mathcal{T}_t = \{\tau : \tau \textit{ stopping time}, t \le \tau \le \tau_{\mathcal{S}} \wedge T \textit{ a.s.}\}.$$

(a) Suppose we can find $\varphi : \bar{\mathcal{S}} \to \mathbb{R}$ such that

(i) $\varphi \in C^1([0,T)\times\mathcal{S}) \cap C(\{T\}\times\bar{\mathcal{S}})$.

(ii) $\varphi \ge \mathcal{M}\varphi$ on $[0,T)\times\mathcal{S}$.

Define the continuation region, i.e. the region inside which no impulses occur, as

$$D = \{(t,x) \in [0,T)\times\mathcal{S} : \varphi(t,x) > \mathcal{M}\varphi(t,x)\},$$

and assume the following:

[2] This case will however be excluded in later chapters.

(iii) *For all* $x \in \mathcal{S}$, $t \in [0,T)$, $\alpha \in \mathcal{V}$,

$$\mathbb{E}^{(t,x)}\left[\int_t^{\tau_{\mathcal{S}}\wedge T} 1_{\{\partial D\}}\big(s, X^{(\alpha)}(s)\big)\,\mathrm{d}s\right],$$

(iv) ∂D *is a Lipschitz surface,*

(v) $\varphi \in C^{1,2}\big([0,T)\times(\mathcal{S}\backslash\partial D)\big)$,

(vi) $\partial_t\varphi + \mathcal{L}\varphi + f \leq 0$ *on* $[0,T)\times(\mathcal{S}\backslash\partial D)$,

(vii) $\varphi = g$ *on the parabolic boundary, that is,*

$$\varphi(t,x) = g(t,x), \quad \textit{for } t\in[0,T] \textit{ and } x \notin \mathcal{S},$$

(viii) *The family* $\big\{\varphi^-\big(\tau, X^{(\alpha)}(\tau)\big) : \tau \in \mathcal{T}_t\big\}$ *is uniformly integrable, for all* $x \in \mathcal{S}$, $\alpha \in \mathcal{V}$,

(ix) *And lastly,*

$$\mathbb{E}^{(t,x)}\left[\int_t^{\tau_{\mathcal{S}}\wedge T}\Big|\partial_s\varphi(s,X^{(\alpha)}(s)) + \mathcal{L}\varphi(s,X^{(\alpha)}(s))\Big|\,\mathrm{d}s + \Big|\varphi(\tau,X^{(\alpha)}(\tau))\Big|\right] < \infty,$$

for all $\tau \in \mathcal{T}_t$, $\alpha \in \mathcal{V}$, $x \in \mathcal{S}$.

Then,

$$\varphi(t,x) \geq v(t,x) \quad \textit{for all } (t,x)\in[0,T]\times\mathcal{S}. \tag{3.1.12}$$

(b) *Suppose in addition that:*

(x) $\partial_t\varphi + \mathcal{L}\varphi + f = 0$ *inside the continuation region* D,

(xi) *A maximizer* $\hat{\zeta}(t,x)$ *exists, in the sense that*

$$\hat{\zeta}(t,x) = \arg\max\big\{\varphi\big(t,\Gamma(t,x,\cdot)\big) + K(t,x,\cdot)\big\} \in \mathcal{Z}(t,x)$$

exists for all $(t,x)\in[0,T)\times\mathcal{S}$ *and* $\hat{\zeta}(\cdot,\cdot)$ *is a Borel measurable selection.*

Put $\hat{\tau}_0 = s$ *and define* $\hat{\alpha} = (\hat{\tau}_1,\hat{\tau}_2,\cdots;\hat{\zeta}_1,\hat{\zeta}_2,\ldots)$ *inductively by* $\hat{\tau}_{j+1} = \inf\{t > \hat{\tau}_j : \big(t, X^{(\hat{a})} \notin D\big)\}\wedge\tau$ *and* $\hat{\zeta}_{j+1} = \hat{\zeta}(\hat{\tau}_{j+1}, X^{(\hat{a}_j)}(\hat{\tau}_{j+1}^-))$ *if* $\hat{\tau}_{j+1} < \tau$, *where* $X^{\hat{a}_j}$ *is the result of applying* $\hat{a}_j := (\hat{\tau}_1,\ldots,\hat{\tau}_j;\hat{\zeta}_1,\ldots,\hat{\zeta}_j)$ *to* X.

Finally, suppose

(xii) $\hat{a}\in\mathcal{V}$ *and* $\{\varphi(\tau, X^{(\hat{a})}(\tau)) : \tau\in\mathcal{T}_t\}$ *is uniformly integrable.*

Then,

$$\varphi(t,x) = v(t,x) \quad \textit{and } \hat{a} \textit{ is an optimal impulse control.} \tag{3.1.13}$$

Proof. See [84, pp.242-243]. □

3.2 The Dynamic Programming Principle

This section is based on the work of Øksendal and Sulem [84, Chapter 10.1]. As the authors argue, in general, an impulse control problem cannot be reformulated as a single optimal stopping problem, because the first intervention action, i.e. the initial intervention time τ_1 and the corresponding impulse ζ_1, affects the subsequent state evolution and therefore changes the structure of all future decisions.

Nevertheless, if one focuses on controls with at most $n < \infty$ of interventions, the resulting truncated impulse control can be solved by backward induction through an iterative procedure that reduces the problem into n optimal stopping problems. At each intervention time, the corresponding impulse is chosen to be the maximizer of the intervention operator found in Definition 3.7. As discussed in this section, the value function of the restricted problem converges to the value function of the unrestricted problem (3.1.11), as $n \to \infty$. Hence, an impulse problem may be approximated by a sequence of iterated optimal stopping problems.

We therefore beign by recalling the unrestricted impulse control problem (3.1.11), that is,

$$v(t,x) = \sup_{a \in \mathcal{V}} J^{(\alpha)}(t,x) = J^{(\alpha^*)}(t,x),$$

where the performance criterion is defined in Equation (3.1.10), the time horizon $T > 0$ is finite, and the time of insolvency is as in Definition 3.6. The set $\mathcal{V}$ contains all admissible (untruncated) impulse controls α.

For $n = 1, 2, \ldots$, let $\mathcal{V}_n$ denote the set of all $\alpha \in \mathcal{V}$ such that $\alpha_n = \{(\tau_j, \zeta_j)\}_{j=1}^{k}$ with $k \leq n$. In other words, $\mathcal{V}_n$ is the set of all admissible controls *with at most n interventions*. By construction,

$$\mathcal{V}_n \subseteq \mathcal{V}_{n+1} \subseteq \mathcal{V},$$

for all $n \in \mathbb{N}$. This is because truncation removes potential optimal impulses that may have been taken after the n-th intervention and reduces the cardinality of the truncated set. Denote the value function of at most n interventions by v_n and define it as

$$v_n(t,x) = \sup_{\alpha_n \in \mathcal{V}_n} J^{(\alpha)}(t,x), \quad n = 1, 2, \ldots . \tag{3.2.1}$$

Finally, to distinguish the time of insolvency related to the non-truncated control α from the one related to the truncated control α_n, we write denote these stopping times by $\tau_{\mathcal{S}}^{(\alpha)}$ and $\tau_{\mathcal{S}}^{(\alpha_n)}$ respectively, and define $\tau^{(\alpha_n)} := \tau_{\mathcal{S}}^{(\alpha_n)} \wedge T$ and $\tau^{(\alpha)} := \tau_{\mathcal{S}}^{(\alpha)} \wedge T$. The lemma below follows.

Lemma 3.10 ([84, Lemma 10.1]). *For $g \geq 0$, we have*

$$\lim_{n \to \infty} v_n(t,x) = v(t,x),$$

for all $(t,x) \in [0,T] \times \mathcal{S}$.

Proof. The proof can be found in [84], see Lemma 10.1 wherein. We split the proof into two cases:

(a) Show that

$$\lim_{n\to\infty} v_n(t,x) \le v(t,x),$$

for all $(t,x) \in [0,T] \times \mathcal{S}$,

(b) and that

$$\lim_{n\to\infty} v_n(t,x) \ge v(t,x).$$

(a) Let $\mathcal{V}_n$ be the set of all admissible controls with at most n interventions, and let $\mathcal{V}$ denote the set of all admissible controls. Then, as seen earlier,

$$\mathcal{V}_n \subseteq \mathcal{V}_{n+1} \subseteq \mathcal{V},$$

for any $n \in \mathbb{N}$. Now consider the the non-truncated and truncated value functions v and v_n defined in (3.2.1) and (3.1.11) respectively. Since the optimal control α^* is more likely to be in $\mathcal{V}$ than in $\mathcal{V}_n$, we must have

$$v_n(t,x) \le v_{n+1}(t,x) \le v(t,x),$$

for all $(t,x) \in [0,T] \times \mathcal{S}$. Hence, for all $(t,x) \in [0,T] \times \mathcal{S}$,

$$\lim_{n\to\infty} v_n(t,x) \le v(t,x).$$

(b) To get the opposite inequality, let us first assume $v(t,x) < \infty$. Then for each $\varepsilon > 0$, there exists, for some $M \le \infty$ a non-truncated control $\alpha = \{(\tau_j, \zeta_j)\}_{j=1}^{M} \in \mathcal{V}$ such that

$$\begin{aligned} J^{(\alpha)}(t,x) &= \mathbb{E}^{(t,x)}\left[\int_t^{\tau^{(\alpha)}} f(s, X^{(\alpha)}(s))\,\mathrm{d}s + g(\tau^{(\alpha)}, X^{(\alpha)}(\tau^{(\alpha)}))\right] \\ &\quad + \mathbb{E}^{(t,x)}\left[\sum_{t\le\tau_j<\tau^{(\alpha)}} K\big(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j\big)\right] \\ &\ge v(t,x) - \varepsilon, \end{aligned} \tag{3.2.2}$$

where J is the performance functional, see (3.1.10).

For $n = \mathbb{N}$, define the finite double sequence $\alpha_n = (\tau_1, \dots, \tau_n, \tau_{\mathcal{S}}^{(\alpha)}; \zeta_1, \dots, \zeta_n)$. Then,

$$X^{(\alpha_n)}(s) = X^{(\alpha)}(s), \quad \text{for all } s \le \tau_n. \tag{3.2.3}$$

Since by assumption $\tau_j \to \tau_{\mathcal{S}}$ a.s. when $j \to \infty$ (see Equation (3.1.6)), we get (3.1.7) and (3.1.9) that there exists $n \in \mathbb{N}$ such that

$$\begin{aligned} 0 &\le \mathbb{E}^{(t,x)}\left[\int_{\tau_n}^{\tau_{\mathcal{S}}^{(\alpha)}} |f(s, X^{(\alpha)}(s)|\,\mathrm{d}s + \int_{\tau_n}^{\tau_{\mathcal{S}}^{(\alpha_n)}} |f(s, X^{(\alpha_n)}(s)|\,\mathrm{d}s\right] \\ &\le \mathbb{E}^{(t,x)}\left[\int_{t}^{\tau_{\mathcal{S}}^{(\alpha)}} |f(s, X^{(\alpha)}(s)|\,\mathrm{d}s + \int_{t}^{\tau_{\mathcal{S}}^{(\alpha_n)}} |f(s, X^{(\alpha_n)}(s)|\,\mathrm{d}s\right] < \infty, \end{aligned}$$

and that

$$0 \le \mathbb{E}^{(t,x)}\Bigg[\sum_{\substack{j>n\\ t\le\tau_j<\tau^{(\alpha)}}} K^-\big(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j\big)\Bigg] \le \mathbb{E}^{(t,x)}\Bigg[\sum_{t\le\tau_j<\tau^{(\alpha)}} K^-\big(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j\big)\Bigg] < \infty,$$

where $K^-(t,x,\zeta) := \max\{-K(t,x,\zeta)\}$. Consequently, there exists an $\varepsilon > 0$ such that

$$\mathbb{E}\left[\int_{\tau_n}^{\tau_S^{(\alpha)}} |f(s, X^{(\alpha)}(s)|\,\mathrm{d}s + \int_{\tau_n}^{\tau_S^{(\alpha_n)}} |f(s, X^{(\alpha_n)}(s)|\,\mathrm{d}s\right] < \varepsilon \tag{3.2.4}$$

and

$$\mathbb{E}\Bigg[\sum_{\substack{j>n\\ t\le\tau_j<\tau^{(\alpha)}}} K^-\big(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j\big)\Bigg] < \varepsilon. \tag{3.2.5}$$

Combining (3.2.3)–(3.2.5), we obtain

$$\begin{aligned}
&\liminf_{n\to\infty} J^{(\alpha_n)}(t,x)\\
&\quad= \liminf_{n\to\infty} \mathbb{E}^{(t,x)}\left[\int_t^{\tau^{(\alpha_n)}} f(s, X^{(\alpha_n)}(s))\,\mathrm{d}s + g(\tau^{(\alpha_n)}, X^{(\alpha_n)}(\tau^{(\alpha_n)})) + \sum_{j=1}^{n} K(\tau_j, \check{X}^{(\alpha_n)}(\tau_j^-), \zeta_j)\right]\\
&\quad\overset{(3.2.3)}{=} \liminf_{n\to\infty}\Bigg\{\underbrace{\mathbb{E}^{(t,x)}\left[\int_t^{\tau^{(\alpha)}} f(s, X^{(\alpha)}(s))\,\mathrm{d}s + g(\tau^{(\alpha)}, X^{(\alpha)}(\tau^{(\alpha)})) + \sum_{t\le\tau_j<\tau^{(\alpha)}} K(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j)\right]}_{:=J^{(\alpha)}(t,x)}\\
&\qquad+ \mathbb{E}^{(t,x)}\left[\int_{\tau_n}^{\tau^{(\alpha_n)}} f(s, X^{(\alpha_n)}(s))\,\mathrm{d}s - \int_{\tau_n}^{\tau^{(\alpha)}} f(s, X^{(\alpha)}(s))\,\mathrm{d}s\right]\\
&\qquad+ \mathbb{E}^{(t,x)}\Bigg[g(\tau^{(\alpha_n)}, X^{(\alpha_n)}(\tau^{(\alpha_n)})) - g(\tau^{(\alpha)}, X^{(\alpha)}(\tau^{(\alpha)})) - \sum_{\substack{j>n\\ t\le\tau_j<\tau^{(\alpha)}}} K\big(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j\big)\Bigg]\Bigg\}\\
&\qquad+ \liminf_{n\to\infty} \mathbb{E}^{(t,x)}\Bigg[g(\tau^{(\alpha_n)}, X^{(\alpha_n)}(\tau^{(\alpha_n)})) - g(\tau^{(\alpha)}, X^{(\alpha)}(\tau^{(\alpha)})) - \sum_{\substack{j>n\\ t\le\tau_j<\tau^{(\alpha)}}} K\big(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j\big)\Bigg]\\
&\quad\ge J^{(\alpha)}(t,x) + \liminf_{n\to\infty} \mathbb{E}^{(t,x)}\left[-\int_{\tau_n}^{\tau^{(\alpha_n)}} |f(s, X^{(\alpha_n)}(s))|\,\mathrm{d}s - \int_{\tau_n}^{\tau^{(\alpha)}} |f(s, X^{(\alpha)}(s))|\,\mathrm{d}s\right]\\
&\qquad+ \liminf_{n\to\infty} \mathbb{E}^{(t,x)}\left[g(\tau^{(\alpha_n)}), X^{(\alpha_n)}(\tau^{(\alpha_n)})) - g(\tau^{(\alpha)}), X^{(\alpha)}(\tau^{(\alpha)}))\right]\\
&\qquad- \liminf_{n\to\infty} \mathbb{E}^{(t,x)}\Bigg[\sum_{\substack{j>n\\ t\le\tau_j<\tau^{(\alpha)}}} K^-\big(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j\big)\Bigg]\\
&\quad\overset{(3.2.4),(3.2.5)}{\ge} J^{(\alpha)}(t,x) - 2\varepsilon + \liminf_{n\to\infty} \mathbb{E}^{(t,x)}\left[g(\tau^{(\alpha_n)}, X^{(\alpha_n)}(\tau^{(\alpha_n)})) - g(\tau^{(\alpha)}, X^{(\alpha)}(\tau^{(\alpha)}))\right]\\
&\quad= J^{(\alpha)}(t,x) - 2\varepsilon.
\end{aligned} \tag{3.2.6}$$

Hence, with (3.2.2) and (3.2.6), the reverse inequality is obtained

$$\liminf_{n\to\infty} v_n(t,x) \geq \liminf_{n\to\infty} J^{(\alpha_n)}(t,x) \geq J^{(\alpha)}(t,x) - 2\varepsilon \geq v(t,x) - 3\varepsilon,$$

and with $\varepsilon > 0$ arbitrary, it follows that

$$\lim_{n\to\infty} v_n(t,x) \geq v(t,x),$$

for all $(t,x) \in [0,T] \times \mathcal{S}$ and whenever $v(t,x) < \infty$. For the case when $v(t,x) = \infty$, we refer to [84, p.257]

Combining cases (a) and (b) then gives us the desired equality. □

Now that we have established that v_n converges pointwise to v as $n \to \infty$, let $X(t) = X^{(0)}(t)$ denote the uncontrolled process (3.1.1). Consider the following iterative approach

$$\varphi_0(t,x) = \mathbb{E}^{(t,x)}\left[\int_t^{\tau_{\mathcal{S}}\wedge T} f\big(s, X(s)\big)\,\mathrm{d}s + g\big(\tau, X(\tau)\big)\right], \tag{3.2.7}$$

and inductively, for $j = 1, 2, \ldots, n$,

$$\varphi_j(t,x) = \sup_{\tau\in\mathcal{T}_t} \mathbb{E}^{(t,x)}\left[\int_t^{\tau} f\big(s, X(s)\big)\,\mathrm{d}s + \mathcal{M}\varphi_{j-1}\big(\tau, X(\tau)\big)\right], \tag{3.2.8}$$

where $\mathcal{T}_t$ denotes the set of stopping times τ satisfying $\tau \leq \tau_{\mathcal{S}} \wedge T$ and j denotes the number of interventions left. Equations (3.2.7) and (3.2.8) are referred to as the iterated optimal stopping representation. We now show that the truncated impulse control problem may indeed be represented by an iterative optimal stopping representation.

Definition 3.11 ([92, Definition 2.1] Space of Polynomially Bounded Functions)**.** We denote the space of polynomially bounded functions by $\mathcal{PB} = \mathcal{PB}([0,T) \times \mathbb{R}^d)$ and it is the space of all measurable functions $h : [0,T) \times \mathbb{R}^d \to \mathbb{R}$ such that

$$|h(t,x)| \leq C_h(1 + |x|^m)$$

for a time-independent constant $C_h > 0$ and $m \in \mathbb{N}$. If m is fixed, we write $\mathcal{PB}_m$.

Theorem 3.12 (Iterated optimal stopping representation, see [84, Theorem 10.2])**.** *Suppose*

$$f, g \text{ and } \mathcal{M}\varphi_{j-1} \in \mathcal{PB}([0,T) \times \mathbb{R}^d), \quad \text{for } j = 1, 2, \ldots, n. \tag{3.2.9}$$

Then,

$$\varphi_n = v_n.$$

Theorem 3.12 provides a way to formulate the truncated impulse control problem into a sequence of standard optimal stopping problems. In particular, when the controller is restricted to at most n interventions, the value function can be computed via backward recursion. After each intervention, the controller optimally selects when to intervene next and takes the solution from the previous step as the continuation value. This representation lets us replace a single, globally coupled impulse optimisation with an iterated sequence of simpler timing problems. To prove Theorem 3.12, a few more auxiliary results are required. We therefore begin by presenting Lemma 3.13 and Corollary 3.14, in which X denotes an uncontrolled stochastic process.

Lemma 3.13 (Dynamic Programming Principle (DPP), see [84, Lemma 10.3])**.** *Suppose $G \in \mathcal{PB}([0,T) \times \mathbb{R}^d)$. Define*

$$\psi(t,x) = \sup_{\tau \in \mathcal{T}_t} \mathbb{E}^{(t,x)} \left[\int_t^\tau f(s, X(s)) \, \mathrm{d}s + G(\tau, X(\tau)) \right]. \tag{3.2.10}$$

(a) Then, for all finite stopping times β, we have

$$\psi(t,x) = \sup_{\tau \in \mathcal{T}_t} \mathbb{E}^{(t,x)} \left[\int_t^{\tau \wedge \beta} f(s, X(s)) \, \mathrm{d}s + G(\tau, X(\tau)) 1_{\{\tau \le \beta\}} + \psi(\beta, X(\beta)) 1_{\{\tau > \beta\}} \right]. \tag{3.2.11}$$

(b) For $\varepsilon > 0$, define the ε-optimal continuation region as

$$D^{(\varepsilon)} = \{(t,x) \in [0,T) \times \mathcal{S} : \psi(t,x) > G(t,x) + \varepsilon\},$$

and define the ε-optimal stopping time

$$\tau^{(\varepsilon)} = \inf \left\{ s \ge t : (s, X(s)) \notin D^{(\varepsilon)} \right\}.$$

Then, if β is a finite stopping time such that $\beta \le \tau^{(\varepsilon)}$ for some $\varepsilon > 0$, we have

$$\psi(t,x) = \mathbb{E}^{(t,x)} \left[\int_t^\beta f(s, X(s)) \, \mathrm{d}s + \psi(\beta, X(\beta)) \right] \tag{3.2.12}$$

Proof. As the proof of Lemma 3.13 relies on theoretical results that lie beyond the scope of this thesis, we refer the reader to [56, Section 4] for a complete proof. □

Corollary 3.14 ([84, Corollary 10.5])**.** *(a) For each given $\tau \in \mathcal{T}_t$, the process*

$$U(s) := u + \int_t^{s \wedge \tau} f(r, X(r)) \, \mathrm{d}r + \psi(s \wedge \tau, X(s \wedge \tau)), \quad s \ge t$$

is a supermartingale. In particular, if τ_1 and τ_2 are finite stopping times such that $t \le \tau_1 \le \tau_2 \le \tau_{\mathcal{S}} \wedge T$, then

$$\mathbb{E}^{(t,x)} \left[\psi(\tau_1, X(\tau_1)) \right] \ge \mathbb{E}^{(t,x)} \left[\int_{\tau_1}^{\tau_2} f(s, X(s)) \, \mathrm{d}s + \psi(\tau_2, X(\tau_2)) \right].$$

(b) For $\varepsilon > 0$, let $\tau^{(\varepsilon)}$ be as in Lemma 3.13a and let β_1, β_2 be finite stopping times satisfying $t \le \beta_1 \le \beta_2 \le \tau^{(\varepsilon)}$. Then,

$$\mathbb{E}^{(t,x)} \left[\psi(\beta_1, X(\beta_1)) \right] = \mathbb{E}^{(t,x)} \left[\int_{\beta_1}^{\beta_2} f(s, X(s)) \, \mathrm{d}s + \psi(\beta_2, X(\beta_2)) \right].$$

Proof. We follow the same approach as in [84, see Proof of Corollary 10.5] and address points (a) and (b) separately.

(a) Define the random variable

$$R(s) = \left(s \wedge \tau, X(s \wedge \tau), \int_t^{s \wedge \tau} f(r, X(r)) \, \mathrm{d}r \right) \in \mathbb{R}^{d+2}$$

and put

$$H(t,x,u) = \psi(t,x) + u,$$

for $(t,x,u) \in [0,T] \times \mathbb{R}^d \times \mathbb{R}$, where u is deterministic. Lastly, recall the definition of the random variable $U(s)$. Then, the Markov property yields

$$\mathbb{E}\left[U(s) \mid \mathcal{F}_t\right] = \mathbb{E}\left[H(R(s)) \mid \mathcal{F}_t\right] = \mathbb{E}^{(s,R(s))}\left[H(R(s-t))\right].$$

Then, by (3.2.11), applied to $\beta := (s-t) \wedge \tau$ and $(t,x,u) = R(t)$, we obtain

$$\begin{aligned}\mathbb{E}^{(s,R(s))}[H(R(s-t))] &= \mathbb{E}^{(t,x)}\left[\psi(\beta, X(\beta)) + u + \int_t^{\beta} f(r,X(r))\,\mathrm{d}r\right] \\ &= u + \sup_{\tau \geq \beta} \mathbb{E}^{(t,x)}\left[\psi(X(\beta\wedge\tau)) + \int_t^{\beta\wedge\tau} f(r,X(r))\,\mathrm{d}r\right].\end{aligned} \tag{3.2.13}$$

Now recall (3.2.11) in Lemma 3.13a. Restricting the supremum in (3.2.11) to the event $\{\tau \geq \beta\}$, we get

$$\psi(t,x) \geq \sup_{\tau\geq\beta} \mathbb{E}^{(t,x)}\left[\psi(\beta\wedge\tau, X(\beta\wedge\tau)) + \int_t^{\tau\wedge\beta} f(r,X(r))\,\mathrm{d}r\right]. \tag{3.2.14}$$

Combining (3.2.13) and (3.2.14), we obtain

$$\begin{aligned}\mathbb{E}^{(s,R(s))}[H(R(s-t))] &= u + \sup_{\tau\geq\beta} \mathbb{E}^{(t,x)}\left[\psi(X(\beta\wedge\tau)) + \int_t^{\beta\wedge\tau} f(r,X(r))\,\mathrm{d}r\right] \\ &\leq u + \psi(t,x) = U(t).\end{aligned}$$

Thus, for all $s \geq t$,

$$\mathbb{E}[U(s) \mid \mathcal{F}_t] = \mathbb{E}^{(s,R(s))}[H(s-t, R(s-t))] \leq U(t),$$

which shows that $U(s)$ is a supermartingale.

For the proof of the second statement in (a), we may apply Doob's sampling theorem (see Appendix A.1), because U is a supermartingale and the stopping times τ_1 and τ_2 are finite by assumption. Therefore, for some stopping time $\tau \geq \tau_2$,

$$\begin{aligned}&\mathbb{E}^{(t,x)}\left[\int_{\tau_1}^{\tau_2} f(r,X(r))\,\mathrm{d}r + \psi(\tau_2, X(\tau_2))\right] \\ &= \mathbb{E}^{(t,x)}\left[\int_t^{\tau_2\wedge\tau} f(r,X(r))\,\mathrm{d}r + \psi(\tau_2\wedge\tau, X(\tau_2\wedge\tau))\right] - \mathbb{E}^{(t,x)}\left[\int_t^{\tau_1} f(r,X(r))\,\mathrm{d}r\right] \\ &= \mathbb{E}^{(t,x)}\left[\mathbb{E}\left[\int_t^{\tau_2\wedge\tau} f(r,X(r))\,\mathrm{d}r + \psi(\tau_2\wedge\tau, X(\tau_2\wedge\tau)) \,\Big|\, \mathcal{F}_{\tau_1}\right]\right] \\ &\qquad - \mathbb{E}^{(t,x)}\left[\int_t^{\tau_1} f(r,X(r))\,\mathrm{d}r\right] \\ &= \mathbb{E}^{(t,x)}\left[\mathbb{E}[U(\tau_2) \mid \mathcal{F}_{\tau_1}]\right] - \mathbb{E}^{(t,x)}\left[\int_t^{\tau_1} f(r,X(r))\,\mathrm{d}r\right] \\ &\leq \mathbb{E}^{(t,x)}[U(\tau_1)] - \mathbb{E}^{(t,x)}\left[\int_t^{\tau_1} f(r,X(r))\,\mathrm{d}r\right]\end{aligned}$$

$$= \mathbb{E}^{(t,x)}[\psi(\tau_1, X(\tau_1))]$$

(b) Let u be an arbitrary time variable. We have

$$\begin{aligned}
&\mathbb{E}^{(t,x)}\left[\int_{\beta_1}^{\beta_2} f(s,X(s))\,\mathrm{d}s + \psi(\beta_2, X(\beta_2))\right] \\
&\quad= \mathbb{E}^{(t,x)}\left[\int_{t}^{\beta_2} f(s,X(s))\,\mathrm{d}s + \psi(\beta_2, X(\beta_2))\right] - \mathbb{E}^{(t,x)}\left[\int_{t}^{\beta_1} f(s,X(s))\,\mathrm{d}s\right] \\
&\quad= \psi(t,x) - \mathbb{E}^{(t,x)}\left[\int_{t}^{\beta_1} f(s,X(s))\,\mathrm{d}s\right]. \qquad (3.2.15)
\end{aligned}$$

Since $\beta_1 \leq \tau^{(\varepsilon)}$ is a finite, ε-optimal stopping time, we have, by Lemma 3.13b,

$$\psi(t,x) = \mathbb{E}^{(t,x)}\left[\int_{t}^{\beta_1} f(s,X(s))\,\mathrm{d}s + \psi(\beta_1, X(\beta_1))\right].$$

Therefore, (3.2.15) becomes

$$\mathbb{E}^{(t,x)}\left[\int_{\beta_1}^{\beta_2} f(s,X(s))\,\mathrm{d}s + \psi(\beta_2, X(\beta_2))\right] = \mathbb{E}^{(t,x)}\left[\psi(\beta_1, X(\beta_1))\right],$$

which concludes the proof. $\square$

With the help of Lemma 3.13 and Corollary 3.14, we may now prove Theorem 3.12.

Proof of Theorem 3.12. Let the finite time horizon $T > 0$ be fixed and choose an impulse control $\alpha_n = (\tau_1, \ldots, \tau_n; \zeta_1, \ldots, \zeta_n)$ with at most n interventions and with $\tau_n \leq \tau_{\mathcal{S}} \wedge T$. Set $\tau_{n+1} = \tau_{\mathcal{S}} \wedge T$ and note that X is uncontrolled between interventions so that $X^{(\alpha_n)}(s) = X(s)$. Then, by Corollary 3.14a, we have

$$\mathbb{E}^{(t,x)}[\varphi_{n-j}(\tau_j, X(\tau_j))] \geq \mathbb{E}^{(t,x)}\left[\int_{\tau_j}^{\tau_{j+1}} f(s,X(s))\,\mathrm{d}s + \varphi_{n-j}(\tau_{j+1}, \check{X}(\tau_{j+1}))\right]. \quad (3.2.16)$$

Choosing $\tau = t$ in (3.2.8) and considering cases where $n - j \geq 1$, we obtain

$$\begin{aligned}
\varphi_{n-j}(t,x) &= \sup_{\tau \in \mathcal{T}_t} \mathbb{E}^{(t,x)}\left[\int_{t}^{\tau} f(s,X(s))\,\mathrm{d}s + \mathcal{M}\varphi_{n-j-1}(\tau, X(\tau))\right] \\
&\geq \mathbb{E}^{(t,x)}[\mathcal{M}\varphi_{n-j-1}(t,x)] = \mathcal{M}\varphi_{n-j-1}(t,x). \qquad (3.2.17)
\end{aligned}$$

Moreover, from the definition of $\mathcal{M}$ (see Definition 3.7), it follows that

$$\mathcal{M}\varphi_{n-j-1}(\tau_{j+1}, \check{X}(\tau_{j+1}^-)) \geq \varphi_{n-j-1}(\tau_{j+1}, X(\tau_{j+1})) + K(\tau_{j+1}, \check{X}(\tau_{j+1}^-), \zeta_{j+1}) \quad (3.2.18)$$

Combining (3.2.16)–(3.2.18), we obtain

$$\mathbb{E}^{(t,x)}\left[\varphi_{n-j}(\tau_j, X(\tau_j))\right] \geq \mathbb{E}^{(t,x)}\left[\int_{\tau_j}^{\tau_{j+1}} f(s,X(s))\,\mathrm{d}s + \varphi_{n-j}(\tau_{j+1}, \check{X}(\tau_{j+1}^-))\right]$$

$$\geq \mathbb{E}^{(t,x)}\left[\int_{\tau_j}^{\tau_{j+1}} f(s,X(s))\,\mathrm{d}s + \varphi_{n-j-1}(\tau_{j+1},X(\tau_{j+1})) + K(\tau_{j+1},\check{X}(\tau_{j+1}^-),\zeta_{j+1})\right], \tag{3.2.19}$$

for $j=0,1,\ldots,n-1$, where we set $\tau_0 := t$. Summing the inequality above from $j=0$ to $j=n-1$, we find that

$$\begin{aligned}
&\sum_{j=0}^{n-1}\mathbb{E}^{(t,x)}\left[\varphi_{n-j}(\tau_j,X(\tau_j)) - \varphi_{n-j-1}(\tau_{j+1},X(\tau_{j+1}))\right] \\
&\qquad \geq \sum_{j=0}^{n-1}\mathbb{E}^{(t,x)}\left[\int_{\tau_j}^{\tau_{j+1}} f(s,X(s))\,\mathrm{d}s + K(\tau_{j+1},\check{X}(\tau_{j+1}^-),\zeta_{j+1})\right] \\
&\qquad = \mathbb{E}^{(t,x)}\left[\int_t^{\tau_n} f(s,X(s))\,\mathrm{d}s\right] + \sum_{j=0}^{n-1}\mathbb{E}^{(t,x)}\left[K(\tau_{j+1},\check{X}(\tau_{j+1}^-),\zeta_{j+1})\right]
\end{aligned} \tag{3.2.20}$$

Noting that the left-hand side in the above inequality simplifies to

$$\sum_{j=0}^{n-1}\mathbb{E}^{(t,x)}\left[\varphi_{n-j}(\tau_j,X(\tau_j)) - \varphi_{n-j-1}(\tau_{j+1},X(\tau_{j+1}))\right] = \varphi_n(t,x) - \mathbb{E}^{(t,x)}\left[\varphi_0(\tau_n,X(\tau_n))\right],$$

since (t,x) is deterministic, the inequality (3.2.20) simplifies to

$$\varphi_n(t,x) - \mathbb{E}^{(t,x)}\left[\varphi_0(\tau_n,X(\tau_n))\right] \geq \mathbb{E}^{(t,x)}\left[\int_t^{\tau_n} f(s,X(s))\,\mathrm{d}s + \sum_{j=0}^{n-1} K(\tau_{j+1},\check{X}(\tau_{j+1}^-),\zeta_{j+1})\right]. \tag{3.2.21}$$

By the strong Markov property, note that

$$\begin{aligned}
\mathbb{E}^{(t,x)}[\varphi_0(\tau_n,X(\tau_n))] &= \mathbb{E}^{(t,x)}\left[\mathbb{E}^{(\tau_n,X(\tau_n))}\left[\int_t^{\tau_{\mathcal{S}}\wedge T} f(s,X(s))\,\mathrm{d}s + g(\tau_{\mathcal{S}}\wedge T, X(\tau_{\mathcal{S}}\wedge T))\right]\right] \\
&= \mathbb{E}^{(t,x)}\left[\int_{\tau_n}^{\tau_{\mathcal{S}}\wedge T} f(s,X(s))\,\mathrm{d}s + g(\tau_{\mathcal{S}}\wedge T, X(\tau_{\mathcal{S}}\wedge T))\right].
\end{aligned} \tag{3.2.22}$$

Combining (3.2.21) and (3.2.22) together with the fact that $\varphi_n(t,x)$ is a deterministic quantity conditional on (t,x), we get

$$\begin{aligned}
\varphi_n(t,x) &\geq \mathbb{E}^{(t,x)}\left[\varphi_0(\tau_n,X(\tau_n))\right] + \mathbb{E}^{(t,x)}\left[\int_t^{\tau_n} f(s,X(s))\,\mathrm{d}s + \sum_{j=1}^{n} K(\tau_j,\check{X}(\tau_j^-),\zeta_j)\right] \\
&= \mathbb{E}^{(t,x)}\left[\int_t^{\tau_{\mathcal{S}}\wedge T} f(s,X(s))\,\mathrm{d}s + g(\tau_{\mathcal{S}}\wedge T, X(\tau_{\mathcal{S}}\wedge T)) + \sum_{j=1}^{n} K(\tau_j,\check{X}(\tau_j^-),\zeta_j)\right] \\
&= J^{(\alpha_n)}(t,x).
\end{aligned} \tag{3.2.23}$$

Since $\alpha_n \in \mathcal{V}_n$ was arbitrary, we conclude that

$$\varphi_n(t,x) \geq v_n(t,x). \tag{3.2.24}$$

It now remains to show that the reverse inequality also holds to complete the proof. To this end, choose an arbitrary $\varepsilon > 0$ and define an increasing sequence of stopping times $t = \hat{\tau}_0 < \hat{\tau}_1 < \cdots < \hat{\tau}_0$ in the following manner: For $j = 1, \ldots, n$, let the continuation region

$$D_j^{(\varepsilon)} = \{(t,x) \in [0,T) \times \mathcal{S} : \varphi_j(t,x) > \mathcal{M}\varphi_{j-1}(t,x) + \varepsilon\}, \tag{3.2.25}$$

and define the first ε-optimal stopping time $\hat{\tau}_1$ by

$$\hat{\tau}_1 = \inf\{s > t : X^{(0)}(s) \notin D_j^{(\varepsilon)}\}, \tag{3.2.26}$$

where $X^{(0)}(s) = X(s)$ is the process without interventions. Then, choose $\hat{\zeta}_1 = \bar{\zeta}_1(\tau_1, X(\tau_1^-))$, where $\bar{\zeta}_1 = \bar{\zeta}_1(t,x) \in \mathcal{Z}(t,x)$ is ε optimal for φ_{n-1} in the sense that

$$\mathcal{M}\varphi_{n-1}(t,x) \leq \varphi_{n-1}(t, \Gamma(t,x,\bar{\zeta}_1)) + K(t,x,\bar{\zeta}_1) + \varepsilon. \tag{3.2.27}$$

Inductively, if $t = \hat{\tau}_0, \ldots, \hat{\tau}_j; \hat{\zeta}_1, \ldots, \hat{\zeta}_j$ have been chosen, where $j \leq n-1$, we let $X^{(j)}(t)$ be the process obtained by applying the ε-optimal impulse $\hat{a}_j = (\hat{\tau}_1, \ldots, \hat{\tau}_j; \hat{\zeta}_1, \ldots, \hat{\zeta}_j)$ to $X(t)$. Define

$$\hat{\tau}_{j+1} = \inf\left\{s \geq \hat{\tau}_j : X^{(j)}(s) \notin D_{n-j}^{(\varepsilon)}\right\} \tag{3.2.28}$$

and $\hat{\zeta}_{j+1} = \bar{\zeta}_{t+1}(\hat{\tau}_{j+1}^-, X^{(j)}(\hat{\tau}_{j+1}^-))$, where $\bar{\zeta}_{j+1} = \bar{\zeta}_{j+1}(t,x) \in \mathcal{Z}(t,x)$ is ε-optimal for φ_{n-j-1} in the sense that

$$\mathcal{M}\varphi_{n-j-1}(t,x) \leq \varphi_{n-j-1}(t, \Gamma(t,x,\bar{\zeta}_{j+1})) + K(t,x,\bar{\zeta}_{j+1}) + \varepsilon \tag{3.2.29}$$

Finally, define the ε-optimal control with at most n interventions

$$\hat{a}_n = (\hat{\tau}_1, \ldots, \hat{\tau}_n; \hat{\zeta}_1, \ldots, \hat{\zeta}_n) \in \mathcal{V}_n.$$

Now apply the argument presented above in (3.2.16)–(3.2.23). By Corollary 3.14b, we have

$$\mathbb{E}^{(t,x)}[\varphi_{n-j}(\hat{\tau}_j, X(\hat{\tau}_j))] \leq \mathbb{E}^{(t,x)}\left[\int_{\hat{\tau}_j}^{\hat{\tau}_{j+1}} f(s, X(s))\,\mathrm{d}s + \varphi_{n-j}(\hat{\tau}_{j+1}^-, \check{X}(\hat{\tau}_{j+1}^-))\right] \tag{3.2.30}$$

Since, by the definition of $\hat{\tau}_{j+1}$, $\check{X}(\hat{\tau}_{j+1}^-) \notin D_{n-j}^{(\varepsilon)}$, we have

$$\varphi_{n-j}(\hat{\tau}_{j+1}^-, \check{X}(\hat{\tau}_{j+1}^-)) \leq \mathcal{M}\varphi_{n-j-1}(\hat{\tau}_{j+1}^-, \check{X}(\hat{\tau}_{j+1}^-)) + \varepsilon, \tag{3.2.31}$$

and by (3.2.29), we get, with $X = X^{(\hat{a}_n)}$,

$$\mathcal{M}\varphi_{n-j-1}(\hat{\tau}_{j+1}^-, \check{X}(\hat{\tau}_{j+1}^-)) \leq \varphi_{n-j-1}(\hat{\tau}_{j+1}, X(\hat{\tau}_{j+1})) + K(\hat{\tau}_{j+1}, \check{X}(\hat{\tau}_{j+1}^-), \hat{\zeta}_{j+1}) + \varepsilon. \tag{3.2.32}$$

Then, combining (3.2.30)–(3.2.32), we obtain

$$\begin{aligned}
&\mathbb{E}^{(t,x)}[\varphi_{n-j}(\hat{\tau}, X(\hat{\tau}))] \\
&\quad \leq \mathbb{E}^{(t,x)}\left[\int_{\hat{\tau}_j}^{\hat{\tau}_{j+1}} f(s, X(s))\,\mathrm{d}s + \varphi_{n-j-1}(\hat{\tau}_{j+1}, X(\hat{\tau}_j)) + K(\hat{\tau}_{j+1}, \check{X}(\hat{\tau}_{j+1}^-), \hat{\zeta}_{j+1})\right] \\
&\qquad + 2\varepsilon.
\end{aligned} \tag{3.2.33}$$

Summing (3.2.33) from $j = 0$ to $j = n-1$ gives

$$\begin{aligned}
&\sum_{j=0}^{n-1} \mathbb{E}^{(t,x)}[\varphi_{n-j}(\hat{\tau}_j, X(\hat{\tau}_j))] \\
&\quad \le \sum_{j=0}^{n-1} \mathbb{E}^{(t,x)}\left[\int_{\hat{\tau}_j}^{\hat{\tau}_{j+1}} f(s, X(s))\,\mathrm{d}s + \varphi_{n-j-1}(\hat{\tau}_{j+1}, X(\hat{\tau}_{j+1})) + K(\hat{\tau}_{j+1}, \check{X}(\hat{\tau}_{j+1}^-), \hat{\zeta}_{j+1}))\right] \\
&\qquad + 2n\varepsilon \\
&\quad \le \mathbb{E}^{(t,x)}\left[\int_t^{\hat{\tau}_n} f(s, X(s))\,\mathrm{d}s + \sum_{j=0}^{n-1} \varphi_{n-j-1}(\hat{\tau}_{j+1}, X(\hat{\tau}_{j+1})) + \sum_{j=0}^{n-1} K(\hat{\tau}_{j+1}, \check{X}(\hat{\tau}_{j+1}^-), \hat{\zeta}_{j+1}))\right] \\
&\qquad + 2n\varepsilon. \qquad (3.2.34)
\end{aligned}$$

Since

$$\sum_{j=0}^{n-1} \mathbb{E}^{(t,x)}[\varphi_{n-j}(\hat{\tau}_j, X(\hat{\tau}_j)) - \varphi_{n-j-1}(\hat{\tau}_{j+1}, X(\hat{\tau}_{j+1}))] = \varphi_n(t,x) - \mathbb{E}^{(t,x)}[\varphi_0(\hat{\tau}_n, X(\hat{\tau}_n))], \tag{3.2.35}$$

we obtain, by combining (3.2.34) and (3.2.35),

$$\begin{aligned}
\varphi_n(t,x) = \mathbb{E}^{(t,x)}&\left[\int_t^{\hat{\tau}_n} f(s, X(s))\,\mathrm{d}s + \varphi_0(\hat{\tau}_n, X(\hat{\tau}_n)) + \sum_{j=0}^{n-1} K(\hat{\tau}_{j+1}, \check{X}(\hat{\tau}_{j+1}^-), \hat{\zeta}_{j+1}))\right] \\
&+ 2n\varepsilon.
\end{aligned}$$

Therefore, by (3.2.22),

$$\begin{aligned}
\varphi_n(t,x) \le \mathbb{E}^{(t,x)}&\left[\int_t^{\hat{\tau}_{\mathcal{S}} \wedge T} f(s, X(s))\,\mathrm{d}s + g(\hat{\tau}_{\mathcal{S}} \wedge T, X(\hat{\tau}_{\mathcal{S}} \wedge T)) + \sum_{j=1}^{n} K(\hat{\tau}_j, \check{X}(\hat{\tau}_j^-), \hat{\zeta}_j)\right] \\
&+ 2n\varepsilon \\
&\le J^{(\hat{a}_n)}(t,x) + 2n\varepsilon.
\end{aligned}$$

Since ε was arbitrary, it follows that

$$\varphi_n(t,x) \le \sup_{\alpha_n \in \mathcal{V}_n} = v_n(y).$$

Combined with (3.2.24), the result above proves Theorem 3.12. □

Theorem 3.12 shows that the truncated impulse-control problem with at most x interventions can be represented by the iterated optimal stopping sequence $\{\varphi_j\}_{j=0}^n$. More precisely, the iterative optimal stopping approach produces exactly the value function v_n of the truncated problem (3.2.1).

It remains to connect the truncated impulse control problem with the unrestricted impulse control problem, which is done through the following two corollaries. Corollary 3.15 combines Lemma 3.10 with Theorem 3.12 to show that the iterated optimal stopping functions (3.2.7)–(3.2.8) monotonically converge to the value function v. Finally, Corollary 3.17 translates the impulse control problem (4.1.1) into a non-linear optimal stopping problem.

Corollary 3.15 ([84, Corollary 10.7]). *Assume $g \geq 0$. Moreover, suppose that φ_n, defined in* (3.2.7)–(3.2.8)*, satisfies condition* (3.2.9)*. Then*

$$\varphi_n \uparrow v \quad \text{as } n \to \infty \quad \text{for all } (t,x) \in [0,T) \times \mathbb{R}^d.$$

Proof. Let $\mathcal{V}_n$ denote the set of all admissible controls with at most n interventions, where we denote by α_n an impulse control with at most n controls, and let $\mathcal{V}$ denote the set of all admissible controls α. Then, by construction,

$$\mathcal{V}_n \subseteq \mathcal{V}_{n+1} \subseteq \mathcal{V},$$

as previously seen. Hence, for $(t,x) \in [0,T) \times \mathcal{S}$, the value function

$$v_n(t,x) := \sup_{a_n \in \mathcal{V}_n} J^{(\alpha_n)}(t,x)$$

satisfies

$$v_n(t,x) \leq v_{n+1}(t,x) \leq v(t,x).$$

Therefore, with Lemma 3.10,

$$v_n(t,x) \uparrow \lim_{n\to\infty} v_n(t,x) = v(t,x). \tag{3.2.36}$$

Moreover, by Theorem 3.12,

$$\varphi_n(t,x) = v_n(t,x) \tag{3.2.37}$$

for all $(t,x) \in [0,T) \times \mathcal{S}$. Combining (3.2.36) and (3.2.37), we then obtain

$$\varphi_n(t,x) \uparrow v(t,x)$$

as $n \to \infty$ and for all $(t,x) \in [0,T) \times \mathbb{R}^d$. □

Remark 3.16. It is worth mentioning that the proof above was not shown in [84].

Corollary 3.17 ([84, Corollary 10.8]). *Assume $g \geq 0$. Moreover, suppose that φ_n, defined in* (3.2.7)–(3.2.8)*, satisfies condition* (3.2.9)*. Then, v is a solution of the following non-linear optimal stopping problem*

$$v(t,x) = \sup_{\tau \in \mathcal{T}_t} \mathbb{E}^{(t,x)} \left[\int_t^\tau f(s, X(s))\,\mathrm{d}s + \mathcal{M}v(\tau, X(\tau)) \right].$$

Proof. Let $\{\varphi_n\}_{n=0}^\infty$ be as in (3.2.7) and (3.2.8). Then, $\varphi_n \uparrow v_n$ as $n \to \infty$ by Corollary 3.15. Therefore,

$$\varphi_n(t,x) \leq \sup_{\tau \in \mathcal{T}_t} \mathbb{E}^{(t,x)} \left[\int_t^\tau f(s, X(s))\,\mathrm{d}s + \mathcal{M}v(\tau, X(\tau)) \right],$$

for all $n \in \mathbb{N}$ and hence

$$v(t,x) \leq \sup_{\tau \in \mathcal{T}_t} \mathbb{E}^{(t,x)} \left[\int_t^\tau f(s, X(s))\,\mathrm{d}s + \mathcal{M}v(\tau, X(\tau)) \right].$$

For the opposite inequality, choose some arbitrary $\varepsilon > 0$ and let $\hat{\tau} \in \mathcal{T}_t$ be a stopping time such that

$$\mathbb{E}^{(t,x)}\left[\int_t^{\hat{\tau}} f(s,X(s))\,\mathrm{d}s + \mathcal{M}v(\hat{\tau},X(\hat{\tau}))\right] \geq \mathbb{E}^{(t,x)}\left[\int_t^{\tau} f(s,X(s))\,\mathrm{d}s + \mathcal{M}v(\tau,X(\tau))\right] - \varepsilon.$$

Then, by monotone convergence, we have

$$\begin{aligned}
v(t,x) &= \lim_{n\to\infty} \varphi_n(t,x) \\
&\geq \lim_{n\to\infty} \sup_{\tau\in\mathcal{T}_t} \mathbb{E}^{(t,x)}\left[\int_t^{\tau} f(s,X(s))\,\mathrm{d}s + \mathcal{M}\varphi_{n-1}(\tau,X(\tau))\right] \\
&\geq \lim_{n\to\infty} \mathbb{E}^{(t,x)}\left[\int_t^{\hat{\tau}} f(s,X(s))\,\mathrm{d}s + \mathcal{M}\varphi_{n-1}(\tau,X(\tau))\right] \\
&= \mathbb{E}^{(t,x)}\left[\int_t^{\hat{\tau}} f(s,X(s))\,\mathrm{d}s + \mathcal{M}v(\tau,X(\tau))\right] \\
&\geq \sup_{\tau\in\mathcal{T}_t} \mathbb{E}^{(t,x)}\left[\int_t^{\tau} f(s,X(s))\,\mathrm{d}s + \mathcal{M}v(\hat{\tau},X(\hat{\tau}))\right] - \varepsilon.
\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, combining both inequalities, we obtain

$$v(t,x) = \sup_{\tau\in\mathcal{T}_t} \mathbb{E}^{(t,x)}\left[\int_t^{\tau} f(s,X(s))\,\mathrm{d}s + \mathcal{M}v(\tau,X(\tau))\right]. \qquad \square$$

We conclude this chapter by noting that, after time discretisation, the dynamic programming principle may be applied to Markov decision process and, consequently, to reinforcement learning (see Chapter 7). As such, the results shown in this section represent the core idea behind large parts of this thesis.

CHAPTER

FOUR

EXISTENCE OF VISCOSITY SOLUTIONS FOR THE IMPULSE-CONTROL QVI

In this chapter, we aim to transition from the probabilistic, dynamic programming formulation presented in Chapter 3 to the Quasi-Variational Inequality (QVI) formulation. Heuristically, Theorem 3.9 suggests that within the continuation region, the value function is determined through the PIDE

$$\partial_t u + \mathcal{L}u + f = 0.$$

On the other hand, if immediate intervention is optimal, then

$$u = \mathcal{M}u$$

must be satisfied. Combining these two choices yields QVI

$$\min\big\{ -(\partial_t u + \mathcal{L}u + f), u - \mathcal{M}u\big\} = 0.$$

However, we need to show that the solution u to the above QVI is indeed the value function, i.e., $u = v$. Moreover, the solution u is not expected to be sufficiently smooth for a classical interpretation of the QVI[1]. This is because the intervention operator produces kinks between the continuation and intervention regions, which results in the value function being non-differentiable on these points or regions. In addition, since the underlying state process is a jump-diffusion, the QVI contains a generator with a nonlocal integral term. In light of this, a weaker solution in the form of *viscosity solutions* is required to treat the QVI.

Roughly speaking, viscosity solutions allow the QVI to be interpreted through smooth test functions without requiring the value function itself to be classically differentiable. In this chapter, we formulate the QVI and show that a viscosity solution to the QVI exists. We rely on the theory provided by Seydel [92] and, as such, the structure of this chapter is identical to Section 4 in Seydel with the exception that we do not consider continuous controls. Nevertheless, some of the proofs presented in this chapter have, to some extent, been extended for clarity.

[1] Or, at least, not everywhere in $[0, T] \times \mathbb{R}^d$.

4.1 Impulse control via QVI and further assumptions

4.1.1 Impulse control via QVI

Recall the finite-horizon impulse control problem (3.1.11), that is,

$$v(t,x)=\sup_{\alpha\in\mathcal{V}} J^{(\alpha)}(t,x),\quad \text{for } (t,x)\in[0,T]\times\mathcal{S}. \tag{4.1.1}$$

For a stopping time $\tau' := \tau_{\mathcal{S}}\wedge T$, where $T<\infty$ the time horizon, the performance criterion J is given by

$$J^{(\alpha)}(t,x)=\mathbb{E}^{(t,x)}\left[\int_t^{\tau'} f(s,X^{(\alpha)}(s))\,\mathrm{d}s+g(\tau',X^{(\alpha)}(\tau'))+\sum_{t\le\tau_j<\tau'}K(\tau_j,\check{X}^{(\alpha)}(\tau_j^-),\zeta_j)\right].$$

The functions f, g, and K are assumed to satisfy the integrability conditions in (3.1.7)–(3.1.9). These conditions ensure that the performance criterion is finite for every admissible control $\alpha\in\mathcal{V}$ and every $(t,x)\in[0,T]\times\mathcal{S}$. The uncontrolled process X follows the SDE (3.1.1), with drift, diffusion, and jump-amplitude functions satisfying (2.3.4)–(2.3.5), and the controlled process $X^{(\alpha)}$ is given by (3.1.3)–(3.1.5). Furthermore, recall that the intervention operator $\mathcal{M}$, introduced in Definition 3.7, is given by

$$\mathcal{M}v(t,x)=\sup_{\zeta\in\mathcal{Z}}\left\{v\big(t,\Gamma(t,x,\zeta)\big)+K(t,x,\zeta)\right\}$$

and that the infinitesimal generator of the uncontrolled process X is given by

$$\begin{aligned}\mathcal{L}u(t,x)&=\langle\mu(t,x),\nabla_x u(t,x)\rangle+\frac{1}{2}\operatorname{Tr}(\sigma(t,x)\sigma^\top(t,x)H_x u(t,x))\\&\quad+\int_{\mathbb{R}^\ell}\{u(t,x+\gamma(t,x,z))-u(t,x)-\langle\nabla u(t,x),\gamma(t,x,z)\rangle 1_{\{|z|<1\}}\}\nu(\,\mathrm{d}z).\end{aligned} \tag{4.1.2}$$

Here, H_x denotes the Hessian operator in x and is defined as

$$H_x u(t,x)=(\partial_{x_i x_j}u(t,x))_{i,j=1}^d\in\mathbb{R}^{d\times d}.$$

We consider the parabolic QVI given by

$$\min\{-(\partial_t u+\mathcal{L}u+f),u-\mathcal{M}u\}=0 \quad \text{in } [0,T)\times\mathcal{S} \tag{4.1.3a}$$
$$\min\{u-g,u-\mathcal{M}u\}=0 \quad \text{in } [0,T)\times(\mathbb{R}^d\backslash\mathcal{S}) \tag{4.1.3b}$$
$$u=\max\{g,\mathcal{M}g\} \quad \text{on } \{T\}\times\mathbb{R}^d \tag{4.1.3c}$$

The hope is to find the value function by solving the (4.1.3).

We now provide an interpretation for the QVI (4.1.3). Consider (4.1.3a), where the Hamiltonian $-(\partial_t u+\mathcal{L}u+f)$ is compared against the intervention gain $u-\mathcal{M}u$. We note that by condition (x) in Theorem 3.9, the Hamiltonian must be zero inside the continuation region

$$D=\{(t,x)\in[0,T)\times\mathcal{S}:u(t,x)>\mathcal{M}u(t,x)\}.$$

Otherwise, the Hamiltonian is positive outside of the continuation region and the controller intervenes, see conditions (ii) and (vi) in Theorem 3.9.

The boundary condition (4.1.3b) compares the value of intervening outside of the solvency region $\mathcal{S}$ versus immediately collecting the reward g. The reason we consider v on $\mathbb{R}^d$, as opposed to the region of interest $\mathcal{S}$, is because the process X has jumps. As Seydel [92] discusses, it is not possible to stay a positive amount of time outside $\mathcal{S}$ since the problem is stopped at $\tau_{\mathcal{S}}$. However, if we consider the infinitesimally small amount of time between an exogenous (Poisson) jump and an intervention, the controller can push the controlled state back into $\mathcal{S}$ through an impulse before the stopping time $\tau_{\mathcal{S}}$ takes notice. This is illustrated in Example 4.1. Hence, defining v outside of $\mathcal{S}$ is needed to be able to decide whether a jump back to solvency is worthwhile.

Finally, condition (4.1.3c) formalises the terminal condition, where one either collects a terminal reward g, or undertakes an impulse and collects $\mathcal{M}g$.

Example 4.1. Let $X^{(\alpha)}$ be a controlled process and recall that impulses are allowed to be taken only after the exogenous Poisson jump takes place, see Definition 3.2. Before intervening, the state we consider is therefore the post-jump, pre-intervention state $\check{X}^{(\alpha)}$. At an intervention time τ, the controlled process is moved to the post-intervention state $\Gamma(\tau, \check{X}^{(\alpha)}(\tau^-), \zeta) = \check{X}^{(\alpha)}(\tau^-) + \zeta$.

Now suppose that $X^{(\alpha)}$ models the cash account of a profitable business. Let $\mathcal{S} = (0, \infty)$ and $X^{(\alpha)}(\tau^-) = 1 \in \mathcal{S}$ be such that the firm begins in a solvent state. If a negative size shock -4 occurs at time τ, the post-jump pre-intervention state is $\check{X}^{(\alpha)}(\tau^-) = 1 - 4 = -3 \notin \mathcal{S}$ and the profitable business has run out of capital. To push the state back into the solvency region, investors may inject capital into the firm through an impulse, say $\zeta = 5$. Then the state jumps back into solvency, i.e. $\Gamma(\tau, \check{X}^{(\alpha)}, \zeta) = -3 + 5 = 2 \in \mathcal{S}$. Thus, the business becomes solvent again.

4.1.2 Further assumptions

Let us now consider the conditions under which the terms $\mathcal{L}u$ and $\mathcal{M}u$ in the QVI (4.1.3) are well-defined. We begin by noting that the integral operator $\mathcal{L}$ (4.1.2) behaves like a differential operator of up to second order, given that ν is singular. This behaviour can be observed by subtracting the first-order compensation term and then applying the Taylor expansion on the function $u \in C^{1,2}([0,T) \times \mathbb{R}^d)$ for some $0 < \delta < 1$. In other words,

$$\begin{aligned} \int_{|z|<\delta} &\big| u\big(t, x + \gamma(t,x,z)\big) - u(t,x) - \langle \nabla u(t,x), \gamma(t,x,z) 1_{\{|z|<1\}} \rangle \big| \nu(\,\mathrm{d}z) \\ &\leq \int_{|z|<\delta} \big|\gamma(t,x,z)\big|^2 \big| H_x u(t,\tilde{x})\big| \nu(\,\mathrm{d}z) \end{aligned} \tag{4.1.4}$$

for some $\tilde{x} \in B(x, \sup_{|z|<\delta} \gamma(t,x,z))$. Here, $B(x,\rho)$ denotes a (open) ball of radius $\rho > 0$ around the point x.

As discussed in Chapter 2, we assume that the basic regularity conditions (2.2.1) and (2.2.2) are satisfied. Then, with (2.2.2) and according to [92], the Hamiltonian $\partial_t u + \mathcal{L}u + f$ is well-defined if, for instance, the following conditions are satisfied

1. $u \in C^{1,2}([0,T) \times \mathbb{R}^d)$, and
2. One of the following inequalities hold locally in $(t,x) \in [0,T) \times \mathcal{S}$ for a constant $C_{t,x}$ dependent on t and x:
   (a) $|\gamma(t,x,z)| \leq C_{t,x}$ if $\nu(\mathbb{R}^d) < \infty$
   (b) $|\gamma(t,x,z)| \leq C_{t,x}|z|$ and $|u(t,z)| \leq C(1+|z|^{p^*})$, where C is independent of t
   (c) Or more generally, for $a, b > 0$ such that $ab \leq p^*$: $|\gamma(t,x,z)| \leq C_{t,x}(1+|z|^a)$, $|u(t,z)| \leq C(1+|z|^b)$ on $|z| \geq 1$. Furthermore, $|\gamma(t,x,z)| \leq C_{t,x}|z|^{q^*/2}$ on $|z| < 1$.

For the rest of the thesis, we assume that one of the above conditions on γ is met. Furthermore, recall Definition 3.11, where the space of polynomially bounded functions $\mathcal{PB}$ is defined. In the following, we formalise the assumptions necessary for the existence proof[2].

Assumption 4.1 ([92, Assumption 2.1])**.** (V1) The functions Γ and K are continuous.

(V2) The set of admissible impulses $\mathcal{Z}(t,x)$ is non-empty and compact for each $(t,x) \in [0,T] \times \mathbb{R}^d$. For a converging sequence $(t_n,x_n) \to (t,x)$ in $[0,T] \times \mathbb{R}^d$ (with $\mathcal{Z}(t_n,x_n)$ non-empty), $\mathcal{Z}(t_n,x_n)$ converges to $\mathcal{Z}(t,x)$ in the Hausdorff metric.

(V3) μ, σ, γ, f are continuous.

(V4) γ satisfies one of the conditions detailed above, and $\mathcal{PB}$ is fixed accordingly.

Remark 4.1. Note that conditions (V1)–(V3) have, in a form or another, been addressed in Chapters 2 and 3.

Let the parabolic, non-local "boundary" be defined as $\partial^+\mathcal{S}_T := \big([0,T) \times (\mathbb{R}^d \backslash \mathcal{S})\big) \cup \big(\{T\} \times \mathbb{R}^d\big)$, and define $\partial^*\mathcal{S}_T := ([0,T) \times \mathcal{S}) \cup \big(\{T\} \times \bar{\mathcal{S}}\big)$. In addition to conditions (V1)–(V4), Assumption 4.2 is required for the existence proof.

Assumption 4.2 ([92, Assumption 2.2])**.** (E1) The value function v is in $\mathcal{PB}([0,T] \times \mathbb{R}^d)$.

(E2) g is continuous.

(E3) For every $(t,x) \in \partial^*\mathcal{S}_T$ and all sequences $(t_n,x_n)_n \subset [0,T) \times \mathcal{S}$ converging to (t,x), the value function v satisfies

$$\liminf_{n \to \infty} v(t_n,x_n) \geq g(t,x).$$

If $v(t_n,x_n) > \mathcal{M}v(t_n,x_n)$ for all $n \in \mathbb{N}$, the value function satisfies

$$\limsup_{n \to \infty} v(t_n,x_n) \leq g(t,x).$$

Remark 4.2. Condition (E3) is typically satisfied if the stochastic process is regular at the boundary $\partial\mathcal{S}$ (see [92, Section 4.1]).

[2] Note that some of these assumptions will also be used in the uniqueness proof as well.

Before moving on to the existence proof, let us briefly discuss Assumptions 4.1 and 4.2. Assumptions (V1) and (V2) concern the intervention operator $\mathcal{M}$. As Γ and K are continuous, and the set $\mathcal{Z}(t,x)$ is compact and continuous in the Hausdorff metric, $\mathcal{M}$ maps locally bounded functions to locally bounded functions and preserves continuity whenever the function, on which $\mathcal{M}$ is applied, is continuous. This will be shown in Lemma 4.7 later in this chapter.

Assumptions (V3) and (V4) concern the Hamiltonian $\partial_t u + \mathcal{L}u + f$ in the QVI (4.1.3). Specifically, for every test function $\varphi \in C^{1,2}([0,T) \times \mathbb{R}^d)$, (V3) gives the required continuity of the functions μ, σ, γ, and f, while (V4) fixes $\mathcal{PB}$ so that the Lévy integral in (4.1.2) is finite. Hence, the Hamiltonian is well-defined and continuous in $[0,T) \times \mathcal{S}$.

Finally, condition (E3) imposes a compatibility condition between the value function inside the solvency region and the boundary payoff. If a sequence of interior points $(t_n, x_n) \in [0,T) \times \mathcal{S}$ approaches a boundary point $(t,x) \in \partial^* \mathcal{S}_T$, then the value should not fall below the exit payoff $g(t,x)$. Moreover, if immediate intervention is not optimal along the sequence, i.e., if $v(t_n,x_n) > \mathcal{M}v(t_n,x_n)$, then there are no impulses near the boundary and the problem locally reduces to a stochastic control problem with exit payoff g. In that case boundary regularity of the underlying process implies that the exit time tends to t and the value converges to $g(t,x)$. Thus, (E3) rules out potential discontinuities caused by boundary behaviour in the continuation region.

4.2 Existence result

We now aim to define viscosity solutions and show that there exists a viscosity solution that solves the QVI (4.1.3). As the definition of the viscosity solution requires the use of lower and upper semi-continuous functions, concepts not yet defined in this thesis, we provide these notions in Definitions 4.3 and 4.4.

Definition 4.3 (Lower Semi-Continuous Functions)**.** Let $T > 0$ be a finite time horizon, and let $u : [0,T] \times \mathbb{R}^d \to \mathbb{R}$ be a function. We say u is lower semi-continuous on $[0,T] \times \mathbb{R}^d$, if for every $(t,x) \in [0,T] \times \mathbb{R}^d$ and every sequence $(t_n,x_n) \to (t,x)$ in $(t,x) \in [0,T] \times \mathbb{R}^d$,

$$u(t,x) \leq \liminf_{n\to\infty} u(t_n,x_n)$$

The lower envelope is denoted by $u_*(t,x) = \liminf_{n\to\infty} u(t_n,x_n)$ for any $(t,x) \in [0,T] \times \mathbb{R}^d$.

Definition 4.4 (Upper Semi-Continuous Functions)**.** Let $T > 0$ be a finite time horizon, and let $u : [0,T] \times \mathbb{R}^d \to \mathbb{R}$ be a function. We say u is upper semi-continuous on $[0,T] \times \mathbb{R}^d$, if for every $(t,x) \in [0,T] \times \mathbb{R}^d$ and every sequence $(t_n,x_n) \to (t,x)$ in $(t,x) \in [0,T] \times \mathbb{R}^d$,

$$u(t,x) \geq \limsup_{n\to\infty} u(t_n,x_n)$$

The upper envelope is denoted by $u_*(t,x) = \limsup_{n\to\infty} u(t_n,x_n)$ for any $(t,x) \in [0,T] \times \mathbb{R}^d$.

Throughout, we denote by $\mathrm{LSC}(\mathcal{C})$ and $\mathrm{USC}(\mathcal{C})$ the set of measurable functions on the set $\mathcal{C}$ that are, respectively, lower semi-continuous and upper semi-continuous.

Furthermore, denote by $\mathcal{PB} := \mathcal{PB}([0,T] \times \mathbb{R}^d)$ the family of bounded polynomial functions, see Definition 3.11. The definition of viscosity solutions is provided in Definition 4.5.

Definition 4.5 ([92, Definition 4.1] Viscosity Solution)**.** Denote $\mathcal{PB} = \mathcal{PB}([0,T] \times \mathbb{R}^d)$. A function $u \in \mathcal{PB}$ is a viscosity subsolution of (4.1.3) if for all $(t_0, x_0) \in [0,T] \times \mathbb{R}^d$ and $\varphi \in \mathcal{PB} \cap C^{1,2}([0,T] \times \mathbb{R}^d)$ with $\varphi(t_0, x_0) = u^*(t_0, x_0)$, where u^* is the upper semi-continuous envelope of u (see Definition 4.4), and with $\varphi \geq u^*$ on $[0,T] \times \mathbb{R}^d$, we have

$$\begin{aligned} \min\big\{ -(\partial_t \varphi + \mathcal{L}\varphi + f), u^* - \mathcal{M}u^* \big\} \leq 0 \quad & \text{in } (t_0, x_0) \in [0,T) \times \mathcal{S} \\ \min\{u^* - g, u^* - \mathcal{M}u^*\} \leq 0 \quad & \text{in } (t_0, x_0) \in \partial^+ \mathcal{S}_T. \end{aligned}$$

A function $u \in \mathcal{PB}$ is a (viscosity) supersolution of (4.1.3) if for all $(t_0, x_0) \in [0,T] \times \mathbb{R}^d$ and $\varphi \in \mathcal{PB} \cap C^{1,2}([0,T) \in \mathbb{R}^d)$ with $\varphi(t_0, x_0) = u_*(t_0, x_0)$, where u_* denotes the lower semi-continuous envelope of u (see Definition 4.3), and with $\varphi \leq u_*$ on $[0,T] \times \mathbb{R}^d$, we have

$$\begin{aligned} \min\big\{ -(\partial_t \varphi + \mathcal{L}\varphi + f), u_* - \mathcal{M}u_* \big\} \geq 0 \quad & \text{in } (t_0, x_0) \in [0,T) \times \mathcal{S} \\ \min\{u_* - g, u_* - \mathcal{M}u_*\} \geq 0 \quad & \text{in } (t_0, x_0) \in \partial^+ \mathcal{S}_T. \end{aligned}$$

A function u is a viscosity solution if it is sub- and supersolution.

Remark 4.6. In Seydel [92], the author remarks that the conditions on the parabolic boundary are included inside the viscosity solution definition (sometimes called "strong viscosity solution", see, e.g., Ishii [55]) because of the implicit form of this "boundary condition". In T , we choose the implicit form because, otherwise, the comparison result would not hold. The time derivative in $t = 0$ is to be understood as a one-sided derivative.

With the definition of the viscosity solution in place, the below lemma, which is important for the existence proof, follow.

Lemma 4.7 ([92, Lemma 4.3])**.** *Let (V1) and (V2) be satisfied. Let u be a locally bounded function*[3] *on $[0,T] \times \mathbb{R}^d$. Then*

(i) $\mathcal{M}u_* \in \mathrm{LSC}([0,T] \times \mathbb{R}^d)$ *and* $\mathcal{M}u_* \leq (\mathcal{M}u_*)_*$.

(ii) $\mathcal{M}u^* \in \mathrm{USC}([0,T] \times \mathbb{R}^d)$ *and* $\mathcal{M}u^* \leq (\mathcal{M}u^*)^*$.

(iii) If $u \leq \mathcal{M}u$ *on* $[0,T] \times \mathbb{R}^d$, *then* $u^* \leq \mathcal{M}u^*$ *on* $[0,T] \times R^d$.

(iv) For an approximating sequence $(t_n, x_n) \to (t,x)$, $(t_n, x_n) \subset [0,T] \times \mathbb{R}^d$ *with* $u(t_n, x_n) \to u^*(t,x)$:
If $u^*(t,x) > \mathcal{M}u^*(t,x)$, *then there exists* $N \in \mathbb{N}$ *such that*

$$u(t_n, x_n) \to \mathcal{M}u(t_n, x_n) \quad \textit{for all } n \geq N.$$

[3]The function u is locally bounded if for every $x \in \mathbb{R}^d$, there is a neighbourhood U of x and a constant $C > 0$ such that $|f(y)| \leq C$ for all $y \in U$.

(v) *$\mathcal{M}$ is monotonous, that is, for two functions u, w satisfying $u \geq w$, $\mathcal{M}u \geq \mathcal{M}w$. In particular, for the value function v, $\mathcal{M}^n v$ for all $n \geq 1$.*

Proof. We provide a separate proof for each point in the above Lemma.

(i) Let $(t_n, x_n)_{n\geq 1}$ be a sequence in $[0, T] \times \mathbb{R}^d$ converging to (t, x). For an arbitrary $\varepsilon > 0$, select an ε-optimal impulse $\zeta^{(\varepsilon)} \in \mathcal{Z}(t, x)$, i.e. a $\zeta^{(\varepsilon)}$ such that

$$u_*(t, \Gamma(t, x, \zeta^{(\varepsilon)})) + K(t, x, \zeta^{(\varepsilon)}) \geq \mathcal{M}u_*(t, x).$$

Now let $\zeta_n \in \mathcal{Z}(t_n, x_n)$ be a sequence. Since by (V2) $\mathcal{Z}(t, x)$ is non-empty and compact for each $(t, x) \in [0, T] \times \mathbb{R}^d$ and $\mathcal{Z}(t_n, x_n) \to Z(t, x)$ for n large enough, the sequence converges to $\zeta^{(\varepsilon)}$ for all n by the Hausdorff convergence theorem [24, Chapter 7]. Then

$$\begin{aligned}
\liminf_{n\to\infty} \mathcal{M}u_*(t_n, x_n) \liminf_{n\to\infty} \Big\{ \sup_{\zeta\in\mathcal{Z}(t_n,x_n)} \big\{u_*(t, \Gamma(t_n, x_n, \zeta)) + K(t_n, x_n, \zeta)\big\}\Big\} & \\
\geq \liminf_{n\to\infty} \big\{u_*(t, \Gamma(t_n, x_n, \zeta_n)) + K(t_n, x_n, \zeta_n)\big\} & \\
\geq u_*(t, \Gamma(t, x, \zeta^{(\varepsilon)})) + K(t, x, \zeta^{(\varepsilon)}) & \\
\mathcal{M}u_*(t, x) - \varepsilon. &
\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, it follows by the definition of lower semi-continuous functions (see Definition 4.3), that

$$\mathcal{M}u_* \in \mathrm{LSC}([0, T] \times \mathbb{R}^d).$$

For the second assertion in (i), we begin by noting that

$$\mathcal{M}u(t, x) \geq \mathcal{M}u_*(t, x) \quad \text{for all } (t, x) \in [0, T] \times \mathbb{R}^d,$$

because $u \geq u_*$ by Definition 4.3. Thus, for any $(t, x) \in [0, T] \times \mathbb{R}^d$

$$\begin{aligned}
(\mathcal{M}u(t, x))_* &\geq (\mathcal{M}u_*(t, x))_* \\
&= \Big(\sup_{\zeta\in\mathcal{Z}(t,x)} \big\{u_*(t, \Gamma(t, x, \zeta)) + K(t, x, \zeta)\big\}\Big)_* \\
&= \sup_{\zeta\in\mathcal{Z}(t,x)} \big\{u_*(t, x\Gamma(t, x, \zeta)) + K(t, x, \zeta)\big\} \\
&= \mathcal{M}u_*(t, x),
\end{aligned}$$

as K is continuous by (V1).

(ii) Fix some $(t, x) \in [0, T] \times \mathbb{R}^d$ and, for $n \in \mathbb{N}$, let the sequence $(t_n, x_n) \in [0, T] \times \mathbb{R}^d$ converge to (t, x). Because u^* is upper semicontinuous and the functions K and Γ are continuous by (V1), the maximum in $\mathcal{M}u^*(t_n, x_n)$ is achieved for each fixed n. That is, there exists $\zeta_n \in \mathcal{Z}(t_n, x_n)$ such that

$$\mathcal{M}u^*(t_n, x_n) = u^*(t_n, \Gamma(t_n, x_n, \zeta_n)) + K(t_n, x_n, \zeta_n).$$

Since $\mathcal{Z}(\cdot, \cdot)$ is a compact set by (V2), $\mathcal{Z}(\cdot, \cdot)$ is bounded and closed. Hence, $\zeta_n \in \mathcal{Z}(t_n, x_n)$ is contained inside a bounded set for all $n \in \mathbb{N}$. In other words, the sequence $\{\zeta_n\}_{n\in\mathbb{N}}$ has a convergent subsequence with limit $\hat{\zeta} \in \mathcal{Z}(t,x)$ (this can be shown by the Hausdorff convergence theorem, see [24, Chapter 7]). Note that the limit $\hat{\zeta}$ is not necessarily optimal, so

$$\begin{aligned}\mathcal{M}u^*(t,x) &= u^*(t,\Gamma(t,x,\hat{\zeta})) + K(t,x,\hat{\zeta}) \\ &\geq \limsup_{n\to\infty} \left\{u^*(t_n,\Gamma(t_n,x_n,\zeta_n)) + K(t_n,x_n,\zeta_n)\right\} \\ &= \limsup_{n\to\infty} \mathcal{M}u^*(t_n,x_n),\end{aligned}$$

which shows that $\mathcal{M}u^* \in \mathrm{USC}([0,T]\times\mathbb{R}^d)$. Moreover, for the second assertion, we note that since

$$\mathcal{M}u \leq \mathcal{M}u^* \quad \text{in } [0,T]\times\mathbb{R}^d,$$

we get, in a similar manner as in (ii),

$$\mathcal{M}u^* \leq (\mathcal{M}u^*)^* = \mathcal{M}u^* \quad \text{in } [0,T]\times\mathbb{R}^d.$$

(iii) This result follows from (ii). If $u \leq \mathcal{M}u$ on $[0,T]\times\mathbb{R}^d$, then

$$\mathcal{M}u^* \leq (\mathcal{M}u^*)^* = \mathcal{M}u^* \quad \text{in } [0,T]\times\mathbb{R}^d.$$

(iv) This results is shown by contradiction. To this end, assume that

$$u(t_n,x_n) \leq \mathcal{M}u(t_n,x_n) \tag{4.2.1}$$

for infinitely many n. Then, by convergence along a subsequence,

$$u^*(t,x) \leq \limsup_{n\to\infty} \mathcal{M}u(t_n,x_n) \leq (\mathcal{M}u)^*(t,x) \leq \mathcal{M}u^*(t,x).$$

However, this is is a contradiction since the above inequality forces

$$u^*(t,x) \leq \mathcal{M}u^*(t,x).$$

Therefore, the assumption (4.2.1) does not hold, i.e. there must exist N such that for all $n \geq N$,

$$u(t_n,x_n) > \mathcal{M}u(t_n,x_n).$$

(v) The monotonicity follows directly from the definition of $\mathcal{M}$. $\mathcal{M}v \leq v$ is necessary for the value function v, because v is already optimal, and $\mathcal{M}^n v \leq v$ for all $n \geq 1$ the follows by induction.

With (i)–(v) proven, the proof is concluded. □

Using Lemma 4.7, we now prove the existence of a viscosity solution in the form of Theorem 4.8. However, before turning to the existence proof, we note that the existence proof frequently makes use of localisation by stopping times to ensure that a stochastic process X, started at $X(t^-) = x$, is contained in some (small) set. A standard choice of such a stopping time is, for radius $\rho_1 > 0$ and a (sufficiently) small time cut-off $\rho_2 > 0$,

$$\tau := \{s \geq t : |X(s) - x| \geq \rho_1\} \wedge (t + \rho_2).$$

τ is to be understood as the minimum between the first exit time from the spatial ball $B(x, \rho_1)$ and the time $t + \rho_2$. If X has continuous paths (i.e. no jumps), then

$$|X(\tau) - x| \le \rho_1 \quad \text{a.s.},$$

and, in fact, $|X(\tau) - x| = \rho_1$ on the event $\{\tau < t + \rho_2\}$. However, if X has non-predictable jumps, one may think that the exit from the ball $B(x, \rho_1)$ may occur by a jump, leading to an overshoot $|X(\tau) - x| > \rho_1$. Fortunately, a key property of Lévy processes is that they are stochastically continuous, see Definition 2.5. Hence, for each fixed $\rho_1 > 0$,

$$\mathbb{P}\big(|X(\tau) - x| > \rho_1\big) \to 0 \quad \text{as } \rho_2 \to 0.$$

Thus, localisation can be enforced with probability arbitrarily close to one by choosing ρ_2 sufficiently small. To use this uniformly in the proof of Theorem 4.8, we follow Seydel [92] and make use of the fact that stochastic continuity on a compact time interval is uniform, see Lemma B.2.

Theorem 4.8 ([92, Theorem 4.2] Existence of the Viscosity Solution)**.** *Let Assumptions 4.1 and 4.2 be satisfied. Then, the value function v is a viscosity solution of* (4.1.3) *as defined above.*

Proof. The proof is split in two parts: v is a supersolution and v is a subsolution. If v is both super- and subsolution, then, by Definition 4.5, the value function is a viscosity solution.

v **is a supersolution**. Recall that for any $(t_0, x_0) \in [0, T] \times \mathbb{R}^d$, the inequality

$$v(t_0, x_0) \ge \mathcal{M}v(t_0, x_0)$$

must hold because $v(t_0, x_0)$ is the supremum over all admissible impulse controls, i.e.

$$v(t_0, x_0) = \sup_{\alpha \mathcal{V}} J^{(\alpha)}(t_0, x_0).$$

By Lemma 4.7(i), we then have

$$\mathcal{M}v_*(t_0, x_0) \le (\mathcal{M}v(t_0, x_0))_* \le v_*(t_0, x_0). \tag{4.2.2}$$

With this in mind, we now verify the condition on the parabolic boundary. We know that on the boundary, we can either jump back to $\mathcal{S}$ or do nothing, in which case we collect the reward g. Since v is optimal,

$$v \ge g \quad \text{on } \partial^+\mathcal{S}_T.$$

Together with (E2) and (E3) in Assumption 4.2, we therefore have that

$$v_* \ge g \quad \text{on } \partial^+\mathcal{S}_T. \tag{4.2.3}$$

It therefore remains to show that

$$\partial_t\varphi + \mathcal{L}\varphi + f \le 0$$

in a fixed $(t_0, x_0) \in [0, T) \times \mathcal{S}$ and for $\varphi \in \mathcal{PB} \cap C^{1,2}([0, T] \times \mathbb{R}^d)$, where φ satisfies $\varphi(t_0, x_0) = v_*(t_, x_0)$ and $\varphi \le v_*$ on $[0, T] \times \mathbb{R}^d$.

From the definition of the lower semi-continuous function v_*, there exists a sequence $(t_n, x_n) \in [0,T) \times \mathcal{S}$ such that $(t_n, x_n) \to (t_0, x_0)$, $\varphi(t_n, x_n) \to \varphi_*(t_0, x_0)$ for $n \to \infty$. By continuity of φ, the sequence

$$\delta_n := v(t_n, x_n) - \varphi(t_n, x_n) \tag{4.2.4}$$

converges from above to zero as $n \to \infty$. In addition, because $t_0, x_0) \in [0,T) \times \mathcal{S}$, there exists some $\rho > 0$ such that for n large enough, $t_n < T$ and $B(x_n, \rho) \subset B(x_0, 2\rho) = \{|y - x_0| < 2\rho\} \subset \mathcal{S}$. Here, $B(x_0, \rho)$ denotes an open ball of radius $\rho > 0$ around x.

Let us now consider an uncontrolled process $X^{(t_n, x_n)}$ starting in $t_n, x_n)$. Furthermore, choose a strictly positive sequence $\{h_n\}_{n\in\mathbb{N}}$ such that $h_n \to 0$ and $\delta_n / h_n \to 0$ as $n \to \infty$, i.e. $\delta_n \to 0$ faster than $h_n \to 0$ as $n \to \infty$. Then, for the stopping time

$$\bar{\tau}_n := \inf\{s \geq t_n : |X^{(t_n, x_n)} - x_n| \geq \rho\} \wedge (t_n + h_n) \wedge T,$$

we get by the strong Makrov property and the Dynkin formula (see Theorem 2.8), for ρ sufficiently small,

$$\begin{aligned}
v(t_n, x_n) &\geq \mathbb{E}^{(t_n, x_n)}\left[\int_{t_n}^{\bar{\tau}_n} f(s, X(s))\,\mathrm{d}s + v(\bar{\tau}_n, \check{X}(\tau_n^-))\right] \\
&\geq \mathbb{E}^{(t_n, x_n)}\left[\int_{t_n}^{\bar{\tau}_n} f(s, X(s))\,\mathrm{d}s + \varphi(\bar{\tau}_n, \check{X}(\tau_n^-))\right] \\
&\overset{\text{Dynkin}}{=} \varphi(t_n, x_n) + \mathbb{E}^{(t_n, x_n)}\left[\int_{t_n}^{\bar{\tau}_n} \{f(s, X(s)) + \partial_s \varphi(s, X(s)) + \mathcal{L}\varphi(s, X(s))\}\,\mathrm{d}s\right],
\end{aligned} \tag{4.2.5}$$

Combining (4.2.4) with (4.2.5), we then obtain

$$\frac{\delta_n}{h_n} \geq E^{(t_n, x_n)}\left[\frac{1}{h_n}\int_{t_n}^{\bar{\tau}_n} \{f(s, X(s)) + \partial_s \varphi(s, X(s)) + \mathcal{L}\varphi(s, X(s))\}\,\mathrm{d}s\right]. \tag{4.2.6}$$

We now aim to let $n \to \infty$ in (4.2.6). However, note that it is not possible to apply the mean value theorem as, for some fixed event $\omega \in \Omega$, the map $s \mapsto f(s, X(s))$ is in general not continuous as X is a jump-diffusion process. We therefore proceed by choosing some $\varepsilon \in (0, \rho)$. Then, by Lemma B.2,

$$\mathbb{P}\left(\sup_{t_n \leq s \leq r} |X^{(t_n, x_n)}(s) - x_n| > \varepsilon\right) \to 0 \quad \text{for } r \downarrow t_n.$$

Define the set

$$A_{n,\varepsilon} := \{\omega : \sup_{t_n \leq s \leq t_n + h_n} |X^{(t_n, x_n)}(s) - x_n| \leq \varepsilon\},$$

as well as its complement

$$A_{n,\varepsilon}^c := \{\omega : \sup_{t_n \leq s \leq t_n + h_n} |X^{(t_n, x_n)}(s) - x_n| > \varepsilon\}.$$

Using the sets $A_{n,\varepsilon}$ and $A_{n,\varepsilon}^c$, the integral in (4.2.6) is split with

$$\left(\int_{t_n}^{t_n + h_n}\right) = \left(\int_{t_n}^{t_n + h_n}\right) 1_{\{A_{n,\varepsilon}\}} + \left(\int_{t_n}^{\bar{\tau}_n}\right) 1_{\{A_{n,\varepsilon}^c\}}. \tag{4.2.7}$$

On the set $A_{n,\varepsilon}$, for the integrand $G(t,x) := f(t,x) + \partial_t\varphi(t,x) + \mathcal{L}\varphi(t,x)$ on the right-hand side of (4.2.6), we obtain

$$\begin{aligned}\left| G(t_0,x_0) - \frac{1}{h_n}\int_{t_n}^{t_n+h_n} G(s,X(s))\,\mathrm{d}s \right| &\le \frac{1}{h_n}\int_{t_n}^{t_n+h_n} \left|G(t_0,x_0) - G(s,X(s))\right|\mathrm{d}s \\ &\le \left|G(t_0,x_0) - G(\hat{t}_{n,\varepsilon},\hat{x}_{n,\varepsilon})\right|, \qquad (4.2.8)\end{aligned}$$

the latter of which is because the integrand G is continuous by (V3) and the assumption on $\varphi \in \mathcal{PB} \cap C^{1,2}([0,T]\times\mathbb{R}^d)$, and the maximum distance of $|G(t_0,x_0) - G(\cdot,\cdot)|$ is assumed in a $(\hat{t}_{n,\varepsilon},\hat{x}_{n,\varepsilon}) \in [t_n, t_n+h_n] \times \overline{B(x_n,\varepsilon)}$[4].

On the set $A^c_{n,\varepsilon}$,

$$\begin{aligned}&\frac{1}{h_n}\left(\int_{t_n}^{\bar{\tau}_n} \left\{f(s,X(s)) + \partial_s\varphi(s,X(s)) + \mathcal{L}\varphi(s,X(s))\right\}\mathrm{d}s\right) 1_{\{A^c_{n,\varepsilon}\}} \\ &\quad \le \operatorname*{ess\,sup}_{t_n\le s\le\bar{\tau}_n} \left| f(s,X(s)) + \partial_s\varphi(s,X(s)) + \mathcal{L}\varphi(s,X(s))\right| 1_{\{A^c_{n,\varepsilon}\}},\end{aligned}$$

which is bounded by the same arguments as above and because a jump in $\bar{\tau}_n$ does not affect the essential supremum.

Now because $h_n \to 0$ and $(t_n,x_n) \to (t_0,x_0)$ for $n\to\infty$ and by stochastic continuity,

$$\mathbb{P}(A_{n,\varepsilon}) \to 1 \quad \text{as } n\to\infty$$

for any $\varepsilon > 0$, or, equivalently,

$$1_{\{A_{n,\varepsilon}\}} \to 1 \quad \text{a.s.,} \quad \text{as } n\to\infty, \qquad (4.2.9)$$

for any $\varepsilon > 0$. (4.2.9) also implies that on the complementary set $A^c_{n,\varepsilon}$,

$$1_{\{A^c_{n,\varepsilon}\}} \to 0 \quad \text{a.s.,} \quad \text{as } n\to\infty. \qquad (4.2.10)$$

Hence by (4.2.6) and combining (4.2.7)–(4.2.10), we obtain

$$\begin{aligned}0 &= \lim_{n\to\infty}\frac{\delta_n}{h_n} \\ &\ge \lim_{n\to\infty}\mathbb{E}^{(t_n,x_n)}\left[\frac{1}{h_n}\int_{t_n}^{\bar{\tau}_n}\left\{f(s,X(s)) + \partial_s\varphi(s,X(s)) + \mathcal{L}\varphi(s,X(s))\right\}\mathrm{d}s\right] \\ &= \lim_{n\to\infty}\mathbb{E}^{(t_n,x_n)}\left[\left\{\frac{1}{h_n}\int_{t_n}^{t_n+h_n}\left\{f(s,X(s)) + \partial_s\varphi(s,X(s)) + \mathcal{L}\varphi(s,X(s))\right\}\right\} 1_{\{A_{n,\varepsilon}\}}\right] \\ &\quad + \lim_{n\to\infty}\mathbb{E}^{(t_n,x_n)}\left[\left\{\frac{1}{h_n}\int_{t_n}^{\bar{\tau}_n}\left\{f(s,X(s)) + \partial_s\varphi(s,X(s)) + \mathcal{L}\varphi(s,X(s))\right\}\right\} 1_{\{A^c_{n,\varepsilon}\}}\right] \\ &\overset{\text{DCT}}{=} \lim_{n\to\infty}\mathbb{E}^{(t_0,x_0)}\left[\lim_{n\to\infty}\left\{\frac{1}{h_n}\int_{t_n}^{t_n+h_n}\left\{f(s,X(s)) + \partial_s\varphi(s,X(s)) + \mathcal{L}\varphi(s,X(s))\right\}\right\} 1_{\{A_{n,\varepsilon}\}}\right] \\ &\quad + \mathbb{E}^{(t_0,x_0)}\left[\lim_{n\to\infty}\left\{\frac{1}{h_n}\int_{t_n}^{\bar{\tau}_n}\left\{f(s,X(s)) + \partial_s\varphi(s,X(s)) + \mathcal{L}\varphi(s,X(s))\right\}\right\} 1_{\{A^c_{n,\varepsilon}\}}\right]\end{aligned}$$

[4] Note that the set $[t_n, t_n+h_n] \times \overline{B(x_n,\varepsilon)}$ is compact and Weierstrass' theorem, see [86, Theorem 3.12 (Extreme-value theorem)], implies that the supremum is attained within the compact set.

$$= f(t_0, x_0) + \partial_t \varphi(t_0, x_0) + \mathcal{L}\varphi(t_0, x_0),$$

by the dominated convergence theorem (DCT). Thus,

$$\partial_t \varphi + \mathcal{L}\varphi + f \le 0 \quad \text{in } [0, T) \times \mathcal{S}. \tag{4.2.11}$$

Combining (4.2.2)–(4.2.11), we see that the supersolution criterion in Definition 4.5 is met and v is therefore a supersolution.

v **is a subsolution**. Let $[0, T] \times \mathbb{R}^d$ and $\varphi \in \mathcal{PB} \cap C^{1,2}([0, T] \times \mathbb{R}^d)$ such that $v^*(t_0, x_0) = \varphi(t_0, x_0)$ and $\varphi \ge v^*$ on $[0, T] \times \mathbb{R}^d$.

We begin by noting that in the case when

$$v^*(t_0, x_0) \le \mathcal{M}v^*(t_0, x_0),$$

then the subsolution inequality in Definition 4.5

$$\begin{aligned} \min\big\{ -(\partial_t \varphi + \mathcal{L}\varphi + f), v^* - \mathcal{M}v^*\big\} &\le 0 \quad \text{in } [0, T) \times \mathcal{S} \\ \min\{v^* - g, v^* - \mathcal{M}v^*\} &\le 0 \quad \text{in } \partial^+ \mathcal{S}_T, \end{aligned}$$

holds trivially. Therefore, consider from now on the case when

$$v^*(t_0, x_0) > \mathcal{M}v^*(t_0, x_0).$$

Consider $(t_0, x_0) \in \partial^+ \mathcal{S}_T$. For an approximating sequence $(t_n, x_n) \to (t_0, x_0)$ in $[0, T] \times \mathbb{R}^d$ with $v(t_n, x_n) \to v^*(t_0, x_0)$, the relation

$$v(t_n, x_n) > \mathcal{M}v(t_n, x_n)$$

holds by Lemma 4.7(iv). Thus, by the continuity of g outside $\bar{\mathcal{S}}$ and, in the case when $x_0 \in \partial \mathcal{S}$ or $t_0 = T$, Assumption (E3),

$$g(t_0, x_0) = \lim_{n \to \infty} g(t_n, x_n) = \lim_{n \to \infty} v(t_n, x_n) = v^*(t_0, x_0).$$

Let us now show that for

$$v^*(t_0, x_0) > \mathcal{M}v^*(t_0, x_0), \tag{4.2.12}$$

one obtains

$$-(\partial_t \varphi + \mathcal{L}\varphi + f) \le 0 \quad \text{in } (t_0, x_0) \in [0, T) \times \mathcal{S},$$

where $\varphi \in \mathcal{PB} \cap C^{1,2}([0, T) \times \mathbb{R}^d)$. To this end, we argue by contradiction and assume that there is an $\eta > 0$ such that

$$\partial_t \varphi(t_0, x_0) + \mathcal{L}\varphi(t_0, x_0) + f(t_0, x_0) < -\eta. \tag{4.2.13}$$

Because $\varphi \in \mathcal{PB} \cap C^{1,2}([0, T] \times \mathbb{R}^d)$ and $\partial_t \varphi + \mathcal{L}\varphi + f$ is continuous in (t, x) by (V3), we can choose some $\delta > 0$ such that

$$\Big|\partial_t \varphi(t, x) + \mathcal{L}\varphi(t, x) + f(t, x) - \big(\partial_t \varphi(t_0, x_0) + \mathcal{L}\varphi(t_0, x_0) + f(t_0, x_0)\big)\Big| < \frac{\eta}{2}$$

whenever $|(t,x)-(t_0,x_0)|<\delta$. By Assumption (4.2.13), for such (t,x) we have

$$\begin{aligned}\partial_t\varphi(t,x)+\mathcal{L}\varphi(t,x)+f(t,x) &< \partial_t\varphi(t_0,x_0)+\mathcal{L}\varphi(t_0,x_0)+f(t_0,x_0)+\frac{\eta}{2}\\ &\overset{(4.2.13)}{<} -\eta+\frac{\eta}{2}\\ &= -\frac{\eta}{2},\end{aligned}$$

so there exists an open set $\mathcal{O}$ surrounding $(t_0,x_0)\in[0,T)\times\mathcal{S}$ such that

$$\partial_t\varphi(t,x)+\mathcal{L}\varphi(t,x)+f(t,x)<-\frac{\eta}{2}\quad\text{for } (t,x)\in\mathcal{O}. \tag{4.2.14}$$

From the definition of v^*, there exists a sequence $(t_n,x_n)\in([0,T)\times\mathcal{S})\cap\mathcal{O}$ such that $(t_n,x_n)\to(t_0,x_0)$, $v(t_n,x_n)\to v^*(t_0,x_0)$ for $n\to\infty$. By continuity of φ, we observe that

$$\delta_n := v(t_n,x_n)-\varphi(t_n,x_n)\to 0\quad\text{as } n\to\infty$$

since $v(t_n,x_n)\to v^*(t_0,x_0)$ and $\varphi(t_n,x_n)\to\varphi(t_0,x_0)=v^*(t_0,x_0)$ as $n\to\infty$. By the definition of the value funciton, there exists for all n and $\varepsilon>0$ ($\varepsilon=\varepsilon_n$ with $\varepsilon_n\downarrow 0$) a sequence of admissible impulse controls $\alpha^n=(\tau_j^n,\zeta_j^n)_{j\geq 0}$ such that

$$v(t_n,x_n)\leq J^{(a^n)}(t_n,x_n)+\varepsilon. \tag{4.2.15}$$

For a $\rho>0$ chosen suitably small, i.e. such that $B(x_n,\rho)\subset B(x_0,2\rho)\subset\mathcal{S}$ and $t_n+\rho<t_0+2\rho<T$ for large n, we define the stopping time $\bar{\tau}_n:=\bar{\tau}_n^\rho\wedge\tau_1^n$, where

$$\bar{\tau}_n^\rho:=\inf\left\{s\geq t_n : |X^{(a^n,t_n,x_n)}-x_n|\right\}\wedge(t_n+\rho),$$

for $X^{(a^n,t_n,x_n)}$ denoting the controlled process starting at (t_n,x_n), and τ_1^n is the first time of intervention in the impulse control α^n.

With this in mind, we now aim to show that $\bar{\tau}_n\to t_0$ in probability. Therefore, we begin by observing that, for some $t,x)\in[0,T]\times\mathcal{S}$, $\tau=\tau_{\mathcal{S}}\wedge T$, and $\tilde{\tau}\leq\tau$, the performance criterion satisfies the following inequality

$$\begin{aligned}J^{(\alpha)}(t,x) &= \mathbb{E}^{(t,x)}\left[\int_t^\tau f(s,X^{(\alpha)}(s))\,\mathrm{d}s+g(\tau,X^{(\alpha)}(\tau))+\sum_{\tau_j\leq\tau}K(\tau_j,\check{X}^{(\alpha)}(\tau_j^-),\zeta_j)\right]\\ &= \mathbb{E}^{(t,x)}\Bigg[\int_t^{\tilde{\tau}} f(s,X^{(\alpha)}(s))\,\mathrm{d}s+\sum_{\tau_j\leq\tau}K(\tau_j,\check{X}^{(\alpha)}(\tau_j^-),\zeta_j)\\ &\qquad+\mathbb{E}^{(\tilde{\tau},\check{X}^{(\alpha)}(\tilde{\tau}^-))}\left[\int_{\tilde{\tau}}^\tau f(s,X^{(\alpha)}(s))\,\mathrm{d}s+g(\tau,X^{(\alpha)}(\tau))+\sum_{\tilde{\tau}\leq\tau_j\leq\tau}K(\tau_j,\check{X}^{(\alpha)}(\tau_j^-),\zeta_j)\right]\Bigg]\\ &= \mathbb{E}^{(t,x)}\left[\int_t^{\tilde{\tau}} f(s,X^{(\alpha)}(s))\,\mathrm{d}s+\sum_{\tau_j<\tilde{\tau}}K(\tau_j,\check{X}^{(\alpha)}(\tau_j^-),\zeta_j)+J^{(\alpha)}(\tilde{\tau},\check{X}^{(\alpha)}(\tilde{\tau}^-))\right]\\ &\leq \mathbb{E}^{(t,x)}\left[\int_t^{\tilde{\tau}} f(s,X^{(\alpha)}(s))\,\mathrm{d}s+\sum_{\tilde{\tau}_j\leq\tau}K(\tau_j,\check{X}^{(\alpha)}(\tau_j^-),\zeta_j)+c(\tilde{\tau},X^{(\alpha)}(\tilde{\tau}^-)\right].\end{aligned} \tag{4.2.16}$$

Hence, from (4.2.15) and (4.2.16), it follows that

$$\begin{aligned} v(t_n,x_n) &\leq E^{(t_n,x_n)}\left[\int_{t_n}^{\bar{\tau}_n} f(s,X(s))\,\mathrm{d}s + v(\bar{\tau}_n, X(\bar{\tau}_n^-))\right] + \varepsilon_n \\ &\leq \mathbb{E}^{(t_n,x_n)}\left[\int_{t_n}^{\bar{\tau}_n} f(s,X(s))\,\mathrm{d}s + \varphi(\bar{\tau}_n, X(\bar{\tau}_n^-))\right] + \varepsilon_n, \end{aligned}$$

where X is the uncontrolled process. By Dynkin's formula on $\varphi(\bar{\tau}_n, X(\bar{\tau}_n^-)$, using $v(t_n,x_n)0 = \varphi(t_n,x_n) + \delta_n$, and with (4.2.14) for $\rho > 0$ small enough, we get

$$\begin{aligned} \delta_n &\leq \mathbb{E}^{(t_n,x_n)}\left[\int_{t_n}^{\bar{\tau}_n} \left\{f(s,X(s)) + \partial_s\varphi(s,X(s)) + \mathcal{L}\varphi(s,X(s))\right\}\mathrm{d}s\right] + \varepsilon_n \\ &\overset{(4.2.14)}{\leq} \mathbb{E}^{(t_n,x_n)}\left[\int_{t_n}^{-\bar{\tau}_n} \left(-\frac{\eta}{2}\right)\,\mathrm{d}s\right] + \varepsilon_n \\ &\leq -\frac{\eta}{2}\mathbb{E}^{(t_n,x_n)}[\bar{\tau}_n - t_n] + \varepsilon_n. \end{aligned}$$

Since $\delta_n \to 0$ as $n \to \infty$ and ε_n is arbitrary, this implies that

$$\lim_{n\to\infty} \mathbb{E}[\bar{\tau}_n] = \lim_{n\to\infty} t_n = t_0,$$

which is equivalent to $\bar{\tau}_n \to t_0$ in probability, see Chebyshev's inequality in Appendix A.2, as $\bar{\tau}_n$ is bounded both from below and above.

Now let us continue with our proof. In the following, we again make use of the stochastic continuity of $X^{(t_n,x_n)}$ up until the first impulse. We define

$$A_n(\rho) := \left\{\omega\in\Omega : \sup_{t_n\leq s\leq\bar{\tau}_n} |X^{(t_n,x_n)}(s) - x_n| \leq \rho\right\}.$$

(4.2.15) combined with the Markov property (4.2.16) gives us

$$\begin{aligned} v(t_n,x_n) \leq \mathbb{E}^{(t_n,x_n)}&\left[\int_{t_n}^{\bar{\tau}_n} f(s,X(s))\,\mathrm{d}s + K(\tau_1^n, \check{X}(\tau_1^{n-}), \zeta_1^n)1_{\{\bar{\tau}_n^\rho\geq\tau_1^n\}} + v(\bar{\tau}_n, X^{(\alpha)}(\bar{\tau}_n))\right] \\ &+ \varepsilon_n. \end{aligned} \tag{4.2.17}$$

and we now aim to find the upper bound for $v(t_n,x_n)$ in (4.2.17). To do so, we make use of indicator functions for three different cases:

$$\begin{aligned} &(I) : \{\bar{\tau}_n^\rho < \tau_1^n\} \\ &(II) : \{\bar{\tau}_n^\rho \geq \tau_1^n\} \cap A_n(\rho)^c \\ &(III) : \{\bar{\tau}_n^\rho \geq \tau_1^n\} \cap A_n(\rho) \end{aligned}$$

We note that (III) is the predominant set, i.e. it has maximum possible cardinality. To see this, note that

$$\left(\{\bar{\tau}_n^\rho \geq \tau_1^n\} \cap A_n(\rho)\right)^c \subseteq \{\tau_n^\rho < \tau_1^n\} \cup A_n(\rho)^c,$$

such that

$$\mathbb{P}(III) = \mathbb{P}\left(\{\bar{\tau}_n^\rho \geq \tau_1^n\} \cap A_n(\rho)\right)$$

$$= 1 - \mathbb{P}\big((\{\bar{\tau}_n^\rho \geq \tau_n^1\} \cup A_n(\rho))^c\big)$$
$$\geq 1 - \mathbb{P}\big(\{\bar{\tau}_n^\rho < \tau_n^1\}\big) - \mathbb{P}\big(A_n(\rho)^c\big) \tag{4.2.18}$$

Then, by Lemma B.2(iii),

$$\mathbb{P}(A_n(\rho)^c) = \mathbb{P}(\{\omega \in \Omega : \sup_{t_n \leq s \leq \bar{\tau}_n} |X^{(t_n,x_n)}(s) - x_n| > \rho\}) \to 0 \quad \text{as } n \to \infty,$$

because $(t_n, x_n) \to (t_0, x_0)$, $\bar{\tau}_n \to t_0$ in probability as $n \to \infty$, and $\bar{\tau}_n \geq t_n$ by definition. Moreover, on the event $\{\bar{\tau}_n^\rho < \tau_1^n\}$, let $\{\hat{\varepsilon}_n\}_{n \geq 0}$ be any time sequence such that

$$\{\bar{\tau}_n^\rho < \tau_1^n\} \subseteq \{\bar{\tau}_n^\rho < \hat{\varepsilon}_n\} \cup \{\tau_1^n \geq \hat{\varepsilon}_n\}.$$

Then,

$$1 - \mathbb{P}(\bar{\tau}_n^\rho < \tau_1^n) \geq \mathbb{P}(\bar{\tau}_n^\rho < \hat{\varepsilon}_n) - \mathbb{P}(\tau_1^n \geq \hat{\varepsilon}_n).$$

Choosing $\hat{\varepsilon}_n \downarrow t_0$ and recalling that $\bar{\tau}_n \to t_0$ in probability as $n \to \infty$, we note that

$$\mathbb{P}(\tau_1^n \geq \hat{\varepsilon}_n) \to 0 \quad \text{as } n \to \infty$$

and that, for $\rho' < \rho$,

$$\{\bar{\tau}_n^\rho < \hat{\varepsilon}\} \subset \Big\{ \sup_{t_n \leq s \leq \varepsilon_n} |X^{(t_n,x_n)} - x_n| > \rho' \Big\},$$

where the inclusion comes from the definition of $\bar{\tau}_n^\rho$ as the first exit from the ball $B(x_n, \rho)$ truncated by $t_n + \rho$. Since $\hat{\varepsilon} \downarrow t_0$ and $t_n \leq \bar{\tau}_n^\rho \leq \bar{\tau}_n \to t_0$ in probability as $n \to \infty$, we have by Lemma B.2(iii), that

$$\mathbb{P}\big(\sup_{t_n \leq s \leq \varepsilon_n} |X^{(t_n,x_n)} - x_n| > \rho' \big) \to 0 \quad \text{as } n \to \infty.$$

Hence,

$$\mathbb{P}(\bar{\tau}_n^\rho < \hat{\varepsilon}_n) \to 0 \quad \text{as } n \to \infty$$

and (4.2.18) becomes

$$\begin{aligned} \mathbb{P}(III) &\geq 1 - \mathbb{P}(\{\bar{\tau}_n^\rho < \tau_1^n\}) - \mathbb{P}(A_n(\rho)^c) \\ &\geq 1 - \mathbb{P}(\bar{\tau}_n^\rho < \hat{\varepsilon}_n) - \mathbb{P}(\tau_1^n \geq \hat{\varepsilon}_n) - \mathbb{P}(A_n(\rho)^c) \longrightarrow 1 \quad \text{as } n \to \infty. \end{aligned}$$

In particular,

$$1_{\{III\}} \to 1 \quad \text{a.s.} \quad \text{as } n \to \infty.$$

Thus, using that if there is an impulse in $\bar{\tau}_n$, i.e. $\bar{\tau}_n^\rho \geq \tau_1^n$, then

$$v(\bar{\tau}_n, X(\bar{\tau}_n)) + K(\bar{\tau}_n, \check{X}(\bar{\tau}_n^-)) \leq \mathcal{M}v(\bar{\tau}_n, \check{X}(\bar{\tau}_n^-)),$$

and

$$\begin{aligned} v(t_n, x_n) \leq & \sup_{\substack{|t'-t_0|<\rho \\ |y'-x_0|<\rho}} f(t', y')\mathbb{E}[\bar{\tau}_n - t_0] + \mathbb{E}\big[|v(\bar{\tau}_n, X(\bar{\tau}_n)|1_{\{I\}}\big] \\ & + \mathbb{E}\big[|\mathcal{M}v(\bar{\tau}_n, \check{X}(\bar{\tau}_n^-))|1_{\{II\}}\big] + \sup_{\substack{|t'-t_0|<\rho \\ |y'-x_0|<\rho}} \mathcal{M}v(t', y')\mathbb{E}[1_{\{III\}}]. \end{aligned}$$

To prove the boundedness in term (I) (uniform in n), we can assume w.l.o.g. that $\nu(\mathbb{R}^\ell) = 1$, and consider only jumps bounded away from 0 for $x_0 > 0$. Then, by Assumption (E1),

$$\mathbb{E}\big[\sup_n |v(\bar{\tau}_n, X(\bar{\tau}_n))|\big] \le C\mathbb{E}[1 + (x_0, 2\rho + Y)^m]$$

for a jump Y with distribution $\nu^{\gamma(t_0+2\rho, x_0+2\rho, \cdot)}$, which is finite by the definition of $\mathcal{PB}$. The same is true for $\mathcal{M}v(\bar{\tau}_n, X(\bar{\tau}_n^-)$.

Sending $n \to \infty$, the terms containing f and (I) in (4.2.18) converge to 0 by the dominated convergence theorem. For the term containing (II) in (4.2.18), a general version of the dominated convergence theorem shows that it is bounded by

$$\mathbb{E}\big[\limsup_{n\to\infty} |\mathcal{M}v(\bar{\tau}_n, \check{X}(\tau_n^-))| 1_{\{A_n(\rho)^c\}}\big],$$

and the term with (III) in (4.2.18) becomes $\sup \mathcal{M}v(t', y')$. Now we let $\rho \to 0$ and obtain using (4.2.18) and Lemma 4.7(ii):

$$v^*(t_0, x_0) \le \lim_{\rho \downarrow 0} \sup_{\substack{|t'-t_0|<\rho \\ |y'-x_0|<\rho}} \mathcal{M}v(t', y') = (\mathcal{M}v)^*(t_0, x_0) \le \mathcal{M}v^*(t_0, x_0).$$

However, this contradicts (4.2.12). As such (4.2.13) must be false and (4.2) must hold. Hence, by Definition 4.5, the value function v is a viscosity solution. □

Thus, with Theorem 4.8, we have established the existence of a viscosity solution to the QVI without relying on classical $C^{1,2}$ regularity. In the next chapter, we show that this viscosity solution is, indeed, unique.

CHAPTER

FIVE

COMPARISON AND UNIQUENESS FOR THE IMPULSE-CONTROL QVI

With the proof of the existence in the previous chapter, we now aim to prove that the previously established viscosity solution is unique. The main result of this chapter is the *Comparison Principle*, which subsequently lets us prove the uniqueness of the viscosity solution. However, the proof of the Comparison Principle requires a few additional assumptions, a transition from the original defintion of the viscosity solution given by Definition 4.5, and a maximum principle. These topics are treated in Chapters 5.1, 5.2, and 5.3 respectively. Chapter 5.4 then proves the Comparison Principle and the uniqueness result.

As in Chapter 4, this chapter closely follows the work of Seydel [92, Section 5]. In addition, we also use [7]. It should be noted that although many of the results provided in this chapter are taken directly from [92], some proofs were not as detailed as the ones we provide. Moreover, the parabolic, non-local Jensen-Ishii Lemma 5.11 we apply to prove the Comparison Principle may be seen as a combination of the works of Barles and Imbert [7, Lemma 1], Jakobsen and Karlsen [60], and Crandall, Ishii and Lions [31, Theorem 8.3].

5.1 Preliminary results and further assumptions

Throughout this chapter, v no longer denotes the value function, and some other variables may serve new purposes as well. With this in mind, recall the parabolic QVI (4.1.3) and the infinitesimal generator (4.1.2). As we study the infinitesimal generator $\mathcal{L}$ more closely in this chapter, we introduce, for $0 < \delta < 1$, $t \in [0, T)$, $x, p \in \mathbb{R}^d$, $X \in \mathbb{R}^{d\times d}$, and $r, l \in \mathbb{R}$, the following functionals:

$$
\begin{aligned}
F(t,x,p,X,l) &= -\frac{1}{2}\operatorname{Tr}\{\sigma(t,x)\sigma^\top(t,x)X\} - \langle\mu(t,x),p\rangle - f(t,x) - l \\
\mathcal{I}^{1,\delta}[t,x,\varphi] &= \int_{|z|<\delta} \left[\varphi(t,x+\gamma(t,x,z)) - \varphi(t,x) - \langle\nabla_x\varphi(t,x),\gamma(t,x,z)\rangle\right]\nu(\,\mathrm{d}z)
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{I}^{2,\delta}[t,x,p,\varphi] &= \int\limits_{|z|\geq\delta} \left[\varphi(t,x+\gamma(t,x,z)) - \varphi(t,x) - \langle p, \gamma(t,x,z)\rangle 1_{\{|z|<1\}}\right]\nu(\,\mathrm{d}z) \\
\mathcal{I}[t,x,\varphi] &= \mathcal{I}^{1,\delta}[t,x,\varphi] + \mathcal{I}^{2,\delta}[t,x,\nabla_x\varphi(t,x),\varphi].
\end{aligned}
$$

Then, the problem 4.1.3 now reads as

$$
\begin{aligned}
&\min\Big\{F\big(t,x,\nabla_x u(t,x), H_x u(t,x), \mathcal{I}[t,x,u(t,\cdot)]\big), \\
&\qquad\qquad u(t,x) - \mathcal{M}u(t,x)\Big\} = 0, \quad \text{for } (t,x)\in\mathcal{S} \\
&\min\big\{u(t,x)-g(t,x), u(t,x)-\mathcal{M}u(t,x)\big\} = 0, \quad \text{for } (t,x)\in\partial^+\mathcal{S}_T, \qquad (5.1.1)
\end{aligned}
$$

where the notation $u(t,\cdot)$ in the integral means that nonlocal terms different from x are used on u. Note that the functional F does not directly depend on the zeroth order derivative (i.e. the function u itself) in our case. However, results provided in this chapter are still applicable to cases when the zeroth order derivative is present in the functional F, see [92, Section 5]. Properties of the functionals F and $\mathcal{I}$ can be found below.

Remark 5.1 ([92, Remark 5.1]). The following properties hold for the functionals F and $\mathcal{I}$:

(P1) Ellipticity of F: $F(t,x,p,X,l) \leq F(t,x,p,Y,k)$ if $X \geq Y$ and $l \geq k$.

(P2) Translation invariance: For $l\in\mathbb{R}$ constant, we have $u-\mathcal{M}u = (u+l)-\mathcal{M}(u+l)$ and $\mathcal{I}[t,x,\varphi] = \mathcal{I}[t,x,\varphi+l]$.

(P3) The map $l\mapsto F(t,x,p,X,l)$ is continuous in the sense that

$$
\big|F(t,x,p,X,l) - F(t,x,p,X,k)\big| \leq |l-k|.
$$

For $(t,x)\in[0,T)\times\mathcal{S}$ and $\delta>0$, define the region $U_\delta := U_\delta(t,x) = \{\gamma(t,x,z) : |z|<\delta\}$. We note that the set U_δ facilitates the splitting of the integral $\mathcal{I}[t,x,\varphi]$ in the sense that changing φ on U_δ affects $\mathcal{I}^{1,\delta}[t,x,\varphi]$ only and vice versa. In addition, we consider the following assumptions for some arbitrary $\delta>0$:

Assumption 5.1 ([92, Assumption 2.4])**.** (B1) For $t\in[0,T)$ and $x,y\in\mathbb{R}^d$, the following inequalities hold:

$$
\begin{aligned}
&\int\limits_{\mathbb{R}^\ell} |\gamma(t,x,z)-\gamma(t,y,z)|^2\nu(\,\mathrm{d}z) < C|x-y|^2, \\
&\int\limits_{|z|\geq 1} |\gamma(t,x,z)-\gamma(t,y,z)|\nu(\,\mathrm{d}z) < C|x-y|.
\end{aligned}
$$

(B2) Let $f(\cdot,\cdot)$, $\mu(\cdot,\cdot)$, and $\sigma(\cdot,\cdot)$ be locally Lipschitz continuous, i.e. for each point $(t_0,x_0)\in[0,T)\times\mathcal{S}$, there is neighbourhood $U\ni(t_0,x_0)$, where U is open in $[0,T)\times\mathbb{R}^d$, and a constant C such that

$$
|f(t,x)-f(t,y)| \leq C|x-y| \quad \text{for all } (t,x),(t,y)\in U,
$$

and likewise for μ and σ.

Let us briefly discuss conditions (B1) and (B2). (B1) imposes an "integrated" Lipschitz condition on the jump amplitude γ, which ensures that the nonlocal Lévy integral varies continuously when comparing it to nearby points. (B2) is the corresponding local Lipschitz condition for the diffusion, drift, and running payoff functions. Together, these assumptions provide the regularity needed to apply the doubling-of-variables argument in the proof of the comparison principle, both for the local differential terms and for the nonlocal jump term in the infinitesimal generator $\mathcal{L}$.

make the nonlocal and local parts of the QVI stable under the doubling-of-variables argument used to prove the Comparison Principle.

We now consider the following result, which will be useful for the maximum principle in Chapter 5.3.

Proposition 5.2 ([92, Proposition 5.1])**.** *Let $\{t_k\}_{k\geq 0} \subset [0,T)$ and $\{x_k\}_{k\geq 0}, \{p_k\}_{k\geq 0} \subset \mathbb{R}^d$, with $t_k \to t \in [0,T)$, $x_k \to x \in \mathbb{R}^d$, $p_k \to p$.*

(i) If $u \in \mathcal{PB} \cap USC([0,T) \times \mathbb{R}^d)$ and $v \in \mathcal{PB} \cap LSC([0,T) \times \mathbb{R}^d)$, with $u(t_k, x_k) \to u(t,x)$ and $v(t_k, x_k) \to v(t,x)$, then

$$\limsup_{k\to\infty} \mathcal{I}^{2,\delta}[t_k, x_k, p_k, u(t_k \cdot)] \leq \mathcal{I}^{2,\delta}[t,x,p,u(t,\cdot)],$$
$$\liminf_{k\to\infty} \mathcal{I}^{2,\delta}[t_k, x_k, p_k, v(t_k \cdot)] \geq \mathcal{I}^{2,\delta}[t,x,p,v(t,\cdot)].$$

Moreover, for $\{\varphi\}_{k\geq 0}, \{\psi_k\}_{k\geq 0} \subset \mathcal{PB} \cap C^{1,2,}([0,T) \times \mathbb{R}^d)$ with $\varphi_k \to u$ and $\psi_k \to v$ monotonously, $\varphi_k(t_k, x_k) \to u(t,x)$, and $\psi_k(t_k, x_k) \to v(t,x)$,

$$\limsup_{k\to\infty} \mathcal{I}^{2,\delta}[t_k, x_k, p_k, \varphi(\cdot)] \leq \mathcal{I}^{2,\delta}[t,x,p,u(t,\cdot)],$$
$$\liminf_{k\to\infty} \mathcal{I}^{2,\delta}[t_k, x_k, p_k, \psi_k(\cdot)] \geq \mathcal{I}^{2,\delta}[t,x,p,\varphi(\cdot)].$$

(ii) If $\varphi \in C^{1,2}([0,T) \times \mathbb{R}^d)$, then $(t,x) \mapsto \mathcal{I}^{1,\delta}[t,x,\varphi(\cdot)]$ is continuous. Moreover, for $\{\varphi\}_{k\geq 0} \subset C^{1,2}([0,T) \times \mathbb{R}^d)$ with $\varphi_k \to \varphi$ monontously, $\varphi_k = \varphi$ in a neighbourhood of (t,x), then

$$\lim_{k\to\infty} \mathcal{I}^{1,\delta}[t_k, x_k, \varphi_k(\cdot)] = \mathcal{I}^{1,\delta}[t_k, x_k, \varphi(\cdot)].$$

(iii) If $u \in \mathcal{PB} \cap \mathrm{USC}([0,T) \times \mathbb{R}^d)$ and $v \in \mathcal{PB} \cap \mathrm{LSC}([0,T) \times \mathbb{R}^d)$, $\{\varphi_k\}_{k\geq 0}, \{\psi_k\}_{k\geq 0} \subset \mathcal{PB} \cap C^{1,2}([0,T) \times \mathbb{R}^d)$ with $\varphi_k \to u$ and $\psi_k \to v$ monotonously, $\varphi_k = u \in C^{1,2}$ and $\psi_k = v \in C^{1,2}$ in an environment of (t,x), then

$$\liminf_{k\to\infty} F\big(t,x,p,X,\mathcal{I}[t_k,x_k,\varphi_k(\cdot)]\big) \geq F\big(t,x,p,X,\mathcal{I}[t,x,u(t,\cdot)]\big) \tag{5.1.2}$$

$$\limsup_{k\to\infty} F\big(t,x,p,X,\mathcal{I}[t_k,x_k,\psi_k(\cdot)]\big) \leq F\big(t,x,p,X,\mathcal{I}[t,x,v(t,\cdot)]\big). \tag{5.1.3}$$

Proof. We consider cases (i)-(iii) separately.

(i) For $u \in$ USC and (t_k, x_k) fixed for k, define the sequence of functions $\{h_k\}_{k\geq 0}$ by

$$h_k(z) := u(t_k, x_k + \gamma(t_k, x_k, z)) - u(t_k, x_k) - \langle p_k, \gamma(t_k, x_k, z)\rangle 1_{\{|z|<1\}}.$$

Since $u(t_k, x_k) \to u(t, x)$ and γ is assumed to be continuous by (V3), it follows that $h_k(z) \to h(z)$ for every $z \in \mathbb{R}^d$. Moreover, as $u \in \mathcal{PB}$, there exists some function $h' \in \mathcal{PB}$ such that h is dominated, i.e.

$$h_k(z) \leq h'(z)$$

for all $k \in \mathbb{N}$ and all $z \in \mathbb{R}^d$. Then, by Fatou's Lemma, see Appendix A.4,

$$\begin{aligned}
\int\limits_{|z|\geq\delta} \big(h'(z) - h(z)\big)\nu(\,\mathrm{d}z) &= \int\limits_{|z|\geq\delta} \liminf_{k\to\infty} \big(h'(z) - h_k(z)\big)\nu(\,\mathrm{d}z) \\
&\leq \liminf_{k\to\infty} \int\limits_{|z|\geq\delta} \big(h'(z) - h_k(z)\big)\nu(\,\mathrm{d}z) \\
&= \int\limits_{|z|\geq\delta} h'(z)\nu(\,\mathrm{d}z) - \limsup_{k\to\infty} \int\limits_{|z|\geq\delta} h_k(z)\nu(\,\mathrm{d}z),
\end{aligned}$$

or, equivalently,

$$\limsup_{k\to\infty} \int\limits_{|z|\geq\delta} h_k(z)\nu(\,\mathrm{d}z) \leq \int\limits_{|z|\geq\delta} h(z)\nu(\,\mathrm{d}z).$$

Hence,

$$\begin{aligned}
&\limsup_{k\to\infty} \mathcal{I}^{2,\delta}[t_k, x_k, p_k, u(t_k, \cdot)] \\
&\quad = \limsup_{k\to\infty} \int\limits_{|z|\geq\delta} \big(u(t_k, x_k + \gamma(t_k, x_k, z)) - u(t_k, x_k) - \langle p_k, \gamma(t_k, x_k, z)\rangle 1_{\{|z|<1\}}\big)\nu(\,\mathrm{d}z) \\
&\quad \leq \int\limits_{|z|\geq\delta} \big(u(t, x + \gamma(x, z)) - u(t, x) - \langle p, \gamma(t, x, z)\rangle 1_{\{|z|<1\}}\big)\nu(\,\mathrm{d}z) \\
&\quad = \mathcal{I}^{2,\delta}[t, x, p, u(t_k, \cdot)].
\end{aligned}$$

For the LSC case, we have

$$\begin{aligned}
&\liminf_{k\to\infty} \mathcal{I}^{2,\delta}[t_k, x_k, p_k, v(t_k, \cdot)] \\
&\quad = \liminf_{k\to\infty} \int\limits_{|z|\geq\delta} \big(v(t_k, x_k + \gamma(t_k, x_k, z)) - v(t_k, x_k) - \langle p_k, \gamma(t_k, x_k, z)\rangle 1_{\{|z|<1\}}\big)\nu(\,\mathrm{d}z) \\
&\quad \geq \int\limits_{|z|\geq\delta} \liminf_{k\to\infty} \big(v(t_k, x_k + \gamma(t_k, x_k, z)) - v(t_k, x_k) - \langle p_k, \gamma(t_k, x_k, z)\rangle 1_{\{|z|<1\}}\big)\nu(\,\mathrm{d}z) \\
&\quad = \int\limits_{|z|\geq\delta} \big(v(t, x + \gamma(t, x, z)) - v(t, z) - \langle p, \gamma(t, x, z)\rangle 1_{\{|z|>1\}}\big)\nu(\,\mathrm{d}z)
\end{aligned}$$

$$= \mathcal{I}^{2,\delta}[t, x, p, v(t, \cdot)].$$

where the inequality results from the direct application of Fatou's Lemma.

For the second assertion, the result for φ_k follows from the monotonous property of $\{\varphi_k\}_{k\geq 0}$ and the assumption that φ_k converges pointwise to u. Applying Dini's Theorem, see [38, Theorem 7.3], one finds that the convergence is uniform, which results in

$$\limsup_{k\to\infty} \varphi_k(t_k, x_k) \leq u(t, x),$$

and

$$\limsup_{k\to\infty} \varphi_k(t_k, x_k + \gamma(t_k, x_k, z)) \leq u(t, x).$$

Hence (5.1.2) holds. A similar logic is applied to obtain (5.1.3).

(ii) Outside of the singularity of ν, the result follows from the first result (i). In the neighbourhood of (t, x), the Taylor expansion (4.1.4) gives the upper bound for the application of DCT. Note that the local boundedness of $U_\delta(t, x)$ holds by (V4) which states that for each fixed $\delta > 0$, the set of small jump sizes is uniformly bounded locally in (t, x).

(iii) This is a direct result from (i), (ii), and (P1).

□

5.2 Equivalent definitions of viscosity solutions

Recall the functionals F, $\mathcal{I}^{1,\delta}$, $\mathcal{I}^{2,\delta}$, and $\mathcal{I}$ defined in the previous section. We now restate Definition 4.5 (Viscosity Solution) using the aforementioned functionals.

Definition 5.3 ([92, Definition 5.2] Viscosity solution 1)**.** A function $u \in \mathcal{PB}$ is a (viscosity) subsolution of (5.1.1) if for all $(t_0, x_0) \in [0, T) \times \mathbb{R}^d$ and $\varphi \in \mathcal{PB} \cap C^{1,2}([0, T) \times \mathbb{R}^d)$ such that $u^* - \varphi$ has a global maximum in (t_0, x_0),

$$\min\Big\{ - \partial_t\varphi + F\big(t_0, x_0, \nabla_x\varphi, H_x\varphi, \mathcal{I}[t_0, x_0, \varphi(t_0, \cdot)]\big),$$
$$u^* - \mathcal{M}u^*\Big\} \leq 0 \quad \text{in } (t_0, x_0) \in [0, T) \times \mathcal{S}$$
$$\min\big\{u^* - g, u^* - \mathcal{M}u^*\big\} \leq 0 \quad \text{in } (t_0, x_0) \in \partial^+\mathcal{S}_T.$$

A function $u \in \mathcal{PB}$) is a (viscosity) supersolution of (5.1.1) if for all $(t_0, x_0) \in [0, T) \times \mathbb{R}^d$ and $\varphi \in \mathcal{PB} \cap C^{1,2}([0, T) \times \mathbb{R}^d)$ such that $u_* - \varphi$ has a global minimum in (t_0, x_0),

$$\min\Big\{ - \partial_t\varphi + F\big(t_0, x_0, \nabla_x\varphi, H_x\varphi, \mathcal{I}[t_0, x_0, \varphi(t_0, \cdot)]\big),$$
$$u_* - \mathcal{M}u_*\Big\} \geq 0 \quad \text{in } (t_0, x_0) \in [0, T) \times \mathcal{S}$$
$$\min\big\{u_* - g, u_* - \mathcal{M}u_*\big\} \geq 0 \quad \text{in } (t_0, x_0) \in \partial^+\mathcal{S}_T.$$

A function u is a viscosity solution if it is sub- and supersolution.

We now introduce another definition of the viscosity solution. We motivate this by noting that the Lévy measure is typically singular near the origin, so we split $\mathcal{I}$ into a small-jump part $\mathcal{I}^{1,\delta}$, evaluated on a smooth test function φ on points $(t,x) \in [0,T] \times U_\delta(t,x)$, and a large-jump part $\mathcal{I}^{2,\delta}$, evaluated on u and preserving monotonicity and semincontinuity under limits.

Definition 5.4 ([92, Definition 5.3] Viscosity solution 2)**.** A function $u \in \mathcal{PB}$ is a (viscosity) subsolution of (5.1.1) if for all $(t_0,x_0) \in [0,T) \times \mathbb{R}^d$ and $\varphi \in \mathcal{PB} \cap C^{1,2}([0,T) \times \mathbb{R}^d)$ such that $u^* - \varphi$ has a maximum in (t_0,x_0) on $[0,T) \times U_\delta(t_0,x_0)$,

$$\begin{aligned}\min\Big\{ -\partial_t\varphi(t_0,x_0) + F\big(t_0,x_0,\nabla_x\varphi, H_x\varphi, \mathcal{I}^{1,\delta}[t_0,x_0,\varphi(t_0,\cdot)] + \mathcal{I}^{2,\delta}[t_0,x_0,\nabla_x\varphi(t_0,\cdot),u^*(\cdot)]\big),&\\ u^* - \mathcal{M}u^*\Big\} \le 0 \quad &\text{in } (t_0,x_0) \in [0,T) \times \mathcal{S}\\ \min\big\{u^* - g, u^* - \mathcal{M}u^*\big\} \le 0 \quad &\text{in } (t_0,x_0) \in \partial^+\mathcal{S}_T\end{aligned}$$

A function $u \in \mathcal{PB}$ is a (viscosity) supersolution of (5.1.1) if for all $(t_0,x_0) \in [0,T) \times \mathbb{R}^d$ and $\varphi \in \mathcal{PB} \cap C^{1,2}([0,T) \times \mathbb{R}^d)$ such that $u_* - \varphi$ has a minimum in (t_0,x_0) on $[0,T) \times U_\delta(t_0,x_0)$,

$$\begin{aligned}\min\Big\{ -\partial_t\varphi(t_0,x_0) + F\big(t_0,x_0,\nabla_x\varphi, H_x\varphi, \mathcal{I}^{1,\delta}[t_0,x_0,\varphi(t_0,\cdot)] + \mathcal{I}^{2,\delta}[t_0,x_0,\nabla_x\varphi(t_0,\cdot),u_*(\cdot)]\big),&\\ u_* - \mathcal{M}u_*\Big\} \ge 0 \quad &\text{in } (t_0,x_0) \in [0,T) \times \mathcal{S}\\ \min\big\{u_* - g, u_* - \mathcal{M}u_*\big\} \ge 0 \quad &\text{in } (t_0,x_0) \in \partial^+\mathcal{S}_T\end{aligned}$$

A function u is a viscosity solution if it is sub- and supersolution.

With the second definition of viscosity solutions in place, we aim to find a third way to define viscosity solutions in order to apply the parabolic, nonlocal Jensen-Ishii Lemma presented in the subchapter to follow. We therefore begin by considering parabolic semijets and their limiting counterparts in Definitions 5.5 and 5.6.

Definition 5.5 (Set of Parabolic Semijets)**.** Let $u : [0,T] \times \mathbb{R}^d \to \mathbb{R}$ be an USC function and $v : [0,T] \times \mathbb{R}^d \to \mathbb{R}$ be a LSC function. We define the set of parabolic superjets of u at $(t,x) \in [0,T] \times \mathbb{R}^d$ with

$$\begin{aligned}J^+u(t,x) = \Big\{&(a,p,X) \in \mathbb{R} \times \mathbb{R}^d \times \mathbb{S}^d : u(t+s,x+z) \le u(t,x) + a \cdot s + \langle p,z\rangle\\ &+ \frac{1}{2}\langle Xz,z\rangle + o(|s| + |z|^2), \text{ as } s,z \to 0, (t+s,z) \in [0,T) \times \mathbb{R}^d\Big\},\end{aligned}$$

and the set of parabolic subjets of v at $(t,x) \in [0,T] \times \mathbb{R}^d$ with

$$\begin{aligned}J^-u(t,x) = \Big\{&(a,p,X) \in \mathbb{R} \times \mathbb{R}^d \times \mathbb{S}^d :\ u(t+s,x+z) \ge u(t,x) + a \cdot s + \langle p,z\rangle\\ &+ \frac{1}{2}\langle Xz,z\rangle + o(|s| + |z|^2), \text{ as } s,z \to 0, (t+s,z) \in [0,T) \times \mathbb{R}^d\Big\}.\end{aligned}$$

Definition 5.6 (Limiting parabolic semijets)**.** Recall the definition of parabolic super- and subjets (see Definition 5.5. Let $u : [0,T] \times \mathbb{R}^d \to \mathbb{R}$ be an USC function and $v : [0,T] \times \mathbb{R}^d \to \mathbb{R}$ be a LSC function. We define the set of limiting superjets of u at $(t,x) \in [0,T] \times \mathbb{R}^d$ with

$$\bar{J}^+u(t,x) = \Big\{(a,p,X) \in \mathbb{R} \times \mathbb{R}^d \times \mathbb{S}^d :\ \text{there exist } (t_k,x_k,a_k,p_k,X_k) \to (t,x,a,p,X),$$

$$(a_k, p_k, X_k) \in J^+u(t_k, x_k) \text{ such that } u(t_k, x_k) \to u(t, x)\Big\},$$

and the set of limiting subjets of u at $(t,x) \in [0,T] \times \mathbb{R}^d$ with

$$\bar{J}^-u(t,x) = \Big\{(a,p,X) \in \mathbb{R} \times \mathbb{R}^d \times \mathbb{S}^d : \text{ there exist } (t_k, x_k, a_k, p_k, X_k) \to (t, x, a, p, X),$$
$$(a_k, p_k, X_k) \in J^-u(t_k, x_k) \text{ such that } u(t_k, x_k) \to u(t, x)\Big\}.$$

The introduction of semijets provides a convenient way to generalise Taylor derivatives and allows us to describe the local behaviour of continuous, but potentially non-differentiable functions by acting as upper or lower parabolic approximations. While semijets decribe local lower and upper approximations at a given point, limiting semijets enlarge this notion by allowing limits of such approximations along convergent sequences, and are therefore useful when it comes to upper and lower semicontinuous envelopes. This brings us to the last definition of the viscosity solution in the form of Definition 5.7, which, as we will see in Proposition 5.8, is equivalent to all previous definitions of the viscosity solution. Moreover, Definition 5.7 will play a big part in the proof of uniqueness in Chapter 5.4.

Definition 5.7 ([92, Definition 5.5] Viscosity solution 3)**.** A function $u \in \mathcal{PB}$ is a viscosity subsolution of (5.1.1) if for all $(t_0, x_0) \in [0,T) \times \mathbb{R}^d$ and $\varphi \in C^{1,2}([0,T) \times \mathbb{R}^d)$ such that $u^* - \varphi$ has a maximum in (t_0, x_0) on $[0,T) \times U_\delta(t_0, x_0)$ and for $(a,p,X) \in P^+u(t_0,x_0)$ with $p = \nabla_x\varphi(t_0,x_0)$ and $X \le H_x\varphi(t_0,x_0)$,

$$\min\Big\{ -a + F\big(t_0, x_0, p, X, \mathcal{I}^{1,\delta}[t_0, x_0, \varphi(t_0,\cdot)] + \mathcal{I}^{2,\delta}[t_0, x_0, p, u^*(t_0,\cdot)]\big),$$
$$u^* - \mathcal{M}u^*\Big\} \le 0 \quad \text{in } (t_0,x_0) \in [0,T) \times \mathcal{S}$$
$$\min\big\{u^* - g, u^* - \mathcal{M}u^*\big\} \le 0 \quad \text{in } (t_0,x_0) \in \partial^+\mathcal{S}_T$$

A function $u \in \mathcal{PB}$ is a (viscosity) supersolution of (5.1.1) if for all $(t_0, x_0) \in [0,T) \times \mathbb{R}^d$ and $\varphi \in C^{1,2}([0,T) \times \mathbb{R}^d)$ such that $u_* - \varphi$ has a minimum in (t_0,x_0) on $[0,T) \times U_\delta(t_0,x_0)$ and for $(b,q,Y) \in P^-u(t_0,x_0)$ with $q = \nabla_x\varphi(t_0,x_0)$ and $Y \ge H_x\varphi(t_0,x_0)$,

$$\min\Big\{ -b + F\big(t_0, x_0, q, Y, \mathcal{I}^{1,\delta}[t_0, x_0, \varphi(t_0,\cdot)] + \mathcal{I}^{2,\delta}[t_0, x_0, q, u_*(t_0,\cdot)]\big),$$
$$u_* - \mathcal{M}u_*\Big\} \ge 0 \quad \text{in } (t_0,x_0) \in [0,T) \times \mathcal{S}$$
$$\min\big\{u_* - g, u_* - \mathcal{M}u_*\big\} \ge 0 \quad \text{in } (t_0,x_0) \in \partial^+\mathcal{S}_T$$

A function u is a viscosity solution if it is sub- and supersolution.

Proposition 5.8 ([92, Propositions 5.4 and 5.6])**.** *The Definitions 5.3, 5.4, and 5.7 are equivalent.*

Proof. For the proof of the equivalence of Definitions 5.3 and 5.4, we refer to [92, Proposition 5.4]. For the proof of the equivalence of Definitions 5.4 and 5.7, we refer to [92, Proposition 5.6] and [7]. □

Remark 5.9. Note that Definitions 5.3, 5.4, and 5.7 in Seydel [92] are all parabolic versions of Definitions 5.2, 5.3, and 5.5. However, the proof remains somewhat unchanged. Moreover, we note that Barles and Imbert [7, Chapter 1] also cover the different definitions of the viscosity solution.

5.3 The maximum principle

We now introduce a concept that plays a central role in the proof of the Comparison Principle, namely the *Maximum Principle.* In the context of local, fully nonlinear degenerate parabolic PDEs, this principle was first developed by Crandall and Ishii [30]. The maximum principle was then extended for nonlocal PIDEs by Jakobsen and Karlsen [60] and later by Barles and Imbert [7]. In what follows, we refer to this result as the parabolic, nonlocal Jensen–Ishii lemma.

We begin by considering a few properties of the intervention operator $\mathcal{M}$, which will be needed for the Comparison Principle.

Lemma 5.10 ([92, Lemma 5.7])**.** *Recall the intervention operator $\mathcal{M}$ in Definition 3.7. We have that*

(i) $\mathcal{M}$ is convex, i.e. for $\lambda \in [0,1]$ and functions $a, b \in \mathcal{PB}$,

$$\mathcal{M}\big(\lambda a(t,x) + (1-\lambda)b(t,x)\big) \leq \lambda \mathcal{M}a(t,x) + (1-\lambda)\mathcal{M}b(t,x),$$

for $(t,x) \in [0,T) \times \mathbb{R}^d$.

(ii) For $\lambda > 0$ and functions $a, b \in \mathcal{PB}$,

$$\mathcal{M}\big(- \lambda(t,x) + (1-\lambda)b(t,x)\big) \leq \lambda \mathcal{M}a(t,x) + (1+\lambda)\mathcal{M}b(t,x).$$

Proof. The proofs for (i) and (ii) are provided separately.

(i) For $(t,x) \in [0,T) \times \mathbb{R}^d$,

$$\begin{aligned}
&\mathcal{M}\big(\lambda a(t,x) + (1-\lambda)b(t,x)\big) \\
&\quad = \sup_{\zeta \in \mathcal{Z}(t,x)} \Big\{\lambda\big(a(t,\Gamma(x,\zeta)) + K(t,x,\zeta)\big) + (1-\lambda)\big(b(t,\Gamma(x,\zeta)) + K(t,x,\zeta)\big)\Big\} \\
&\quad \leq \lambda \sup_{\zeta \in \mathcal{Z}(t,x)} \Big\{\big(a(t,x,\Gamma(x,\zeta)) + K(t,x,\zeta)\big\} \\
&\qquad\quad + (1-\lambda) \sup_{\zeta \in \mathcal{Z}(t,x)} \Big\{\big(b(t,x,\Gamma(x,\zeta)) + K(t,x,\zeta)\big\} \\
&\quad = \lambda \mathcal{M}a(t,x) + (1-\lambda)\mathcal{M}b(t,x).
\end{aligned}$$

(ii) For $(t,x) \in [0,T) \times \mathbb{R}^d$,

$$\begin{aligned}
&\mathcal{M}\big(- \lambda a(t,x) + (1+\lambda)b(t,x)\big) \\
&\quad = \sup_{\zeta \in \mathcal{Z}(t,x)} \Big\{ - \big(\lambda a(t,\Gamma(x,\zeta)) + K(t,x,\zeta)\big) + (1+\lambda)\big(b(t,\Gamma(x,\zeta)) + K(t,x,\zeta)\big)\Big\} \\
&\quad \geq -\lambda \sup_{\zeta \in \mathcal{Z}(t,x)} \big\{a(t,\Gamma(x,\zeta)) + K(t,x,\zeta)\big\} \\
&\qquad\quad + (1+\lambda) \sup_{\zeta \in \mathcal{Z}(t,x)} \big\{b(t,\Gamma(x,\zeta)) + K(t,x,\zeta)\big\} \\
&\quad = -\lambda\, \mathcal{M}a(t,x) + (1+\lambda)\mathcal{M}b(t,x).
\end{aligned}$$

□

Let us now consider the parabolic, nonlocal Jensen-Ishii Lemma.

Lemma 5.11 (Parabolic, nonlocal Jensen-Ishii Lemma)**.** *Let $u \in \mathrm{USC}([0,T) \times \mathbb{R}^d)$, $v \in \mathrm{LSC}([0,T) \times \mathbb{R}^d)$, and $\varphi \in C^{1,2}([0,T) \times \mathbb{R}^{2d})$. If $(t_0, x_0, y_0) \in [0,T) \times \mathbb{R}^{2d}$ is a zero global maximum point of $u(t,x) - v(t,y) - \varphi(t,x,y)$ and if $p = \nabla_x\varphi(t_0, x_0, y_0)$, $q = \nabla_y\varphi(t_0, x_0, y_0)$, then, for any $K > 0$, there exists $\bar{\alpha}(K) > 0$ such that for any $0 < \alpha < \bar{\alpha}(K)$, we have: There exist sequences $t_k \to t_0$, $x_k \to x_0$, $y_k \to y_0$, $p_k \to p$, $q_k \to q$, matrices X_k and Y_k, and a sequence of functions $\{\varphi_k\}_{k\geq 0}$, converging to the function*

$$\varphi_\alpha(t,x,y) := R^\alpha[\varphi]\big(t,(x,y),(p,q)\big)$$

uniformly in $[0,T) \times \mathbb{R}^{2d}$ and in $C^{1,2}\big([0,T) \times B((x_0,y_0),K)\big)$ such that

$$u(t_k, x_k) \to u(t_0, x_0) \quad \text{and} \quad v(t_k, y_k) \to v(t_0, y_0) \tag{5.3.1}$$

$$(t_k, x_k, y_k) \text{ is a global maximum point of } u - v - \varphi_k \tag{5.3.2}$$

$$(a_k, p_k, X_k) \in J^+u(t_k, x_k) \tag{5.3.3}$$

$$(b_k, -q_k, Y_k) \in J^-v(t_k, y_k) \tag{5.3.4}$$

$$a_k - b_k = \partial_t\varphi_k(t_k, x_k, y_k) \tag{5.3.5}$$

$$-\frac{1}{\alpha}I \leq \begin{bmatrix} X_k & 0 \\ 0 & -Y_k \end{bmatrix} \leq H_{(x,y)}\varphi_k(t_k, x_k, y_k). \tag{5.3.6}$$

Here, $p_k = \nabla_x\varphi_k(t_k, x_k, y_k)$, $q_k = \nabla_y\varphi_k(t_k, x_k, y_k)$, and $\nabla_{(x,y)}\varphi_\alpha(t_0, x_0, y_0) = \nabla_{(x,y)}\varphi(t_0, x_0, y_0)$.

Remark 5.12 ([92, Remark 5.4]). The expression $\varphi_\alpha(t,x,y) = R^\alpha[\varphi]\big((x,y),(p,q)\big)$ is the "modified sup-convolution" as used by Barles and Imbert [7]. For all compacts C, φ_α converges uniformly to φ in $C^{1,2}(C)$ as $\alpha \to 0$.

Remark 5.13. Note that the above lemma is taken in the sense of Barles and Imbert [7]. More precisely, the nonlocal part, that is the block-matrix estimate with $\left(\begin{smallmatrix} X & 0 \\ 0 & -Y \end{smallmatrix}\right)$, and the sign convention is given by Lemma 1 [7] $(b, -q, Y) \in J^-v)$ is given by *Lemma 1* in [7]. The corresponding parabolic extension is stated explicitly as *Corollary 2* in [7]. For the underlying parabolic semijet theory and the local Theorem of Sums used in doubling-of-variables arguments, we refer to Crandall–Ishii–Lions [31, Theorem 8.3].

We also note that as an alternative, one may also apply the Maximum Principle for Semicontinuous Functions for PIDEs, see Jakobsen and Karlsen [60], which plays an analogous role in comparison proofs for viscosity solutions of fully nonlinear, nonlocal equations.

Having established the relevant maximum principle, we now aim to apply it to our QVI problem. The following corollary shows how the preceding result applies to the present QVI.

Corollary 5.14 ([92, Corollary 5.13], [7, Corollary 2])**.** *Assume conditions (V1) and (V2) are satisfied. Let u be a viscosity subsolution and v a viscosity supersolution of (5.1.1). Furthermore, let $\varphi \in C^{1,2}([0,T) \times \mathbb{R}^{2d})$ be a test function. If $(t_0, x_0, y_0) \in [0,T) \times \mathbb{R}^{2d}$ is a global maximum point of $u^*(t,x) - v_*(t,y) - \varphi(t,x,y)$ on $[0,T) \times \mathbb{R}^{2d}$, then, for any $\delta > 0$, there exists $\bar{\alpha}$ such that for $0 < \alpha < \bar{\alpha}$, there are $(a,p,X) \in \bar{J}^+u^*(t_0,x_0)$ and $(b,q,Y) \in \bar{J}^-v_*(t_0,y_0)$ with*

$$\min\Big\{ -a + F\big(t_0, x_0, p, X, \mathcal{I}^{1,\delta}[t_0, x_0, \varphi_\alpha(t_0, \cdot, y_0)] + \mathcal{I}^{2,\delta}[t_0, x_0, p, u^*(t_0, \cdot)]\big),$$

$$u^*(t_0,x_0) - \mathcal{M}u^*(t_0,x_0)\Big\} \leq 0,$$

and

$$\min\Big\{ -b + F\big(t_0, y_0, q, Y, \mathcal{I}^{1,\delta}[t_0, y_0, -\varphi_\alpha(t_0,x_0,\cdot)] + \mathcal{I}^{2,\delta}[t_0, y_0, q, v_*(t_0,\cdot)]\big), \\ v_*(t_0,y_0) - \mathcal{M}v_*(t_0,y_0)\Big\} \geq 0,$$

if $x_0 \in \mathcal{S}$ *or* $y_0 \in \mathcal{S}$ *respectively. Here,* $a - b = \partial_t\varphi(t_0,x_0,y_0)$, $p = \nabla_x\varphi(t_0,x_0,y_0) = \nabla_x\varphi_\alpha(t_0,x_0,y_0)$, $q = -\nabla_y\varphi(t_0,x_0,y_0) = -\nabla_y\varphi_\alpha(t_0,x_0,y_0)$, *and furthermore,*

$$-\frac{1}{\alpha}I \leq \begin{bmatrix} X & 0 \\ 0 & -Y \end{bmatrix} \leq H_{(x,y)}\varphi_\alpha(t_0,x_0,y_0) = H_{(x,y)}\varphi(t_0,x_0,y_0) + o_\alpha(1) \tag{5.3.7}$$

Proof. We begin by noting that because of the translation invariance, see $P2$ in Remark 5.1, we can assume WLOG that

$$u^*(t_0,x_0) - v_*(t_0,x_0) - \varphi(t_0,x_0,y_0) = 0.$$

Choose sequences according to Lemma 5.11 (applied for u^* and v_*), and

$$K := \max\big\{\mathrm{dist}(x_0, U_\delta(x_0)), \mathrm{dist}(y_0, U_\delta(y_0))\big\} + 1 > 0.$$

Furthermore, fix $\alpha \in (0, \bar{\alpha}(K))$. With this in mind, we split the proof into two cases: The subsolution and The supersolution cases.

Subsolution case: We begin by noting that since $x_k \to x_0$ as $k \to \infty$, $x_0 \in \mathcal{S}$, and $\mathcal{S}$ is an open set, then $x_k \in \mathcal{S}$ for k large enough.

Fix the y-coordinate to some $y_{k'}$ and note that from (5.3.2), for every $(t,x) \in [0,T) \times \mathcal{S}$,

$$u(t,x) - v(t,y_{k'}) - \varphi_k(x,y_{k'}) \leq u(t_k,x_k) - v(t_k,y_k) - \varphi_k(t_k,x_k,y_k).$$

In particular, (t_k,x_k) is a local maximiser on the neighbourhood $U_\delta(t_k,x_k)$. With $(a_k,p_k,X_k) \in J^+u(t_k,x_k)$, where J^+ is the parabolic semijet of u on (t_k,x_k), we obtain

$$p_k = \nabla_x\varphi(t_k,x_k,y_{k'}),$$

and, with (5.3.6),

$$X_k \leq H_x\varphi_k(t_k,x_k,y_{k'}).$$

As $y_{k'}$ was fixed, we may apply Definition 5.7, and, together with the translation invariance (see (P2) in Remark 5.1), we obtain

$$\min\big\{ -a_k + F\big(t_k, x_k, p_k, X_k, \mathcal{I}^{1,\delta}[t_k, x_k, \varphi(t_k,\cdot,y_k)] + \mathcal{I}^{2,\delta}[t_k, x_k, p_k, u^*(t_k,\cdot)]\big), \\ u^*(t_k,x_k) - \mathcal{M}u^*(t_k,x_k)\big\} \leq 0 \tag{5.3.8}$$

We now aim to prove the convergence of the PIDE part F in (5.3.8) as $k \to \infty$. Recall that by Lemma 5.11, $x_k \to x_0$, $u^*(t_k,x_k) \to u^*(t_0,x_0)$, $a_k \to a$, and $p_k \to p$, as $k \to \infty$. Also, the matrix sequence $\{X_k\}_{k\geq 0}$ is contained in a compact set in $\mathbb{R}^{d\times d}$ by (5.3.6) because $\varphi_k \in C^{1,2}([0,T) \times \mathbb{R}^{2d})$, so it admits a convergent subsequence to an

X satisfying (5.3.7). Moreover, by Lemma 5.11, $u^*(t_k,x_k)\to u^*(t_0,x_0)$, so it therefore remains to show that the integral part also converges as $k\to\infty$.

For the convergence of the integral term $\mathcal{I}^{1,\delta}$, by DCT, (V3), and uniform convergence,

$$\begin{aligned}
\lim_{k\to\infty}\mathcal{I}^{1,\delta}[t_k,x_k,\varphi_k] &= \lim_{k\to\infty}\int\limits_{|z|<\delta}\Big(\varphi_k\big(t_k,x_k+\gamma(t_k,x_k,z),y_k\big)-\varphi_k(t_k,x_k,y_k)\\
&\qquad\qquad -\langle\nabla_x\varphi(t_k,x_k,y_k),\gamma(t_k,x_k,z)\rangle\Big)\nu(\,\mathrm{d}z)\\
&= \int\limits_{|z|<\delta}\lim_{k\to\infty}\Big(\varphi_k\big(t_k,x_k+\gamma(t_k,x_k,z),y_k\big)-\varphi_k(t_k,x_k,y_k)\\
&\qquad\qquad -\langle\nabla_x\varphi(t_k,x_k,y_k),\gamma(t_k,x_k,z)\rangle\Big)\nu(\,\mathrm{d}z)\\
&= \int\limits_{|z|<\delta}\Big(\varphi_\alpha\big(t_0,x_0+\gamma(t_0,x_0,z),y_0\big)-\varphi(t_0,x_0,y_0)\\
&\qquad\qquad -\langle\nabla_x\varphi_\alpha(t_0,x_0,y_0),\gamma(t_0,x_0,z)\rangle\Big)\nu(\,\mathrm{d}z).
\end{aligned}$$

Furthermore, with the application of the Taylor expansion on $\varphi_k\in C^{1,2}([0,T)\times\mathbb{R}^{2d})$ about (t_k,x_k,y_k), we find the the ν-integrable upper estimate

$$\begin{aligned}
&\varphi_k\big(t_k,x_k+\gamma(t_k,x_k,z),y_k\big)-\varphi_k(t_k,x_k,y_k)-\langle\nabla_x\varphi_k(t_k,x_k,y_k),\gamma(t_k,x_k,z)\rangle\\
&\overset{\text{Taylor}}{=}\varphi_k(t_k,x_k,y_k)+\langle\nabla_x\varphi_k(t_k,x_k,y_k),\gamma(t_k,x_k,y_k)\rangle\\
&\qquad+\frac{1}{2}\gamma(t_k,x_k,z)^\top H_x\varphi_k(t_k,x_k,y_k)\,\gamma(t_k,x_k,z)\\
&\qquad-\varphi_k(t_k,x_k,y_k)-\langle\nabla_x\varphi_k(t_k,x_k,y_k),\gamma(t_k,x_k,y_k)\rangle\\
&\le\frac{1}{2}|\gamma(t_k,x_k,z)|^2|H_x\varphi_k(t_k,x_k,y_k)|,
\end{aligned}$$

where, for k large enough and some $\kappa_1,\kappa_2>0$,

$$\sup_{|(s,\xi)-(t,x)|<\kappa_1}|H_x\varphi_k(t,\xi,y)|\le\sup_{|(s,\xi)-(t,x)|<\kappa_1}|H_x\varphi_\alpha(s,\xi,y)|+\kappa_2.$$

Recalling that $\int_C|z|^2\nu(\,\mathrm{d}z)$ for all compact sets C, and that $U_\delta(t_0,x_0))\downarrow 0$ for singular $\nu(z)$, we see that $\mathcal{I}^{1,\delta}$ is bounded uniformly from above. For the second integral term $\mathcal{I}^{2,\delta}$ the upper limit is provided by Proposition 5.2(i).

Lastly, for the impulse part, we know by Lemma 4.7(ii) that $\mathcal{M}u^*$ is USC, so

$$\begin{aligned}
\liminf_{k\to\infty}\big\{u^*(t_k,x_k)-\mathcal{M}u^*(t_k,x_k)\big\}&=u^*(t_0,x_0)-\limsup_{k\to\infty}\mathcal{M}u^*(t_k,x_k)\\
&\ge u^*(t_0,x_0)-\mathcal{M}u^*(t_0,x_0).
\end{aligned}$$

Hence, combining the results thus far under the subsolution case, taking the subsequences iteratively using (5.1.2) in Proposition 5.2(iii), and taking into account the property (P2) (see Remark 5.1), we get

$$\begin{aligned}
0\ge\liminf_{k\to\infty}\min\Big\{&-a_k+F\big(t_k,x_k,p_k,X_k,\mathcal{I}^{1,\delta}[t_k,x_k,\varphi_k(t_k,\cdot,y_k)]+\mathcal{I}^{2,\delta}[t_k,x_k,p_k,u^*(t_k,\cdot)]\big),\\
&u^*(t_k,x_k)-\mathcal{M}u^*(t_k,x_k)\Big\}
\end{aligned}$$

$$
\begin{aligned}
&= \min\Big\{ \liminf_{k\to\infty} \big\{ -a_k + F\big(t_k, x_k, p_k, X_k, \mathcal{I}^{1,\delta}[t_k, x_k, \varphi_k(t_k, \cdot, y_k)] + \mathcal{I}^{2,\delta}[t_k, x_k, p_k, u^*(t_k, \cdot)]\big) \big\}, \\
&\qquad \liminf_{k\to\infty} \big\{ u^*(t_k, x_k) - \mathcal{M}u^*(t_k, x_k) \big\} \Big\} \\
&\geq \min\Big\{ -a + F\big(t_0, x_0, p, X, \mathcal{I}^{1,\delta}[t_0, x_0, \varphi(t_0, \cdot, y_0)] + \mathcal{I}^{2,\delta}[t_0, x_0, p, u^*(t_0, \cdot)]\big), \\
&\qquad u^*(t_0, x_0) - \mathcal{M}u^*(t_0, x_0) \Big\},
\end{aligned}
$$

which, by Definition 5.7, is a subsolution.

Supersolution case: The supersolution case is shown in an analogous way to the subsolution case, except that in this case we use (5.1.3) for the convergence of the PIDE part. For the impulse part, we use the fact that $\mathcal{M}v_*$ is LSC by Lemma 4.7 (i) so that

$$
\limsup_{k\to\infty} \big\{ v_*(t_k, y_k) - \mathcal{M}v_*(t_k, y_k) \big\} = v_*(t_0, y_0) - \liminf_{k\to\infty} \mathcal{M}v_*(t_k, y_k) \leq v_* - \mathcal{M}v_*(t_0, y_0).
$$

Hence,

$$
\begin{aligned}
0 \leq \min\Big\{ -b + F\big(t_0, y_0, q, Y, \mathcal{I}^{1,\delta}[t_0, y_0, -\varphi_\alpha(t_0, x_0, \cdot)] + \mathcal{I}^{2,\delta}[t_0, y_0, q, v_*(t_0, \cdot)]\big), \\
v_*(t_0, y_0) - \mathcal{M}v_*(t_0, y_0) \Big\}.
\end{aligned}
$$

□

5.4 Comparison principle and uniqueness

Before considering the Comparison Principle, consider Lemma 5.15 the following result, which is a perturbation technique needed for the proof of the Comparison Principle.

Lemma 5.15. *Assume that conditions (V1) and (V2) be satisfied. Let u be a subsolution and v a supersolution of (5.1.1). Assume that there is a function $w \in \mathcal{PB} \cap C^{1,2}([0,T) \times \mathbb{R}^d)$ and a positive function $\kappa : [0,T] \times \mathbb{R}^d$ such that*

$$
\begin{aligned}
\min\Big\{ -\big(\partial_t w - \mathcal{L}w - f, w - \mathcal{M}w\big) \Big\} &\geq \kappa \quad \text{in } [0,T) \times \mathcal{S}, \\
\min\big\{ w - g, w - \mathcal{M}w \big\} &\geq \kappa \quad \text{in } \partial^+\mathcal{S}
\end{aligned}
$$

Then, for $m \in \mathbb{N}$,

$$
v_m := \left(1 - \frac{1}{m}\right) v_* + \frac{1}{m} w
$$

is a supersolution of

$$
\begin{aligned}
\min\Big\{ -\big(\partial_t \psi - \mathcal{L}\psi - f\big), \psi - \mathcal{M}\psi \Big\} - \frac{\kappa}{m} &= 0 \qquad \text{in } [0,T) \times \mathcal{S}, \\
\min\big\{ \psi - g, \psi - \mathcal{M}\psi \big\} - \frac{\kappa}{m} &= 0 \qquad \text{in } \partial^+\mathcal{S}.
\end{aligned}
\tag{5.4.1}
$$

Similarly,

$$
u_m := \left(1 + \frac{1}{m}\right) u^* - \frac{1}{m} w
$$

is a subsolution of

$$
\begin{aligned}
\min\Big\{-\big(\partial_t\psi+\mathcal{L}\psi+f\big),\psi-\mathcal{M}\psi\Big\}+\frac{\kappa}{m}&=0 \qquad \textit{in } [0,T)\times\mathcal{S},\\
\min\big\{\psi-g,\psi-\mathcal{M}\psi\big\}+\frac{\kappa}{m}&=0 \qquad \textit{in } \partial^+\mathcal{S}.
\end{aligned}
\tag{5.4.2}
$$

Remark 5.16. Note that this is the parabolic extension to [92, Lemma 5.10]. However, the proof remains the same as in [92].

Proof of Lemma 5.15. The proof is based on the first definition of the viscosity solution, see Definition 4.5. We split the proof into two parts, corresponding to the supersolution and subsolution cases.

Supersolution case: Let $\varphi\in\mathcal{PB}\cap C^{1,2}([0,T)\times\mathbb{R}^d)$ such that $\varphi_m(t,x)=u_m(t,x)$, $(t_0,x_0)\in[0,T)\times\mathcal{S}$ such that $\varphi_m(t_0,x_0)=v_m(t_0,x_0)$ and $\varphi_m\geq v_m$ everywhere else on $[0,T)\times\mathbb{R}^d$. Choose

$$
\varphi:=\frac{m}{m-1}\left(\varphi_m-\frac{1}{m}w\right),
$$

and note that

$$
\varphi(t_0,x_0)=v_*(t_0,x_0)\quad\text{and}\quad\varphi\leq v_*.
$$

Since v is assumed to be a supersolution, we know by Definition 4.5 that

$$
-\big(\partial_t\varphi+\mathcal{L}\varphi+f\big)\geq 0\quad\text{on } [0,T)\times\mathcal{S},
$$

and it is sufficient to show that

$$
-\big(\partial_t\varphi+\mathcal{L}\varphi+f\big)\geq\frac{\kappa}{m}
$$

holds on $[0,T)\times\mathcal{S}$. Using the linearity of ∂_t and $\mathcal{L}$, we obtain

$$
\begin{aligned}
0&\geq\partial_t\varphi(t,x)+\mathcal{L}\varphi(t,x)+f\\
&=\partial_t\Big(\frac{m-1}{m}\big(\varphi_m(t,x)-\frac{1}{m}w(t,x)\big)\Big)+\mathcal{L}\Big(\frac{m-1}{m}\big(\varphi_m(t,x)-\frac{1}{m}w(t,x)\big)\Big)+f(t,x)\\
&=\frac{m-1}{m}\Big(\partial_t\varphi_m(t,x)-\frac{1}{m}w(t,x)\Big)+\frac{m-1}{m}\Big(\mathcal{L}\varphi_m(t,x)-\frac{1}{m}\mathcal{L}w(t,x)\Big)+f(t,x)
\end{aligned}
$$

or, equivalently,

$$
\begin{aligned}
0&\geq\partial_t\varphi(t,x)-\frac{1}{m}\partial_t w(t,x)+\mathcal{L}\varphi_m(t,x)-\frac{1}{m}\mathcal{L}w(t,x)+\frac{m}{m-1}f(t,x)\\
&=\partial_t\varphi_m(t,x)+\mathcal{L}\varphi_m(t,x)+f(t,x)-\frac{1}{m}\big(\underbrace{\partial_t w(t,x)+\mathcal{L}w(t,x)+f(t,x)}_{\leq-\kappa(t,x)}\big)\\
&\geq\partial_t\varphi_m(t,x)+\mathcal{L}\varphi_m(t,x)+f(t,x)+\frac{\kappa(t,x)}{m},
\end{aligned}
$$

for $(t,x)\in[0,T)\times\mathcal{S}$. Hence,

$$
-\big(\partial_t\varphi_m-\mathcal{L}\varphi_m-f\big)\geq\frac{\kappa}{m}\quad\text{on } (t,x)\in[0,T)\times\mathcal{S}.
\tag{5.4.3}
$$

For the intervention part, because $\mathcal{M}$ is convex by Lemma 5.10(i), in any point $(t,x) \in [0,T) \times \mathbb{R}^d$,

$$\begin{aligned} v_m(t,x) - \mathcal{M}v_m(t,x) &= v_m(t,x) - \mathcal{M}\left(\frac{m-1}{m}v_*(t,x) + \frac{1}{m}w(t,x)\right) \\ &\geq v_m(t,x) - \frac{m-1}{m}\mathcal{M}v_*(t,x) - \frac{1}{m}\mathcal{M}w(t,x) \\ &= \frac{m-1}{m}\big(\underbrace{v_*(t,x) - \mathcal{M}v_*(t,x)}_{\geq 0}\big) + \frac{1}{m}\big(\underbrace{w(t,x) - \mathcal{M}w(t,x)}_{\geq \kappa(t,x)}\big) \\ &\geq \frac{\kappa(t,x)}{m}. \end{aligned} \tag{5.4.4}$$

Finally, with inequality (5.4.4), we get

$$\begin{aligned} v_m(t,x) - g(t,x) &= \frac{m-1}{m}v_*(t,x) + \frac{1}{m}w(t,x) - g(t,x) \\ &= \frac{m-1}{m}\big(v_*(t,x) - g(t,x)\big) + \frac{1}{m}\big(w(t,x) - g(t,x)\big) \\ &\geq \frac{\kappa(t,x)}{m} \end{aligned} \tag{5.4.5}$$

for $(t,x) \in [0,T) \times \mathbb{R}^d$. Combining inequalities (5.4.3)–(5.4.5), we obtain (5.4.1).

Subsolution case: The proof for the subsolution u is not different from the supersolution case, except that we use Lemma 5.10(ii) to show the anticonvexity of $\mathcal{M}$. To this end, let $\varphi \in \mathcal{PB} \cap C^{1,2}([0,T) \times \mathbb{R}^d)$, such that $\varphi_m(t_0,x_0) = u_m(t_0,x_0)$ and $\varphi_m \geq u_m$ on $[0,T) \times \mathbb{R}^d$. Then, for

$$\varphi := \frac{m}{m+1}\left(\varphi_m + \frac{1}{m}w\right),$$

we have $\varphi(t_0,x_0) = u^*(t_0,x_0)$ and $\varphi \geq u^*$ everywhere else. For any $(t,x) \in [0,T) \times \mathcal{S}$ and by the linearity of ∂_t and $\mathcal{L}$,

$$\begin{aligned} 0 &\leq \frac{m+1}{m}\big(\partial_t\varphi(t,x) + \mathcal{L}\varphi(t,x) + f(t,x)\big) \\ &= \partial_t\varphi_m(t,x) + \mathcal{L}\varphi_m(t,x) + f(t,x) + \frac{1}{m}\big(\underbrace{\partial_t w(t,x) + \mathcal{L}w(t,x) + f(t,x)}_{\leq -\kappa(t,x)}\big) \\ &\leq \partial_t\varphi_m(t,x) + \mathcal{L}\varphi_m(t,x) + f(t,x) - \frac{\kappa(t,x)}{m}, \end{aligned}$$

which is equivalent to

$$-\big(\partial_t\varphi_m(t,x) + \mathcal{L}\varphi_m(t,x) + f(t,x)\big) \leq -\frac{\kappa(t,x)}{m}. \tag{5.4.6}$$

Furthermore, and now for $(t,x) \in [0,T) \times \mathbb{R}^d$,

$$u_m(t,x) - \mathcal{M}u_m(t,x) = u_m(t,x) - \mathcal{M}\left(\frac{m+1}{m}u^*(t,x) - \frac{1}{m}w(t,x)\right)$$

$$
\begin{aligned}
&\le u_m(t,x) - \frac{m+1}{m}\mathcal{M}u^*(t,x) + \frac{1}{m}\mathcal{M}w(t,x) \\
&\le u_m(t,x) - \frac{m+1}{m}u^*(t,x) + \frac{1}{m}\mathcal{M}w(t,x) \\
&= -\frac{1}{m}\big(w(t,x) - \mathcal{M}w(t,x)\big) \\
&\le -\frac{\kappa(t,x)}{m}.
\end{aligned} \tag{5.4.7}
$$

Finally, by a similar argument as in the supersolution case,

$$
u_m(t,x) - g(t,x) \le -\frac{\kappa(t,x)}{m}. \tag{5.4.8}
$$

Hence, with (5.4.6)–(5.4.8), we obtain (5.4.2) and have that u_m is a subsolution according to Definition 4.5. $\square$

Lemma 5.15 is used to perturb the sub- and supersolutions in order to ensure that the maximum of $u_m - v_m$ is attained in the Comparison Principle.

Recall that $\mathcal{PB}_m := \mathcal{PB}_m([0,T]\times\mathbb{R}^d)$ denotes the set of all function $u \in \mathcal{PB}$ for which there is a time-dependent constant C_t such that

$$
|u(t,x)| \le C_t(1+|x|^m) \quad \text{for all } (t,x)\in[0,T]\times\mathbb{R}^d.
$$

The lower and upper semicontinuous envelopes, v_* and u^* respectively, are taken from within $[0,T]\times\mathbb{R}^d$.

By (B2), the functions f, μ, and σ are locally Lipschitz continuous. Hence, by classical results, see [42, Lemma V.7.1], our functional F has the property:

> For any $R>0$, there exist a modulus of continuity ω_R such that, for any $|x|,|y|,|v|\le R$, $l\in\mathbb{R}$ and for any $X,Y\in\mathbb{S}^d$ satisfying
>
> $$
> \begin{bmatrix} X & 0 \\ 0 & -Y \end{bmatrix} \le \frac{1}{\varepsilon}\begin{bmatrix} I & -I \\ -I & I \end{bmatrix} + o_\alpha(1)
> $$
>
> for some $\varepsilon>0$ and $o_\alpha(1)$ independent on ε, then
>
> $$
> \begin{aligned}
> F(t,x,\varepsilon^{-1}(x-y),X,l) &- F(t,y,\varepsilon^{-1}(x-y),Y,l) \\
> &\le \omega_R(|x-y| + \varepsilon^{-1}|x-y|^2) + o_\alpha(1),
> \end{aligned}
> $$
>
> where $o_\alpha(1)$ is once again independent of ε, and the first term is independent of α.

In the proof of Theorem 5.17, x_ε and y_ε converge to the same limit x_0, so the requirement can be relaxed to local Lipschitz condition, as required by Assumption 5.1, and the property holds only for x and y in a suitable neighbourhood of x_0.

Theorem 5.17 ([92, Theorem 5.14], Comparison Principle)**.** *Let Assumptions 4.1 and 5.1 be satisfied. Assume further that, for a constant $\kappa>0$, there is a $w\ge 0$ as in Lemma 5.15 with $|w(t,x)|/|x|^m \to \infty$ for $|x|\to\infty$ uniformly in t. If $u\in\mathcal{PB}_m([0,T]\times\mathbb{R}^d)$ is a subsolution and $v\in\mathcal{PB}_m([0,T]\times\mathbb{R}^d)$ a supersolution of (5.1.1), then $u^*\le v_*$ on $[0,T]\times\mathbb{R}^d$.*

Proof. Recall Lemma 5.15, where we defined the sub-and supersolutions

$$u_m := \left(1 + \frac{1}{m}\right) u^* - \frac{1}{m} w \quad \text{and} \quad v_m := \left(1 - \frac{1}{m}\right) v_* + \frac{1}{m} w$$

respectively, for subsolution u and supersolution v. Noting that

$$\lim_{m\to\infty} u_m = u \quad \text{and} \quad \lim_{m\to\infty} v_m = v,$$

we observe that if we can show that $u_m \geq v_m$ for m large enough, then we also must have $u^* \leq v^*$. To this end, let m be fixed and define

$$M := \sup_{t\in[0,T], x\in\mathbb{R}^d} \left\{ u_m(t,x) - v_m(t,x) \right\}.$$

We prove this theorem by means of contradiction, so assume from now on that $M > 0$. We now consider two steps:

Step 1: In the first step, we show that M cannot be strictly positive on $\partial^+\mathcal{S}_T$.

Step 2: Then, we show that M cannot be strictly positive on $[0,T) \times \mathcal{S}$ by employing the doubling of variables as seen in Corollary 5.14.

Together, these two steps contradict the assumption where $M > 0$, i.e. $u^* \leq v_*$ must hold on $[0,T] \times \mathbb{R}^d$.

Step 1: We want to show that the supremum in the expression of M cannot be approximated from within $\partial^+\mathcal{S}_T$. Assume that for each $\varepsilon_1 > 0$, we can find an $(\hat{t},\hat{x}) = (\hat{t}_{\varepsilon_1}, x_{\varepsilon_1}) \in \partial^+\mathcal{S}_T$ such that

$$u_m(\hat{t},\hat{x}) - v_m(\hat{t},\hat{x}) + \varepsilon_1 > M$$

and WLOG

$$u_m(\hat{t},\hat{x}) - v_m(\hat{t},\hat{x}) > 0.$$

By Lemma 5.15, we have

$$\min\left\{u_m(\hat{t},\hat{x}) - g(\hat{t},\hat{x}), u_m(\hat{t},\hat{x}) - \mathcal{M}u_m(\hat{t},\hat{x})\right\} \leq 0$$
$$\min\left\{v_m(\hat{t},\hat{x}) - g(\hat{t},\hat{x}), v_m(\hat{t},\hat{x}) - \mathcal{M}v_m(\hat{t},\hat{x})\right\} \geq \frac{\kappa}{m}.$$

If $u_m(\hat{t},\hat{x}) - g(\hat{t},\hat{x}) \leq$, then

$$\begin{aligned} u_m(\hat{t},\hat{x}) - v_m(\hat{t},\hat{x}) + \frac{\kappa}{m} &= u_m(\hat{t},\hat{x}) \underbrace{- v_m(\hat{t},\hat{x}) + g(\hat{t},\hat{x}) + \frac{\kappa}{m}}_{\leq 0} - g(\hat{t},\hat{x}) \\ &\leq u_m(\hat{t},\hat{x}) - g(\hat{t},\hat{x}) \leq 0, \end{aligned}$$

which is a contradiction as we assume that $M > 0$. Furthermore, if $u_m(\hat{t},\hat{x}) \leq \mathcal{M}u_m(\hat{t},\hat{x})$, then select for $\varepsilon_2 > 0$ an impulse $\hat{\zeta} = \hat{\zeta}_{\varepsilon_1,\varepsilon_2}$ such that

$$u_m\big(\hat{t}, \Gamma(\hat{x},\hat{\zeta})\big) + K(\hat{x},\hat{\zeta}) + \varepsilon_2 > \mathcal{M}u_m(\hat{t},\hat{x}).$$

Then, with $-v_m + \mathcal{M}v_m \leq -\frac{\kappa}{m}$,

$$\begin{aligned}
M - \varepsilon_1 &= \sup_{t\in[0,T],x\in\mathbb{R}^d} \big\{u_m(t,x) - v_m(t,x)\big\} - \varepsilon_1 \\
&< u_m(\hat{t},\hat{x}) - v_m(\hat{t},\hat{x}) \\
&\leq u_m\big(\hat{t},\Gamma(\hat{x},\hat{\zeta})\big) + K(\hat{x},\hat{\zeta}) + \varepsilon_2 - \frac{\kappa}{m} - \mathcal{M}v_m(\hat{t},\hat{x}) \\
&\leq u_m\big(\hat{t},\Gamma(\hat{x},\hat{\zeta})\big) + K(\hat{;}\hat{\zeta}) + \varepsilon_2 - \frac{\kappa}{m} - v_m\big(\hat{t},\Gamma(\hat{;}\hat{\zeta})\big) - K(\hat{x},\hat{\zeta}) \\
&\leq M + \varepsilon_2 - \frac{\kappa}{m},
\end{aligned}$$

which, for sufficiently small $\varepsilon_1, \varepsilon_2 > 0$ is a contradiction as $\frac{\kappa}{m} > 0$. Therefore, the supremum M cannot be attained in $\partial^+\mathcal{S}_T$, neither can it be approached from within $\partial^+\mathcal{S}_T$.

Step 2: Now that we are certain that we need not to take into account the boundary conditions, we employ the doubling of variables. We define, for $\varepsilon > 0$,

$$M_\varepsilon = \sup_{t\in[0,T],x,y\in\mathbb{R}^d} \big\{u_m(t,x) - v_m(t,y) - \frac{1}{2\varepsilon}|x-y|^2\big\},$$

with u_m and v_m as defined in Lemma 5.15. In view of the definiiton of u_m, v_m, and w, we find that

$$\begin{aligned}
&u_m(t,x) - v_m(t,y) - \frac{1}{2\varepsilon} \\
&= \frac{m+1}{m}u(t,x) - \frac{1}{m}w(t,x) - \frac{m-1}{m}v(t,y) - \frac{1}{m}w(t,y) - \frac{1}{2\varepsilon}|x-y|^2 \\
&\leq C_{t,m}(1+|x|^m) + C'_{t,m}(1+|y|^m) - \frac{1}{m}\big(w(t,x)+w(t,y)\big) - \frac{1}{2\varepsilon}|x-y|^2
\end{aligned}$$

for time-dependent constants $C_{t,m}, C'_{t,m} > 0$. Furthermore, for fixed $t < T < \infty$ and $y \in \mathbb{R}^d$, note that

$$\frac{1}{|x|^m}\left(C_{t,m}(1+|x|^m) + C'_{t,m}(1+|y|^m) - \frac{1}{m}\big(w(t,x)+w(t,y)\big) - \frac{1}{2\varepsilon}|x-y|^2\right) \to -\infty,$$

as $|x| \to \infty$, because $w(t,x)/|x|^m \to \infty$ as $|x| \to \infty$ by assumption. Therefore, the maximum of $u_(t,x) - v_m(t,y) - \frac{1}{2\varepsilon}|x-y|^2$ is attained in a compact set C independent of small ε.

Now pick a point $(t_\varepsilon, x_\varepsilon, y_\varepsilon)$, where for small ε, we know from the previous step that $t_\varepsilon < T$ and $x_\varepsilon, y_\varepsilon \in \mathcal{S}$. We assume that the supremum of M_ε is attained at $(t_\varepsilon, x_\varepsilon, y_\varepsilon)$. Then, by Lemma 3.1 in [7] and with $u_m \in \mathrm{USC}([0,T)\times\mathcal{S})$ and $v_m \in \mathrm{LSC}([0,T)\times\mathcal{S})$, we obtain

$$\frac{1}{2\varepsilon}|x_\varepsilon - y_\varepsilon|^2 \quad \text{as } \varepsilon \to 0,$$

and

$$M_\varepsilon \to M = u_m(t_0,x_0) - v_m(t_0,x_0)$$

for all limit points x_0 of the sequence $\{x_\varepsilon\}$. We assume from now on that we have chosen a convergent subsequence of $\{x_\varepsilon\}$ and $\{y_\varepsilon\}$ converging to the same limit $x_0 \in C$. Let ε be small enough such that $x_0, y_0 \in \mathcal{S}$, and that all local estimates in (B1) and (B2) hold. Moreover, assume that $\{t_\varepsilon\}$ converges to $t_0 < T$.

With the above assumptions in mind, we may apply Corollary 5.14 in $(t_\varepsilon, x_\varepsilon, y_\varepsilon)$ for $\varphi(x,y) = \frac{1}{2\varepsilon}|x-y|^2$: For any $\delta > 0$, there is a range of $\alpha > 0$ for which there are matrices X, Y satisfying (5.3.7), $(p, -q) = \nabla\varphi(x_\varepsilon, y_\varepsilon)$, so that $p = q = \frac{1}{\varepsilon}(x_\varepsilon - y_\varepsilon)$, and $a + b = \partial_t \varphi(x_0, y_0) = 0$ such that

$$
\begin{aligned}
\min\Big\{ -a + F\big(t_\varepsilon, x_\varepsilon, p, X, \mathcal{I}^{1,\delta}[t_\varepsilon, x_\varepsilon, \varphi_\alpha(\cdot, y_\varepsilon)] + \mathcal{I}^{2,\delta}[t_\varepsilon, x_\varepsilon, p, u_m(t_\varepsilon, \cdot)]\big),& \\
u_m(t_\varepsilon, x_\varepsilon) - \mathcal{M}u_m(t_\varepsilon, x_\varepsilon)\Big\} &\le 0 \\
\min\Big\{ -b + F\big(t_\varepsilon, y_\varepsilon, q, Y, \mathcal{I}^{1,\delta}[t_\varepsilon, y_\varepsilon, -\varphi_\alpha(x_\varepsilon, \cdot)] + \mathcal{I}^{2,\delta}[t_\varepsilon, y_\varepsilon, q, v_m(t_\varepsilon, \cdot)]\big),& \\
v_m(t_\varepsilon, y_\varepsilon) - \mathcal{M}v_m(t_\varepsilon, y_\varepsilon)\Big\} &\ge \frac{\kappa}{m}.
\end{aligned}
$$

This representation lets us split the proof into 2 cases that we need to consider for the proof:

Case 2a: If $u_m(t_\varepsilon, x_\varepsilon) - \mathcal{M}u_m(t_\varepsilon, x_\varepsilon) \le 0$.

Case 2b: If $u_m(t_\varepsilon, x_\varepsilon) - \mathcal{M}u_m(t_\varepsilon, x_\varepsilon) > 0$.

Case 2a: Assume that

$$u_m(t_\varepsilon, x_\varepsilon) - \mathcal{M}u_m(t_\varepsilon, x_\varepsilon) \le 0.$$

Using

$$v_m(t_\varepsilon, y_\varepsilon) - \mathcal{M}v_m(t_\varepsilon, y_\varepsilon) \ge \frac{\kappa}{m}$$

for $\varepsilon > 0$ small enough, we have

$$
\begin{aligned}
M &= \limsup_{\varepsilon\to 0} \big\{u_m(t_\varepsilon, x_\varepsilon) - v_m(t_\varepsilon, y_\varepsilon)\big\} \\
&\le \limsup_{\varepsilon\to 0} \mathcal{M}u_m(t_\varepsilon, x_\varepsilon) - \liminf_{\varepsilon\to 0} \mathcal{M}v_m(t_\varepsilon, y_\varepsilon) - \frac{\kappa}{m} \\
&\le \mathcal{M}u_m(t_0, x_0) - \mathcal{M}v_m(t_0, x_0) - \frac{\kappa}{m}.
\end{aligned}
$$

Applying the same approach as in **Step 1**, select for $\hat\varepsilon > 0$ a $\hat\zeta = \hat\zeta_{\hat\varepsilon}$ such that

$$u_m\big(t_0, \Gamma(x_0, \hat\zeta)\big) + K(x_0, \hat\zeta) + \hat\varepsilon > \mathcal{M}u_m(t_0, x_0).$$

Then,

$$
\begin{aligned}
M &\le \mathcal{M}u_m(t_0, x_0) - \mathcal{M}v_m(t_0, x_0) \\
&< u_m\big(t_0, \Gamma(x_0, \hat\zeta)\big) + K(x_0, \hat\zeta) + \hat\varepsilon - v_m(t_0, x_0) - K(x_0, \hat\zeta) - \frac{\kappa}{m} \\
&\le M + \hat\varepsilon - \frac{\kappa}{m},
\end{aligned}
$$

which is a contradiction for sufficiently small $\hat\varepsilon$ since $\kappa/m > 0$. This implies that the assumption

$$u_m(t_\varepsilon, x_\varepsilon) - \mathcal{M}u_m(t_\varepsilon, x_\varepsilon) \le 0$$

cannot hold if we want $M > 0$.

Case 2b: Assume now that $u_m(t_\varepsilon, x_\varepsilon) - \mathcal{M}u_m(t_\varepsilon, x_\varepsilon) > 0$. Then, we must have

$$F\big(t_\varepsilon, x_\varepsilon, p, X, \mathcal{I}^{1,\delta}[t_\varepsilon, x_\varepsilon, \varphi_\alpha(\cdot, y_\varepsilon)] + \mathcal{I}^{2,\delta}[t_\varepsilon, x_\varepsilon, p, u_m(t_\varepsilon, \cdot)]\big) \le 0 \tag{5.4.9}$$

$$F\big(t_\varepsilon, y_\varepsilon, q, Y, \mathcal{I}^{1,\delta}[t_\varepsilon, y_\varepsilon, \varphi_\alpha(x_\varepsilon, \cdot)] + \mathcal{I}^{2,\delta}[t_\varepsilon, y_\varepsilon, q, v_m(t_\varepsilon, \cdot)]\big) \ge \frac{\kappa}{m} \tag{5.4.10}$$

Before we can proceed, we have to compare the integral terms in both inequalities. First, note that because

$$|x + \gamma(t, x, z) - y|^2 = |x - y|^2 + 2\langle x - y, \gamma(t, x, z)\rangle + |\gamma(t, x, z)|^2,$$

we have, for the test function $\varphi(x, y)\frac{1}{2\varepsilon}|x - y|^2$ and fixed $t_\varepsilon < T$,

$$\begin{aligned}
&\mathcal{I}^{1,\delta}[t_\varepsilon, x_\varepsilon, \varphi(\cdot, y_\varepsilon)] \\
&\quad= \int\limits_{|z|<\delta} \Big(\varphi\big(x_\varepsilon + \gamma(t_\varepsilon, x_\varepsilon, z)\big) - \varphi(x_\varepsilon, y_\varepsilon) - \langle \nabla_x \varphi(x_\varepsilon, y_\varepsilon), \gamma(t_\varepsilon, x_\varepsilon, z)\rangle\Big)\nu(\,\mathrm{d}z) \\
&\quad= \int\limits_{|z|<\delta} \left(\frac{1}{2\varepsilon}|x_\varepsilon + \gamma(t_\varepsilon, x_\varepsilon, z) - y_\varepsilon|^2 - \frac{1}{2\varepsilon}|x_\varepsilon - y_\varepsilon|^2 - \left\langle \frac{1}{\varepsilon}(x_\varepsilon - y_\varepsilon), \gamma(t_\varepsilon, x_\varepsilon, z)\right\rangle\right)\nu(\,\mathrm{d}z) \\
&\quad= \int\limits_{|z|<\delta} \bigg(\frac{1}{2\varepsilon}|x_\varepsilon - y_\varepsilon|^2 - \frac{1}{\varepsilon}\langle x_\varepsilon - y_\varepsilon, \gamma(t_\varepsilon, x_\varepsilon, z)\rangle + \frac{1}{2\varepsilon}|\gamma(t_\varepsilon, x_\varepsilon, z)|^2 - \frac{1}{2\varepsilon}|x_\varepsilon - y_\varepsilon|^2 \\
&\qquad\qquad - \frac{1}{\varepsilon}\langle (x_\varepsilon - y_\varepsilon), \gamma(t_\varepsilon, x_\varepsilon, z)\rangle\bigg)\nu(\,\mathrm{d}z) \\
&\quad= \frac{1}{2\varepsilon}\int\limits_{|z|<\delta} |\gamma(t_\varepsilon, x_\varepsilon, z)|^2\nu(\,\mathrm{d}z).
\end{aligned}$$

Similarly,

$$\mathcal{I}^{1,\delta}[t_\varepsilon, y_\varepsilon, -\varphi(\cdot, y_\varepsilon)] = -\frac{1}{2\varepsilon}\int\limits_{|z|<\delta} |\gamma(t_\varepsilon, y_\varepsilon, z)|^2\nu(\,\mathrm{d}z).$$

We also note that $\mathcal{I}^{1,\delta}[t_\varepsilon, x_\varepsilon, \varphi(\cdot, y_\varepsilon)] < \infty$ and $\mathcal{I}^{1,\delta}[t_\varepsilon, y_\varepsilon, -\varphi(\cdot, y_\varepsilon)] < \infty$ by (V4) and by the definition of $\mathcal{PB}$, see Definition 3.11. Define

$$o_\delta(1) := \int\limits_{|z|<\delta} \big(|\gamma(t_\varepsilon, x_\varepsilon, z)|^2 + |\gamma(t_\varepsilon, x_\varepsilon, z)|^2\big)\nu(\,\mathrm{d}z).$$

Then, we have

$$\begin{aligned}
\mathcal{I}^{1,\delta}[t_\varepsilon, x_\varepsilon, \varphi(\cdot, y_\varepsilon)] &= \mathcal{I}^{1,\delta}[t_\varepsilon, y_\varepsilon, -\varphi(\cdot, y_\varepsilon)] + \frac{1}{2\varepsilon}o_\delta(1) \\
&\le \mathcal{I}^{1,\delta}[t_\varepsilon, y_\varepsilon, -\varphi(\cdot, y_\varepsilon)] + \frac{1}{\varepsilon}o_\delta(1).
\end{aligned}$$

Because we know that φ_α converges to φ unifromly in $C^{1,2}(C)$ for any compact C, we have that

$$\mathcal{I}^{1,\delta}[t_\varepsilon, x_\varepsilon, \varphi_\alpha(\cdot, y_\varepsilon)] \le \mathcal{I}^{1,\delta}[t_\varepsilon, y_\varepsilon, -\varphi_\alpha(\cdot, y_\varepsilon)] + \frac{1}{\varepsilon}o_\delta(1) + o_\alpha(1),$$

where $o_\alpha(1)$ may depend on ε, but is independent of small δ.

Now for the other integral term, that is, $\mathcal{I}^{2,\delta}$, we use that $(t_\varepsilon, x_\varepsilon, y_\varepsilon)$ is a maximum point to find that

$$u_m(t_\varepsilon, x_\varepsilon+h)-v_m(t_\varepsilon, y_\varepsilon+h')-\frac{1}{2\varepsilon}|(x_\varepsilon+h)-(y_\varepsilon+h')|^2 \leq u_m(t_\varepsilon, x_\varepsilon)-v_m(t_\varepsilon, y_\varepsilon)-\frac{1}{2\varepsilon}|x_\varepsilon-y_\varepsilon|^2 \tag{5.4.11}$$

for arbitrary vectors $h, h' \in \mathbb{R}^d$. Then, with the identity

$$|x+y|^2 = |x|^2 + 2\langle x, y\rangle + |y|^2,$$

and rearranging (5.4.11), we obtain

$$\begin{aligned}
&u_m(t_\varepsilon, x_\varepsilon + d) - u_m(t_\varepsilon, x_\varepsilon) \\
&\quad \leq v_m(t_\varepsilon, y_\varepsilon + d') - v_m(t_\varepsilon, y_\varepsilon) + \frac{1}{2\varepsilon}|(x_\varepsilon + d) - (y_\varepsilon + d')|^2 - \frac{1}{2\varepsilon}|x_\varepsilon - y_\varepsilon|^2 \\
&\quad = v_m(t_\varepsilon, y_\varepsilon + d') - v_m(t_\varepsilon, y_\varepsilon) + \frac{1}{2\varepsilon}\big(|x_\varepsilon - y_\varepsilon|^2 + 2\langle x_\varepsilon - y_\varepsilon, d - d'\rangle + |d - d'|^2\big) \\
&\qquad - \frac{1}{2\varepsilon}|x_\varepsilon - y_\varepsilon|^2 \\
&\quad = v_m(t_\varepsilon, y_\varepsilon + d') - v_m(t_\varepsilon, y_\varepsilon) + \frac{1}{\varepsilon}\langle x_\varepsilon - y_\varepsilon, d - d'\rangle + \frac{1}{2\varepsilon}|d - d'|^2.
\end{aligned} \tag{5.4.12}$$

Since $x_\varepsilon, y_\varepsilon, d, d' \in \mathbb{R}^d$,

$$\langle x_\varepsilon - y_\varepsilon, d - d'\rangle = \langle x_\varepsilon - y_\varepsilon, d\rangle - \langle x_\varepsilon - y_\varepsilon, d'\rangle,$$

we obtain, by rearranging terms in (5.4.12),

$$\begin{aligned}
&u_m(t_\varepsilon, x_\varepsilon + d) - u_m(t_\varepsilon, x_\varepsilon) - \frac{1}{\varepsilon}\langle x_\varepsilon - y_\varepsilon, d\rangle \\
&\qquad \leq v_m(t_\varepsilon, y_\varepsilon + d') - v_m(y_\varepsilon) - \frac{1}{\varepsilon}\langle x_\varepsilon - y_\varepsilon, d'\rangle + \frac{1}{2\varepsilon}|d - d'|^2.
\end{aligned} \tag{5.4.13}$$

Then, with $p = q = \frac{1}{\varepsilon}(x_\varepsilon - y_\varepsilon)$, $d = \gamma(t_\varepsilon, x_\varepsilon, z)$, and $d' = \gamma(t_\varepsilon, y_\varepsilon, z)$, integrating (5.4.13) yields

$$\begin{aligned}
&\mathcal{I}^{2,\delta}[t_\varepsilon, x_\varepsilon, p, u_m(\cdot)] \\
&\quad = \int\limits_{|z|\geq\delta} \Big(u_m\big((t_\varepsilon, x_\varepsilon + \gamma(t_\varepsilon, x_\varepsilon, z)\big) - u_m(t_\varepsilon, x_\varepsilon) - \langle p, \gamma(t_\varepsilon, x_\varepsilon, z)\rangle 1_{\{|z|<1\}}\Big)\nu(\mathrm{d}z) \\
&\qquad - \int\limits_{|z|\geq\delta} \langle p, \gamma(t_\varepsilon, x_\varepsilon)\rangle 1_{\{|z|\geq 1\}}\nu(\mathrm{d}z) + \int\limits_{|z|\geq\delta} \langle p, \gamma(t_\varepsilon, x_\varepsilon)\rangle 1_{\{|z|\geq 1\}}\nu(\mathrm{d}z) \\
&\quad = \int\limits_{|z|\geq\delta} \Big(u_m(t_\varepsilon, x_\varepsilon + \gamma\big(t_\varepsilon, x_\varepsilon, z\big) - u_m(t_\varepsilon, x_\varepsilon) - \langle p, \gamma(t_\varepsilon, x_\varepsilon, z)\rangle\Big)\nu(\mathrm{d}z) \\
&\qquad + \int\limits_{|z|\geq 1} \langle p, \gamma(t_\varepsilon, x_\varepsilon, z)\rangle\nu(\mathrm{d}z) \\
&\quad \leq \int\limits_{|z|\geq\delta} \Big(v_m\big(t_\varepsilon, y_\varepsilon + \gamma(t_\varepsilon, y_\varepsilon, z)\big) - v_m(t_\varepsilon, y_\varepsilon) - \langle q, \gamma(t_\varepsilon, y_\varepsilon, z)\rangle\Big)\nu(\mathrm{d}z)
\end{aligned}$$

$$
\begin{aligned}
&+\frac{1}{2\varepsilon}\int\limits_{|z|\geq\delta}|\gamma(t_\varepsilon,x_\varepsilon,z)-\gamma(t_\varepsilon,y_\varepsilon,z)|^2\nu(\,\mathrm{d}z)+\int\limits_{|z|\geq 1}\langle p,\gamma(t_\varepsilon,x_\varepsilon,z)\rangle\nu(\,\mathrm{d}z)\\
&=\mathcal{I}^{2,\delta}[t_\varepsilon,y_\varepsilon,q,v_m(t_\varepsilon,\cdot)]+\frac{1}{2\varepsilon}\int\limits_{|z|\geq\delta}|\gamma(t_\varepsilon,x_\varepsilon,z)-\gamma(t_\varepsilon,y_\varepsilon,z)|^2\nu(\,\mathrm{d}z)\\
&+\int\limits_{|z|\geq 1}\langle p,\gamma(t_\varepsilon,x_\varepsilon,z)-\gamma(t_\varepsilon,y_\varepsilon,z)\rangle\nu(\,\mathrm{d}z).
\end{aligned}
$$

Now denote

$$
\begin{aligned}
l^1&=\mathcal{I}^{1,\delta}[t_\varepsilon,x_\varepsilon,\varphi(\cdot,y_\varepsilon)]+\mathcal{I}^{2,\delta}[t_\varepsilon,x_\varepsilon,p,u_m(t_\varepsilon,\cdot)]\\
l^2&=\mathcal{I}[t_\varepsilon,y_\varepsilon,-\varphi(x_\varepsilon,\cdot)]+\mathcal{I}^{2,\delta}[t_\varepsilon,y_\varepsilon,q,v_m(t_\varepsilon,\cdot)].
\end{aligned}
$$

By (B1), we have, for $\varepsilon>0$ small enough,

$$
l^1\leq l^2+\mathcal{O}\left(\frac{1}{\varepsilon}|x_\varepsilon-y_\varepsilon|^2\right)+\frac{1}{\varepsilon}o_\delta(1)+o_\alpha(1),
$$

where the $\mathcal{O}$ term is independent of α and δ. Thus, by (5.4.9) and (5.4.10), as well as (P1) and (P3),

$$
\begin{aligned}
\frac{\kappa}{m}&\leq F\big(t_\varepsilon,y_\varepsilon,q,Y,l^2\big)-F\big(t_\varepsilon,x_\varepsilon,p,X,l^1\big)\\
&\overset{(P1),(P3)}{\leq} F\big(t_\varepsilon,y_\varepsilon,q,Y,l^1\big)-F\big(t_\varepsilon,x_\varepsilon,v_m(t_\varepsilon,x_\varepsilon),p,X,l^1\big)\\
&\qquad+\mathcal{O}\left(\frac{1}{\varepsilon}|x_\varepsilon-y_\varepsilon|^2\right)+\frac{1}{\varepsilon}o_\delta(1)+o_\alpha(1),
\end{aligned}
$$

Furthermore, with $\varphi(x,y):=\frac{1}{2\varepsilon}|x-y|^2$, the matrix inequality in (5.3.7) in Corollary 5.14 becomes

$$
-\frac{1}{\alpha}I\leq\begin{bmatrix}X&0\\0&-Y\end{bmatrix}\leq H_{(x,y)}\varphi(x,y)=\frac{1}{\varepsilon}\begin{bmatrix}I&-I\\-I&I\end{bmatrix}\leq\frac{1}{\varepsilon}\begin{bmatrix}I&-I\\-I&I\end{bmatrix}+o_\alpha(1)
$$

By (B2), for $R>0$ large enough and ε small enough, we have

$$
\begin{aligned}
\frac{\kappa}{m}&\leq F\big(t_\varepsilon,y_\varepsilon,q,Y,l^2\big)-F\big(t_\varepsilon,x_\varepsilon,p,X,l^1\big)\\
&\qquad+\mathcal{O}\left(\frac{1}{\varepsilon}|x_\varepsilon-y_\varepsilon|^2\right)+\frac{1}{\varepsilon}o_\delta(1)+o_\alpha(1)\\
&\leq\omega_R\left(|x_\varepsilon-y_\varepsilon|+\frac{1}{\varepsilon}|x_\varepsilon-y_\varepsilon|^2\right)\mathcal{O}\left(\frac{1}{\varepsilon}|x_\varepsilon-y_\varepsilon|^2\right)+\frac{1}{\varepsilon}o_\delta(1)+o_\alpha(1).
\end{aligned}
$$

Letting $\delta\to 0$, the integral truncation error, $o_\delta(1)$ vanishes and so does $o_\alpha(1)$ as $\alpha\to 0$. Finally, as $\varepsilon\to 0$, $\frac{1}{\varepsilon}|x_\varepsilon-y_\varepsilon|\to 0$ and $|x_\varepsilon-y_\varepsilon|\to 0$ by assumption. This leaves us with the contradiction

$$
\frac{\kappa}{m}\leq 0.
$$

Hence, the assumption $M:=\sup_{t\in[0,T],x\in\mathbb{R}^d}\{u_m(x)-v_m(x)\}>0$ does not hold on the interior $[0,T)\times\mathcal{S}$ either. We therefore conclude that we must have $u_m\leq v_m$ and for m large enough, we obtain $u^*\leq v^*$. $\square$

Remark 5.18. At the end of **Case 2b** of the above proof, we note that α depends on δ. However, letting $\delta \downarrow 0$ does not affect α. As we let $\delta \downarrow 0$, the admissible region for α becomes larger, i.e. $\bar{\alpha}(K)$ grows, but we still may fix some $\alpha > 0$ and the proof still holds.

Remark 5.19. The above proof relies on the same technique used by Seydel for the proof of Theorem 5.11 in [92].

Finally, and with the help of Theorem 5.17, we may prove the uniqueness of the viscosity solution.

Corollary 5.20 ([92, Corollary 5.15] Viscosity Solution: Uniqueness)**.** *Under the same assumptions as in Theorem 5.17, there is at most one viscosity solution of* (5.1.1)*, and it is continuous on* $[0,T] \times \mathbb{R}^d$.

Proof. Let u and v be viscosity solutions of (5.1.1). By Theorem 5.17, we must have

$$u^* \leq v_* \quad \text{and} \quad v^* \leq u_* \quad \text{on } [0,T] \times \mathbb{R}^d. \tag{5.4.14}$$

Also, for any function h, we always have $h_* \leq h^*$. Hence,

$$v_* \leq v^* \quad \text{and} \quad u_* \leq u^* \quad \text{on } [0,T] \times \mathbb{R}^d. \tag{5.4.15}$$

Combining (5.4.14) and (5.4.15), we obtain

$$u^* \leq v_* \leq v^* \tag{5.4.16}$$

and

$$v^* \leq u_* \leq u^*. \tag{5.4.17}$$

Putting together (5.4.16) and (5.4.17), we obtain

$$u^* = v_*, \quad u_* = v_*, \quad \text{and} \quad u^* = v_* \quad \text{on } [0,T] \times \mathbb{R}^d.$$

The lower and upper envelopes of u and v coincide. Hence, $u = v$ on $[0,T] \times \mathbb{R}^d$, i.e. u is unique.

Moreover, because u is a viscosity solution, i.e. it is both sub- and supersolution, applying Theorem 5.17 yields

$$u^* \leq u_* \quad \text{on} [0,T] \times \mathbb{R}^d.$$

Trivially, we also have $u_* \leq u^*$, so $u_* = u^*$. This means that u is both LSC and USC, which means that u is continuous. Therefore,

$$u = u_* = u^* \quad \text{on } [0,T] \times \mathbb{R}^d,$$

which shows that there exist at most one viscosity solution of (5.1.1). □

Taken together, the existence result in Chapter 4 and the uniqueness result in this chapter, we may conclude that the QVI (5.1.1) admits a unique viscosity solution. Moreover, since the value function (4.1.1) satisfies the QVI in the viscosity sense, it follows that the value function is uniquely determined by the QVI.

However, as it is often the case with (HJB)QVIs, a closed-form solution may not exist and often depends on the choice of the functions f, g, and K. This motivates us to apply numerical schemes to approximate the solution of (5.1.1) in the chapters that follow.

CHAPTER

SIX

NUMERICAL SOLUTION OF THE QVI

This chapter describes the finite-difference scheme used to approximate the QVI introduced in Chapters 4 and 5. We closely follow [3] and, under additional assumptions, derive the explicit-impulse scheme, which is a finite-difference scheme, used to solve the QVI in lower dimensions. Because finite-difference schemes for QVIs require convergence arguments slightly different from those found in Barles and Souganidis [8], we do not prove convergence, monotonicity, stability, and nonlocal consistency of the scheme. We instead refer the interested reader to [3, 5, 6] for a thorough analysis of the explicit-impulse scheme.

6.1 Preliminaries

Recall the QVI (4.1.3). For the reader's convenience, we write it again as

$$\begin{aligned} \min\{-(\partial_t v + \mathcal{L}v + f), v - \mathcal{M}v\} &= 0 \quad \text{in } [0,T)\times\mathcal{S} \\ \min\{v - g, v - \mathcal{M}v\} &= 0 \quad \text{in } [0,T)\times(\mathbb{R}^d\backslash\mathcal{S}) \\ v &= \max\{g, \mathcal{M}g\} \quad \text{on } \{T\}\times\mathbb{R}^d, \end{aligned} \tag{QVI}$$

where v is the value function and the intervention operator is defined as

$$\mathcal{M}v(t,x) := \sup_{\zeta\in\mathcal{Z}(t,x)} \{v(t,\Gamma(t,x,\zeta)) + K(t,x,\zeta)\}.$$

Here Γ is the post-intervention state (see Definition 3.4) and $K(t,x,\zeta)$ is the cost of intervening.

6.1.1 Simplification of the QVI

To apply the numerical scheme presented in [3], we reduce (QVI) to a one-dimensional, finite-horizon QVI. The simplification consists of the following points: First, the controlled process $X^{(\alpha)}$ is one-dimensional, so the spatial variable is denoted by $x \in \mathbb{R}$.

Secondly, there is no jump term in the diffusion dynamics. Accordingly, the Lévy measure is zero, i.e. $\nu \equiv 0$, and the integral term in the infinitesimal generator $\mathcal{L}$ disappears. Thirdly, the numerical problem is posed on a fixed and truncated spatial domain. As a consequence, we only consider the QVI on $\mathcal{S}$ (or on a truncated domain inside $\mathcal{S}$) and apply an artificial Neumann boundary condition. We will revisit this new boundary condition later. Finally, the bequest function is taken to be a function solely depending on the state, that is $g(t,x) \equiv g(x)$.

With these conventions, the QVI simplifies to

$$\min\{-(\partial_t v + \mathcal{L}v + f), v - \mathcal{M}v\} = 0 \quad \text{in } [0,T) \times \mathcal{S} \tag{6.1.1}$$

$$\min\{v - g, v - \mathcal{M}v\} = 0 \quad \text{in } \{T\} \times \mathcal{S}. \tag{6.1.2}$$

The infinitesimal generator is now the local second-order operator

$$\mathcal{L}v(t,x) = \mu(t,x)\partial_x v(t,x) + \frac{1}{2}\sigma^2(t,x)\partial_{xx}v(t,x),$$

for $(t,x) \in [0,T] \times \mathcal{S}$.

6.1.2 Terminal condition

The terminal condition (6.1.2) can be written as

$$v(T,x) = \max\{g(T,x), \mathcal{M}v(T,x)\}, \quad \text{for } x \in \mathcal{S}, \tag{6.1.3}$$

which is an implicit terminal condition, because $v(T,\cdot)$ appears on both sides. For a numerical scheme that proceeds backward in time (beginning at the terminal condition), it is preferable to start from a known terminal condition rather than solving an additional nonlinear problem at time T. A simplification of the terminal condition is therefore needed.

For a function g depending on the spatial variable only, a time-T intervention reads as

$$\mathcal{M}g(x) = \sup_{\zeta \in \mathcal{Z}(T,x)} \big\{g(\Gamma(T,x,\zeta)) + K(T,x,\zeta)\big\}.$$

We assume the pointwise inequality

$$\mathcal{M}g(x) \le g, \quad \text{for } x \in \mathcal{S}. \tag{6.1.4}$$

Then, $v(T,x) = g(x)$ solves (6.1.3), and the terminal condition reduces to

$$v(T,x) = g(x), \quad \text{for } x \in \mathcal{S}.$$

Assumption (6.1.4) means that an intervention at the final time cannot improve the terminal payoff. In financial problems with transaction costs, this assumption can be interpreted as follows: a last-instant transaction pays its cost but leaves no remaining time in which the new position can generate additional value.

6.1.3 Numerical grid

As the known initial condition in the QVI is the terminal condition at time T, it is natural to let the scheme evolve backwards in time, beginning at $t = T$. Let $N_t \geq 2$ be a positive integer and define the uniform grid

$$\Delta t := \frac{T-t}{N_t}, \quad t_n := T - n\Delta t, \quad \text{for } n = 0, 1, \ldots, N_t,$$

so that $t_0 = T$ and $t_{N_t} = t$.

Assume first that the computational spatial domain is the bounded interval $\mathcal{S} = (a, b)$[1]. Let $N_x \geq 2$ be a positive integer and define the uniform spatial grid

$$\Delta x := \frac{b-a}{N_x}, \quad x_i := a + i\Delta x, \quad \text{for } i = 0, 1, \ldots, N_x.$$

We write

$$V_i^n \approx v(t_n, x_i), \quad V^n := (V_0^n, \ldots, V_{N_x}^n)^\top.$$

For brevity, we also use the notation

$$\mu_i^n := \mu(t_n, x_i), \quad \sigma_i^n := \sigma(t_n, x_i), \quad f_i^n := f(t_n, x_i).$$

Finally, the terminal condition is imposed by

$$V_i^0 = g(x_i), \quad \text{for } i = 0, 1, \ldots, N_x.$$

Remark 6.1. Throughout this chapter, we have kept the numerical grid uniform. It is worth noting, however, that the same construction can be extended to non-uniform grids, provided that the finite-difference stencils and interpolation weights are modified accordingly.

6.1.4 Stencils used

The time derivative $\partial_t v$ in (6.1.1) is discretised by

$$\partial_t v(t_n, x_i) \approx \frac{v(t_n + \Delta t, x_i) - v(t_n, x_i)}{\Delta t} = \frac{v(t_{n-1}, x_i) - v(t_n, x_i)}{\Delta t} \approx \frac{V_i^{n-1} - V_i^n}{\Delta t}. \tag{6.1.5}$$

For the second derivative $\partial_{xx} v$, we write $\partial_{xx} v(t_n, x_i) \approx (\mathcal{D}_2 V^n)_i$, where $\mathcal{D}_2$ is the centered three-point approximation. We impose the artificial Neumann boundary condition $\partial_{xx} v(t, x_0) = \partial_{xx} v(t, x_{N_x}) = 0$, so that, for $U = (U_0, \ldots, U_{N_x})^\top$,

$$(\mathcal{D}_2 U)_i = \begin{cases} \dfrac{U_{i+1} - 2U_i + U_{i-1}}{(\Delta x)^2}, & \text{if } 0 < i < M, \\ 0, & \text{otherwise.} \end{cases} \tag{6.1.6}$$

In many implicit schemes, the discretisation of the first-order spatial derivative $\partial_x v$ is done via an upwind stencil. However, in the explicit-impulse scheme, rather than using an upwind scheme, the first-order spatial derivative $\partial_x v$ and the drift term are handled through a semi-Lagrangian approach. A more detailed derivation of this approach can be found in Chapter 6.2.

[1] If the original solvency set is unbounded, this open interval is to be understood as a truncation.

6.1.5 Intervention operator

We begin by recalling that the intervention operator is non-local as the the value of $\mathcal{M}v(t,x)$ depends on the value of v at the post-intervention point $\Gamma(t,x,\zeta)$, which need not be a grid point. In light of this, we therefore combine a finite approximation of the impulse with spatial interpolation.

For a vector $U = (U_0, \ldots, U_M)^\top$, define the piecewise linear interpolation operator $\mathcal{I}_{\Delta x}$ by

$$(\mathcal{I}_{\Delta x}U)(x) = \begin{cases} U_0, & \text{if } x \le x_0, \\ w_k\, U_{k+1} + (1-w_k)U_k, & \text{if } x_k < x < x_{k+1} \quad \text{for } k = 0, \ldots, N_x, \\ U_M, & \text{if } x \ge x_{N_x}, \end{cases} \tag{6.1.7}$$

where $w_k \in (0,1)$ denotes the one-dimensional interpolation weight and is given by

$$w_k := \frac{x - x_k}{x_{k+1} - x_k}.$$

The first and last cases in (6.1.7) are included to prevent extrapolation beyond the domain boundary.

Let $\mathcal{Z}_{\Delta x}(t_n, x_i)$ be a finite, non-empty subset of $\mathcal{Z}(t_n, x_i)$ used to approximate the admissible impulse set. The discrete intervention operator is then defined by

$$(\mathcal{M}_n U)_i := \max_{\zeta \in \mathcal{Z}_{\Delta x}(t_n, x_i)} \left\{ (\mathcal{I}_{\Delta x}U)\big(\Gamma(t_n, x_i, \zeta)\big) + K(t_n, x_i, \zeta) \right\}. \tag{6.1.8}$$

The maximum is well-defined here because $\mathcal{Z}_{\Delta x}(t_n, x_i)$ is finite by assumption. The admissible-impulse set is usually chosen so that it becomes denser as the grid is refined. If $\Gamma(t_n, x_i, \zeta)$ lies between two grid points, (6.1.7) gives the interpolated continuation value. If it lies outside the domain of interest, the value is imposed to be the nearest boundary value.

6.2 The explicit-impulse scheme

In [3], the author introduces three schemes for HJBVQI's: the direct control scheme, the penalty scheme, and the explicit-impulse scheme. The first two schemes lead to nonlinear systems and are usually solved by policy iteration, for instance Howard's policy iteration algorithm [20, 52], or the ϵ-policy iteration algorithm [3, Chapter 3]. However, for the finite-horizon QVI system in (6.1.1) and (6.1.2), applying the explicit-impulse scheme is preferable because each time step reduces to a linear system and because we omitted continuous stochastic controls.

Recall (6.1.1) with terminal condition $v(T,x) = g(x)$. For some smooth function $\varphi \in C^{1,2}([0,T] \times \mathbb{R}^d)$, a Taylor expansion about (t,x) gives

$$\varphi(t + \Delta t, x + \mu(t,x)\Delta t) = \varphi(t,x) + \partial_t \varphi(t,x)\Delta t + \mu(t,x)\partial_x \varphi(t,x)\Delta t + \mathcal{O}\big((\Delta t)^2\big).$$

Equivalently,

$$\partial_t\varphi(t,x)+\mu(t,x)\partial_x\varphi(t,x)=\frac{\varphi(t+\Delta t,x+\mu(t,x)\Delta t)-\varphi(t,x)}{\Delta t}+\mathcal{O}(\Delta t).$$

On the uniform numerical grid proposed in Chapter 6.1.3, this motivates the approximation

$$\begin{aligned}\partial_t v(t_n,x_i)+\mu(t_n,x_i)\partial_x v(t_n,x_i)&\approx\frac{v(t_n+\Delta t,x_i+\mu_i^n\Delta t)-v(t_n,x_i)}{\Delta t}\\&=\frac{v(t_{n-1},x_i+\mu_i^n\Delta t)-v(t_n,x_i)}{\Delta t}.\end{aligned}\tag{6.2.1}$$

Since the spatial point $x_i+\mu_i^n\Delta t$ is generally not a grid point, we use the interpolant from (6.1.7), so that (6.2.1) becomes

$$\partial_t v(t_n,x_i)+\mu(t_n,x_i)\partial_x v(t_n,x_i)\approx\frac{(\mathcal{I}_{\Delta x}V^{n-1})(x_i+\mu_i^n\Delta t)-V_i^n}{\Delta t}.$$

The second derivative in the QVI is approximated by the finite-difference operator $\mathcal{D}_2$ defined in (6.1.6). The impulse term is treated explicitly in time, so the intervention operator is applied to V^{n-1} rather than to the unknown vector V^n. A substitution into (6.1.1) gives

$$\begin{aligned}0=\min\Bigg\{&-\left(\frac{(\mathcal{I}_{\Delta x}V^{n-1})(x_i+\mu_i^n\Delta t)-V_i^n}{\Delta t}+\frac{1}{2}(\sigma_i^n)^2(\mathcal{D}_2V^n)_i+f_i^n\right),\\&V_i^n-(\mathcal{M}_nV^{n-1})_i\Bigg\},\end{aligned}$$

which is equivalent to

$$\begin{aligned}0=\max\Bigg\{&(\mathcal{I}_{\Delta x}V^{n-1})(x_i+\mu_i^n\Delta t)-V_i^n+\frac{1}{2}(\sigma_i^n)^2(\mathcal{D}_2V^n)_i\Delta t+f_i^n\Delta t,\\&(\mathcal{M}_nV^{n-1})_i-V_i^n\Bigg\},\end{aligned}\tag{6.2.2}$$

because $\Delta t>0$.

Now the vector V^n in (6.2.2) is unknown and to isolate it on the left-hand side, we add an error term $\mathcal{O}(\Delta t)$ in the form of $\frac{1}{2}(\sigma_i^n)^2(\mathcal{D}_2V^n)_i\Delta t$ to the impulse branch in (6.2.2). In other words,

$$\begin{aligned}0&=\max\Bigg\{(\mathcal{I}_{\Delta x}V^{n-1})(x_i+\mu_i^n\Delta t)-V_i^n+\frac{1}{2}(\sigma_i^n)^2(\mathcal{D}_2V^n)_i\Delta t+f_i^n\Delta t,\\&\qquad(\mathcal{M}_nV^{n-1})_i-V_i^n+\frac{1}{2}(\sigma_i^n)^2(\mathcal{D}_2V^n)_i\Delta t\Bigg\}\\&=\max\left\{(\mathcal{I}_{\Delta x}V^{n-1})(x_i+\mu_i^n\Delta t)+f_i^n\Delta t,(\mathcal{M}_nV^{n-1})_i-V_i^n\right\}\\&\quad-V_i^n+\frac{1}{2}(\sigma_i^n)^2(\mathcal{D}_2V^n)_i\Delta t,\end{aligned}$$

Hence, the explicit-impulse scheme is

$$V_i^n-\frac{1}{2}(\sigma_i^n)^2(\mathcal{D}_2V^n)_i\Delta t=\max\left\{(\mathcal{I}_{\Delta x}V^{n-1})(x_i+\mu_i^n\Delta t)+f_i^n\Delta t,(\mathcal{M}_nV^{n-1})_i-V_i^n\right\},\tag{6.2.3}$$

for $n = 1, \ldots, N_t$ and $i = 0, 1, \ldots, N_x$.

Finally, note that (6.2.3) is the i-th row of the linear system

$$A^n V^n = y^n, \tag{6.2.4}$$

where y_i^n is given by

$$y_i^n := \max\left\{(\mathcal{I}_{\Delta x} V^{n-1})(x_i + \mu_i^n \Delta t) + f_i^n \Delta t, (\mathcal{M}_n V^{n-1})_i\right\},$$

and the matrix A^n by

$$A^n = I + \frac{\Delta t}{2(\Delta x)^2} \begin{pmatrix} 0 & & & & & & \\ -(\sigma_1^n)^2 & 2(\sigma_1^n)^2 & -(\sigma_1^n)^2 & & & & \\ & -(\sigma_2^n)^2 & 2(\sigma_2^n)^2 & -(\sigma_2^n)^2 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & -(\sigma_{M-1}^n)^2 & 2(\sigma_{M-1}^n)^2 & -(\sigma_{M-1}^n)^2 \\ & & & & & 0 \end{pmatrix}$$

on a uniform grid. The matrix A^n is strictly diagonally dominant, i.e.

$$|a_{jj}| > \sum_{k \neq j} |a_{jk}| \quad \text{for all } j = 1, \ldots, N_x + 1.$$

Hence, by [96, Theorem 1.21], A^n by is non-singular for every $n = 1, \ldots, N_t$ and, as such, invertible, meaning that the linear system in (6.2.4) has a solution and we can recover V^n. Moreover, the right-hand side y^n depends only on the known vector V^{n-1}. Therefore, each time step consists of computing the semi-Lagrangian continuation values, evaluating the discrete intervention operator, and solving the linear system (6.2.4).

CHAPTER

SEVEN

DEEP REINFORCEMENT LEARNING FOR IMPULSE CONTROL PROBLEMS

This chapter introduces deep reinforcement learning (DRL) as an alternative approach to solving impulse control problems. In the previous chapters, the value function was characterised as the unique viscosity solution of the associated QVI. However, solving the QVI by finite-difference methods can become computationally expensive as the dimension of the state space increases. DRL methods are therefore considered as they provide a mesh-free alternative by approximating a dynamic programming problem. As such, and after time discretisation, the approaches presented in this chapter are closely related to the iterative optimal stopping approach presented in Chapter 3.2.

We begin with a brief overview of reinforcement learning and its mathematical foundations, highlighting the connection between Markov decision processes, dynamic programming, and neural networks in Chapter 7.1. Chapter 7.2 considers the use of Double Deep Q-Networks, which is a method that directly approximates the value function, while a policy-based method, the Proximal Policy Optimsation algorithm, is considered in Chapter 7.3.

7.1 A Primer on deep reinforcement learning

Reinforcement Learning (RL) is an area of machine learning and optimal control. The objective is to learn, from repeated interaction or simulated experience, how actions affect future rewards. The main decision-making entity, a so-called *agent*, interacts with an environment through trial and error, using observed rewards to improve its decisions over time.

7.1.1 A brief overview of MPDs

In essence, reinforcement learning problems involve making a sequence of decisions over time while interacting with a stochastic environment. The decision-making process can be modeled by MDPs, who describe how certain actions influence future states and rewards [13]. We therefore begin by considering the components that constitute an MDP.

A *state* $x \in \mathcal{X}$ represents the current situation of the environment as observed by the agent. It contains all the relevant information the agent needs in order to make a decision. For instance, in chess, a state corresponds to a particular configuration of pieces on the board. A state is either discrete or continuous.

An *action* $a \in \mathcal{A}$ represents the choice an agent can make in a given state. Actions determine how the agent interacts with the environment, for example, by moving a chess piece. Like states, actions may be discrete or continuous.

A *policy* π specifies how the agent chooses actions based on the current state. In this work, we focus on stochastic policies, where actions are sampled according to a probability distribution $\pi(\cdot|x)$ rather than deterministically.

Rewards $r \in \mathcal{R}$ provide feedback to the agent by quantifying how good or bad an action is. They guide learning by encouraging desirable behavior over time and may depend on the state, action, and time. We model rewards using a function $r : [0, T] \times \mathcal{X} \times \mathcal{A} \to \mathbb{R}$.

Finally, the transition dynamics describe how the stochastic environment evolves. They therefore give the probability of moving to a new state x' given the current state x and the action a.

Together, these elements form an MDP. We formalise MDPs and stochastic policies Definitions 7.1 and 7.3.

Definition 7.1 (MDP)**.** An MDP is a tuple

$$(\mathcal{X}, \mathcal{A}, P, \mathcal{R}),$$

consisting of the state space $\mathcal{X}$, the action space $\mathcal{A}$, the transition dynamics $P : \mathcal{X} \to [0, 1]$, and the reward space $\mathcal{R}$.

Remark 7.2. In reinforcement learning, future rewards are often discounted to balance immediate and long-term outcomes, which is why MDPs are sometimes written with an explicit discount factor δ, i.e. $(\mathcal{X}, \mathcal{A}, P, \mathcal{R}, \delta)$. However, in financial applications, discounting is applied directly to monetary quantities through the functions f, g, and K. For this reason, we set $\delta = 1$, and the MDP reduces to the form given in Definition 7.1.

Definition 7.3 (Stochastic policy function)**.** The policy function, $\pi : \mathcal{X} \times \mathcal{A} \to [0, 1]$, is a probability density of taking an action $a \in \mathcal{A}$ at some time t given a state $x \in \mathcal{X}$ at time t, that is,

$$\pi(a\,|x) = \mathbb{P}(A_t = a|X_t = x).$$

In particular, for every fixed $x \in \mathcal{X}$, the mapping $a \mapsto \pi(a|x)$ is a probability distribution on $\mathcal{A}$, so that the properties

$$\pi(a\,|x) \geq 0, \text{ for every } a \in \mathcal{A}, \quad \text{and} \quad \int_{\mathcal{A}} \pi(a|x)\,\mathrm{d}a = 1.$$

In many reinforcement learning models, time is often discretised in order to mirror the step-by-step nature of decision-making. Moreover, a discrete-time setup often simplifies computation and training procedures. We therefore consider a uniform time grid, where the indexing is now forward in time. Specifically, let $[t, T]$ be the time interval of interest and discretise it through

$$\Delta t := \frac{T-t}{N}, \quad t_n := t + n\Delta t, \quad \text{for } n = 0, 1, \dots, N_t.$$

We denote by $\mathcal{T}$ the partition of the time interval $[t, T]$ and define it as

$$\mathcal{T} := \{t = t_0 < t_1 < \dots < t_N = T\}. \tag{7.1.1}$$

For any policy π, let A_n denote the random action taken at time step t_n under the policy π. Then, conditional on the random state $X_n \in \mathcal{X}$, the action $A_n \in \mathcal{A}$ is sampled according to the distribution $A_n \sim \pi(\cdot, X_n)$. to evaluate the policy π, we introduce the policy-value and state-action value function in the definitions that follow.

Definition 7.4 (Policy-value function)**.** We define the policy value function at step n under policy π to be the expected sum of future rewards under policy π starting from $X_n = x$ in time step $t_n \in \mathcal{T}$ until the end of the episode, which we denote $V^\pi : \mathcal{T} \times \mathcal{X} \mapsto \mathbb{R}$ and define as

$$V^\pi(t_n, x) := \mathbb{E}_\pi^{(t_n,x)} \left[\sum_{m=n}^{N} r(X_m, A_m) \right],$$

where the notation $\mathbb{E}_\pi[\cdot]$ means that the expectation is taken over actions sampled by the policy π.

Definition 7.5 (State-action value function)**.** We define the state-action value function $Q^\pi : \mathcal{T} \times \mathcal{X} \mapsto \mathbb{R}$ as the sum of expected rewards received after taking a random action $A_n = a$ at time step $t_n \in \mathcal{T}$ given state $X_n = x$, and the following policy π in all subsequent time steps. Formally,

$$Q^\pi(t_n, x, a) := r(t_n, x, a) + \mathbb{E}_\pi^{(t_n,x,a)} \left[\sum_{m=n+1}^{N} r(t_m, X_m, A_m) \right],$$

where the notation $\mathbb{E}_\pi[\cdot]$ means that the expectation is taken over actions sampled by the policy π.

The state-action value function Q^π is often referred to as the Quality-function or simply the Q-function. The Q-function measures the value of taking an action given a current state. To quantify how advantageous it is to take a stochastic action A_n at time t_n and given state X_n, we define the advantage function as follows:

Definition 7.6 (Advantage function)**.** We define the advantage function at time step t_n under policy π as the expected excess return obtained by taking action $A_n = a$ in state $X_n = x$, relative to the expected return obtained by following policy π from state $X_n = x$. Specifically, denote the advantage function by $G^\pi : \mathcal{T} \times \mathcal{X} \times \mathcal{A} \to \mathbb{R}$ and define it as

$$G^\pi(t_n, x, a) := Q^\pi(t_n, x, a) - V^\pi(t_n, x).$$

Under suitable assumptions, it can be shown that there exists an optimal stochastic policy π^* such that an an optimal value function is attained, i.e.

$$V^{\pi^*}(t,x) = \sup_{\pi} V^{\pi}(t,x), \quad \text{for } (t,x) \in \mathcal{T} \times \mathcal{X},$$

see [19, 74, 43, 87, 15]. For ease of notation, we write $V^* = V^{\pi^*}$ and $Q^* = Q^{\pi^*}$. Then, under the stochastic policy setting, the Bellman equations state that

$$\begin{aligned} &V^{\pi}(t_n,x) = \mathbb{E}_{\pi}^{(t_n,x)}\big[Q^{\pi}(t_n,X_n,A_n)\big] && \text{for all } x \in \mathcal{X}, \\ &Q^{\pi}(t_n,x,a) = r(t_n,x,a) + \mathbb{E}_{\pi}^{(t_n,x)}\big[V^{\pi}(t_{n+1},X_n,A_n)\big] && \text{for all } (x,a) \in \mathcal{X} \times \mathcal{A} \\ &V^{\pi}(t_{N+1},x) = 0 && \text{for all } x \in \mathcal{X}. \end{aligned} \tag{7.1.2}$$

For the optimal policy π^*, it additionally holds that $V^*(t_n,x) = \max_{a\in\mathcal{A}} Q^*(x,a)$.

7.1.2 Neural networks as function approximators

In reinforcement learning, the functions of interest are the value, the Q-, and the policy functions. When the state and action spaces are small and discrete, these functions can be represented explicitly using tabular (matrix) methods. However, in many practical settings, such as financial decision-making, the state and action spaces are continuous and high-dimensional. This makes tabular representations infeasible.

To address this challenge, modern reinforcement learning methods rely on neural networks as function approximators. Neural networks provide a flexible and mesh-free parametric class of functions that are useful for approximating maps of the form

$$(t,x) \mapsto V^{\pi}(t,x), \quad (t,x,a) \mapsto Q^{\pi}(t,x,a), \quad \text{and} \quad x \mapsto \pi(a|x).$$

The ability of neural networks to approximate nonlinear functions allows agents to operate effectively in large and continuous environments. As such, the combination of neural networks and reinforcement learning enables the agent to learn effective decision rules from collected past experience. The combination of reinforcement learning with neural networks is commonly referred to as Deep Reinforcement Learning (DRL).

To formally define neural networks, we begin by considering activation functions.

Definition 7.7 (Activation function)**.** An activation function is a measurable, nonlinear map $\phi : \mathbb{R} \to \mathbb{R}$, applied componentwise to the affine pre-activation so of a neural network layer. Given an input vector $x \in \mathbb{R}^d$, we write the vectorised input as

$$\phi(x) = (\phi(x_1), \ldots, \phi(x_d)).$$

Remark 7.8. For approximations of functions with neural networks, one often assumes that the activation function ϕ is continuous and non-polynomial. Under these conditions, it can be shows that certain types of neural networks can approximate a function arbitrarily well (see [68]).

The specific form of the activation function ϕ can be quite general. In this thesis, we use the following activation functions:

- The rectified linear unit (ReLU) function, defined as

$$\phi(x) := \max\{x, 0\}. \tag{7.1.3}$$

- The hyperbolic tangent function, defined as

$$\phi(x) := \frac{e^x - e^{-x}}{e^x + e^{-x}}. \tag{7.1.4}$$

With Definition 7.7 in mind, let us now consider feedforward neural networks.

Definition 7.9 ([47], Feedforward neural networks)**.** Let L and $n_0, \ldots, n_L \in \mathbb{N}$ be respectively the number of layers and neurons in each layer, with n_0 being the size of the input vector. The activation function of layer $\ell = 0, 1, \ldots, L$ is denoted by $\phi_\ell(\cdot) : \mathbb{R} \mapsto \mathbb{R}$. In addition, let $C_1 \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_0}$, $\mathbf{c}_1 \in \mathbb{R}^{n_1}$, $C_\ell \in \mathbb{R}^{n_\ell} \times \mathbb{R}^{n_0+n_\ell-1}$, $\mathbf{c}_\ell \in \mathbb{R}^{n_\ell}$ for $\ell = 2, \ldots, L-1$, $C_L \in \mathbb{R}^{n_L} \times \mathbb{R}^{n_{L-1}}$, $\mathbf{c}_L \in \mathbb{R}^{n_L}$ be neural weights defining the input, intermediate, and output layers. Finally, define the functions

$$\begin{aligned} &\psi_\ell = \phi_\ell(C_\ell z + \mathbf{c}_\ell), && \ell = 1, L, \\ &\psi_\ell(z, x) = \phi_\ell(C_k (z, x)^\top + \mathbf{c}_\ell), && \ell = 2, \ldots, L-1, \end{aligned}$$

where the actiation functions $\phi_\ell(\cdot)$ are applied componentwise, $x \in \mathbb{R}^{n_0}$ is the input variable, and $z \in \mathbb{R}^{n_{\ell-1}}$ are the current hidden activations coming from the previous neural network layer. Then the neural network is a function $F : \mathbb{R}^{n_0} \mapsto \mathbb{R}^{n_L}$ with parameters $\theta = \{(C_\ell, \mathbf{c}_\ell)\}_{\ell=1,\ldots,L}$, and is defined as

$$F(z) = \psi_L(x) \circ \psi_{L-1}(\cdot, x) \circ \cdots \circ \psi_2(\cdot, x) \circ \psi_1(x).$$

Remark 7.10. While the definition does not exclude the use of different activation functions across layers[1], many practical implementations rely on a single activation function WLOG. Consequently, we adopt this simplified setting and consider networks with one fixed activation function.

When motivating the use of neural networks in reinforcement learning, we highlighted their ability to approximate nonlinear functions. The following result, known as the *Universal Approximation Theorem*, provides a theoretical proof for this approximation ability.

Theorem 7.11 ([68], Universal approximation theorem)**.** *Let $B \subset \mathbb{R}^d$ be a compact set and ϕ be a non-polynomial activation function. Then, given any function $u : B \mapsto \mathbb{R}$ and $\varepsilon > 0$, there exists a neural network F with one hidden layer and activation function ϕ such that*

$$\sup_{x \in B} |u(x) - F(x)| < \varepsilon.$$

Proof. We refer the interested reader to [68] for the proof of the above result. □

Remark 7.12. Even though Theorem 7.11 assumes that the neural network consists of one hidden layer, later studies prove that neural networks with arbitrarily many width and hidden layers with suitable non-polynomial activation functions are universal approximations as well (see [85, 45]).

[1] For instance, it is in theory possible to use a ReLU activation function on one layer and then the sigmoid activation function on the next layer.

In addition to the universal approximation result, it can be shown that neural networks are statistically consistent. Roughly speaking, under suitable conditions and with sufficient data, neural network approximations converge to the target function under the mean-squared error metric. For further details around the theorem and its proof, we refer to [9, 102].

Taken together, the universal approximation and statistical consistency results motivate the use of neural networks as parametric representations of the value, Q-, and policy functions. Hence, we approximate

$$V_\theta(t,x) \approx V^\pi(t,x), \quad Q_\theta(t,x,a) \approx Q^\pi(t,x,a), \quad \text{and } \pi_\theta(a|x) \approx \pi(a|x),$$

for all $(x,a) \in \mathcal{X} \times \mathcal{A}$, where θ denotes the network parameters learned by the agent from interactions with the environment.

Remark 7.13 (DRL as an approximation of the viscosity solution). Recall Chapters 4 and 5, where we proved that the value function v is a viscosity solution of the QVI (4.1.3), and that this viscosity solution is unique. This let us conclude that the QVI characterizes the impulse-control value function v.

While the DRL methods considered in this chapter do not solve the QVI directly like the explicit-impulse scheme in Chapter 6.2, DRL uses dynamic programming and neural networks to approximate the value function. More precisely, after time discretization, the impulse-control problem can be interpreted as a Markov decision problem in which the agent learns when to continue and when to intervene. Depending on the DRL architecture, the value function is approximated using either a Q-network or through a method that learns both the value and the policy functions. The resulting policy, whether represented explicitly by an actor network or induced implicitly by maximizing the learned Q-function, determines the approximate continuation-versus-intervention strategy. If this learned policy is close to optimal and the discretization error is small, the learned value approximation can therefore be interpreted as a numerical approximation of v, and hence of the viscosity solution.

This interpretation is motivated by universal approximation theorems such as Theorem 7.11, as well as by the statistical consistency of neural networks. However, this should be understood as a heuristic approximation argument rather than a convergence proof. For this argument to hold, monotonicity, stability, and consistency results for the DRL schemes are needed. These results are not provided in this thesis and, to the best of our knowledge, are yet to be derived. A starting point for such a venture can be found in [11].

7.2 Value-based approximation of impulse control problems

Motivated by the empirical results found in [1], we consider our first DRL algorithm in the form of the Double Deep Q-Network (DDQN). The DDQN algorithm is a value-based DRL method that learns an approximation of the Q-function by letting the agent interact with an (stochastic) environment. The method is well-suited for problems with a discrete action space and can therefore be applied impulse control problems in which the set of admissible impulses $\mathcal{Z}$ is discrete. Moreover, the algorithm learns by using data stored inside a replay buffer[2] and does not require a policy π. The double Q-learning approach was first introduced in [49], while the deep version, the DDQN, was developed in [50].

The motivation for learning the Q-function, rather than the value function, follows from the Bellman equations (7.1.2). For a fixed policy, the value function is the expected Q-value over all admissible actions. Consequently, once the Q-function is known, both the value function and the optimal policy can be recovered. To apply the DDQN model, we first rewrite the QVI (4.1.3) into a dynamic programming form. We note that this form resembles the one derived in Chapter 3.2.

7.2.1 From QVI to dynamic programming

Recall the QVI (4.1.3), that is,

$$\min\big\{-(\partial_t v + \mathcal{L}v + f), v - \mathcal{M}v\big\} = 0 \quad \text{in } [0,T)\times\mathcal{S} \tag{7.2.1}$$
$$\min\{v - g, v - \mathcal{M}v\} = 0 \quad \text{in } [0,T)\times(\mathbb{R}^d\backslash\mathcal{S}) \tag{7.2.2}$$
$$v = \max\{g, \mathcal{M}g\} \quad \text{on } \{T\}\times\mathbb{R}^d, \tag{7.2.3}$$

where the intervention operator $\mathcal{M}$ is given by

$$\mathcal{M}v(t,x) = \sup_{\zeta\in\mathcal{Z}(t,x)}\{v(t,\Gamma(x,\zeta)) + K(t,x,\zeta)\}.$$

As in Chapter 6, we assume that intervening at time $t = T$ does not add any economic value and focus on the solvency region $\mathcal{S}$ only. Under these assumptions, the boundary and terminal conditions (7.2.2) and (7.2.3) simplify to the explicit terminal condition

$$v(T,x) = g(x), \quad x\in\mathbb{R}^d.$$

As previously discussed, inside $\mathcal{S}$, (7.2.1) is a comparison between to alternatives: Continue without intervention, or intervene immediately. To make this comparison explicit, we use the identity

$$\min\{-a,-b\} = -\max\{a,b\},$$

together with a few additional assumptions in Appendix C.2 and Proposition C.2 to obtain the approximation

$$v(t,x) \approx \max\left\{\mathbb{E}^{(t,x)}\left[\int_t^{t+h} f\big(s,X(s)\big)\,\mathrm{d}s + v\big(t+h, X(t+h)\big)\right], \mathcal{M}v(t,x)\right\}, \tag{7.2.4}$$

for $(t,x)\in[0,T)\times\mathcal{S}$.

---

[2] A replay buffer is a data structure that stores the agent's past experiences.

Now let $t = t_0 < t_1 < \cdots < t_{N_t} = T$ be a uniform time grid, with $\Delta t := t_n - t_{n-1}$. Then, with $h := \Delta t$, (7.2.4) becomes

$$v(t_n, x) = \max\left\{\mathbb{E}^{(t,x)}\left[\int_{t_n}^{t_{n+1}} f\big(s, X(s)\big)\,\mathrm{d}s + v\big(t_{n+1}, X(t_{n+1})\big)\right], \mathcal{M}v(t_n, x)\right\}, \tag{7.2.5}$$

for $n = 0, 1, \ldots, N_t - 2$ and $x \in \mathcal{S}$. At the final time step, the value function is given by the terminal payoff

$$v(t_{N_t}, x) = g(x) \quad \text{for } x \in \mathcal{S}.$$

Since (7.2.5) restricts intervention decisions to a finite grid with at most N_t time points, it can be interpreted as a discretized analogue of the iterated optimal-stopping approach in Chapter 3.2. This correspondence becomes more accurate as the grid is refined.

7.2.2 Double Deep-Q Networks and their application for impulse control problems

The dynamic programming formula (7.2.5) motivates the use of DRL methods for impulse control problems. Recall that in continuous time an impulse control is defined as a possibly infinite double-sequence

$$\alpha = \{(\tau_j, \zeta_j)\}_{j=1}^{M}, \quad M \leq \infty,$$

where τ_j is an intervention time and ζ_j is the corresponding impulse at τ_j. In the discrete-time DRL setting, interventions can only happen at grid points t_n. We therefore replace α by a finite action sequence

$$\beta = (a_0, a_1, \ldots, a_{N_t-1}).$$

Here, a_n is the action chosen by the DRL algorithm at time t_n. If $a_n = 0$, the agent does not intervene, and if $a_n \neq 0$ an intervention take place.

As in Chapter 6.1.5, let $\mathcal{Z}_{N_\zeta} := \{\zeta^{(1)}, \ldots, \zeta^{(N_\zeta)}\}$ be a finite, non-empty subset of $\mathcal{Z}$. We define set of admissible actions that an agent can take as

$$\mathcal{A} := \{0\} \cup \mathcal{Z}_{N_\zeta},$$

where the action 0 represents waiting (i.e. no intervention). Thus, the action set in the DDQN setting is the discrete impulse set $\mathcal{Z}_{N_\zeta}$, which is extended with a impulse of size zero. For $n = 0, 1, \ldots, N_t - 1$, the process $X^{(\beta)}$ controlled by the agent then reads as

$$X^{(\beta)}(t_n) = \begin{cases} X(t_n), & \text{if } a_n = 0 \\ \Gamma(t_n, \check{X}^{(\beta)}(t_n), a_n), & \text{if } a_n \neq 0, \end{cases}$$

where the uncontrolled process X evolves according to the jump-diffusion SDE (3.1.1) on the time interval $[t_n, t_{n+1})$.

Let Q_θ denote the Q-function approximated by a neural network with parameters θ. Inside the continuation region, the Q-value is given by

$$Q_\theta(t_n, x, 0) = \mathbb{E}^{(t_n,x)}\left[\int_{t_n}^{t_{n+1}} f(s, X(s))\,\mathrm{d}s + v(t_{n+1}, X(t_{n+1}))\right] \tag{7.2.6}$$

Using a left-point approximation of the integral, (7.2.6) becomes

$$Q_\theta(t_n, x, 0) \approx f(t_n, x)\Delta t + \mathbb{E}^{(t_n,x)}\left[V(t_{n+1}, X(t_{n+1}))\right]. \tag{7.2.7}$$

For an action $a_n \neq 0$, the Q-value is given by

$$Q(t_n, x, a_n) = K(t_n, x, a_n) + V(t_n, \Gamma(t_n, x, a_n)), \tag{7.2.8}$$

where K is the cost function and Γ is the post-intervention function defined in Chapter 3.1. Combining (7.2.7) and (7.2.8), the Q-value function reads as

$$Q_\theta(t_n, x, a) \approx r(t_n, x, a_n) + \mathbb{E}^{(t_n,x,a_n)}\left[v(t_{n+1}, X^{(\beta)}(t_{n+1}))\right],$$

where the reward function r is defined as

$$r(t_n, x, a_n) = f(t_n, x)\Delta t + 1_{\{a_n \neq 0\}} K(t_n, x, a_n).$$

At each time step t_n and state x, the value function V is then approximated by taking the best possible action, that is,

$$V(t_n, x) = \max_{a_n \in \mathcal{A}} Q_\theta(t_n, x, a_n). \tag{7.2.9}$$

(7.2.9) is a greedy approach.

It can be understood that the Q-function directly outputs the continuation and intervention regions. Indeed, if $a_n^* = \arg\max_{a \in \mathcal{A}} Q_\theta(t_n, x, a)$ is the optimal action at time step t_n, the learned continuation region is

$$\hat{D} = \left\{(t_n, x) : Q_\theta(t_n, x, 0) \geq \max_{\zeta \in \mathcal{Z}_{N_\zeta}} Q_\theta(t_n, x, \zeta)\right\}.$$

Hence, the neural network learns the free boundary between waiting and intervening.

Because neither the value function nor the Q-function are known, supervised regression cannot be used. Therefore, to approximate the Q-function, we use the DDQN algorithm. DDQN uses two neural networks: online and target networks. The online network Q_θ, with parameters θ, chooses the next action, whereas the target network $Q_{\bar\theta}$, with parameters $\bar\theta \neq \theta$, evaluates that action. Given a transition

$$(t_n, X_n^{(\beta)}, a_n, r_n, X_{n+1}^{(\beta)}),$$

the DDQN target is defined as

$$Y_n := r_n + Q_{\bar\theta}\left(t_{n+1}, x', \arg\max_{a' \in \mathcal{A}} Q_\theta(t_{n+1}, x', a')\right), \tag{7.2.10}$$

where we set $x' = X^{(\beta)}(t_{n+1})$ and $r_n = r(t_n, x, a_n)$ for brevity.

The online network, with parameters θ, is trained by minimising the Huber loss across all time steps, that is,

$$L^{\mathrm{DDQN}}(\theta) = \sum_{n=0}^{N_t-1} \mathbb{E}^{(t_0,x)} \left[\ell \left(Q_\theta(t_n, X^{(\beta)}(t_n), a_n) - Y_n \right) \right], \tag{7.2.11}$$

where $X(t_0) = x$ and the function ℓ is the Huber loss function defined as

$$\ell(z) = \begin{cases} \frac{1}{2} z^2, & \text{if } |z| \leq \delta \\ \delta \left(|z| - \frac{1}{2}\delta \right) & \text{otherwise.} \end{cases}$$

The threshold $\delta > 0$ is usually fixed. We set $\delta = 1$ which is common in DDQN approaches. The Huber loss is preferred over the mean-squared error as it improves training stability by being less sensitive to large errors and outliers.

In theory, the expectation in (7.2.11) can be approximated by Monte Carlo simulation. In practice, however, this approximation is performed by sampling a small random subset of previously observed transitions from the replay buffer $\mathcal{D}$. This random subset is called a *mini-batch*. Let M denote the cardinality of the mini-batch. Then, the loss (7.2.11) is approximated via

$$L^{\mathrm{DDQN}}(\theta) \approx \frac{1}{M} \sum_{m=1}^{M} \sum_{n=0}^{N_t-1} \ell \left(Q_\theta(t_n, X^{(\beta^m,m)}(t_n), a_n^m) - Y_n^m \right), \tag{7.2.12}$$

where $X^{(\beta^m,m)}$ denotes the m-th controlled path sampled from the mini-batch, generated under the sequence $\beta^m = (a_0^m, a_1^m, \ldots, a_{N_t-1}^m)$. The use of mini-batches was first introduced by Mnih et al. [79].

Using mini-batches during training instead of naive Monte Carlo simulations helps stabilize learning and improves efficiency by reducing variance in the updates. Instead of updating the network from a single transition, training on small batches of experiences sampled from a replay buffer reduces the variance of gradient updates and breaks correlations between consecutive samples. This leads to faster learning, more efficient use of collected data, and faster convergence of the Q-function.

7.2.3 Implementation and training procedure

As explained previously, the DDQN is implemented with two feedforward neural networks. Both networks use the ReLU function (7.1.3) as the activation function to capture kinks at the free boundary. The online network, denoted by Q_θ, has its parameters θ updated at the end of each training iteration. Specifically, the agent first simulates one step, stores the resulting transition within a replay buffer, and then updates the parameters θ based on the DDQN loss in (7.2.12). On the other hand, the target network $Q_{\bar{\theta}}$, which has the same neural network architecture as the online network and starts with the same initial parameters as the online network, does not have its parameters updated at every iteration. Instead, they are updated only after a fixed number of training iterations. This makes the target values more stable and predictable during training.

The input to the neural network is the current time step and the current state. For example, for the m-th simulated controlled path, the input reads as

$$(t_n, X^{(\beta^m,m)}(t_n)).$$

Before these values are passed to the network they are scaled. Specifically, we use the scaled time t_n/T and the scaled state x/R, where $R > 0$ is fixed in advance. This is done to improve numerical stability and to prevent inputs with large magnitudes from dominating the learning process.

The output of the neural network has size $N_\zeta + 1$, where N_ζ is the predetermined size of the subset $\mathcal{Z}_{N_\zeta}$. The first output corresponds to the waiting action $a = 0$. The remaining outputs correspond to all possible impulse actions. Thus, for a given time-state pair (t_n, x), the output of the online network is

$$Q_\theta(t_n, x, \cdot) = \big(Q_\theta\big(t_n, x, 0\big), Q_\theta\big(t_n, x, a^{(1)}\big), \ldots, Q_\theta\big(t_n, x, a^{(N_\zeta)}\big)\big)^\top \in \mathbb{R}^{N_\zeta+1}. \quad (7.2.13)$$

As in [1], the algorithm uses an ε-greedy exploration rule. With probability $1-\varepsilon$, where $\varepsilon \in (0,1]$, the agent chooses the admissible action with the highest Q-value. With probability $1-\varepsilon$, it chooses a random admissible action. This encourages the agent to explore alternative actions. The value of ε is then reduced gradually during training. Hence, the agent explores more in the beginning and becomes more greedy towards the end.

Each interaction with the environment gives one transition

$$(t_n, X_n^{(\beta)}, a_n, r_n, t_{n+1}, X_{n+1}^{\beta}),$$

where X_{n+1} denotes the uncontrolled path. These transitions are then stored in a replay buffer $\mathcal{D}$. Because we need to be able to sample at least M transitions from the replay buffer at the beginning of the first training iteration, the replay buffer is filled with a minimum number M of transitions. The algorithm then samples mini-batches from the replay buffer and computes the loss (7.2.12).

The online network is trained by backpropagation. First, the loss is computed from the sampled mini-bach. Then automatic differentiation is used to compute the gradient $\nabla_\theta L^{\mathrm{DDQN}}$. For details on backpropagation and automatic differentiation, see [33] and [10], respectively. The parameter set θ is then updated with gradient descent and the AdamW optimiser. The AdamW optimiser is a variant of Adam known for improved regularisation and training stability [69].

Finally, the target network is updated only at fixed intervals. The new weights are then computed using a soft update, i.e.

$$\bar{\theta} \leftarrow (1-\lambda)\bar{\theta} + \lambda\theta,$$

where $\lambda \in (0,1]$. The complete training procedure is summarised in Algorithm 1.

7.3 Policy-based approximation of impulse control problems

Thus far, we have considered a value-based approach in the form of the DDQN model, where the DRL model directly learns the Q-function. However, a limitation of value-based methods is that the policy is only obtained indirectly by first estimating a

Algorithm 1 DDQN Training procedure

Initialize Q-network Q_θ with random weights
Initialize target network $Q_{\bar{\theta}}$ with weights $\bar{\theta} \leftarrow \theta$
Initialize replay buffer $\mathcal{D}$ with capacity $N_{\mathcal{D}}$
Sample initial uncontrolled time-state pairs (t_n, X_n^m) for $m = 1, \dots, M$
for iteration $k = 1$ to N_{iter} **do**
    Set exploration rate $\varepsilon \leftarrow \varepsilon_k$
    for mini-batch $m = 1$ to M **do**
        Observe current state pair (t_n, X_n^m).
        Select action $a_n^* = \begin{cases} \text{random in } \mathcal{A}, & \text{w.p. } \epsilon \\ \arg\max_{a_n \in \mathcal{A}} Q_\theta(t, \Gamma(t_n, X_n^m, a_n), a_n), & \text{otherwise} \end{cases}$
        Execute action, observe r_n and $X_{n+1}^{(\beta^m, m)} = \Gamma(t_{n+1}, X_{n+1}^m, a_{n+1})$
        Store $(t_n, X_n^{(\beta^m, m)}, r_n, t_{n+1}, X_{n+1}^{(\beta^m, m)})$ in $\mathcal{D}$
        Sample minibatch from $\mathcal{D}$
        Compute the target Y_n with (7.2.10)
        Compute the loss L^{DDQN} with (7.2.12) and update θ by gradient descent (AdamW).
        if $k \mod 500 = 0$ **then**
            $\bar{\theta} \leftarrow \theta$
        end if
    end for
    Decay ε_k
end for

Q-function and then selecting actions that maximise the Q-estimate. While this approach is effective in discrete action spaces, it can struggle when action spaces are large and continuous, stochastic policies are desirable, or small errors in the Q-function can lead to poor greedy decisions.

With this in mind, we turn to actor-critic methods. In an actor-critic method, the actor selects actions according to a policy π, while the critic evaluates these actions by estimating the corresponding value function in Definition 7.4. Although the actor and the critic have distinct roles, their objectives are often related. As a result, it is often advantageous for them to use a single neural network with shared parameters and two outputs (one for the policy and one for the value function) rather than being trained on separate neural networks. This shared-network structure often improves computational efficiency and can lead to faster and more stable training.

A prominent actor–critic algorithm that commonly employs a shared network is the Proximal Policy Optimisation (PPO) algorithm, which we further augment with Self-imitation Learning (SIL) and entropy regularisation. As discussed later in this chapter, these augmentations provide additional stability and encourage effective exploration. The combination of PPO with SIL is inspired by the work of Oh et al. [83], where it is shown to improve policy learning. Furthermore, the PPO-SIL approach was first applied on impulse control problems in finance by Jain et al. [59], where it was empirically shown to be effective in market making applications.

7.3.1 From the performance criterion to the PPO surrogate

As with many DRL models, it is common to work on a discrete time grid. As such, recall the partition $\mathcal{T}$ on the time interval $[t, T]$ (7.1.1). Also, recall the performance criterion J, which is given by

$$J^{(\alpha)}(t,x) = \mathbb{E}^{(t,x)}\left[\int_t^T f(s, X^{(\alpha)}(s))\,\mathrm{d}s + g(T, X^{(\alpha)}(T)) + \sum_{t \le \tau_j < T} K(\tau_j, \check{X}^{(\alpha)}(\tau_j^-), \zeta_j)\right].$$

Similarly as in Chapter 7.2, we denote by $\beta := (a_0, a_1, \dots, a_{N_t})$ the sequence of actions taken by the reinforcement learning model on the time partition $\mathcal{T}$. However, the main difference now is that the action a_n is allowed to be continuous. Hence, with $\mathcal{A} := \{0\} \cup \mathcal{Z}$, $a_n \in \mathcal{A}$ is the continuous action taken at time step n. Moreover, we assume that the policy π can be approximated by a neural network with parameters θ and denote by π_θ the neural network approximation of the (true) policy π. Every action chosen by the agent is therefore sampled from π_θ.

Using the previously defined time grid and the Riemann sum to approximate the integral in J, the reinforcement learning performance criterion reads as

$$J^{(\beta)}(x,t) = \sum_{n=0}^{N_t} \mathbb{E}_{\pi_\theta}^{(t_0,x)}\left[r\left(t_n, X^{(\beta)}(t_n), a_n\right)\right],$$

where the reward function r is given by

$$r(t,x,\zeta) = \begin{cases} f(t,x)\Delta t + 1_{\{a \neq 0\}} K(t,x,a) & \text{if } t < T \\ g(t,x) & \text{if } t = T \\ 0 & \text{otherwise.} \end{cases} \tag{7.3.1}$$

For t, x, and T fixed, it can then be shown that the training objective of the deep reinforcement learning model takes the form

$$L(\theta) = \sum_{n=0}^{N_t} \mathbb{E}_{\pi_{\theta_{\text{old}}}}^{(t_0,x)}\left[\frac{\pi_\theta\left(a_n | X^{(\beta)}(t_n)\right)}{\pi_{\theta_{\text{old}}}\left(a_n | X^{(\beta)}(t_n)\right)} G\left(t_n, X^{(\beta)}(t_n), a_n\right)\right] + \mathcal{O}(\|\theta - \theta_{\text{old}}\|^2), \tag{7.3.2}$$

where θ_{old} denotes the neural network parameter set from the previous training iteration, θ denotes the new (unknown) neural network parameters from the current training iteration, and G is the advantage function (see Definition 7.6). The main objective of the DRL algorithm is therefore to find some optimal parameters θ that maximise the sum of expected rewards, i.e.

$$\theta^* = \arg\max_\theta L(\theta; t, x, T),$$

which is done by means of policy gradients [94]. DRL methods with an objective of maximising (7.3.2) are commonly referred to as the *Trust Region Policy Optimisation* (TRPO) [89].

However, the TRPO objective L has a caveat. In particular, it does not explicitly constrain the updated parameter vector θ to remain close to the previous parameter set

θ_{old}. As a result, the quadratic error term in (7.3.2) can become too large. Therefore, maximising L with gradient-based methods can lead to overly large parameter updates that lead to local maxima rather than the desired global solution.

As such, let $X^{(\beta)}(t_n) := X_n^{(\beta)}$ for brevity and define the generalised advantage estimate (GAE) $\hat{G}_n$ as

$$\hat{G}_n = \sum_{j=n}^{N_t} \lambda_{\text{GAE}}^{j-n} \big(r\big(t_j, X_j^{(\beta)}, a_j\big) + V_\theta\big(t_{j+1}, X_{j+1}^{(\beta)}\big) - V_\theta\big(t_{j+1}, X_{j+1}^{(\beta)}\big)\big), \tag{7.3.3}$$

where $\lambda \in (0,1)$ is a hyperparameter and V_θ is a neural network estimate of the value function. In essence, the advantage estimate represents a bias-variance trade-off. Lower values for λ_{GAE} reduce variance but introduce bias, whereas higher values decrease bias but increase variance, see [90].

The issue around the TRPO objective is then addressed by modifying the objective as follows

$$\tilde{L}^{\text{PPO}}(\theta) = \sum_{n=0}^{N_t} \mathbb{E}_{\pi_{\theta_{\text{old}}}}^{(t_0,x)} \left[\min \left\{ \frac{\pi_\theta(a_n|X_n^{(\beta)})}{\pi_{\theta_{\text{old}}}(a_n|X_n^{(\beta)})} \hat{G}_n, \text{clip}\left(\frac{\pi_\theta(a_n|X_n^{(\beta)})}{\pi_{\theta_{\text{old}}}(a_n|X_n^{(\beta)})}, 1-\varepsilon, 1+\varepsilon \right) \right\} \right],$$

where $\varepsilon > 0$ is a hyperparameter and the function clip is defined as

$$\text{clip}(y; b, c) = \begin{cases} b, & \text{if } x \le b \\ y, & \text{if } b < y < c \\ c, & \text{if } y \ge c. \end{cases}$$

For practical implementations, it is more convenient to minimise the training objective rather than maximising it. As such, we define

$$L^{\text{PPO}}(\theta) = -\tilde{L}^{\text{PPO}}(\theta). \tag{7.3.4}$$

Then the objective of the PPO algorithm is to find some optimal parameter set θ^* that minimises the loss L^{PPO} given the initial time t, initial state x, and a finite time horizon $T > t$. Here as well the minimisation is done through policy gradients. The PPO approach was first introduced by Schulman et al. [91].

Remark 7.14. In RL literature, the term *clipped surrogate* is used to describe the training objective L^{PPO}. In our case, this can be understood as a substitute objective that we optimise instead of the performance criterion J. The hope is that the model learns the actions optimising J via L^{PPO}.

7.3.2 Self-imitation learning and entropy regularisation

As Jain et al. [59] note, standard gradient policy methods, including the PPO algorithm, suffer from high variance and poor sample efficiency when applied to financial trading problems. This inefficiency can be attributed to the sparsity of high-reward trajectories in the policy distribution, particularly during early training phases when the agent has not yet discovered profitable patterns.

To address these challenges, we add to the PPO loss a self-imitation learning (SIL) loss. Briefly speaking, the objective of SIL is to imitate the agent's past good experiences. As such, one augments the PPO-surrogate loss (7.3.4) by adding a cross-entropy term that encourages the current policy to replicate actions from a replay buffer of high-performing trajectories, i.e.

$$L^{\mathrm{SIL}}(\theta) = -\mathbb{E}_{(a_n, X_n^\beta, R_n)\sim\mathcal{B}_{\mathrm{good}}}\left[1_{\left\{R_n > V_\theta(t_n, X_n^{(\beta)})\right\}} \log \pi_\theta(a_n | X_n^{(\beta)})\right],$$

where $\mathcal{B}_{\mathrm{good}} = \{(a_n, X_n^\beta, R_n)_{n\geq 0}\}$ denotes the SIL replay buffer containing good past experiences, R_n is the remaining sum of returns starting from time step t_n, and V_θ is the neural network approximation of the value function. It is worth noting that the SIL buffer stores individual state-action-return transitions and not complete paths. The theoretical justification for the SIL algorithm can be found in Oh et al. [83].

As pointed out by Jain et al. [59], the application of self-imitation learning provides several advantages in our impulse control setting. Specifically,

1. **Avoidance of catastrophic forgetting:** By explicitly maintaining and imitating past successes, SIL ensures that profitable patterns, once discovered, remain accessible to the policy.
2. **Accelerated convergence:** The sparsity of profitable trajectories means that policy gradients provide weak learning signals, particularly during early training when the agent's exploration is largely undirected. SIL effectively amplifies the learning signal from rare successful experiences by repeatedly reinforcing the actions that led to those outcomes, thereby accelerating the convergence to profitable strategies.
3. **Variance reduction:** By focusing policy updates on trajectories with high returns rather than the full distribution of explored behaviors, SIL reduces the variance of gradient estimates.

In addition to the self-imitation buffer, we add an entropy regularisation term to the total actor loss. Specifically, we define entropy as

$$\mathcal{H}(\pi_\theta) := \mathcal{H}(\pi_\theta(\cdot|x)) = -\int_{\mathcal{A}} \pi_\theta(a|x) \log \pi_\theta(a|x)\, \mathrm{d}a. \tag{7.3.5}$$

Entropy regularisation is commonly used in RL to prevent the policy from becoming too deterministic too early. For instance, during training, an agent may initially assign high probability to actions that only appear good because of limited experience. By adding an entropy term to the objective, the agent is encouraged to keep its action distribution sufficiently spread out, encouraging continued exploration of alternative behaviours. This helps to reduce premature convergence to suboptimal policies and improves the chance of discovering actions that lead to higher long-term rewards. Hence, entropy regularisation balances the exploitation of currently promising actions and the exploration of uncertain ones.

The combined actor loss of the actor becomes

$$L^{\mathrm{actor}}(\theta) = L^{\mathrm{PPO}}(\theta) + \lambda_{\mathrm{SIL}} L^{\mathrm{SIL}}(\theta) - \lambda_{\mathcal{H}} \mathcal{H}(\pi_\theta), \tag{7.3.6}$$

where $\lambda_{\mathrm{SIL}}, \lambda_{\mathcal{H}} > 0$ are hyperparameters that control the SIL-loss and entropy regularisation, respectively. The task of finding the optimal parameter set θ^* is translated into a minimisation problem, where the goal is to minimise the total actor loss (7.3.6) by means of gradient descent with respect to θ.

7.3.3 Critic and total losses

Let R_n denote the pathwise return-to-go in the interval $[t_n, T]$ and define it by

$$R_n := \sum_{m=n}^{N_t} r(t_n, X_n^{(\beta)}, a_n). \tag{7.3.7}$$

We let the critic-loss be the mean-squared error between the value function approximation V_θ and the total future rewards, i.e.

$$L^{\text{critic}}(\theta) = \mathbb{E}_{\pi_\theta}^{(t_0,x)}\left[\left(V_\theta(t_n, X_n^{(\beta)}) - R_n\right)^2\right], \tag{7.3.8}$$

which we aim to minimise. Together with the actor loss (7.3.6), the overall actor–critic objective is defined as the weiighted sum of the actor and critic losses, namely

$$L^{\text{total}}(\theta) = \lambda_{\text{actor}} L^{\text{actor}}(\theta) + L^{\text{critic}}(\theta),$$

where $\lambda_{\text{actor}} > 0$ is a hyperparameter that scales the actor loss.

7.3.4 Decomposition of the Impulse Control Problem

Inspired by the auxiliary control formulations found in [59, 77], and by the fact that an impulse control consists of both an intervention time and an impulse size, we approximate the impulse control problem by breaking it down into two components: when to intervene and, conditional on intervention, what impulse to apply. Specifically, we introduce two actor-critic networks:

- **A decision network d_χ (timing):** A network with parameters χ that determines *when* to intervene, and;
- **An impulse-size network u_ξ (impulse selection):** A network with parameters ξ that determines *what* impulse size ζ to execute, conditional on the decision to intervene.

Let us now recall that the intervention operation on the value function v is given by

$$\mathcal{M}v(t,x) = \sup_{\zeta \in \mathcal{Z}} \left\{ v(t, \Gamma(t,x,\zeta)) + K(t,x,\zeta) \right\}.$$

The decision network is meant to approximate the intervention region. Heuristically, the optimal binary decision satisfies

$$d^*(t,x) \approx 1_{\{\mathcal{M}v(t,x) \geq v(t,x)\}}.$$

As the decision is binary, the network outputs an intervention probability

$$p_\chi(t,x) := \pi_\chi\left(d = 1|x\right),$$

where $\pi_\chi(d|x)$ for $d \in \{0,1\}$ is the decision policy and $d := d(t,x)$. The binary decision d is sampled through

$$d_n \sim \text{Bernoulli}\left(p_\chi(t,x)\right).$$

Thus, $d = 0$ means continuation, while $d = 1$ means intervention.

On the other hand, the impulse-size network samples, conditionally on $d = 1$, an impulse

$$\zeta \sim \pi_\xi\left(\cdot|x\right).$$

If $d = 0$, no impulse is applied. Using the critic $V_{\chi,\xi}$, which is the approximation of the policy-value function under the current joint policy $\pi_{\chi,\xi}$ (see Definition 7.4), the actor network approximates

$$u_\xi(t,x) \approx \arg\max_{\zeta\in\mathcal{Z}} \left\{V_{\chi,\xi}(t,\Gamma(t,x,\zeta)) + K(t,x,\zeta)\right\}.$$

Now because action the action a essentially contains the outputs of two neural networks, we define the action to be a pair

$$a := \begin{cases} (0,0) & \text{when } d = 0 \\ (1,\zeta) & \text{when } d = 1 \end{cases}$$

The joint policy is then

$$\pi_{\chi,\xi}(a|x) = \begin{cases} \pi_\chi\big(0|x\big), & \text{when } d = 0, \\ \pi_\chi\big(1|x\big)\pi_\xi(\zeta|x), & \text{when } d = 1. \end{cases} \tag{7.3.9}$$

When it comes to the neural network parameter updates, we note that the impulse-size network u_ξ is updated using only time steps at which the switch d was active (i.e. when intervention was made). In contrast, the decision network d_χ is updated at every time step because every time step requires a decision between continuation and intervention.

The PPO loss (7.3.4) for the two-headed policy is then given by

$$L^{\text{PPO}}(\chi,\xi) = -\sum_{n=0}^{N_t} \mathbb{E}^{(t_0,x)}_{\pi_{\chi_{\text{old}},\xi_{\text{old}}}} \Bigg[\hat{G}_n \min\Bigg\{\frac{\pi_{\chi,\xi}(a_n|X_n^{(\beta)})}{\pi_{\chi_{\text{old}},\xi_{\text{old}}}(a_n|X_n^{(\beta)})}, \\ \text{clip}\left(\frac{\pi_{\chi,\xi}(a_n|X_n^{(\beta)})}{\pi_{\chi_{\text{old}},\xi_{\text{old}}}(a_n|X_n^{(\beta)})}, 1-\varepsilon, 1+\varepsilon\right)\Bigg\}\Bigg], \tag{7.3.10}$$

where $\hat{G}_n$ is given by (7.3.3).

The SIL objective is

$$L^{\mathrm{SIL}}(\chi,\xi) = -\sum_{n=0}^{N_t} \mathbb{E}_{(a_n, X_n^{\beta}, R_n)\sim \mathcal{B}_{\mathrm{good}}} \left[1_{\left\{R_n > V_{\chi,\xi}(t_n, X_n^{(\beta)})\right\}} \log \pi_{\chi,\xi}(a_n | X_n^{(\beta)}) \right]. \tag{7.3.11}$$

Finally, for a fixed time t and state x, the entropy (7.3.5) is derived via

$$\begin{aligned}
\mathcal{H}(\pi_{\chi,\xi}) &:= -\int_{\mathcal{A}} \pi_{\chi,\xi}(a|x) \log \pi_{\chi,\xi}(a|x)\, \mathrm{d}a \\
&= -\sum_{d_\chi(t,x)\in\{0,1\}} \int_{\mathcal{Z}} \pi_{\chi,\xi}\big((d_\chi(t,x),\zeta)|x\big) \log \pi_{\chi,\xi}\big((d_\chi(t,x),\zeta)|x\big)\, \mathrm{d}\zeta \\
&= \underbrace{-\sum_{d_\chi(t,x)\in\{0,1\}} \pi_\chi\big(d_\chi(t,x)|x\big) \log \pi_\chi\big(d_\chi(t,x)|x\big)}_{=:\mathcal{H}(\pi_\chi)} \\
&\quad \underbrace{- \pi_\chi\big(d_\chi(t,x)=1|x\big) \int_{\mathcal{Z}} \pi_\xi\big(\zeta|x\big) \log \pi_\xi\big(\zeta|x\big)\, \mathrm{d}\zeta}_{=p_\chi(t,x)\mathcal{H}(\pi_\xi)} \\
&= \mathcal{H}(\pi_\chi) + p_\chi(t,x)\mathcal{H}(\pi_\xi).
\end{aligned} \tag{7.3.12}$$

The entropies $\mathcal{H}(\pi_\chi)$ and $\mathcal{H}(\pi_\xi)$ in (7.3.12) are computed using (7.3.5).

Combining (7.3.10)–(7.3.12), the actor loss of the two-headed PPO-SIL policy is given by

$$L^{\mathrm{actor}}(\chi,\xi) = L^{\mathrm{PPO}}(\chi,\xi) + \lambda_{\mathrm{SIL}} L^{\mathrm{SIL}}(\chi,\xi) - \lambda_{\mathcal{H}}\mathcal{H}(\pi_{\chi,\xi}),$$

with $\lambda_{\mathrm{SIL}}, \lambda_{\mathcal{H}} > 0$ as fixed hyperparameters. The critic is trained separately by minimising

$$L^{\mathrm{critic}}(\chi,\xi) = \sum_{n=0}^{N_t} \mathbb{E}^{(t_0,x)}_{\pi_{\chi,\xi}} \left[\left(V_{\chi,\xi}(t_n, X_n^{(\beta)}) - R_n \right)^2 \right].$$

Hence, the total loss of the actor-critic is

$$L^{\mathrm{total}}(\chi,\xi) = L^{\mathrm{actor}}(\chi,\xi) + L^{\mathrm{critic}}(\chi,\xi). \tag{7.3.13}$$

7.3.5 Implementation and training procedure

As in the DDQN model discussed in Chapter 7.2, we use two feedforward neural networks with identical architecture and initial weights χ_{init} and ξ_{init}, respectively. However, the main difference is now that one network outputs the decision $d_\chi \in \{0,1\}$ and the other network outputs the impulse-size $u_\xi \approx \zeta$. Both networks use the hyperbolic tangent (7.1.4) as activation function. For the implementation, the code written by Jain et al. [59] is used, see [58].

The PPO-SIL algorithm is trained using simulated rollouts[3]. Each rollout produces a full path of states, actions, rewards, and pointwise value function estimates. Then, at the beginning of each training iteration, a sample of paths of the uncontrolled process

[3]A rollout is one simulation of the controlled process.

X is generated. Conditional on these paths, the current joint policy $\pi_{\chi,\xi}$, which is given by (7.3.9), is applied step by step along the time grid. That is, at each time step t_n, the policy observes the current controlled state $X_n^{(\beta)}$ and samples an action a_n. If intervention is chosen, the state is moved from $X_n^{(\beta)}$ to $\Gamma(t_n, \check{X}_n^{(\beta)}, \zeta_n)$. Otherwise, the process evolves without intervention until the next decision time. This approach results in a pathwise realisation of the controlled process, and the corresponding rewards, actions, value estimates, and log-probabilities are collected and stored inside the SIL buffer.

Let the superscript m denote the m-th rollout path of the controlled process $X^{(\beta)}$ and let $\beta^m = (a_0^m, a_1^m, \dots, a_{N_t}^m)$ denote the corresponding strategy. Return-to-go (7.3.7) and generalised advantage estimate (7.3.3) are computed for each simulated path. The PPO loss is then approximated using

$$L^{\text{PPO}}(\chi,\xi) \approx -\frac{1}{N_{\text{rollout}}} \sum_{n=0}^{N_t} \sum_{m=1}^{N_{\text{rollout}}} \hat{G}_n^m \min\left\{ \frac{\pi_{\chi,\xi}(a_n^m | X_n^{(\beta^m,m)})}{\pi_{\chi_{\text{old}},\xi_{\text{old}}}(a_n^m | X_n^{(\beta^m,m)})}, \right.$$
$$\left. \text{clip}\left(\frac{\pi_{\chi,\xi}(a_n^m | X_n^{(\beta^m,m)})}{\pi_{\chi_{\text{old}},\xi_{\text{old}}}(a_n^m | X_n^{(\beta^m,m)})}, 1-\varepsilon, 1+\varepsilon \right) \right\}. \tag{7.3.14}$$

The SIL loss is given by

$$L^{\text{SIL}}(\chi,\xi) \approx -\frac{1}{N_{\text{batch}}} \sum_{n=0}^{N_t} \sum_{m=1}^{N_{\text{batch}}} 1_{\left\{R_n^m > V_{\chi,\xi}(t_n)\right\}} \log \pi_{\chi,\xi}(a_n^m, X_n^{(\beta^m,m)}), \tag{7.3.15}$$

where $a_n, X_n^{(\beta^m)}, R_n^m$ are sampled from the SIL replay buffer $\mathcal{B}_{\text{good}}$, and the critic loss by

$$L^{\text{critic}}(\chi,\xi) = \frac{1}{N_{\text{rollout}}} \sum_{n=0}^{N_t} \sum_{j=1}^{N_{\text{rollout}}} \left(V_{\chi,\xi}(t_n, X_n^{(\beta^m,m)}) - R_n^m \right)^2. \tag{7.3.16}$$

After each training iteration, the SIL replay buffer $\mathcal{B}_{\text{good}}$ is updated with state-action-return tuples, $(a_n^m, X_n^{(\beta^m,m)}, R_n^m)$, for which the realised return exceeds the current value estimate. As mentioned previously, this encourages the policy to imitate past successful decisions.

Finally, the total loss (7.3.13) is obtained by combining the PPO loss (7.3.14), the SIL loss (7.3.15), the critic loss (7.3.16), and the entropy regularisation term (7.3.12). Gradients with respect to the parameters χ and ξ are then computed, and the network is updated using backpropagation with the Adam optimiser. A simplified overview of the training procedure is provided in Algorithm 2.

Algorithm 2 PPO-SIL Training procedure

Initialize decision network d_χ, impulse network u_ξ, and value function $V_{\chi,\xi}$
Initialize SIL buffer $\mathcal{B}_{\text{good}}$ with capacity $N_{\mathcal{B}_{\text{good}}}$
for iteration $k = 1$ to N_{iter} **do**
  Set old policy parameters $\chi_{\text{old}} \leftarrow \chi$ and $\xi_{\text{old}} \leftarrow \xi$
  Sample N_{rollout} uncontrolled paths on the time grid $\{t_0, \dots, t_{N_t}\}$
  for rollout path $m = 1$ to N_{rollout} **do**
    for time step $n = 0$ to $N_t - 1$ **do**
      Observe current state $S_n^m = (t_n, X_n^{(\beta^m,m)})$
      Sample intervention decision $d_n^m \sim \pi_{\chi_{\text{old}}}(\cdot \mid S_n^m)$
      if $d_n^m = 1$ **then**
        Sample impulse $\zeta_n^m \sim \pi_{\xi_{\text{old}}}(\cdot \mid s_n^m)$ and set $a_n^m = (1, \zeta_n^m)$
        Apply intervention $X_n^{(\beta^m,m)} \leftarrow \Gamma(t_n, \check{X}_n^{(\beta^m,m)}, \zeta_n^m)$
      else
        Set $a_n^m = (0, 0)$
      end if
    end for
    Compute returns-to-go R_n^m and the advantage estimate $\hat{G}_n^m$ using (7.3.7) and (7.3.3), respectively, for $n = 0, \dots, N_t - 1$
  end for
  Add all transitions (s_n^m, a_n^m, R_n^m) with $R_n^m > V_{\chi,\xi}(s_n^m)$ to $\mathcal{B}_{\text{good}}$
  if $|\mathcal{B}_{\text{good}}| > N_{\mathcal{B}_{\text{good}}}$ **then**
    Remove the oldest transitions from $\mathcal{B}_{\text{good}}$
  end if
  Compute the total loss using (7.3.13)
  Update (χ, ξ) by gradient descent using Adam
end for

CHAPTER

EIGHT

NUMERICAL EXPERIMENT I: IMPULSE CONTROL OF THE FOREIGN EXCHANGE RATE

The optimal impulse control of foreign exchange (FEX) rates constitutes one of the simplest and earliest financial applications of impulse control, with research made in the late 1990s, see [81, 25, 26]. In our case, we were inspired by [3, Chapter 5]. In this chapter, we apply the deep reinforcement learning methods discussed in Chapter 7 to approximate the solution of a simplified version of the FEX rate problem. We use the explicit-impulse scheme derived in Chapter 6 as a benchmark solution to compare with the DRL methods. The Python code used in this experiment can be found in [21].

8.1 Problem formulation

Consider a central bank that can intervene in the FEX market by buying or selling foreign currency in large quantities. Let w be a constant and denote the difference between the domestic and foreign interest rates over the time interval $[t, T]$. In addition, let $t \leq \tau_1 \leq \tau_2 \leq \ldots \leq \infty$ be the times when the central bank intervenes in the FEX market with the corresponding intervention sizes $\zeta_1, \zeta_2, \ldots$, with $\zeta_j \in \mathbb{R}$ for every $j = 1, 2, \ldots$. We denote impulse controls using the sequence $\alpha = \{(\tau_j, \zeta_j)_{j=1}^{M}\}$ for $M \leq \infty$. The solvency and the admissible impulse sets are given by $\mathcal{S} := \mathbb{R}$ and $\mathcal{Z}(t, x) := \mathbb{R}$, respectively.

In this model, a positive value for ζ represents the purchase of foreign currency, while a negative value for ζ indicates the sale of a foreign currency. We assume that the market is *liquid*, roughly meaning that there will always be a buyer (seller) side willing to execute trades at the prevailing market price without causing significant price impact.

Let X and $X^{(\alpha)}$ respectively denote the uncontrolled and controlled FEX rate process in the logarithmic scale, with both processes starting at $X(t) = x \in \mathbb{R}$. We assume that $X^{(\alpha)}$ evolves according to

$$X^{(\alpha)}(0) = x \quad \text{and} \quad X^{(\alpha)}(t) = X(t), \qquad \text{for } t < \tau_1 \tag{8.1.1a}$$
$$X^{(\alpha)}(\tau_j) = X(\tau_j^-) + \zeta_j, \qquad \text{for } j = 1, 2, \ldots, \tag{8.1.1b}$$
$$\mathrm{d}X(t) = -\mu w \,\mathrm{d}t + \sigma \,\mathrm{d}W(t), \qquad \tau_j \leq t < \tau_{j+1} \wedge T. \tag{8.1.1c}$$

where μ and σ are nonnegative, constant drift and volatility terms of X, respectively. In particular, (8.1.1c) has the solution

$$X(t) - X(\tau_j) = -\mu w t + \sigma W(t), \quad \text{for } \tau_j \leq t < \tau_{j+1} \wedge T. \tag{8.1.2}$$

Furthermore, we infer from (8.1.1b) that the post-intervention function takes the form $\Gamma(x, \zeta) := x + \zeta$.

The central bank's objective is to keep the FEX rate as close as possible to some target m, where trading results in intervention costs of the form

$$K(t, x, \zeta) = -e^{-\delta t}(c + \kappa|\zeta|) \tag{8.1.3}$$

Here, $\delta \in \mathbb{R}$ denotes the constant, risk-free rate, and $c > 0$ and $\kappa > 0$ are the fixed and proportional transaction costs respectively. The profit and bequest functions, f and g respectively, are defined as

$$f(t, x) := e^{-\delta t}\left(-(x - m)^2 + \gamma w^2\right) \quad \text{and} \quad g(t, x) := 0,$$

where $\gamma \geq 0$ denotes the effect of interest rate differential. The performance criterion then reads

$$J^{(\alpha)}(t, x) = \mathbb{E}^{(t,x)}\left[-\int_t^T e^{-\delta s}\left((X^{(\alpha)}(s) - m)^2 + \gamma w^2\right) \mathrm{d}s - \sum_{t \leq \tau_j < T} e^{-\delta \tau_j}\left(c + \kappa|\zeta_j|\right)\right]. \tag{8.1.4}$$

Here, the term $(x - m)^2$ penalises the central bank from straying from the target m, while the term γw^2 penalises the central bank for having chosen an interest rate that is either too low or too high with respect to the foreign interest rate in the interval $[t, T]$.

The definition of the value function v remains unchanged and is given by

$$v(t, x) = \sup_\alpha J^{(\alpha)}(t, x).$$

In this problem, the value function is understood to be the present value of the total costs a central bank must pay in total in an effort to keep the FEX rate as close to the target m as possible.

For the parameters in the optimal FEX impulse control problem, we use the values reported in Table 1. These parameter values coincide with those used in [3].

	Parameter	Value
Interest rate differential	w	0.05
Drift speed	μ	0.25
Volatility	σ	0.3
Target	m	0.02
Effect of interest rate differential	γ	3
Proportional transaction cost	κ	1
Fixed transaction cost	c	0.1
Discount factor	δ	0.02
Time horizon (years)	T	1
Truncated domain boundary	R	2

Table 1: Parameters for FEX rate problem

8.2 Implementation details

8.2.1 Explicit-impulse scheme

For the optimal FEX rate problem, the QVI reads as

$$\begin{aligned}\min\Big\{ -\Big(\partial_t v - \mu w \partial_x v + \frac{1}{2}\sigma^2 \partial_{xx} v - e^{-\delta t}\big((x-m)^2 + \gamma w^2\big)\Big), \\ v - \mathcal{M}v\Big\} = 0 \quad \text{on } (t,x) \in [0,T) \times \mathcal{S} \\ v = 0 \quad \text{on } (t,x) \in \{T\} \times \mathcal{S},\end{aligned}$$

where the intervention operator is now

$$\mathcal{M}v(t,x) = \sup_{\zeta \in \mathbb{R}} \left\{ v(t, x+\zeta) - e^{-\delta t}(c + \kappa|\zeta|) \right\}.$$

Recall that we set $\mathcal{S} := \mathbb{R}$ and $\mathcal{Z} := \mathbb{R}$. This is numerically infeasible, so we truncate $\mathcal{S}$ to $(-R, R)$ and $\mathcal{Z}(t,x)$ to $(-R-x, R-x)$ for $R > 0$. Specifically, let $\Omega = [0,T) \times (-R,R)$, $\partial_R^+\Omega = [0,T) \times \{-R, R\}$, and $\partial_T^+\Omega = \{T\} \times [-R,R]$. The truncated QVI is

$$\begin{aligned}\min\Big\{ -\Big(\partial_t v - \mu w \partial_x v + \frac{1}{2}\sigma^2 \partial_{xx} v - e^{-\delta t}\big((x-m)^2 + \gamma w^2\big)\Big), \\ v - \mathcal{M}v\Big\} = 0 \qquad \text{on } \Omega \qquad (8.2.1a) \\ \min\Big\{ -\Big(\partial_t v - e^{-\delta t}\big((x-m)^2 + \gamma w^2\big)\Big), \\ v - \mathcal{M}v\Big\} = 0 \qquad \text{on } \partial_R^+\Omega \qquad (8.2.1b) \\ v = 0 \qquad \text{on } \partial_T^+\Omega \qquad (8.2.1c)\end{aligned}$$

where (8.2.1b) corresponds to imposing artificial Neumann boundary conditions

$$\partial_x v(t, \pm R) = 0 \quad \text{and} \quad \partial_{xx} v(t, \pm R) = 0.$$

Convergence arguments for the explicit-impulse scheme for the truncated QVI can be found in [3, Appendix C].

To solve the QVI on a numerical grid with N_x spatial and N_t time grid points, we discretise the set of admissible impulses as follows

$$\mathcal{Z}_{\Delta x}(t,x) = \{x_0 - x, x_1 - x, \ldots, x_{N_x} - x\},$$

where x_i, $i = 0, 1, \ldots, M$, is a grid point. Given a numerical solution $V^n = (V_0^n, V_1^n, \ldots, V_M^n)$ for timestep n, the intervention operator becomes

$$\begin{aligned}
(\mathcal{M}_n V^n)_i &= \max_{\zeta_i \in \{x_0 - x_i, \ldots, x_M - x_i\}} \left\{ (\mathcal{I}_{\Delta x} V^n)(x_i + \zeta_i) - e^{-\delta t_n}(\kappa|\zeta_i| + c) \right\} \\
&= \max_{0 \le j \le M} \left\{ (\mathcal{I}_{\Delta x} V^n)(x_j) - e^{-\delta t_n}(\kappa|x_j - x_i| + c) \right\} \\
&= \max_{0 \le j \le M} \left\{ V_j^n - e^{-\delta t_n}(\kappa|x_j - x_i| + c) \right\}.
\end{aligned}$$

Thus, the value function at time n, denoted by V^n, is obtained by solving the linear system

$$AV^n = y^n,$$

where the matrix A now reads as

$$A = I + \frac{\Delta t}{2(\Delta x)^2} \begin{pmatrix}
0 & & & & & \\
-\sigma^2 & 2\sigma^2 & -\sigma^2 & & & \\
& -\sigma^2 & 2\sigma^2 & -\sigma^2 & & \\
& & \ddots & \ddots & \ddots & \\
& & & -\sigma^2 & 2\sigma^2 & -\sigma^2 \\
& & & & & 0
\end{pmatrix}$$

and y_i^n is given by

$$y_i^n := \max \left\{ (\mathcal{I}_{\Delta x} V^{n-1})(x_i + \mu w \Delta t) - e^{-\delta t_n} \left(-(x_i - m)^2 + \gamma w^2 \right) \Delta t, (\mathcal{M}_n V^{n-1})_i \right\}.$$

8.2.2 DDQN

The DDQN algorithm uses two fully connected feedforward neural networks, one for the target network and one for the online network, with identical architecture

$$\text{Input: } 2 \to 512 \to 64 \to 101 \text{: Output}$$

per network. As with the explicit-impulse scheme, the admissible impulse set is discretised. Specifically, we use $N_\zeta = 100$ candidate impulses. Since the DDQN algorithm evaluates both the no-intervention action and each admissible impulse candidate, the network must output $N_\zeta + 1 = 101$ Q-values. Hence, the output layer has 101 neurons. The two layers in-between the input and output layers have hidden dimensions 512 and 64, respectively, and the ReLU function is used for activation. While such a bottleneck neural network architecture may seem unusual, the reduction in dimensions limits the amount of information that can pass through. This forces the network to retain only the most crucial information. Moreover, as the number of trainable parameters decreases with the number of neurons, the training time decreases in such an approach, see [51, 54, 57].

In order to compare the value approximation of the DDQN algorithm across different initial FEX rates $X(0) = x_0$ at time $t = 0$, the controlled FEX rate $X^{(\beta)}$ starts various initial conditions in the interval $x \in [-R, R]$, where $R = 2$. Simulation data is generated according to (8.1.2). Furthermore, the reward function r is given by

$$r(t, x, a) = e^{-\delta t} \left(-(x-m)^2 + \gamma w^2\right) \Delta t + 1_{\{a \neq 0\}} K(t, x, a),$$

where K is specified in (8.1.3).

The exploration rate ε_k is decreased at each iteration $k = 1, \dots, N_{\text{iter}}$. Specifically, we set

$$\varepsilon_k = \varepsilon_{\max} - \min\left\{1, \frac{k}{\max\{1, 0.8 N_{\text{iter}}\}}\right\} (\varepsilon_{\max} - \varepsilon_{\min}) \quad \text{for } k = 1, \dots, N_{\text{iter}}.$$

Figure 1 shows how the exploration rate ε_k is set over N_{iter} =1'000'000 iterations.

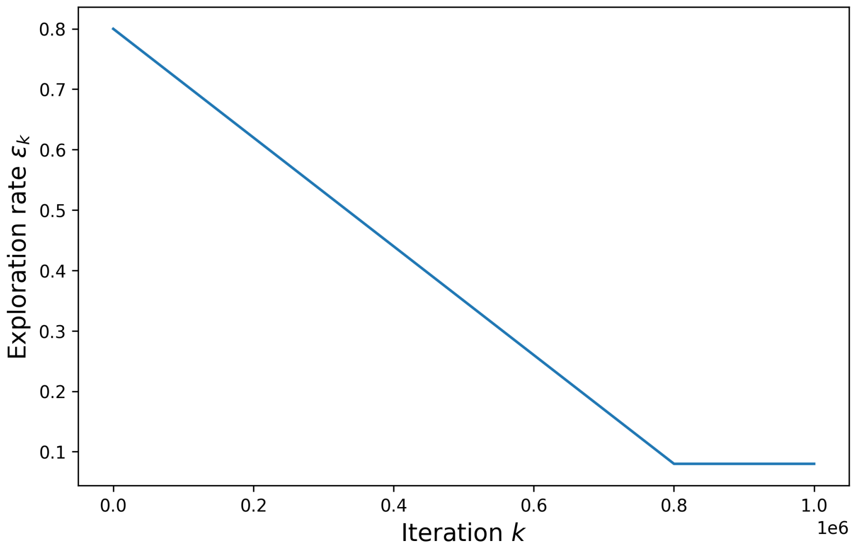

Figure 1: Exploration rate per iteration k.

Finally, the replay buffer has a capacity of 1'000'000 transitions and, at each iteration, it is filled with $M = 1'024$ batches. As such, the buffer is filled up after less than 1'000 iterations. We train the DDQN algorithm on 1'000'000 iterations. Table 2 gives an overview of parameters used to train the DDQN algorithm.

8.2.3 PPO-SIL

The PPO-SIL algorithm uses two fully connected feedforward neural networks, one for the decision network and one for the impulse network, with architecture

$$\text{Input: } 2 \to 512 \to 512 \to 512 \to 2 \text{: Output}$$

	Parameter	Value
Number of time steps	N_t	252
Number of possible impulses	N_ζ	100
Buffer capacity	$N_{\mathcal{D}}$	1'000'000
Number of batches	M	1024
Number of iterations	N_{iter}	1'000'000
Initial spatial value	x_0	$[-2, 2]$
Huber loss truncation	δ	1.0
Minimum exploration rate	ε_{min}	0.08
Maximum exploration rate	ε_{max}	0.8
Scaling parameter	R	2
Neural network width	L	4
Neural network hidden dimensions	n_ℓ	$\{2, 512, 64, 101\}$
Learning rate	lr	2e-04

Table 2: Parameters used for the training of the DDQN algorithm

for each network. The input for each of the networks is the timestep t_n and controlled state $X^{(\beta)}$, while the outputs are the stochastic policies π_χ and π_ξ, as well as the values d_χ and u_ξ for each of the networks.

Rewards are computed via the function

$$r(t, x, \zeta) = e^{-\delta t} \left(-(x - m)^2 + \gamma w^2\right) \Delta t + K(t, x, \zeta),$$

where $\zeta \in \mathcal{Z}$.

The neural networks are trained on Monte Carlo samples of size N_{rollout} and on a time grid made of N_t time steps over a time horizon of a year. Furthermore, the SIL algorithm's buffer size is 400'000, meaning that 400'000 transitions can be stored. An overview of training parameters used for to train the PPO algorithm can be found in Table 3.

8.3 Results

For all numerical results found in this section, we use the parameter values reported in Table 1. We perform a convergence test on the explicit-impulse scheme to validate that the numerical scheme indeed converges to a solution as the numerical grid is refined. The DDQN and PPO algorithm are trained using the hyperparameters found in Tables 2 and 3, respectively. We will compare each of the DRL algorithms against the explicit-impulse control scheme. All numerical experiments were implemented on the same computer, specifically on a MacBook Pro M2, with 10 CPU and 10 GPU cores, and 16GB of unified RAM.

8.3.1 Explicit-impulse scheme and convergence test

Figure 2 shows the numerical approximation V of the value function at time $t = 0$. The numerical grid used for the approximation consists of $N_t = 252$, $N_x = 401$,

	Parameter	Value
Number of time steps	N_t	252
Number of rollout paths	N_{rollout}	4096
Number of training iterations	N_{iter}	300
SIL buffer capacity	$N_{\mathcal{B}_{\text{good}}}$	400'000
SIL loss weight	λ_{SIL}	0.05
Entropy regularisation weight	$\lambda_{\mathcal{H}}$	1e-03
Actor loss weight	λ_{actor}	0.5
PPO clipping parameter	ε	0.3
Decision network width	L_{d_χ}	5
Decision network hidden dimensions	$n_{d_\chi,\ell}$	$\{512, 512, 512\}$
Impulse network width	L_{u_ξ}	5
Impulse network hidden dimensions	$n_{u_\xi,\ell}$	$\{512, 512, 512\}$
Learning rate[1]	lr	2e-04
Generalised advantage estimate parameter	λ_{GAE}	1.0

Table 3: Parameters used for the training of the PPO algorithm

and $N_\zeta = 101$ points. The solution has a curved profile near the target level m, corresponding to the continuation region where no impulse is applied. Outside of this region, the profile becomes approximately affine. This is consistent with immediate intervention. As x is sufficiently far from the target, the optimal impulse moves the state back to the continuation region, and the value is given by the post-intervention value minus the fixed and proportional transaction costs. The proportional term $\kappa|\zeta|$ in the cost function K produces an affine tail. The transition points between the curved part and the affine tails are the free boundaries that separate the continuation and intervention regions.

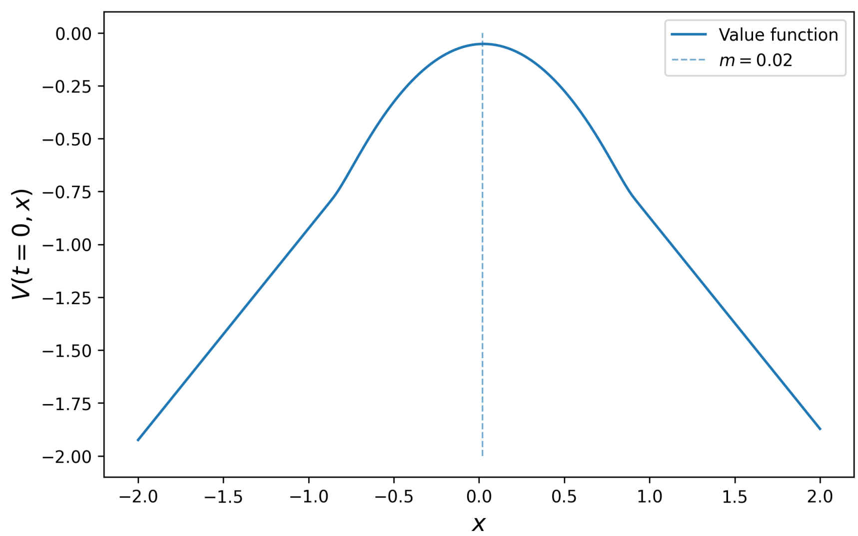

Figure 2: Numerical solution of FEX rate problem at time $t = 0$.

Figure 3 shows the behaviour of the value function across time. Similarly to the results shown in 2, the value function approximation across time is curved around the target m, which corresponds to the continuation region, while for sufficiently large deviations from m, the value becomes approximately affine in x, corresponding to intervention. As t approaches the terminal time, the value function moves towards the terminal condition $g \equiv 0$, reflecting that fewer future running costs remain to be accumulated. Indeed, as Azimzadeh [3, Chapter 5.1.4] notes, the continuation and intervention regions are separated by time-dependent free boundaries and as such can be modelled with some function η.

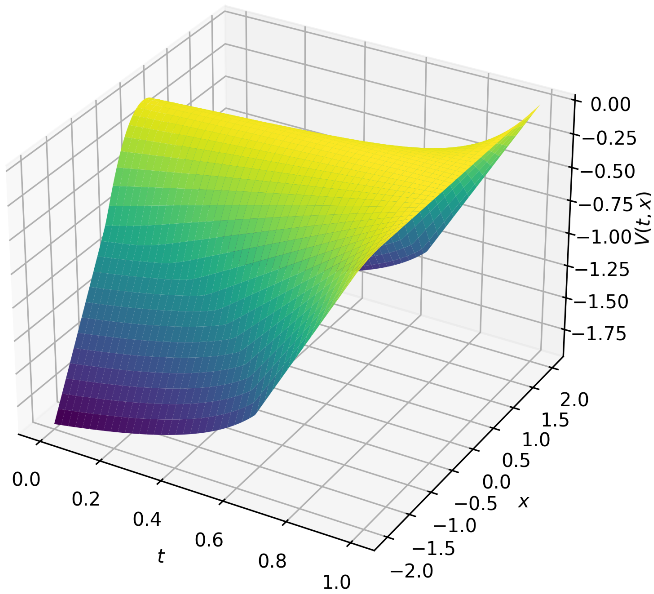

Figure 3: Numerical solution of FEX rate problem at for times $t \in [0, 1]$.

Table 4 presents the approximations of the value function at $t = 0$ and $x = m$ obtained using the explicit-impulse scheme under successive grid refinements. The corresponding numbers of time, spatial, and impulse size discretisation points, denoted by N_t, N_x, and N_ζ respectively, are also reported. The values stabilise around $V(0, m) \approx -0.05129$, showing empirical convergence.

N_t	N_x	N_ζ	$V(t = 0, x = m)$	Ratio	CPU time (s)
16	32	16	-0.057488		2.63e-03
32	64	32	-0.054501		2.68e-03
64	128	64	-0.052996	1.99	5.17e-03
128	256	128	-0.052377	2.43	1.67e-02
256	512	256	-0.052120	2.41	1.05e-01
512	1024	512	-0.051995	2.06	1.67e+00
1024	2048	1024	-0.051938	2.19	1.08e+01

Table 4: Convergence test for the explicit-impulse scheme

Because the convergence test shows stable approximation results, and since the computed value function exhibits the expected continuation and intervention structure, these observations support the use of the explicit-impulse scheme as a numerical benchmark comparison with the DRL algorithms.

8.3.2 DDQN approximation

Comparison plots between the DDQN approximation and the explicit-impulse scheme can be found in Figures 4–6. The explicit-impulse scheme is used as a benchmark.

Consider Figure 4, which compares the approximations at $t = 0$ and for $x \in [-2, 2]$. As one can observe, the DDQN algorithm reproduces a value function approximation comparable to the benchmark. In particular, the DRL algorithm captures the curved continuation region near the target m, as well as the approximately affine tails corresponding to an intervention. The agreement between both schemes is strongest in the central part of the domain, where the value function is smooth and where the simulated training data are more concentrated. Furthermore, the DDQN curve also captures the change in slope near the transition between the continuation and intervention regions. This indicates that the network has learned not only the general shape of the value function, but also the intervention region of the impulse-control problem.

However, there are small visible irregularities in the DDQN curve, with the most visible irregularity being the local oscillations, especially near the transition between the cured and affine parts of the value function. This behaviour is expected, as the plotted DDQN approximation is obtained using the greedy impulse selection in (7.2.9), together with a Monte Carlo approximation of the continuation value. Specifically, the maximisation step in the greedy impulse choice can amplify small sampling errors, particularly near the free boundary. Hence, the online Q-network output (7.2.13) may have similar estimated values for several impulses. As a consequence, small perturbations in the learned value function may lead to different selected impulses, which leads to small local irregularities in the plotted approximation.

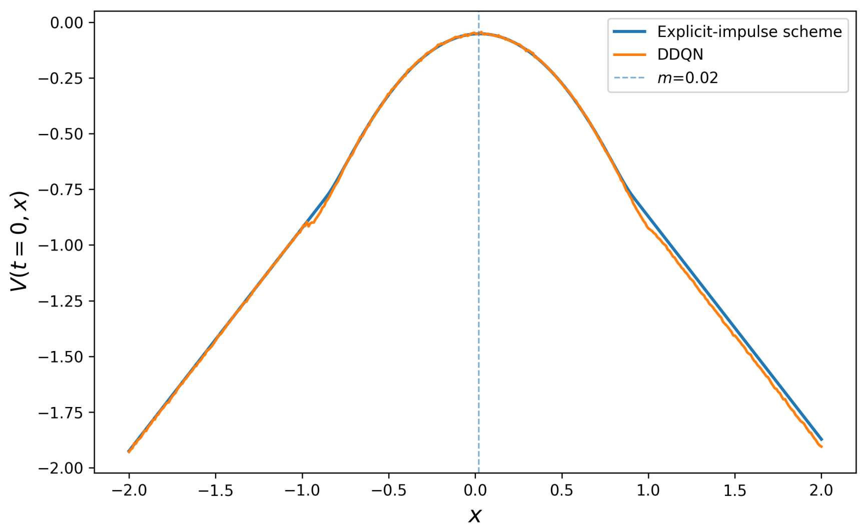

Figure 4: Comparison of DDQN and Explicit-impulse scheme value function approximations

Figure 5 gives the corresponding comparison over the time interval $t \in [0, T]$. The DDQN surface is qualitatively close to the explicit-impulse surface. Both approximations exhibit a continuation region around the target m, affine behaviour for large deviations from the target, and movement towards the terminal condition as t approaches T. This indicates that the DDQN approximation therefore captures the time-dependent structure of the value function, including the evolution of the intervention and continuation regions. Furthermore, the largest visible deviations occur close to the free boundary and in regions that are less frequently visited during simulation. Regions that are visited less frequently correspond to large values of x, as the negative drift reduces the likelihood that the stochastic process X moves into and remains in these regions. In addition, areas near the domain boundaries are also visited less frequently.

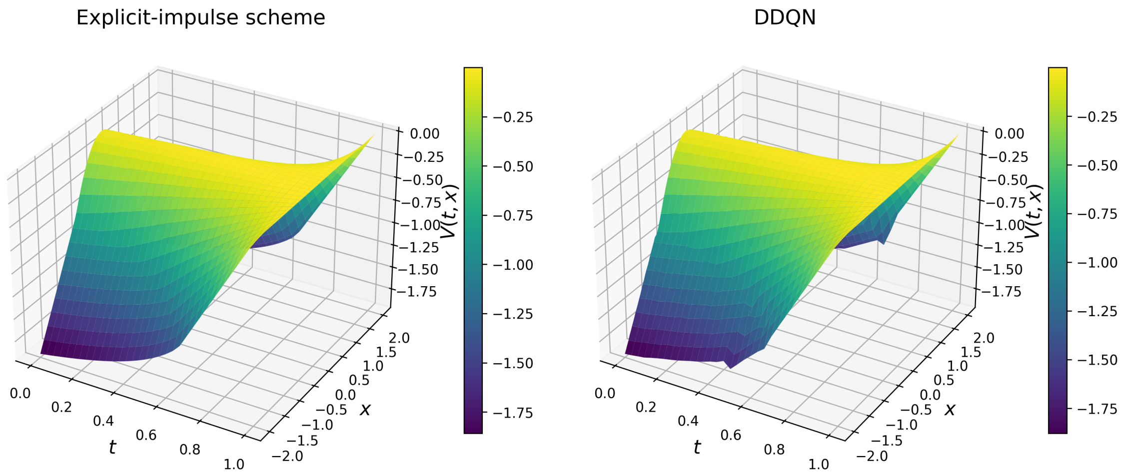

Figure 5: Comparison of DDQN and Explicit-impulse scheme value function approximations for times $t \in [0, 1]$.

Figure 6 shows the pointwise error between the DDQN approximation and the explicit-impulse benchmark. The heatmap confirms that the error is small over most of the computational domain. The largest errors can be found near the free boundary, where the value function transitions between the continuation and intervention regimes. This is also the region where the impulse decision is the most sensitive. Hence, a small error in the learned value function can change whether the greedy policy selects intervention or continuation. The resulting error heatmap is therefore useful for understanding what free boundary the network learned.

Besides the computational correctness, another important aspect to consider is the computation cost, which in our case differs substantially between the two methods. The DDQN training time was approximately 122 minutes, whereas the explicit-impulse scheme only needed 36.6 ms to compute the value function. Another aspect worth mentioning is the training stability of the DDQN algorithm. As the algorithm involves a large number of hyperparameters, the set of possible configurations is extensive. In practice, we therefore selected the first set of hyperparameters that produced satisfactory results, although further tuning could lead to improvements in performance and computational efficiency. We also observe that the DDQN algorithm is highly sensitive to these choices, as even small changes in hyperparameters can result in significantly worse performance.

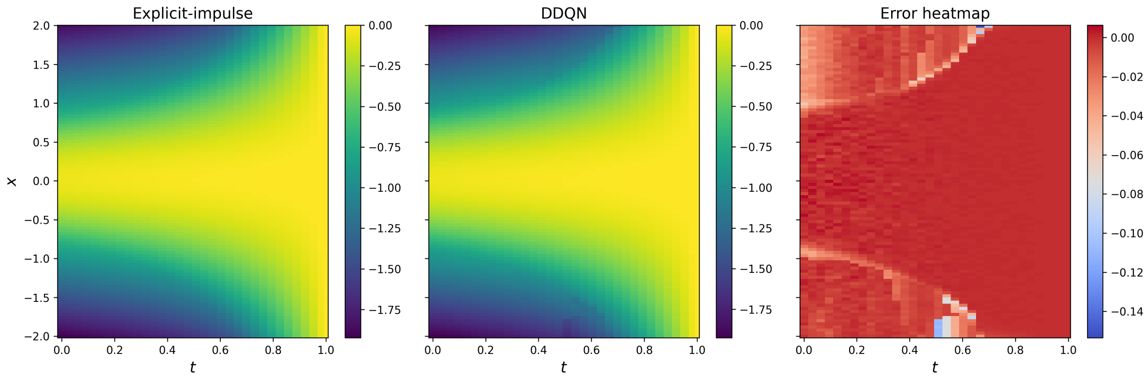

Figure 6: Error heatmap of the DDQN algorithm

Overall, the DDQN approximation is consistent with the explicit-impulse benchmark. The algorithm is able to reproduce the continuation region, the affine intervention tails, the time evolution of the surface, and the approximate location of the free boundary. Although the DDQN is substantially more expensive to train and exhibits larger errors near the boundary and in under-sampled regions, both methods show similar results. As such, we may conclude that the DDQN has learned to approximate the value function well enough.

8.3.3 PPO-SIL approximation

The PPO-SIL approximation is compared with the explicit-impulse scheme in Figures 7–9. As before, the explicit-impulse scheme is used as a benchmark and the reported errors should be interpreted as errors relative to this benchmark rather than errors relative to the exact value function.

Figure 7 compares the approximation at $t = 0$ and for $x \in [-2, 2]$. On the first look, the PPO-SIL algorithm's value function approximation is comparable to the benchmark. Indeed, the central, curved part of the value function, which corresponds to the continuation region, is well approximated. The agreement between both schemes is particularly strong around the target level m, where the value function is smooth and simulated trajectories provide relatively informative training data. In addition, the figure indicates that the learned policy can effectively capture the intervention structure of the problem, as the learned value function stays relatively close to the explicit-impulse approximation.

However, we note that local, small errors, near the free boundary between the continuation and intervention regions, can be observed. These are expected as the value function plotted from the PPO-SIL algorithm is obtained from a learned stochastic policy and a Monte Carlo approximation of the value function. Furthermore, the decision between continuation intervention is sensitive near the free boundary. Hence, small irregularities are observed at the free boundary in Figure 7.

Figure 8 compares the approximation of the surfaces of the value functions of both schemes over the time interval $t \in [0, 1]$. The PPO-SIL surface reproduces the main qualitative structure of the explicit-impulse scheme, with the value being largest near the target m and the tails being approximately affine in x. The surface moves towards

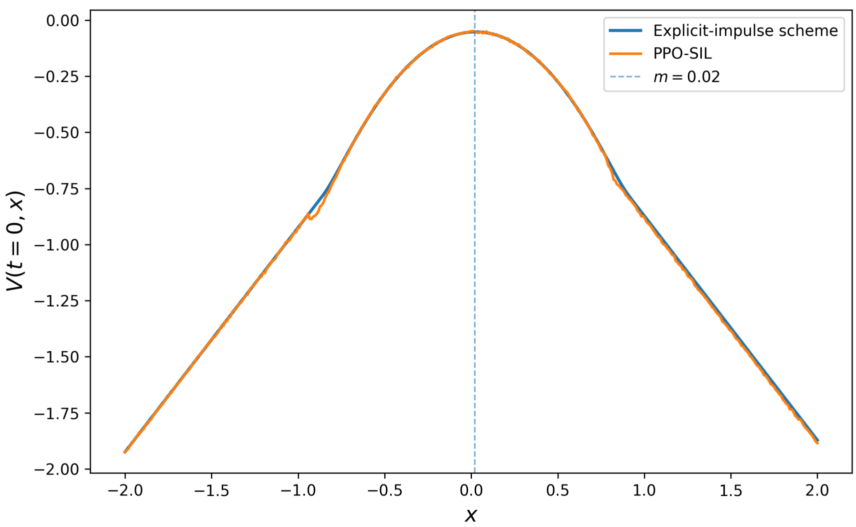

Figure 7: Comparison of PPO-SIL and explicit-impulse scheme value function approximations

the terminal condition as t approaches 1. Thus, the PPO-SIL has learned the broad time-dependent continuation structure of the impulse-control problem.

However, the PPO-SIL approximation does not come without numerical errors, which mostly appear in the outer regions, especially for positive x-values and later times. These errors are most likely caused by the combination of the trajectory-based training, Monte Carlo noise, and under-sampling of parts of the state space. In particular, the FEX process has a negative drift in this experiment, meaning that simulated paths tend to move towards negative x-values. Consequently, the algorithm observes fewer samples with positive x-values, which makes the approximation less reliable in that part of the domain.

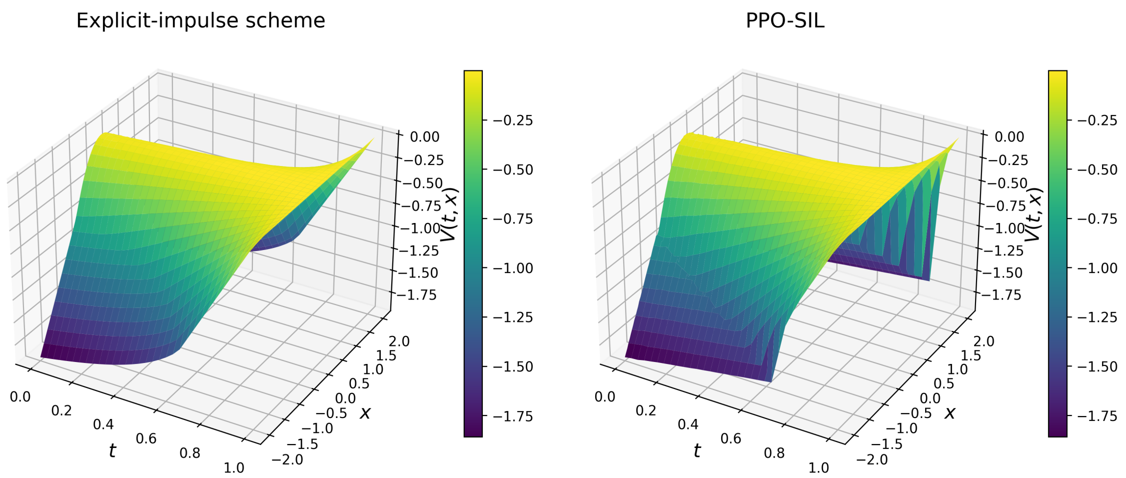

Figure 8: Comparison of PPO-SIL and Explicit-impulse scheme value function approximations for times $t \in [0, 1]$

This interpretation is further suggested by the error plot shown in Figure 9. The error is small for most of the computational domain, especially in the central region around the target. The largest errors can be found near the free boundary and in the under-sampled positive-x region. The free boundary is visible in the heatmap as the region where the error changes more sharply. Because the boundary is relatively smooth, the policy tends to wait rather than intervene immediately, acting only once the boundary becomes more clearly defined. This behaviour is also amplified by small numerical errors in the learned approximation.

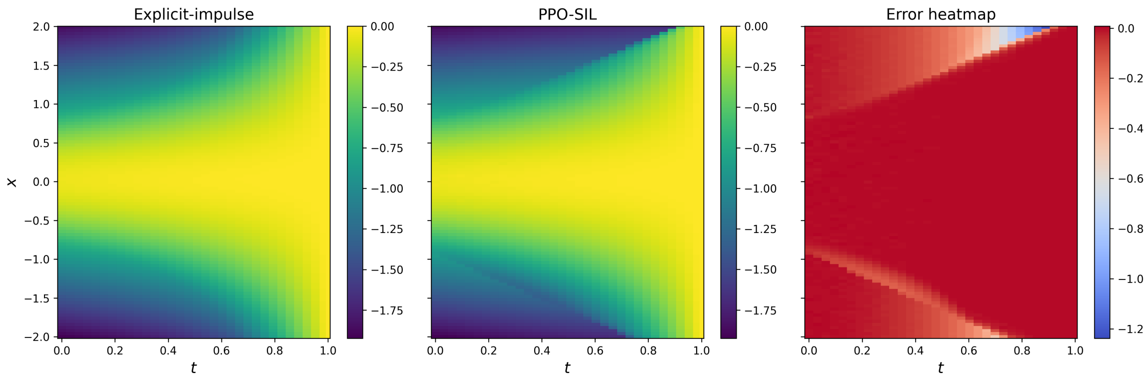

Figure 9: Error heatmap of the PPO-SIL algorithm

Although the PPO–SIL algorithm requires less training time than DDQN, its computational cost still differs significantly from that of the explicit-impulse scheme, with the PPO-SIL training time being approximately 24 minutes. Similarly to the DDQN algorithm, the PPO-SIL is sensitive to hyperparameter changes and we therefore selected the first set of hyperparameters that produced satisfactory results. As such, further tuning may improve the approximation accuracy and computational efficiency.

However, in general, the PPO-SIL algorithm approximates the value function well enough for comparison purposes. It captures the continuation region, the affine intervention tails, the time evolution of the value surface, and the approximate free-boundary structure. However, the error heatmap and the three-dimensional plot show that the approximation is less reliable near the free boundary and in under-sampled regions of the state space. But, rather than being an issue of the algorithm itself, these errors are mostly caused by the simulation-based nature of the method. Hence, this issue can be fixed by means of, for instance, over-sampling the state variable in certain regions.

CHAPTER

NINE

NUMERICAL EXPERIMENT II: DYNAMIC PORTFOLIO ALLOCATION

The second numerical experiment studies a finite-horizon dynamic portfolio allocation problem in a liquid equity market with transaction costs. The state consists of the cash balance, the risky asset prices, and the current holdings. As such, the problem is high-dimensional, and both the explicit-impulse scheme and the DDQN algorithm become computationally expensive in this setting. We therefore restrict the experiment to the PPO-SIL algorithm.

This choice is motivated by three considerations. First, PPO-SIL showed empirical convergence towards the theoretical solution in Chapter 8. Second, the method can be easily scaled to higher-dimensional control problems as noted by Jain et al. [59], whose work considers a 23-dimensional MDP. Finally, the PPO-SIL algorithm learns the stochastic policy directly, which is practical in dynamic portfolio allocation because the policy determines the sequence of trading decisions over time.

9.1 Financial assumptions

We assume that the investment universe consists of d risky assets and one cash account, where the cash account accrues at a constant risk-free rate δ. The market is assumed to be liquid, roughly meaning that there will always be a buyer (seller) side willing to execute trades at the prevailing market price without causing a significant price impact. At each admissible trading time, the investor can buy or sell the chosen quantity at the observed mid-price. Trades have no price impact and are settled immediately. We also assume that there are no bid-ask spreads and market frictions enter through transaction costs only. Additionally, assets are assumed to not pay dividends during the investment horizon. Hence, changes in portfolio value are generated only by price movements, interest on cash, and transaction costs.

Trading is allowed only at the rebalancing dates of the numerical time grid. The continuous-time impulse-control problem is therefore approximated by a finite-time impulse-control problem. Between two rebalancing dates, the risky holdings remain constant. At each rebalancing date, the investor may either choose the no-trade action or submit a vector of buy and sell orders. A positive component of the trading vector represents a purchase, while a negative component represents a sale.

The portfolio is self-financing, that is, no external cash is injected into or withdrawn from the portfolio during the investment horizon. Shares in risky positions can be split into fractions. In addition, we impose no-shorting, i.e no negative positions in risky shares, and no-borrowing constraints. Thus, both the risky holdings and the cash balance must remain non-negative after each trade.

9.2 Dynamic portfolio allocation via impulse control

9.2.1 Financial environment and impulse controls

Let $(\Omega, \mathcal{F}, \{\mathcal{F}\}_{t\geq 0}, \mathbb{P})$ be a filtered probability space with $\mathbb{P}$ denoting the real-world probability measure. The investment horizon is finite and denoted by $T > 0$. All price, cash, and holding processes are assumed to be adapted and càdlàg.

Let $\mathbf{s} = (s_1, \ldots, s_d) \in \mathbb{R}^d_+$ denote the observed risky asset prices at time $t = 0$. For $t > 0$, let $\mathbf{S}(t) = (S_1(t), \ldots, S_d(t)) \in \mathbb{R}^d_+$ denote the risky asset price process. Here $S_i(t) > 0$ is the mid-price of asset i at time t. The price process $\mathbf{S}$ is exogenous. The modelling and simulation of correlated stock prices are discussed in later sections of this chapter.

Let τ_j denote the j-th intervention time and $\zeta(\tau_j) = (\zeta_1(\tau_j), \ldots, \zeta_d(\tau_j)) \in \mathbb{R}^d$ denote the d-dimensional impulse. Not all risky assets present in the portfolio require an impulse at an intervention time. In light of this, we allow at most $d-1$ impulses in $\zeta(\tau_j) \in \mathbb{R}^d$ to be zero. The impulse control is then given by $\alpha = \{(\tau_j, \zeta(\tau_j))\}_{j=0}^{M}$ for $M \leq \infty$.

The controlled process is then the number of risky assets that the investor holds in her portfolio. Formally, let $\mathbf{X}^{(\alpha)}(t) = (X_1^{(\alpha)}, \ldots, X_d^{(\alpha)}) \in \mathbb{R}^d_+$ denote the number of shares held in each asset at time t, where $X_i^{(\alpha)}$ corresponds to the number of shares in asset i with price $S_i(t)$. The initial holdings are $\mathbf{x} = (x_1, \ldots, x_d) \in \mathbb{R}^d$. Because continuous trading is not allowed, the holding process is piecewise constant and changes only when the portfolio manager trades. At the rebalancing time τ_j, the vector of shares held is given by

$$\mathbf{X}^{(\alpha)}(\tau_j) = \mathbf{X}^{(\alpha)}(\tau_j^-) + \zeta_j.$$

Hence, the post-intervention function Γ is defined as

$$\Gamma(\mathbf{x}, \zeta) := \mathbf{x} + \zeta.$$

Let $B^{(\alpha)}(t)$ denote the time-t gross cash balance given a trading strategy α. We denote the initial condition of the cash balance by $b > 0$. $B^{(\alpha)}$ records the cash effect of purchases and sales, but not the transaction costs. At an intervention time τ_j, we have

$$B^{(\alpha)}(\tau_j) = B^{(\alpha)}(\tau_j^-) - \mathbf{S}(\tau_j) \cdot \zeta(\tau_j),$$

where $\cdot$ denotes the Euclidean inner product. Between intervention times, $B^{(\alpha)}$ accrues at the risk-free rate δ. That is, for $t \in [\tau_j, \tau_{j+1})$,

$$B^{(\alpha)}(t) = B(\tau_j)e^{\delta(t-\tau_j)}, \quad \text{for } j = 1, 2, \dots .$$

For any cash amount $\tilde{b}$, the portfolio value function is defined as

$$\pi\big(\tilde{b}, \mathbf{s}, \mathbf{x}\big) = \tilde{b} + \mathbf{s} \cdot \mathbf{x}. \tag{9.2.1}$$

For brevity, we denote by the stochastic process $\Pi^{(\alpha)} = \{\Pi^{(\alpha)}(t)\}_{t\geq 0}$ the portfolio value process, defined by

$$\Pi^{(\alpha)}(t) := B^{(\alpha)}(t) + \mathbf{S}(t) \cdot \mathbf{X}^{(\alpha)}(t) = \pi\big(B^{(\alpha)}(t), \mathbf{S}(t), \mathbf{X}^{(\alpha)}(t)\big). \tag{9.2.2}$$

The initial portfolio value is given by

$$\pi_0 = \pi\big(b, \mathbf{s}, \mathbf{x}\big). \tag{9.2.3}$$

Only non-zero impulses lead to transaction costs. Hence, for a non-zero impulse size ζ_i, we define the discounted transaction cost to be

$$\mathrm{TC}(t, s_i, \zeta_i) = -1_{\{\zeta_i \neq 0\}} e^{-\delta t}(c + \kappa s_i|\zeta_i|),$$

where $c, \kappa > 0$ denote the fixed and proportional costs terms respectively. The total transaction costs K are therefore given by

$$K(t, \mathbf{s}, \zeta) = \sum_{i=1}^{d} \mathrm{TC}(t, s_i, \zeta_i).$$

Since $\mathrm{TC}(t, s_i, \zeta_i)$ is non-positive, it follows that $K(t, \mathbf{s}, \mathbf{x})$ is non-positive as well.

Finally, let $\mathcal{H} \subseteq \mathbb{R}^d_+$ denote the admissible inventory region. This set contains the allowed risky holdings and includes upper position limits. The admissible impulse set is

$$\mathcal{Z}(t, b, \mathbf{s}, \mathbf{x}) = \big\{\zeta \in \mathbb{R}^d : \mathbf{x} + \zeta \in \mathcal{H}, b - \mathbf{s} \cdot \zeta - K(t, \mathbf{s}, \zeta) \geq 0\big\}, \tag{9.2.4}$$

and the solvency region is defined as

$$\mathcal{S} = \big\{(t, b, \mathbf{s}, \mathbf{x}) \in [0, T] \times \mathbb{R}_+ \times \mathbb{R}^d_+ \times \mathcal{H} : \Pi(t) > 0\big\}.$$

9.2.2 Performance criterion and value function

The objective of the investor is to maximise the expected, discounted terminal portfolio value after subtracting the discounted transaction consts incurred along the trading path. At the initial investment time $t = 0$ and for an admissible control α, define the performance criterion

$$J_0^{(\alpha)}(\pi_0, \mathbf{s}, \mathbf{x}) = \mathbb{E}_{\mathbb{P}}^{(\pi_0, \mathbf{s}, \mathbf{x})}\left[e^{-\delta T}\Pi^{(\alpha)}(T) + \sum_{0\leq\tau_j<T} K\big(\tau_j, S(\tau_j), \zeta(\tau_j)\big)\right]. \tag{9.2.5}$$

The portfolio process $\Pi^{(\alpha)}$ is defined as in (9.2.2) and π_0 denotes the initial portfolio value. The value function is

$$v(\pi_0, \mathbf{s}) = \sup_{\alpha} J_0^{(\alpha)}(\pi_0, \mathbf{s}),$$

where the supremum is taken over all admissible impulse controls. The value function is interpreted as the maximal expected discounted terminal portfolio value attainable from the initial portfolio value π_0 and stock prices $\mathbf{s}$.

We note that the criterion (9.2.5) is risk-neutral in the risk preference sense[1], see [23]. Risk control is therefore imposed not through a risk preference parameter, but through the investment universe, the finite horizon, as well as through the cash and inventory constraints.

9.2.3 Rewards and PPO-SIL objective

To be able to use the PPO-SIL algorithm, we discretise the time interval $[0, T]$ using a uniform time grid. Specifically, we consider grid points $0 = t_0 < t_1 < \cdots < t_{N_t} = T$ with time step $\Delta t = T/N_t$. Interventions are allowed only at the grid points $t_0, t_1, \ldots, t_{N_t - 1}$, while no trading is allowed at the terminal time. Finally, let $\beta = \{(t_n, a_n)\}_{n=0}^{N_t}$ denote the PPO-SIL trading strategy on the discrete time grid.

Using this discretisation, the discounted portfolio process can be rewritten in incremental form along each path, i.e.

$$e^{-\delta T}\Pi^{(\alpha)}(T) = \pi_0 + \sum_{n=0}^{N_t - 1} \left(e^{-\delta t_{n+1}}\Pi^{(\alpha)}(t_{n+1}) - e^{-\delta t_n}\Pi^{(\alpha)}(t_n)\right). \tag{9.2.6}$$

Let r denote the expected discounted reward function. We define the time-t_n reward as

$$\begin{aligned} r\big(t_n, \Pi^{(\alpha)}(t_n), \mathbf{S}(t_n), \zeta(t_n)\big) = {} & \underbrace{e^{-\delta t_{n+1}}\mathbb{E}_{\mathbb{P}}^{\left(\Pi^{(\alpha)}(t_n), \mathbf{S}(t_n)\right)}[\Pi^{(\alpha)}(t_{n+1})] - e^{-\delta t_n}\Pi^{(\alpha)}(t_n)}_{\text{Expected profit-and-loss over } [t_n, t_{n+1}]} \\ & + \underbrace{K\big(t_n, \mathbf{S}(t_n), \zeta(t_n)\big)}_{\text{Trading costs}}. \end{aligned}$$

The reward function can thus be interpreted as the expected discounted profit-and-loss over the interval $[t_n, t_{n+1}]$, minus trading costs incurred at t_n.

Then, the performance criterion (9.2.5) is approximated by the sum of these rewards over the time grid, that is,

$$J_0^{(\alpha)}(\pi_0, \mathbf{s}, \mathbf{x}) \approx \pi_0 + \sum_{n=0}^{N_t} \mathbb{E}_{\mathbb{P}}^{(\pi_0, \mathbf{s}, \mathbf{x})}\left[r\big(t_n, \Pi^{(\alpha)}(t_n), \mathbf{S}(t_n), \zeta(t_n)\big)\right].$$

This representation is well-suited for the PPO-SIL algorithm, as it expresses the objective in terms of accumulated rewards along simulated paths. The corresponding PPO-SIL loss functions are computed as described in Chapter 7.3.

[1] This does not mean that we are working under a risk-neutral measure (recall that we are working under the real-world measure $\mathbb{P}$), but that the investor rather maximises expected wealth rather than expected utility.

9.3 Implementation details

This subsection outlines the practical implementation of the PPO-SIL algorithm in the context of dynamic portfolio allocation. In particular, we describe the construction of the initial portfolio, the calibration and simulation of the underlying stock price models, and the training setup for the reinforcement learning agent. Together, these components define the environment in which we apply the PPO-SIL algorithm. The Python code used in this experiment can be found in [21].

9.3.1 Equity universe and initial portfolio

The stock universe is limited to the stocks found in the Exchange-Traded Fund (ETF) *Xtrackers Switzerland ETF 1D fund*, which tracks the Swiss Market Index (SMI) and has an average annual return of 8% over the past three years[2]. The fund consists of 20 stocks and the ETF portfolio weights are used to construct the initial portfolio allocation. We set the initial portfolio value to be $\pi_0 = 50000$ CHF[3]. The fixed part of transaction costs is $c = 3$ CHF and the proportional part is $\kappa = 0.1\%$. The initial portfolio weights, together with the corresponding asset names, tickers, and spot prices can be found in Table 5, while an overview of the initial portfolio composition can be found in Table 6.

The choice behind the aforementioned ETF is due to the fact that the Swiss Average Rate Overnight (SARON) rate is approximately zero, which slightly simplifies the model and its interpretation. The risk-free rate is therefore $\delta = 0$.

9.3.2 Simulation and calibration of stock prices

Risky assets are modelled as correlated pure-jump processes, with marginal log-returns following Variance Gamma (VG) dynamics via multivariate subordination. The dependence between assets is introduced through linear correlation in the Brownian components and nonlinear dependence in the jump components via a common market-jump factor in the multivariate VG subordinator, see [72, 71]. This modelling choice captures the non-Gaussian returns and tail dependence observed in equity returns, which cannot be explained by linear correlation alone. Further theoretical details and the simulation scheme can be found in Appendix D .

The marginal VG parameters σ_i, θ_i, and ν_i are calibrated to European call option prices using the Fast Fourier Transform, as described in Appendix E. Option prices were collected on May 14th 2026, and are as such current market data rather than historical observations. The drift coefficients μ_i are estimated from two years of historical stock price data, with the upper 5% of log-returns trimmed to obtain more robust and conservative estimates. The estimated model parameters can be found in Table 7. An overview of the corresponding data used for calibration can be found in Tables 13 and 14 in Appendix G.1. These parameters are used to simulate the joint paths of the d-dimensional correlated stock price process $\mathbf{S}$ for every time step

[2] I.e. from May 14th 2023 to May 14th 2026.

[3] CHF = Swiss franc

Asset name	Ticker	Spot Price (CHF)	Weight (%)
Cash			0.14
ABB Ltd	ABBN	82.66	7.58
Alcon AG	ALCC	49.61	2.22
Compagnie Financière Richemont SA	CFR	156.55	5.52
Galderma Group AG	GALD	158.35	1.81
Geberit AG	GEBN	503.4	1.32
Givaudan SA	GIVN	2683.00	1.88
Holcim AG	HOLN	76.26	2.47
Lonza Group AG	LONN	474.2	2.61
Nestlé SA	NESN	76.88	15.05
Novartis AG	NOVN	116.80	17.12
Partners Group Holding AG	PGHN	883.60	1.39
Roche Holding AG	ROPC	320.00	16.79
Sandoz Group AG	SDZ	67.76	1.98
SGS SA	SGSN	84.68	1.06
Sika AG	SIKA	140.95	1.49
Swiss Life Holding AG	SLHN	837.60	1.81
Swisscom AG	SCMN	678.00	1.29
Swiss Reinsurance Company AG	SRENH	119.45	3.00
UBS Group AG	UBSG	36.22	7.12
Zurich Insurance Group AG	ZURN	563.00	6.35

Table 5: Summary of the initial portfolio allocation as of May 14th, 2026.

Component	Value (CHF)	Weight (%)
Initial risky asset holdings	49'930	99.86
Initial cash balance	70	0.14
Total initial portfolio value	50'000	100.00

Table 6: Initial portfolio composition.

TKR	μ (%)	σ (%)	θ (%)	ν (%)
ABB	12.71	29.42	-36.63	8.31
ALCC	-8.35	27.31	-28.01	8.87
CFR	0.03	30.20	-0.01	0.94
GALD	18.74	27.61	-84.67	3.64
GEBN	-0.70	21.32	-15.62	1.13
GIVN	-5.89	25.68	-33.67	38.18
HOLN	13.36	25.21	-29.84	4.99
LONN	1.13	27.89	-28.42	5.73
NESN	-3.45	18.48	-31.11	8.21
NOVN	6.16	20.60	-25.11	1.83
PGHN	-6.21	23.76	-79.28	1.19
ROPC	6.76	21.61	-16.70	1.88
SDZ	18.30	28.44	-62.67	3.45
SGSN	3.86	19.74	-42.65	10.52
SIKA	-12.97	30.29	-22.27	9.68
SLHN	6.08	18.93	-0.01	0.66
SCMN	7.67	19.56	-22.00	45.00
SRENH	2.24	22.56	-31.83	15.14
UBSG	8.61	25.65	-44.70	5.18
ZURN	3.86	17.65	-25.12	2.29

Table 7: Estimated model parameters.

t_n in line with Algorithm 3. The theoretical background behind the aforementioned algorithm can be found in Appendix D, and Figure 10 illustrates a simulation of the 20 correlated stocks using the estimated marginal parameters.

9.3.3 PPO-SIL hyperparameters

Similarly to the FEX example in Chapter 8, the PPO-SIL algorithm employs two fully-connected feedforward neural networks with hyperbolic tangent activation functions. One network determines whether to intervene, while the other selects the corresponding impulse size. Both networks share the same architecture, given by

$$\text{Input: } 2 \to 512 \to 256 \to 128 \to 256 \to 512 \to 2\text{: Output}$$

As discussed earlier, the bottleneck neural network architecture helps reduce training time, while retaining the most relevant information.

To ensure realistic trading behaviour and portfolio diversification, we impose several constraints. In particular, the agent must maintain at least 1% of the initial portfolio value in each risky asset. Moreover, the maximum position in each asset is capped at 22% of the initial portfolio value in each risky asset, and at most 25 shares in each equity can be traded in a single transaction[4].

[4] In other words, a total of $25 \times 20 = 500$ shares can at most be traded at each time step.

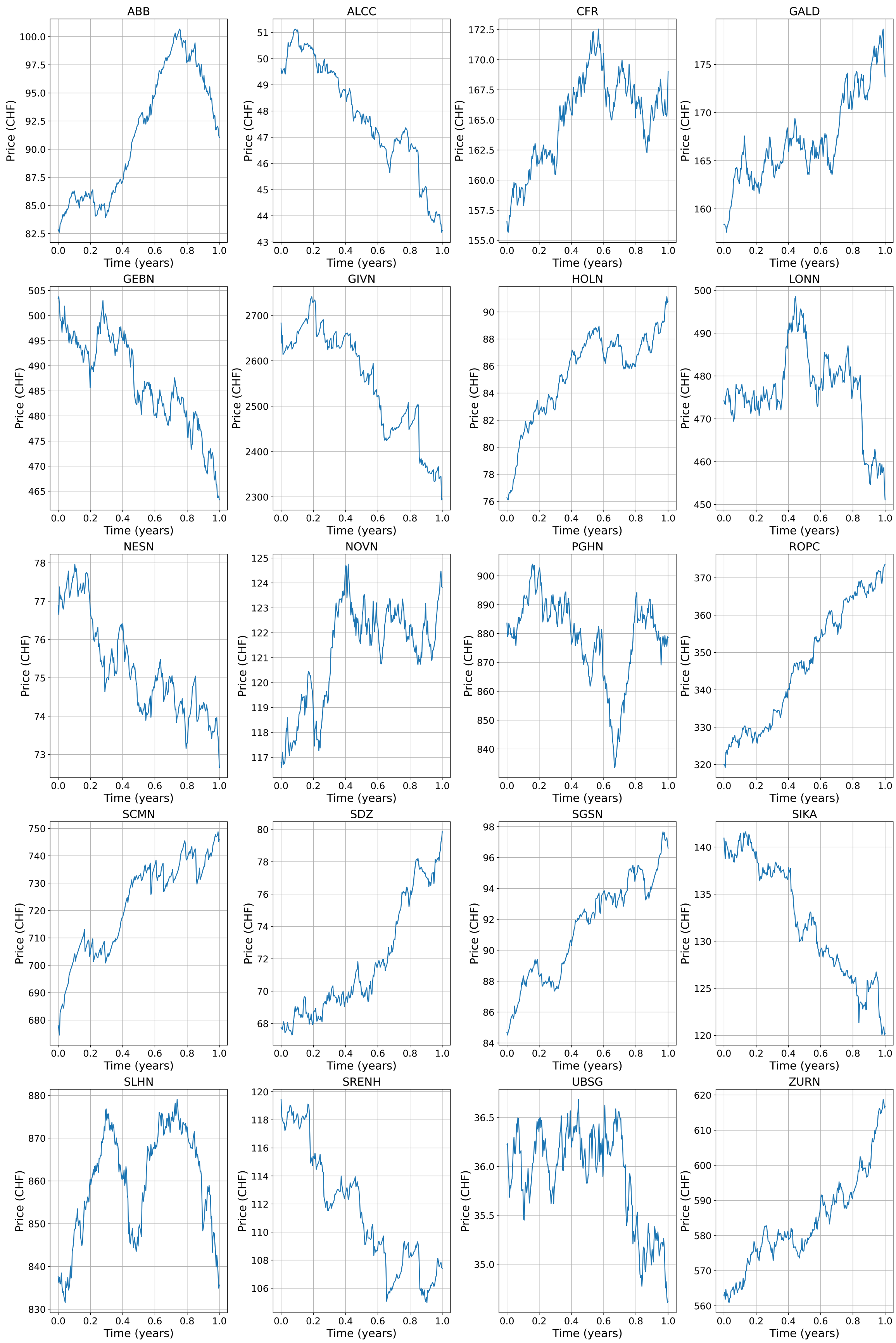

Figure 10: Simulation of a single sample path for each of the 20 portfolio constituents over 252 trading days.

Both neural networks are trained on the same data using $N_{\text{rollout}} = 1024$ rollouts. Since a typical trading year consists of 252 days, we set the time grid size to $N_t = 252$. To encourage sufficient exploration during training, the entropy regularisation weight is chosen as $\lambda_{\mathcal{H}} = 0.5$. Other hyperparameters used for the training of the PPO-SIL algorithm can be found in Table 8. The training and evaluation of the PPO-SIL algorithm is done on a Macbook Pro M2 with a 10-core CPU and a 10-core GPU, as well as 16GB unified RAM. The total training time is about 48 minutes for the PPO-SIL with hyperparameters shown in Table 8.

	Parameter	Value(s)
Number of time steps	N_t	252
Number of rollout paths	N_{rollout}	1024
Number of training iterations	N_{iter}	200
SIL buffer capacity	$N_{\mathcal{B}_{\text{good}}}$	500'000
SIL loss weight	λ_{SIL}	0.05
Entropy regularisation weight	$\lambda_{\mathcal{H}}$	0.5
Actor loss weight	λ_{actor}	0.5
PPO clipping parameter	ε	0.3
Decision network width	L_{d_χ}	7
Decision network hidden dimensions	$n_{d_\chi,\ell}$	$\{512, 256, 128, 256, 256, 512\}$
Impulse network width	L_{u_ξ}	7
Impulse network hidden dimensions	$n_{u_\xi,\ell}$	$\{512, 256, 128, 256, 256, 512\}$
Learning rate[5]	lr	2e-04
Generalised advantage estimate parameter	λ_{GAE}	0.95

Table 8: Parameters used for the training of the PPO algorithm in the dynamic portfolio allocation example.

9.4 Results and evaluation

We focus on training and evaluation phases. In particular, we compare the PPO-SIL algorithm against a buy-and-hold portfolio with weights reported in Table 5. The comparison is performed on both a single path simulation and on 5'000 simulated paths.

9.4.1 Training phase

As shown in Figure 11, the training phase shows a clear improvement in the performance of the learned policy. The mean terminal wealth increases substantially during training, rising from around 50'000 CHF to around 56'000 CHF. The most visible improvement occurs during the first 100 to 150 iterations, after which the rate of increase becomes more gradual. From approximately iteration 150 onward, the curve appears stabilises around a higher wealth level, with only moderate stochastic fluctuations and occasional upward spikes. This suggests that the algorithm can identify progressively better policies before stabilising at around 55'000 CHF.

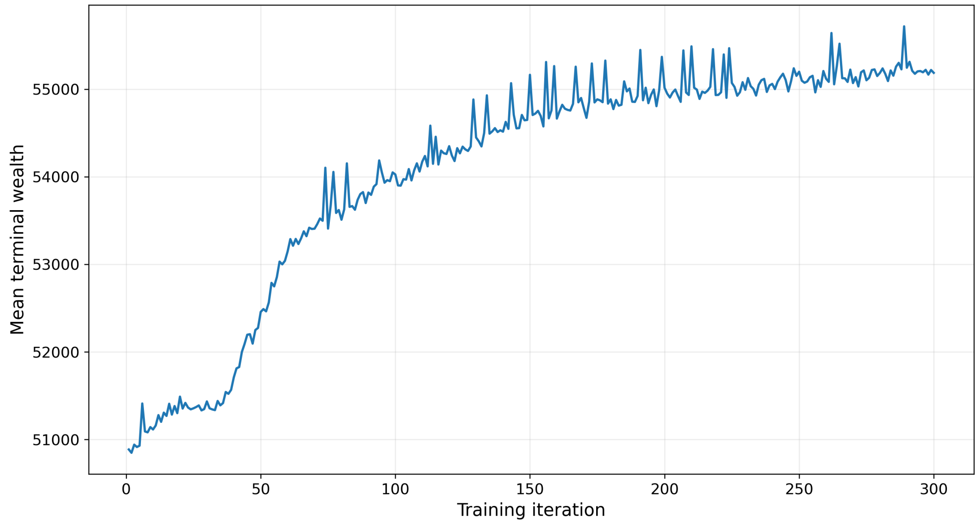

Figure 11: Mean terminal wealth per training iteration.

Figure 12 shows that this improvement is not driven by an increase in trading activity. On the contrary, the total number of non-zero executed trades decreases sharply during the initial training period, falling from roughly 50 to around 40 trades per sampled path. After the initial decline in trading activity, the trading frequency remains relatively stable, fluctuating around a narrow range. The fluctuation can be explained by the relatively large entropy coefficient $\lambda_{\mathcal{H}} = 0.5$, which encourages exploration but also causes sharp fluctuations in the actions. In general, however, the relatively constant amount of trades during training indicates that the policy learns to act more selectively, reducing unnecessary interventions while still improving terminal wealth.

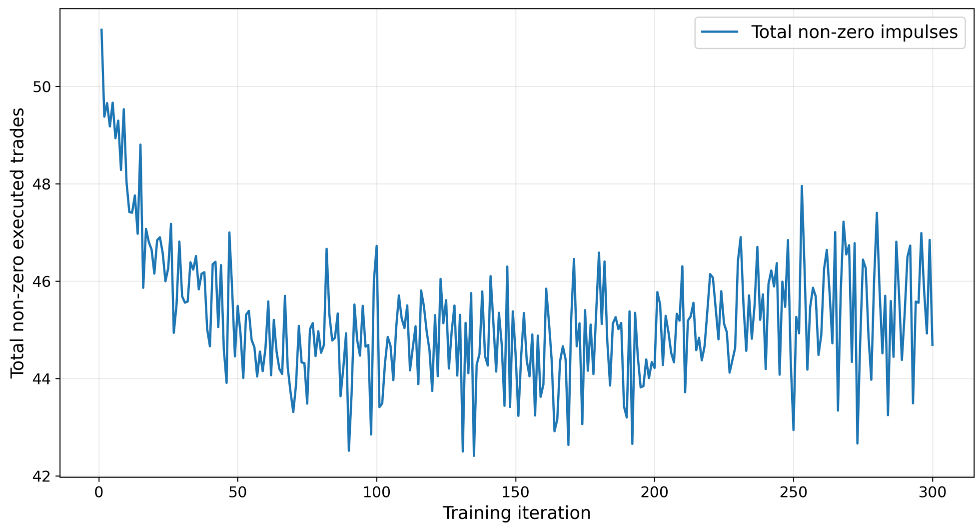

Figure 12: Mean absolute total trades per iteration.

Taken together, Figures 11 and 12 suggest that the trained policy becomes both more profitable and more selective in its use of trades. The simultaneous increase in terminal wealth in terminal wealth and a slight decrease in trading activity is consistent with an improvement in the quality of trading-decisions rather than a simple expansion of market activity. The stabilization of both quantities in the later iterations provides empirical evidence that the training phase approaches a relatively stable policy, although the remaining fluctuations indicate that convergence is approximate rather than exact.

9.4.2 Single-scenario evaluation

For the single-scenario evaluation, we consider one fixed realisation of the stock price process over the trading horizon. The trained PPO-SIL policy is evaluated sequentially in this fixed price scenario. That is, at each time step, the policy observes the current portfolio value as well as current asset prices and decides whether to trade and how much to trade based on that state. We note that it neither observes future stock prices nor uses the SIL buffer to generate future price paths during evaluation.

Figure 13 shows the normalised stock price paths used for the single-scenario evaluation. Each price is divided by its initial value, so the figure illustrates relative price changes over the a year. We note that the assets do not move uniformly and that some assets decline over the horizon, while other appreciate. The realised market environment is therefore heterogeneous across assets, which makes the allocation problem non-trivial and more challenging, especially for static portfolios.

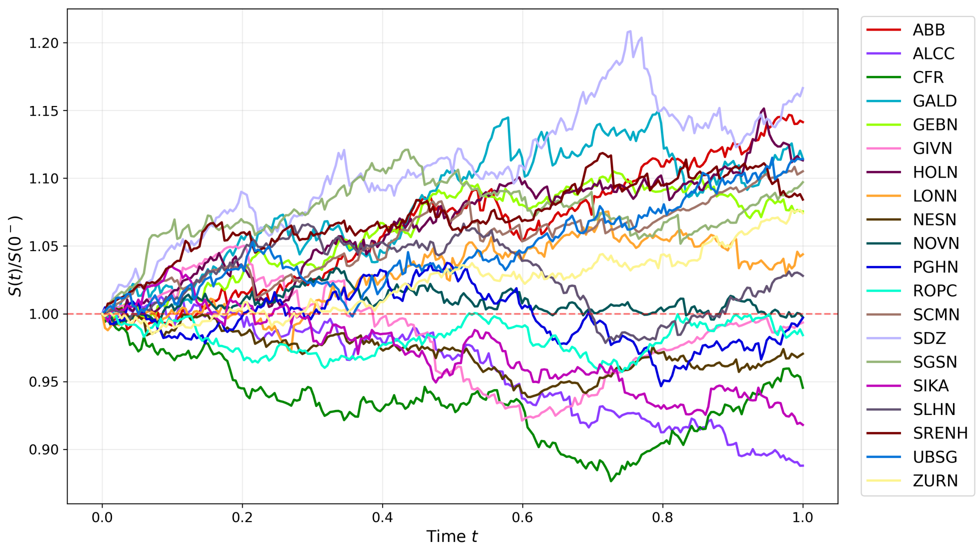

Figure 13: Normalised paths of correlated stocks present in the portfolio.

Figure 14 compares the cumulative portfolio return of the PPO-SIL strategy with the corresponding buy-and-hold benchmark. The buy-and-hold benchmark is constructed from the initial ETF allocation, see Table 5. The number of shares in each risky asset is kept constant throughout the horizon. Hence, the buy-and-hold portfolio does not trade after the initial allocation and does not incur transaction costs during the evaluation period.

In this scenario, the PPO-SIL strategy outperforms the buy-and-hold benchmark. The PPO-SIL portfolio increases relatively steadily and ends with a yearly return of approximately 9-10%. The buy-and-hold portfolio ends with a yearly return of

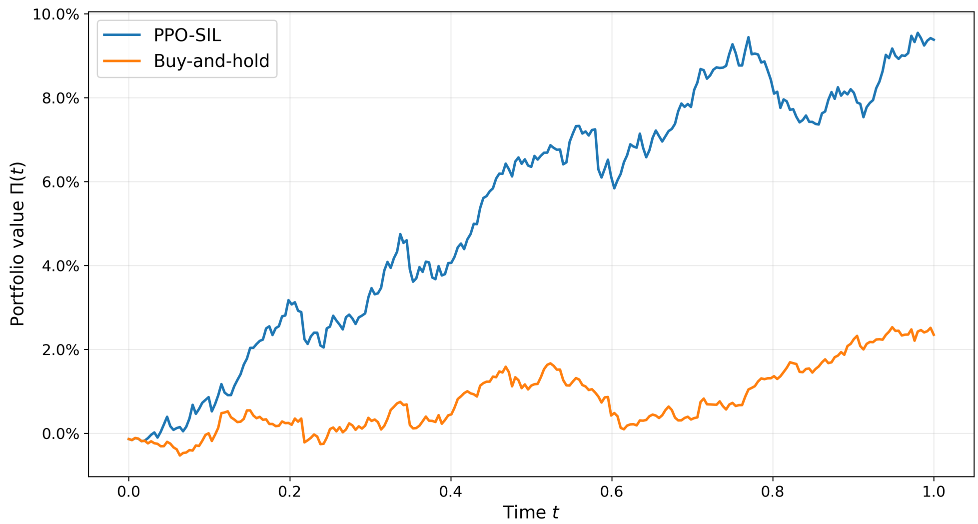

Figure 14: Comparison of the cumulative returns of the PPO-SIL and Buy-and-hold portfolios.

approximately 2-3%. The gap in performance of both portfolios widens during the middle part of the horizon. This suggests that, on this scenario, the learned policy benefits from active reallocation rather than only from the average upward movement of the market. However, we note that this result is path-specific and should not be interpreted as a general performance conclusion.

Table 9 compares the initial and terminal prices, as well as portfolio weights. We observe that the terminal PPO-SIL allocation is concentrated in a small set of assets. Specifically, ABBN, GALD, HOLN, and SDZ receive large final weights. These assets also have relatively large positive drift estimates (see Table 7). As such, this may indicate that the agent favours assets with high model-implied drift. However, this interpretation should be treated with caution and the results shown in this subsection alone do not prove that the agent has learned the true long-term behaviour of the stocks.

Figure 15 gives a more detailed view of the risky allocation over time, with the plot showing the portfolio allocation across risky assets. As one immediately observes, the policy changes the portfolio substantially near the beginning of the horizon. This initial adjustment moves the portfolio away from the ETF allocation and toward the assets that the policy prefers under the learned model. After this initial phase, the allocation becomes more stable, with the policy maintaining a relatively broad exposure to a selected subset of assets. Occasional adjustments are made later in the horizon as well.

We note that this behaviour is consistent with the impulse-control interpretation of the underlying trading problem, implying that the policy does not trade continuously. Instead, it performs a large initial repositioning and then waits for sufficiently meaningful changes before making further adjustments. Moreover, the admissibility constrains also shape the allocation. In particular, risk management constraints, such as maximum position limits, trade-size bounds, and the no-borrowing constraint, prevent the policy from taking too concentrated positions.

Figure 16 shows the impulse controls, i.e. trading sizes, over time. Recall that

Ticker	Initial Price (CHF)	Final Price (CHF)	Initial Weight (%)	Final Weight (%)
Cash			0.14	0.00
ABBN	82.66	94.58	7.58	15.01
ALCC	49.61	44.05	2.22	1.83
CFR	156.55	148.01	5.52	2.18
GALD	158.35	176.49	1.81	20.26
GEBN	503.40	541.40	1.32	0.99
GIVN	2683.00	2665.77	1.88	0.00
HOLN	76.26	84.89	2.47	12.98
LONN	474.20	495.00	2.61	6.43
NESN	76.88	74.61	15.05	1.36
NOVN	116.80	116.51	17.12	1.28
PGHN	883.60	880.92	1.39	1.61
ROPC	320.00	314.97	16.79	1.15
SDZ	67.76	79.04	1.06	21.01
SGSN	84.68	92.90	1.49	2.85
SIKA	140.95	129.40	1.81	1.18
SLHN	837.60	861.06	3.00	3.92
SCMN	678.00	749.28	1.98	1.37
SRENH	119.45	129.52	1.29	1.42
UBSG	36.22	40.35	7.12	2.07
ZURN	563.00	604.75	6.35	1.11

Table 9: Comparison of initial and terminal stock prices and portfolio weights

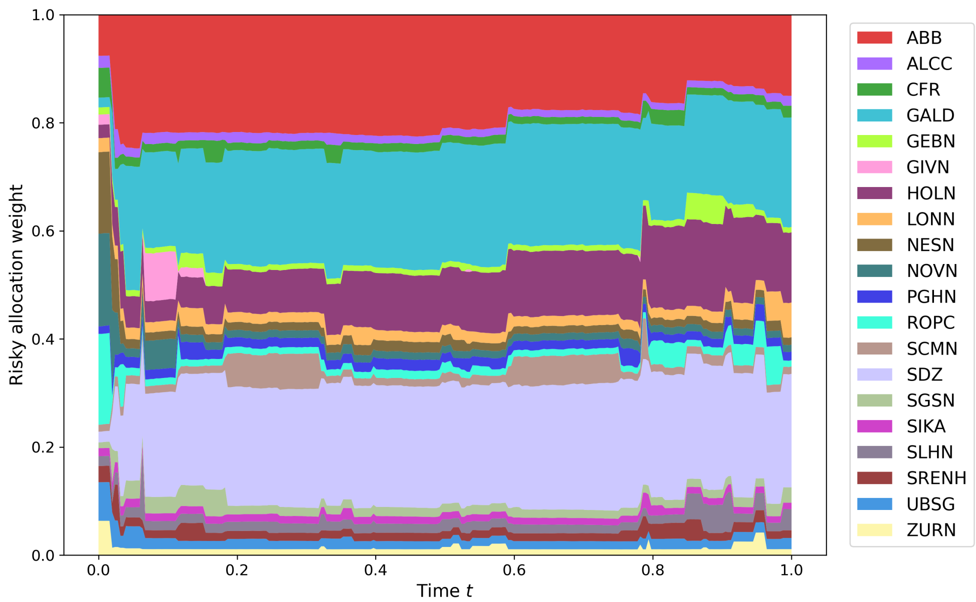

Figure 15: Risky asset allocation of the PPO-SIL algorithm per time step.

positive impulses correspond to a purchase of the corresponding asset, while a negative impulse corresponds to a sale. As previously discussed, the largest impulses occur near the beginning, and after this initial period, most impulses are close to zero, and non-zero trades occur only sparsely. The later trades are smaller and more localised, albeit a few larger adjustments still occur at selected dates, further indicating that frequent trading is not optimal due to transaction costs.

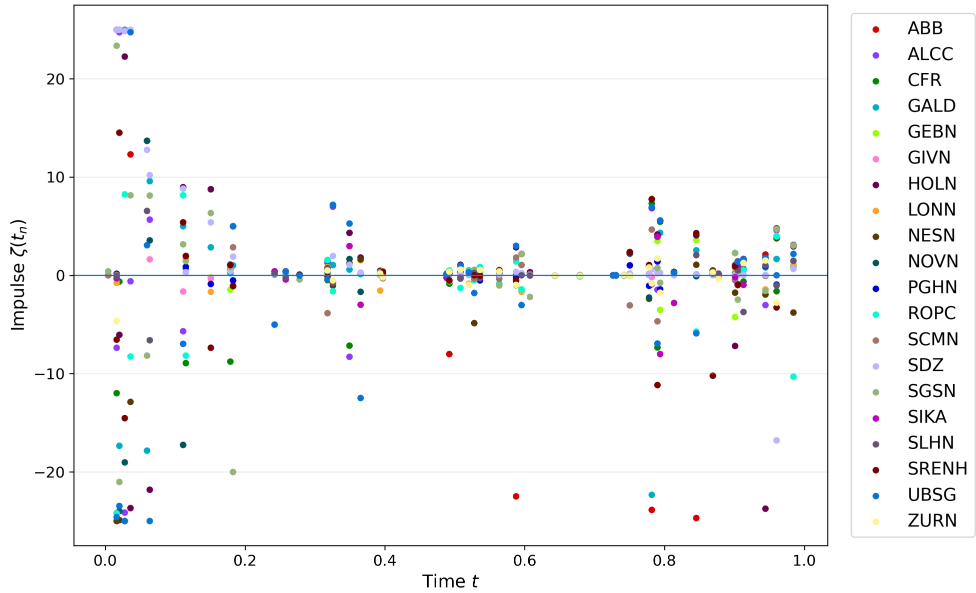

Figure 16: Trade sizes per stock and time step.

Overall, the single-scenario evaluation suggests that the trained PPO-SIL policy effectively combines active allocation with trading sparsity. The policy performs a rapid initial repositioning, maintains exposure to selected assets, and uses later impulses selectively. On this scenario, the policy substantially outperforms the buy-and-hold benchmark while avoiding excessive trading after the initial adjustment.

9.4.3 Multiple-path evaluation

While the single-path evaluation gives a useful illustration of the trading behaviour of the PPO-SIL policy, it is not sufficient to assess the robustness of the strategy. We therefore evaluate the trained policy on 5'000 simulated stock price paths and compare it with the buy-and-hold benchmark using the same simulated price paths for both portfolios.

Based on the simulated stock price paths, the empirical probability of the agent outperforming the buy-and-hold portfolio is 99.32%, meaning that PPO-SIL policy achieves a higher terminal return than the buy-and-hold portfolio in almost all cases. This is a strong indication that the outperformance is not driven by a small number of unusually favourable paths. Instead the active policy appears to shift the overall terminal wealth distribution upward relative to the passive benchmark.

The transaction costs and trade counts of the PPO-SIL strategy can be found in Table 10. The average total transaction costs are 187.36 CHF, which corresponds to approximately 0.37% of the initial portfolio value, and ranges from 85.55 CHF to 342.84 CHF. The average number of total trades is 50.01, with a minimum of 24 and a maximum of 165. Table 11 compares the main performance statistic of the buy-and-

PPO-SIL			
	Average	Smallest	Largest
Transaction costs (CHF)	187.36	85.55	342.84
Trade count	50.01	24.00	165.00

Table 10: Transaction costs and trade counts for the PPO-SIL strategy.

hold and PPO-SIL portfolios. The formulas for each of the performance measures can be found in Appendix F. The annualised return measures the percentage change in total portfolio value over the investment horizon. The buy-and-hold portfolio achieves an average terminal return of 4.44%, while the PPO-SIL portfolio achieves an average terminal return of 10.55%. Hence, the agent delivers an average return that is more than twice as high as the passive benchmark. The best realised PPO-SIL return is also substantially higher, with 24.83% compared with 12.97% for buy-and-hold. The worst PPO-SIL annual return is -2.31%, compared with -3.53% for buy-and-hold.

	Buy-and-hold			PPO-SIL		
	Average	Worst	Best	Average	Worst	Best
Terminal return (%)	4.44	-3.53	12.97	10.55	-2.31	24.83
Volatility (%)	2.21	3.13	1.68	3.12	4.34	2.41
Sharpe ratio	2.01	-1.46	5.98	3.26	-0.73	7.69
Calmar ratio	3.81	-0.84	24.75	7.43	-0.39	39.68

Table 11: Summary performance statistics for the buy-and-hold and PPO-SIL portfolio strategies based on 5,000 simulated stock price paths.

Nevertheless, the high returns of the PPO-SIL portfolio are accompanied with greater annualised portfolio volatility compared to the buy-and-hold portfolio. For instance, the average volatility of the PPO-SIL is 3.12%, while the passive strategy yields a volatility of 2.21%. However, this behaviour is expected as the PPO-SIL strategy actively reallocates capital and takes more concentrated positions. Moreover, the agent's only goal is to maximise the stepwise profit-and-loss, as opposed to minimising the portfolio volatility.

The Sharpe ratio measures excess return per unit of volatility. A higher Sharpe ratio indicates that the portfolio earns more return for each unit of volatility, implying better risk-adjusted performance.The average Sharpe ratio of the buy-and-hold portfolio is 2.01, while the average Sharpe ratio of the PPO-SIL portfolio is 3.26[6]. This shows that the PPO-SIL not only achieves higher returns by taking more volatile positions. The worst Sharpe ratio is -1.46 for buy-and-hold and -0.73 for PPO-SIL, while

[6] According to Investopedia, Sharpe ratios of around 2 are "very good" portfolios, whereas a Sharpe ratio above 3 is counted as "excellent".

the best Sharpe ratio is 5.98 for buy-and-hold and 7.69 for PPO-SIL. Thus, the PPO-SIL shows stronger portfolio performance in both average and extreme Sharpe-ratio outcomes than the passive strategy.

In addition to the Sharpe ratio, consider the Calmar ratio, which measures return relative to the maximum realised drop (drawdown). Hence, a high Calmar ratio indicates that the portfolio generates strong returns without suffering large drawdowns. As we can see, the PPO-SIL strategy yields higher Calmar ratios than the buy-and-hold strategy, both on average as well as in worst and favourable scenarios.

Finally, consider Table 12, where tail-risk metrics are reported. The metrics are computed from the empirical distribution of terminal losses based on the formulas found in Appendix F. Here, terminal loss is defined as initial wealth minus terminal wealth. Positive values therefore represent actual losses, while negative values represent gains. The empirical VaR at the 5% tail gives the loss threshold that is exceeded only in the worst 5% simulated paths. The empirical expected shortfall (ES) gives the average terminal loss inside this worst 5% tail. The buy-and-hold portfolio has

Risk metric	Buy-and-hold	PPO-SIL
$\widehat{\mathrm{VaR}}_{0.05}$ (CHF)	-235.46	-2389.68
$\widehat{\mathrm{ES}}_{0.05}$ (CHF)	269.78	-1615.28
Loss probability (%)	3.28	0.16

Table 12: Risk metrics for both portfolios based on 5'000 simulated stock price paths.

an empirical 5% VaR of -235 CHF. Because this value is negative, it indicates a small terminal gain in portfolio value. However, ES is 269.78 CHF, meaning that the mean 5%-loss quantile is an actual loss of 269 CHF. The empirical loss probability is 3.28%, which means that the buy-and-hold portfolio ends below it initial value in 3.28% of the simulated paths, i.e. 164 out of the 5'000 were unprofitable.

On the other hand, the PPO-SIL portfolio has a 5% VaR of -2'389 CHF and an ES of -1'615.28 CHF. Both values are negative under the terminal-loss convention. This means that even the 5% tail of the PPO-SIL terminal-loss distribution is profitable on average. In particular, ES indicates that the average outcome among the worst 5% of PPO-SIL paths is still a gain of approximately 1'615 CHF, which is an improvement relative to the buy-and-hold portfolio. Moreover, the empirical loss probability for the PPO-SIL policy is only 0.16%, which translates to 8 out of 5'000 Monte Carlo paths.

Overall, the multiple-path evaluation shows that the PPO-SIL outperforms buy-and-hold across almost all reported performance measures, and exhibits higher annualised returns, larger Sharpe and Calmar ratios, as well as better empirical tail-risk, despite trading and paying for transaction costs.

9.5 Remarks on the portfolio experiment

The results shown in Chapter 9.4.3 suggest that the PPO-SIL policy learns that the PPO-SIL policy learns a profitable active allocation strategy under the calibrated simulation time. Nevertheless, the results should be interpreted with care. For instance, the evaluation paths are simulated from the calibrated stock price model and not taken from realised future market data. Hence, the reported performance measures do not prove that the policy would outperform in the real market. The numerical experiment is therefore best understood as a "within-model" test of the PPO-SIL in a high-dimensional portfolio allocation problem with transaction costs and constraints.

Our main concern is that the policy may have learned the asset-specific structure of the simulator, since the drift coefficients were kept fixed during both training and evaluation. Although these coefficients are not explicitly provided as inputs, the ordering assets in the state vector is fixed. For example, ABBN always appears first in the simulated stock price vector, while ZURN appears last. Over many training simulations, the reward signals can therefore reveal which assets tend to generate higher terminal wealth. As a result, the PPO–SIL algorithm may have learned to allocate capital toward assets with higher estimated drift, rather than developing a robust decision rule. At the same time, a wealth-maximising policy would naturally exploit assets with higher expected returns. We therefore note that a limitation of this experiment is the potential sensitivity of the learned policy to misspecification of the drift parameters.

The calibration procedure therefore becomes an important source of model risk. The drift coefficients are estimated using historical data. However, historical data does not always reflect future stock price paths. Moreover, expected future returns are hard to estimate (as otherwise, everyone would be rich by now), and small changes in the estimation window or market regime changes could drastically change the future drifts. In addition, historical option data are not available on the LSEG terminal and, as a result, it was not possible to perform a full backtest using out-of-sample data. We therefore had to rely on future price simulations from a single calibrated parameter set.

Finally, we note that the dynamic portfolio allocation experiment is not limited to equities only, but can instead be combined with a broad universe of securities, such as fixed-income instruments, options, or even DeFi assets.

CHAPTER

TEN

CONCLUDING REMARKS

Conclusion

This thesis studies stochastic impulse control problems on a finite time horizon, with applications to foreign exchange intervention and dynamic portfolio allocation. We perform analysis of the problem from two complementary perspectives. First, a probabilistic approach based on the dynamic programming principle is used to recursively solve the value function with iterated optimal stopping times. Second, a partial integro-differential equation approach characterises the value function as the viscosity solution of the quasi-variational inequality.

The numerical part of the thesis connects reinforcement learning theory with the impulse control problem. We show by means of numerical experiments that deep reinforcement learning can approximate impulse-control problems and compare the results against a finite-difference scheme in the foreign exchange rate example. In the portfolio allocation problem, risky asset parameters were estimated using market data and correlated equity price paths simulated using multivariate subordination. The Proximal Policy Optimisation algorithm was then combined with Self-imitation Learning to trade and maximise portfolio value over a finite time horizon. The results indicate that the reinforcement learning algorithm effectively learns a trading strategy and achieves superior performance compared to a naive buy-and-hold portfolio.

Future research

A first direction for future research is to extend the impulse-control problem beyond risk-neutral objectives. For instance, in the dynamic portfolio allocation experiment, the investor maximises expected discounted terminal wealth, while risk is controlled indirectly through the choice of equity universe, position limits, no-shorting, no-borrowing, and solvency constraints. While such constraints may resemble risk limits used by financial intermediaries[1], they do not directly include risk preferences

[1] I.e. banks, insurance companies, and asset managers.

in the objective. A more realistic formulation could instead use a convex risk measure or a risk-sensitive criterion. For instance, [84, Chapter 4.5.2] discusses the convex dual representation of convex risk measures, which provides a possible starting point for robust impulse-control formulations. Such dual representations can also be related to zero-sum impulse games, where the investor and market can be interpreted as competing decision-makers. Theory on zero-sum games is treated in [4, 29]. Furthermore, finite-horizon risk-sensitive criteria for jump models are studied in [35, 36, 37, 34], while neural-network approaches can be found in [17, 22, 23, 40, 41, 82].

A second direction concerns numerical schemes for HJBQVIs with jump-diffusion dynamics. The explicit-impulse scheme used in this thesis provides a useful benchmark in the one-dimensional diffusion setting, but it does not solve the QVI with generators containing nonlocal terms. An extension would be to split the PIDE into local and nonlocal parts and use an implicit-explicit discretisation of the differential operator. However, because the resulting QVI contains a nonlocal intervention operator, the scheme would require its own convergence proof. In particular, one would need to establish consistence, stability, monotonicity, and convergence for the QVI scheme. Starting points are the Barles-Souganidis convergence result [8], the numerical impulse-control schemes in [6, 3], and finite-difference methods for PIDEs found in [18].

A third direction is to study reinforcement learning algorithms in a more systematic fashion. The DDQN algorithm is well-suited for problems with discrete impulse-sets, but it is sensitive to hyperparameters. Future work should therefore include a systematic hyperparameter study. In addition, other DRL algorithms could be considered, such as the Soft actor-critic (SAC), the Advantage Actor-Critic (A2C), and the Asynchronous Advantage Actor-Critic (A3C) algorithms [78, 46], while deep stopping time approaches also look promising, see [12, 48, 67].

Finally, further work is needed on the portfolio experiment and the economic scenario generator. Currently, we rely on a correlated Variance-Gamma model, calibrated partly to option prices and partly to historical stock data. While this approach captures jump behaviour and non-Gaussian return features of equities, the resulting trading policy may still be sensitive to the estimated drift parameters. Future work should therefore include backtesting and stress testing across different market regimes, together with more robust benchmark portfolios. For example, stress tests based on multidimensional t-copulas, Archimedean copulas, or even Lévy copulas could provide a strong assessment of robustness based on tail dependence of stock prices, co-jumps, and market-wide shocks [76, 95, 63, 44]. From a stock price modelling perspective, regime-switching models [66, 34, 99] and models with mean-reverting drift may better capture long-term market dynamics than the currently used constant-drift coefficients. In addition, enriching the state representation with features such as market sentiment [75], regime indicators, or volatility measures may lead to more robust trading policies.

REFERENCES

[1] Pranay Anchuri. *RAmmStein: Regime Adaptation in Mean-Reverting Markets with Stein Thresholds–Optimal Impulse Control in Concentrated AMMs*. 2026. DOI: 10.48550/arXiv.2602.19419. arXiv: 2602.19419 [cs.LG].

[2] David Applebaum. *Lévy Processes and Stochastic Calculus*. 2nd ed. Vol. 116. Cambridge Studies in Advanced Mathematics. Cambridge: Cambridge University Press, 2009. DOI: 10.1017/CBO9780511809781.

[3] Parsiad Azimzadeh. *Impulse Control in Finance: Numerical Methods and Viscosity Solutions*. 2018. DOI: 10.48550/arXiv.1712.01647. arXiv: 1712.01647 [math.NA].

[4] Parsiad Azimzadeh. 'A zero-sum stochastic differential game with impulses, precommitment, and unrestricted cost functions'. In: *Applied Mathematics & Optimization* 79.2 (2019), pp. 483–514. DOI: 10.1007/s00245-017-9445-x.

[5] Parsiad Azimzadeh, Erhan Bayraktar and George Labahn. 'Convergence of implicit schemes for Hamilton–Jacobi–Bellman quasi-variational inequalities'. In: *SIAM Journal on Control and Optimization* 56.6 (2018), pp. 3994–4016. DOI: 10.1137/18M1171965.

[6] Parsiad Azimzadeh and Peter A. Forsyth. 'Weakly chained matrices, policy iteration, and impulse control'. In: *SIAM Journal on Numerical Analysis* 54.3 (2016), pp. 1341–1364. DOI: 10.1137/15M1043431.

[7] Guy Barles and Cyril Imbert. 'Second-order elliptic integro-differential equations: viscosity solutions' theory revisited'. In: *Annales de l'Institut Henri Poincaré C, Analyse Non Linéaire* 25.3 (2008), pp. 567–585. DOI: 10.1016/j.anihpc.2007.02.007.

[8] Guy Barles and Panagiotis E. Souganidis. 'Convergence of approximation schemes for fully nonlinear second order equations'. In: *Asymptotic Analysis* 4.3 (1991), pp. 271–283. DOI: 10.3233/ASY-1991-4305.

[9] Andrew R. Barron. 'Approximation and estimation bounds for artificial neural networks'. In: *Machine Learning* 14.1 (1994), pp. 115–133. DOI: 10.1007/BF00993164.

[10] Atilim Gunes Baydin, Barak A. Pearlmutter, Alexey Andreyevich Radul and Jeffrey Mark Siskind. 'Automatic differentiation in machine learning: a survey'. In: *Journal of Machine Learning Research* 18.153 (2018), pp. 1–43.

[11] Erhan Bayraktar and Ali Devran Kara. 'Approximate Q-learning for controlled diffusion processes and its near optimality'. In: *SIAM Journal on Mathematics of Data Science* 5.3 (2023), pp. 615–638. DOI: 10.1137/22M1484201.

[12] Sebastian Becker, Patrick Cheridito and Arnulf Jentzen. 'Deep optimal stopping'. In: *Journal of Machine Learning Research* 20.74 (2019), pp. 1–25.

[13] Richard E. Bellman. *Dynamic Programming.* Rand Corporation Research Study. Princeton, NJ: Princeton University Press, 1957.

[14] Alain Bensoussan and Jacques-Louis Lions. 'Nouvelles méthodes en contrôle impulsionnel'. In: *Applied Mathematics and Optimization* 1.4 (1975), pp. 289–312. DOI: 10.1007/BF01447955.

[15] Dimitri P. Bertsekas and Steven E. Shreve. *Stochastic Optimal Control: The Discrete-Time Case.* Vol. 139. Mathematics in Science and Engineering. New York: Academic Press, 1978.

[16] Rabi Bhattacharya and Edward C. Waymire. *A Basic Course in Probability Theory.* Universitext. New York: Springer, 2007. DOI: 10.1007/978-0-387-71939-9.

[17] Tomasz R. Bielecki, Tao Chen and Igor Cialenco. 'Risk-sensitive Markov decision problems under model uncertainty: finite time horizon case'. In: *Stochastic Analysis, Filtering, and Stochastic Optimization: A Commemorative Volume to Honor Mark H. A. Davis's Contributions.* Cham: Springer, 2022, pp. 33–52. DOI: 10.1007/978-3-030-98519-6_2.

[18] Imran H. Biswas, Espen R. Jakobsen and Kenneth H. Karlsen. 'Difference-quadrature schemes for nonlinear degenerate parabolic integro-PDEs'. In: *SIAM Journal on Numerical Analysis* 48.3 (2010), pp. 1110–1135. DOI: 10.1137/090761501.

[19] David Blackwell. 'Discounted Dynamic Programming'. In: *The Annals of Mathematical Statistics* 36.1 (1965), pp. 226–235. DOI: 10.1214/aoms/1177700285.

[20] Olivier Bokanowski, Stefania Maroso and Hasnaa Zidani. 'Some convergence results for Howard's algorithm'. In: *SIAM Journal on Numerical Analysis* 47.4 (2009), pp. 3001–3026. DOI: 10.1137/08073041X.

[21] Gonchigsuren Bor. *Dynamic Portfolio Allocation.* GitHub repository. URL: https://github.com/goshabor/Dynamic-portfolio-allocation (visited on 04/06/2026).

[22] Hans Buehler, Lukas Gonon, Josef Teichmann and Ben Wood. 'Deep hedging'. In: *Quantitative Finance* 19.8 (2019), pp. 1271–1291. DOI: 10.1080/14697688.2019.1571683.

[23] Hans Buehler, Phillip Murray and Ben Wood. *Deep Bellman Hedging.* 2022. DOI: 10.48550/arXiv.2207.00932. arXiv: 2207.00932 [q-fin.CP].

[24] Dmitri Burago, Yuri Burago and Sergei Ivanov. *A Course in Metric Geometry.* Vol. 33. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society, 2001.

[25] Abel Cadenillas and Fernando Zapatero. 'Optimal central bank intervention in the foreign exchange market'. In: *Journal of Economic Theory* 87.1 (1999), pp. 218–242. DOI: 10.1006/jeth.1999.2523.

[26] Abel Cadenillas and Fernando Zapatero. 'Classical and impulse stochastic control of the exchange rate using interest rates and reserves'. In: *Mathematical Finance* 10.2 (2000), pp. 141–156. DOI: 10.1111/1467-9965.00086.

[27] Peter Carr and Dilip B. Madan. 'Option valuation using the fast Fourier transform'. In: *Journal of Computational Finance* 2.4 (1999), pp. 61–73. DOI: 10.21314/JCF.1999.043.

[28] James W. Cooley and John W. Tukey. 'An algorithm for the machine calculation of complex Fourier series'. In: *Mathematics of Computation* 19.90 (1965), pp. 297–301. DOI: 10.1090/S0025-5718-1965-0178586-1.

[29] Andrea Cosso. 'Stochastic differential games involving impulse controls and double-obstacle quasi-variational inequalities'. In: *SIAM Journal on Control and Optimization* 51.3 (2013), pp. 2102–2131. DOI: 10.1137/120880094.

[30] Michael G. Crandall and Hitoshi Ishii. 'The maximum principle for semicontinuous functions'. In: *Differential and Integral Equations* 3.6 (1990), pp. 1001–1014.

[31] Michael G. Crandall, Hitoshi Ishii and Pierre-Louis Lions. 'User's guide to viscosity solutions of second order partial differential equations'. In: *Bulletin of the American Mathematical Society* 27.1 (1992), pp. 1–67. DOI: 10.1090/S0273-0979-1992-00266-5.

[32] Michael G. Crandall and Pierre-Louis Lions. 'Viscosity solutions of Hamilton–Jacobi equations'. In: *Transactions of the American Mathematical Society* 277.1 (1983), pp. 1–42. DOI: 10.1090/S0002-9947-1983-0690039-8.

[33] Saeed Damadi, Golnaz Moharrer, Mostafa Cham and Jinglai Shen. 'The Backpropagation Algorithm for a Math Student'. In: *2023 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2023, pp. 1–9. DOI: 10.1109/IJCNN54540.2023.10191596.

[34] Milan Kumar Das, Anindya Goswami and Nimit Rana. 'Risk sensitive portfolio optimization in a jump diffusion model with regimes'. In: *SIAM Journal on Control and Optimization* 56.2 (2018), pp. 1550–1576. DOI: 10.1137/17M1121809.

[35] Mark H. A. Davis and Sébastien Lleo. *Jump-diffusion risk-sensitive asset management*. 2009. DOI: 10.48550/arXiv.0905.4740. arXiv: 0905.4740 [q-fin.PM].

[36] Mark H. A. Davis and Sébastien Lleo. 'Jump-diffusion risk-sensitive asset management I: diffusion factor model'. In: *SIAM Journal on Financial Mathematics* 2.1 (2011), pp. 22–54. DOI: 10.1137/090760180.

[37] Mark H. A. Davis and Sébastien Lleo. 'Jump-diffusion risk-sensitive asset management II: jump-diffusion factor model'. In: *SIAM Journal on Control and Optimization* 51.2 (2013), pp. 1441–1480. DOI: 10.1137/110825881.

[38] Emmanuele DiBenedetto. *Real Analysis*. Birkhäuser Advanced Texts. Boston, MA: Birkhäuser, 2002. DOI: 10.1007/978-1-4612-0117-5.

[39] Jerome F. Eastham and Kevin J. Hastings. 'Optimal impulse control of portfolios'. In: *Mathematics of Operations Research* 13.4 (1988), pp. 588–605. DOI: 10.1287/moor.13.4.588.

[40] Tobias Enders, James Harrison and Maximilian Schiffer. *Risk-Sensitive Soft Actor-Critic for Robust Deep Reinforcement Learning under Distribution Shifts*. 2024. DOI: 10.48550/arXiv.2402.09992. arXiv: 2402.09992 [cs.LG].

[41] Yingjie Fei, Zhuoran Yang, Yudong Chen and Zhaoran Wang. 'Exponential Bellman Equation and Improved Regret Bounds for Risk-Sensitive Reinforcement Learning'. In: *Advances in Neural Information Processing Systems*. Vol. 34. 2021, pp. 20436–20446.

[42] Wendell H. Fleming and H. Mete Soner. *Controlled Markov Processes and Viscosity Solutions*. 2nd ed. Vol. 25. Stochastic Modelling and Applied Probability. New York: Springer, 2006. DOI: 10.1007/0-387-31071-1.

[43] Nagata Furukawa. 'Markovian Decision Processes with Compact Action Spaces'. In: *The Annals of Mathematical Statistics* 43.5 (1972), pp. 1612–1622. DOI: 10.1214/aoms/1177692393.

[44] Oliver Grothe and Marius Hofert. 'Construction and sampling of Archimedean and nested Archimedean Lévy copulas'. In: *Journal of Multivariate Analysis* 138 (2015). High-Dimensional Dependence and Copulas, pp. 182–198. ISSN: 0047-259X. DOI: 10.1016/j.jmva.2014.12.004.

[45] Namig J. Guliyev and Vugar E. Ismailov. 'On the approximation by single hidden layer feedforward neural networks with fixed weights'. In: *Neural Networks* 98 (2018), pp. 296–304. DOI: 10.1016/j.neunet.2017.12.007.

[46] Tuomas Haarnoja, Aurick Zhou, Kristian Hartikainen, George Tucker, Sehoon Ha, Jie Tan, Vikash Kumar, Henry Zhu, Abhishek Gupta and Pieter Abbeel. *Soft Actor-Critic Algorithms and Applications*. 2018. DOI: 10.48550/arXiv.1812.05905. arXiv: 1812.05905 [cs.LG].

[47] Donatien Hainaut and Alex Casas. 'Option pricing in the Heston model with physics inspired neural networks'. In: *Annals of Finance* 20.3 (2024), pp. 353–376. DOI: 10.1007/s10436-024-00452-7.

[48] Yuecai Han and Nan Li. 'A new deep neural network algorithm for multiple stopping with applications in options pricing'. In: *Communications in Nonlinear Science and Numerical Simulation* 117 (2023), p. 106881. DOI: 10.1016/j.cnsns.2022.106881.

[49] Hado van Hasselt. 'Double Q-learning'. In: *Advances in Neural Information Processing Systems*. Vol. 23. 2010, pp. 2613–2621.

[50] Hado van Hasselt, Arthur Guez and David Silver. 'Deep Reinforcement Learning with Double Q-learning'. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 30. 1. 2016, pp. 2094–2100. DOI: 10.1609/aaai.v30i1.10295.

[51] Geoffrey E. Hinton and Ruslan R. Salakhutdinov. 'Reducing the Dimensionality of Data with Neural Networks'. In: *Science* 313.5786 (2006), pp. 504–507. DOI: 10.1126/science.1127647.

[52] Ronald A. Howard. *Dynamic Programming and Markov Processes*. Cambridge, MA: MIT Press, 1960.

[53] Masashi Ieda. 'An implicit method for the finite time horizon Hamilton–Jacobi–Bellman quasi-variational inequalities'. In: *Applied Mathematics and Computation* 265 (2015), pp. 163–175. DOI: 10.1016/j.amc.2015.04.031.

[54] Maximilian Igl, Kamil Ciosek, Yingzhen Li, Sebastian Tschiatschek, Cheng Zhang, Sam Devlin and Katja Hofmann. 'Generalization in Reinforcement Learning with Selective Noise Injection and Information Bottleneck'. In: *Advances in Neural Information Processing Systems*. Vol. 32. 2019. DOI: 10.48550/arXiv.1910.12911. arXiv: 1910.12911 [cs.LG].

[55] Katsuyuki Ishii. 'Viscosity solutions of nonlinear second order elliptic PDEs associated with impulse control problems'. In: *Funkcialaj Ekvacioj* 36.1 (1993), pp. 123–141.

[56] Yasushi Ishikawa. 'Optimal control problem associated with jump processes'. In: *Applied Mathematics and Optimization* 50.1 (2004), pp. 21–65. DOI: 10.1007/s00245-004-0795-9.

[57] Riashat Islam, Hongyu Zang, Manan Tomar, Aniket Didolkar, Md Mofijul Islam, Samin Yeasar Arnob, Tariq Iqbal, Xin Li, Anirudh Goyal, Nicolas Heess and Alex Lamb. *Representation Learning in Deep RL via Discrete Information Bottleneck*. 2022. DOI: 10.48550/arXiv.2212.13835. arXiv: 2212.13835 [cs.LG].

[58] Konark Jain. *HawkesRLTrading*. Software, GitHub repository. URL: https://github.com/konqr/lobSimulations/tree/master/HawkesRLTrading (visited on 03/06/2026).

[59] Konark Jain, Nick Firoozye, Jonathan Kochems and Philip Treleaven. *An Impulse Control Approach to Market Making in a Hawkes LOB Market*. 2025. DOI: 10.48550/arXiv.2510.26438. arXiv: 2510.26438 [q-fin.TR].

[60] Espen R. Jakobsen and Kenneth H. Karlsen. 'A maximum principle for semicontinuous functions applicable to integro-partial differential equations'. In: *Nonlinear Differential Equations and Applications NoDEA* 13.2 (2006), pp. 137–165. DOI: 10.1007/s00030-005-0031-6.

[61] Robert A. Jarrow. *Continuous-Time Asset Pricing Theory: A Martingale-Based Approach*. Springer Finance Textbooks. Cham: Springer, 2018. DOI: 10.1007/978-3-319-77821-1.

[62] Zhengyao Jiang, Dixing Xu and Jinjun Liang. *A Deep Reinforcement Learning Framework for the Financial Portfolio Management Problem*. 2017. DOI: 10.48550/arXiv.1706.10059. arXiv: 1706.10059 [q-fin.CP].

[63] Jan Kallsen and Peter Tankov. 'Characterization of dependence of multidimensional Lévy processes using Lévy copulas'. In: *Journal of Multivariate Analysis* 97.7 (2006), pp. 1551–1572. ISSN: 0047-259X. DOI: 10.1016/j.jmva.2005.11.001.

[64] Ioannis Karatzas and Steven E. Shreve. *Brownian Motion and Stochastic Calculus*. 2nd ed. Vol. 113. Graduate Texts in Mathematics. New York: Springer, 1991. DOI: 10.1007/978-1-4612-0949-2.

[65] Ralf Korn. 'Portfolio optimisation with strictly positive transaction costs and impulse control'. In: *Finance and Stochastics* 2.2 (1998), pp. 85–114. DOI: 10.1007/s007800050034.

[66] Ralf Korn, Yaroslav Melnyk and Frank Thomas Seifried. 'Stochastic impulse control with regime-switching dynamics'. In: *European Journal of Operational Research* 260.3 (2017), pp. 1024–1042. DOI: 10.1016/j.ejor.2016.12.029.

[67] Mathieu Laurière and Mehdi Talbi. *Deep Learning for the Multiple Optimal Stopping Problem*. 2025. DOI: 10.48550/arXiv.2512.22961. arXiv: 2512.22961 [math.OC].

[68] Moshe Leshno, Vladimir Ya. Lin, Allan Pinkus and Shimon Schocken. 'Multilayer feedforward networks with a nonpolynomial activation function can approximate any function'. In: *Neural Networks* 6.6 (1993), pp. 861–867. DOI: 10.1016/S0893-6080(05)80131-5.

[69] Ilya Loshchilov and Frank Hutter. 'Decoupled Weight Decay Regularization'. In: *International Conference on Learning Representations*. 2019. DOI: 10.48550/arXiv.1711.05101. arXiv: 1711.05101 [cs.LG].

[70] Yaowen Lu and Duy-Minh Dang. 'A semi-Lagrangian ε-monotone Fourier method for continuous withdrawal GMWBs under jump-diffusion with stochastic interest rate'. In: *Numerical Methods for Partial Differential Equations* 40.3 (2024), e23075. DOI: 10.1002/num.23075.

[71] Elisa Luciano, Marina Marena and Patrizia Semeraro. 'Dependence calibration and portfolio fit with factor-based subordinators'. In: *Quantitative Finance* 16.7 (2016), pp. 1037–1052. DOI: 10.1080/14697688.2015.1114661.

[72] Elisa Luciano and Patrizia Semeraro. 'Multivariate time changes for Lévy asset models: Characterization and calibration'. In: *Journal of Computational and Applied Mathematics* 233.8 (2010), pp. 1937–1953. ISSN: 0377-0427. DOI: 10.1016/j.cam.2009.08.119.

[73] Dilip B. Madan, Peter P. Carr and Eric C. Chang. 'The variance gamma process and option pricing'. In: *European Finance Review* 2.1 (1998), pp. 79–105. DOI: 10.1023/A:1009703431535.

[74] Ashok Maitra. 'Discounted Dynamic Programming on Compact Metric Spaces'. In: *Sankhyā: The Indian Journal of Statistics, Series A* 30.2 (1968), pp. 211–216.

[75] Yanying Mao, Qun Liu and Yu Zhang. 'Sentiment analysis methods, applications, and challenges: A systematic literature review'. In: *Journal of King Saud University - Computer and Information Sciences* 36.4 (2024), p. 102048. ISSN: 1319-1578. DOI: 10.1016/j.jksuci.2024.102048.

[76] Alexander J. McNeil, Rüdiger Frey and Paul Embrechts. *Quantitative Risk Management: Concepts, Techniques and Tools*. Revised. Princeton, NJ: Princeton University Press, 2015.

[77] David Mguni, Aivar Sootla, Juliusz Ziomek, Oliver Slumbers, Zipeng Dai, Kun Shao and Jun Wang. 'Timing is Everything: Learning to Act Selectively with Costly Actions and Budgetary Constraints'. In: *International Conference on Learning Representations*. 2023. DOI: 10.48550/arXiv.2205.15953. arXiv: 2205.15953 [cs.LG].

[78] Volodymyr Mnih, Adrià Puigdomènech Badia, Mehdi Mirza, Alex Graves, Timothy Lillicrap, Tim Harley, David Silver and Koray Kavukcuoglu. 'Asynchronous Methods for Deep Reinforcement Learning'. In: *Proceedings of the 33rd International Conference on Machine Learning*. Vol. 48. Proceedings of Machine Learning Research. PMLR, 2016, pp. 1928–1937.

[79] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen, Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King, Dharshan Kumaran, Daan Wierstra, Shane Legg and Demis Hassabis. 'Human-level control through deep reinforcement learning'. In: *Nature* 518.7540 (2015), pp. 529–533. DOI: 10.1038/nature14236.

[80] Andrew J. Morton and Stanley R. Pliska. 'Optimal portfolio management with fixed transaction costs'. In: *Mathematical Finance* 5.4 (1995), pp. 337–356. DOI: 10.1111/j.1467-9965.1995.tb00071.x.

[81] Gabriela Mundaca. 'Optimal stochastic intervention control with application to the exchange rate'. In: *Journal of Mathematical Economics* 29.2 (1998), pp. 225–243. DOI: 10.1016/S0304-4068(97)00013-X.

[82] Erfaun Noorani, Christos N. Mavridis and John S. Baras. 'Risk-Sensitive Reinforcement Learning with Exponential Criteria'. In: *IEEE Transactions on Cybernetics* 55.8 (2025), pp. 3774–3787. DOI: 10.1109/TCYB.2025.3575240.

[83] Junhyuk Oh, Yijie Guo, Satinder Singh and Honglak Lee. 'Self-Imitation Learning'. In: *Proceedings of the 35th International Conference on Machine Learning*. Vol. 80. Proceedings of Machine Learning Research. PMLR, 2018, pp. 3878–3887.

[84] Bernt Øksendal and Agnès Sulem. *Applied Stochastic Control of Jump Diffusions*. 3rd ed. Universitext. Springer, 2019. DOI: 10.1007/978-3-030-02781-0.

[85] Jooyoung Park and Irwin W. Sandberg. 'Universal approximation using radial-basis-function networks'. In: *Neural Computation* 3.2 (1991), pp. 246–257. DOI: 10.1162/neco.1991.3.2.246.

[86] Murray H. Protter and Charles B. Morrey. *A First Course in Real Analysis*. 2nd ed. Undergraduate Texts in Mathematics. New York: Springer, 1991.

[87] Manfred Schäl. 'Conditions for Optimality in Dynamic Programming and for the Limit of n-Stage Optimal Policies to Be Optimal'. In: *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 32 (1975), pp. 179–196. DOI: 10.1007/BF00532612.

[88] Wim Schoutens. *Lévy Processes in Finance: Pricing Financial Derivatives*. Wiley Series in Probability and Statistics. Chichester: John Wiley & Sons, 2003. DOI: 10.1002/0470870230.

[89] John Schulman, Sergey Levine, Pieter Abbeel, Michael I. Jordan and Philipp Moritz. 'Trust Region Policy Optimization'. In: *Proceedings of the 32nd International Conference on Machine Learning*. Vol. 37. Proceedings of Machine Learning Research. PMLR, 2015, pp. 1889–1897.

[90] John Schulman, Philipp Moritz, Sergey Levine, Michael I. Jordan and Pieter Abbeel. 'High-Dimensional Continuous Control Using Generalized Advantage Estimation'. In: *International Conference on Learning Representations*. 2016. DOI: 10.48550/arXiv.1506.02438. arXiv: 1506.02438 [cs.LG].

[91] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford and Oleg Klimov. *Proximal Policy Optimization Algorithms*. 2017. DOI: 10.48550/arXiv.1707.06347. arXiv: 1707.06347 [cs.LG].

[92] Roland C. Seydel. 'Existence and uniqueness of viscosity solutions for QVI associated with impulse control of jump-diffusions'. In: *Stochastic Processes and their Applications* 119.10 (2009), pp. 3719–3748. DOI: 10.1016/j.spa.2009.07.004.

[93] Rainer Storn and Kenneth Price. 'Differential evolution–a simple and efficient heuristic for global optimization over continuous spaces'. In: *Journal of Global Optimization* 11.4 (1997), pp. 341–359. DOI: 10.1023/A:1008202821328.

[94] Richard S. Sutton, David McAllester, Satinder Singh and Yishay Mansour. 'Policy Gradient Methods for Reinforcement Learning with Function Approximation'. In: *Advances in Neural Information Processing Systems*. Vol. 12. 1999, pp. 1057–1063.

[95] Peter Tankov. 'Simulation and option pricing in Lévy copula models'. In: *Risk* 17.5 (2004), pp. 83–87.

[96] Richard S. Varga. *Matrix Iterative Analysis*. 2nd ed. Vol. 27. Springer Series in Computational Mathematics. Berlin, Heidelberg: Springer, 2000. DOI: 10.1007/978-3-642-05156-2.

[97] Martin J. Wainwright. *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge: Cambridge University Press, 2019. DOI: 10.1017/9781108627771.

[98] Xinyao Wang and Lili Liu. 'Risk-sensitive deep reinforcement learning for portfolio optimization'. In: *Journal of Risk and Financial Management* 18.7 (2025), p. 347. DOI: 10.3390/jrfm18070347.

[99] Zhen Wu and Yan Zhang. 'Recursive impulse control problem with Markov-switching and viscosity solution of HJB equation'. In: *Mathematical Control and Related Fields* 17 (2026), pp. 208–236. DOI: 10.3934/mcrf.2025036.

[100] Yuta Yaegashi, Hidekazu Yoshioka, Kentaro Tsugihashi and Masayuki Fujihara. 'An exact viscosity solution to a Hamilton–Jacobi–Bellman quasi-variational inequality for animal population management'. In: *Journal of Mathematics in Industry* 9.1 (2019), p. 5. DOI: 10.1186/s13362-019-0062-y.

[101] Hidekazu Yoshioka, Yuta Yaegashi, Yumi Yoshioka and Kunihiko Hamagami. 'Hamilton–Jacobi–Bellman quasi-variational inequality arising in an environmental problem and its numerical discretization'. In: *Computers & Mathematics with Applications* 77.8 (2019), pp. 2182–2206. DOI: 10.1016/j.camwa.2018.12.004.

[102] Haoran Zhan and Yingcun Xia. *Consistency for Large Neural Networks: Regression and Classification*. 2024. DOI: 10.48550/arXiv.2409.14123. arXiv: 2409.14123 [stat.ML].

APPENDIX

A

SOME USEFUL THEOREMS

A.1 Doob's Sampling Theorem

Theorem A.1 ([16, Theorem 3.6c] Optional Stopping)**.** *Let* $\{X(t)\}_{t\in[0,T]}$ *be a righ-continuous* $\{\mathcal{F}_t\}$*-martingale, where* $T \in [0, \infty)$. *Suppose* τ *is a stopping time such that*

(i)

$$\mathbb{P}(\tau < \infty) = 1, \tag{A.1.1}$$

and

(ii)

$$X(\tau \wedge t) \quad \text{is uniformly integrable for } t \in [0, T]. \tag{A.1.2}$$

Then,

$$\mathbb{E}[X(\tau)] = \mathbb{E}[X(0)].$$

A.2 Chebyshev's Inequality

Theorem A.2 ([97, Chapter 2] Chebyshev's Inequality)**.** *Let* X *be an integrable random variable with finite, nonzero variance* σ^2 *and mean* μ. *Then, for any real number* $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k) \leq \frac{Var[X]}{k^2}.$$

A.3 Markov's Inequality

Theorem A.3 ([97, Chapter 2] Markov's Inequality)**.** *Let* X *be an integrable random variable with finite, nonzero* k*-th order moment. Then, for any real number* k *and* $t > 0$,

$$\mathbb{P}(|X|^k \leq t) \leq \frac{\mathbb{E}[|X|^k]}{t^k}.$$

A.4 Fatou's Lemma

Lemma A.4 ([38, Fatou's Lemma]). *Let $\{X, \mathcal{A}, \mu\}$ be a measure space and E a measurable set. Let $\{f_n\}$ be a sequence of measurable functions and a.e. nonnegative function in E. Then,*

$$\int_E \liminf_{n\to\infty} f_n \,\mathrm{d}\mu \leq \liminf_{n\to\infty} \int_E f_n \,\mathrm{d}\mu.$$

APPENDIX

B

AUXILIARY RESULTS FOR THE PROOF OF THE EXISTENCE OF THE VISCOSITY SOLUTION

Lemma B.1 ([92, Lemma 7.1])**.** *Consider a process $X^{(t_n,x_n)}$ starting at (t_n, x_n) and following the jump-diffusion process (3.1.1). Denote the first exit time for $\rho > 0$ by $\tau_n^\rho :=\in \{t \geq t_n : |X^{t_n,x_n} - x_n| \geq \rho\}$. Suppose that all conditions for existence and uniqueness are satisfied for X and $(t_n, x_n) \to (t_0, x_0)$. Then for all $\delta > 0$, there is a $\hat{\varepsilon} > 0$ such that*

$$\limsup_{n\to\infty} \mathbb{P}(|\tau_n^\rho - t| < \hat{\varepsilon}) < \delta.$$

Proof. We refer to [92, Lemma 7.1]. □

Lemma B.2 ([92, Lemma 7.2])**.** *Let $\{X(t)\}_{t\geq 0}$ be a sequence of random variables, and let $X(t)$ converge to $X(0) = x \in \mathbb{R}^d$ in probability for $t \downarrow 0$. Then:*

(i) For all $\varepsilon > 0$ and for all sequences $t_n \to 0$,

$$\mathbb{P}\left(\sup_{0\leq s\leq t_n} |X(s) - x| > \varepsilon\right) \to 0 \quad \text{as } n \to \infty.$$

(ii) (i) holds true also for a sequence of positive random variables $\{\tau_n\}_{n\in\mathbb{N}}$ converging to 0 in probability (not necessarily independent of X).

(iii) (ii) holds true also in the setting and under the assumptions of Lemma B.1, i.e.

$$\mathbb{P}\left(\sup_{t_n\leq s\leq \tau_n} |X^{t_n,x_n}(s) - x_n| > \varepsilon\right) \to 0$$

for $(t_n, x_n) \to (t_0, x_0)$, random variables $\tau_n \geq t_n$, and $\tau_n \to t_0$ in probability.

Proof. We refer to [92, Lemma 7.2]. □

APPENDIX

C

AUXILIARY RESULTS FOR DYNAMIC PROGRAMMING

The results provided in this chapter are needed for the derivation of the dynamic programming formula used by the DDQN algorithm in Chapter 7.2.

C.1 Brief review of the QVI

Recall the QVI (4.1.3), that is,

$$\begin{aligned} \min\big\{-(\partial_t v+\mathcal{L}v+f), v-\mathcal{M}v\big\} &= 0 \quad \text{in } [0,T)\times\mathcal{S} && \text{(C.1.1)}\\ \min\{v-g, v-\mathcal{M}v\} &= 0 \quad \text{in } [0,T)\times(\mathbb{R}^d\backslash\mathcal{S}) && \text{(C.1.2)}\\ v &= \{g,\mathcal{M}v\} \quad \text{on } \{T\}\times\mathbb{R}^d, && \text{(C.1.3)}\end{aligned}$$

where the intervention operator is defined as

$$\mathcal{M}v(t,x) := \sup_{\zeta\in\mathcal{Z}(t,x)}\{v(t,\Gamma(x,\zeta))+K(t,x,\zeta)\},$$

and the infinitesimal generator $\mathcal{L}$ is given by

$$\begin{aligned}\mathcal{L}\varphi(t,x) &= \langle\mu(t,x),\nabla_x\varphi(t,x)\rangle+\frac{1}{2}\operatorname{Tr}(\sigma(t,x)\sigma^\top(t,x)H_x\varphi(y))\\ &\quad+\int_{\mathbb{R}^\ell}\{\varphi(t,x+\gamma(t,x,z))-\varphi(t,x)-\langle\nabla\varphi(t,x),\gamma(t,x,z)\rangle 1_{\{|z|<1\}}\}\nu(\,\mathrm{d}z)\end{aligned} \tag{C.1.4}$$

for some test function $\varphi\in C^{1,2}([0,T]\times\mathbb{R}^d)\cap\mathcal{PB}([0,T]\times\mathbb{R}^d)$. Here, $\mathcal{PB}$ denotes the family of polynomially bounded functions. For brevity, we write $\mathcal{PB}:=\mathcal{PB}([0,T]\times\mathbb{R}^d)$ in what follows.

In this appendix, the main focus lies on the continuation region D, defined as

$$D := \big\{(t,x)\in[0,T)\times\mathcal{S} : v(t,x)>\mathcal{M}v(t,x)\big\}. \tag{C.1.5}$$

In D, an intervention is suboptimal and the process evolves freely without any intervention. Therefore, there are no obstacles inside D and the PIDE

$$\partial_t v + \mathcal{L}v + f = 0 \quad \text{in } D$$

is the governing equation.

The jump-diffusion process X follows the SDE

$$\mathrm{d}X(s) = \mu(s, X(s^-))\,\mathrm{d}s + \sigma(s, X(s^-))\,\mathrm{d}W(s) + \int_{\mathbb{R}^\ell} \gamma(t, X(s^-), z)\tilde{N}(\,\mathrm{d}z,\,\mathrm{d}s), \quad X(t^-) = x. \tag{C.1.6}$$

Assuming that X starts in D, i.e. $(t, x) \in D$, we define the exit time from D as

$$\tau_D := \inf\big\{s \geq 0 : \big(t+s, X(t+s)\big) \notin D\big\}. \tag{C.1.7}$$

C.2 Assumptions

We assume that the jump-diffusion X is a càdlàg and $\mathcal{F}_t$-adapted Lévy process. The Lévy measure ν satisfies

$$\int_{\mathbb{R}^\ell} (1 \wedge |z|^2)\nu(\,\mathrm{d}z) < \infty$$

and, for some $p^* > 0$ and $q^* \geq 0$,

$$\int_{|z|\geq 1} |z|^{p^*}\nu(\,\mathrm{d}z) < \infty \quad \text{and} \quad \int_{|z|<1} |z|^{q^*}\nu(\,\mathrm{d}z) < \infty. \tag{C.2.1}$$

To ensure that the SDE (C.1.6) admits a unique and stable solution, we assume that the standard (local) Lipschitz continuity and linear growth conditions hold. Furthermore, we note that the Hamiltonian $\partial_t u + \mathcal{L}u + f$ is well-defined if

1. $u \in C^{1,2}([0,T) \times \mathbb{R}^d)$, and
2. One of the following inequalities hold locally in $(t, x) \in [0, T) \times \mathcal{S}$ for a constant $C_{t,x}$ dependent on t and x:
   (a) $|\gamma(t, x, z)| \leq C_{t,x}$ if $\nu(\mathbb{R}^d) < \infty$
   (b) $|\gamma(t, x, z)| \leq C_{t,x}|z|$ and $|u(t, z)| \leq C(1 + |z|^{p^*})$, where C is independent of t
   (c) Or more generally, for $a, b > 0$ such that $ab \leq p^*$: $|\gamma(t, x, z)| \leq C_{t,x}(1+|z|^a)$, $|u(t, z)| \leq C(1 + |z|^b)$ on $|z| \geq 1$. Furthermore, $|\gamma(t, x, z)| \leq C_{t,x}|z|^{q^*/2}$ on $|z| < 1$.

If the conditions above hold, the integral term $\mathcal{L}$ is finite, i.e.

$$\left|\int_{\mathbb{R}^\ell} \big[u(t, x + \gamma(t, x, z)) - u(t, x) - \langle \nabla_x u(t, x), \gamma(t, x, z)\rangle 1_{\{|z|<1\}}\big]\nu(\,\mathrm{d}z)\right| < \infty.$$

For the results derived in this chapter, the assumptions shown below are needed.

Assumption C.1 (Assumptions about the stochastic process X)**.** (S1) μ, σ, and γ are continuous.

(S2) γ satisfies one of conditions (a)–(c)

(S3) For $t \in [0, T)$ and $x, y \in \mathbb{R}^d$, the following inequality holds

$$\int_{\mathbb{R}^\ell} |\gamma(t, x, z) - \gamma(t, y, z)|^2 \nu(\,\mathrm{d}z) < C|x - y|^2$$

(S4) For a fixed time step $h \in (t, T - t)$ sufficiently small the m-th moment of X exists, i.e.

$$\mathbb{E}^{(t,x)}\left[\sup_{t \le s \le t+h} |X(s)|^m\right] < \infty.$$

Assumption C.2 (Assumptions about the intervention operator $\mathcal{M}$)**.** (I1) The functions Γ and K are continuous.

(I2) The set of admissible impulses $\mathcal{Z}(t, x)$ is non-empty and compact for each $(t, x) \in [0, T] \times \mathbb{R}^d$. For a converging sequence $(t_n, x_n) \to (t, x)$ in $[0, T] \times \mathbb{R}^d$ (with $\mathcal{Z}(t_n, x_n)$ non-empty), $\mathcal{Z}(t_n, x_n)$ converges to $\mathcal{Z}(t, x)$ in the Hausdorff metric.

(I3) For the value function v, assume that $v \in C^{1,2}([0, T) \times \mathbb{R}^d)$.

Assumption C.3 (Assumptions about the Hamiltonian)**.** (H1) The function f is continuous and in $\mathcal{PB}$.

(H2) The Hamiltonian is polynomially bounded, i.e.

$$\left|\partial_t v(t, x) + \mathcal{L}v(t, x) + f(t, x)\right| \le C_H\big(1 + |x|^p\big), \quad p \ge 0.$$

We note that with (I1)–(I3), the continuation region D essentially becomes an open set, which we rely on in the proof of Proposition C.2. The assumptions (S2) and (I3) ensure that integral operator in $\mathcal{L}$ is finite, while (S4) ensures that X has finite moments. Moreover, (H1), (H2), and (S4) together ensure that the Dynkin condition (2.4.1) is satisfied.

C.3 Results

Proposition C.1. *Let Assumptions (H1), (H2), and (S4) be satisfied. Furthermore, let $X := X^{t,x}$ be an uncontrolled process started at $X(t^-) = x$ and suppose $(t, x) \in D$, where D is the continuation region (C.1.5). Then, for any time increment $0 < h < T - t$,*

$$v(t, x) = \mathbb{E}^{(t,x)}\left[\int_t^\tau f\big(s, X(s)\big)\,\mathrm{d}s + v\big(\tau, X(\tau)\big)\right],$$

where $\tau := t + h \wedge \tau_D$ and τ_D is the first exit time out of D.

Proof. Fix $(t, x) \in D$ as well as $h > 0$, and note that τ_D is a finite stopping time bounded by $T < \infty$. Since both t and h are finite, $\tau \leq \tau_D$ by construction. Moreover, by assumptions (H1), (H2), and (S4), the Dynkin condition (2.4.1) is satisfied. Hence, applying the Dynkin formula, see Theorem 2.8, gives

$$v(t,x) = \mathbb{E}^{(t,x)} \left[-\int_t^\tau \Big(\partial_s v\big(s, X(s)\big) + \mathcal{L}v\big(s, X(s)\big) \Big) \, \mathrm{d}s + v\big(\tau, X(\tau)\big) \right], \tag{C.3.1}$$

Combining the identity $\partial_t v + \mathcal{L}v = -f$ on D with (C.3.1), we obtain

$$v(t,x) = \mathbb{E}^{(t,x)} \left[\int_t^\tau f\big(s, X(s)\big) \, \mathrm{d}s + v\big(\tau, X(\tau)\big) \right]. \qquad \square$$

Proposition C.2. *Let Assumptions C.1–C.3 be satisfied. Furthermore, let $X := X^{t,x}$ be an uncontrolled process started at $X(t^-) = x$ and suppose $(t, x) \in D$, where D is the continuation region (C.1.5). Then, for $h > 0$ sufficiently small,*

$$v(t,x) = \mathbb{E}^{(t,x)} \left[\int_t^{t+h} f\big(s, X(s)\big) \, \mathrm{d}s + v\big(t+h, X(t+h)\big) \right] + \mathcal{O}(h^{3/2}).$$

Proof. Fix $(t, x) \in D$. From Proposition C.1, we obtain

$$\begin{aligned} v(t,x) &= \mathbb{E}^{(t,x)} \left[\int_t^\tau f\big(s, X(s)\big) \, \mathrm{d}s + v\big(\tau, X(\tau)\big) \right] \\ &= \mathbb{E}^{(t,x)} \left[\int_t^{t+h} f(t, X(t)) + v(t+h, X(t+h)) \right] \\ &- \mathbb{E}^{(t,x)} \left[1_{\{\tau_d \leq h\}} \left(\int_{t+\tau_D}^{t+h} f(s, X(s)) \, \mathrm{d}s - v(t+\tau_D, X(t+\tau_D)) + v(t+h, X(t+h)) \right) \right]. \end{aligned}$$

Let the residual term in the above equation be denoted by R and be defined as

$$R := -\mathbb{E}^{(t,x)} \left[1_{\{\tau_d \leq h\}} \left(\int_{t+\tau_D}^{t+h} f(s, X(s)) \, \mathrm{d}s - v(t+\tau_D, X(t+\tau_D)) + v(t+h, X(t+h)) \right) \right]. \tag{C.3.2}$$

Thus, our main objective is to show that $|R| \leq Ch^{3/2}$. For that, we distinguish two events: $\{\tau_D \leq h\}$ and $\{\tau_D > h\}$. However, note that on the event $\{\tau_D > h\}$, the residual term R is zero, so the upper bound holds trivially. Therefore, consider only the event $\{\tau_D \leq h\}$.

Consider (C.3.2) and recall that the Dynkin condition (2.4.1) is satisfied due to (H1), (H2), and (S4). Moreover, as $h < T - t$ is a finite time increment, we obtain by the Markov property, the tower rule, and the Dynkin formula, see Theorem 2.8,

$$\begin{aligned} R &= -\mathbb{E}^{(t,x)} \Bigg[1_{\{\tau_D \leq h\}} \mathbb{E}^{(t+\tau_D, X(t+\tau_D))} \Bigg[\int_{t+\tau_D}^{t+h} f(s, X(s)) \, \mathrm{d}s - v(t+\tau_D, X(t+\tau_D)) \\ &\qquad\qquad + v(t+h, X(t+h)) \Bigg] \Bigg] \\ &= -\mathbb{E}^{(t,x)} \left[1_{\{\tau_D \leq h\}} \mathbb{E}^{(t+\tau_D, X(t+\tau_D))} \left[\int_{t+\tau_D}^{t+h} \Big(\partial_s v(s, X(s)) + \mathcal{L}v(s, X(s)) + f(s, X(s)) \Big) \, \mathrm{d}s \right] \right] \end{aligned}$$

$$= -\mathbb{E}^{(t,x)}\left[1_{\{\tau_D \leq h\}}\int_{t+\tau_D}^{t+h}\Big(\partial_s v(s,X(s)) + \mathcal{L}v(s,X(s)) + f(s,X(s))\Big)\,\mathrm{d}s\right]. \quad \text{(C.3.3)}$$

Then the Cauchy-Schwarz inequality is applied on expectation in (C.3.3) to separate the indicator term from the integral term. This gives,

$$\begin{aligned} R &\leq \left(\mathbb{E}^{(t,x)}[1_{\{\tau_D \leq h\}}]\right)^{1/2}\left(\mathbb{E}^{(t,x)}\left[\left|\int_{t+\tau_D}^{t+h}\partial_s v(s,X(s)) + \mathcal{L}v(s,X(s)) + f(s,X(s))\,\mathrm{d}s\right|^2\right]\right)^{1/2} \\ &\leq \mathbb{P}(\tau_D \leq h)^{1/2}\left(\mathbb{E}^{(t,x)}\left[\left|\int_{t+\tau_D}^{t+h}\partial_s v(s,X(s)) + \mathcal{L}v(s,X(s)) + f(s,X(s))\,\mathrm{d}s\right|^2\right]\right)^{1/2}, \end{aligned} \quad \text{(C.3.4)}$$

where the probability is conditional on $X(t^-) = x$, i.e, $\mathbb{P}(\cdot) := \mathbb{P}(\cdot|X(t^-) = x)$.

Now, define

$$A_1 := \left(\mathbb{E}^{(t,x)}\left[\left|\int_{t+\tau_D}^{t+h}\partial_s v(s,X(s)) + \mathcal{L}v(s,X(s)) + f(s,X(s))\,\mathrm{d}s\right|^2\right]\right)^{1/2},$$

and

$$A_2 := \mathbb{P}(\tau_D \leq h)^{1/2}.$$

To determine an upper bound for the residual term, we consider the terms A_1 and A_2 separately.

Let us begin with the term A_1. Recall that by (S1), (S2), and (I3), the Hamiltonian is well-defined. Furthermore, the Hamiltonian is polynomially bounded by (H2), while (S4) ensures that X has a bounded p-th moment. Applying these assumptions on A_1 therefore yields

$$\begin{aligned} A_1 &= \left(\mathbb{E}^{(t,x)}\left[\left|\int_{t+\tau_D}^{t+h}\partial_s v(s,X(s)) + \mathcal{L}v(s,X(s)) + f(s,X(s))\,\mathrm{d}s\right|^2\right]\right)^{1/2} \\ &\leq h\left(\mathbb{E}^{(t,x)}\left[\sup_{t\leq s\leq t+h}\left|\partial_s v(s,X(s)) + \mathcal{L}v(s,X(s)) + f(s,X(s))\right|^2\right]\right)^{1/2} \\ &\leq Ch\left(1 + \mathbb{E}^{(t,x)}\left[\sup_{t\leq s\leq t+h}\left|X(s)\right|^{2m}\right]\right) \end{aligned}$$

where $C > 0$. Because X has bounded moments of order $p := 2m$ for some h sufficiently small, we obtain that the upper bound for A_1 is finite. Hence, for $C_1 > 0$, we have

$$|A_1| \leq C_1 h. \quad \text{(C.3.5)}$$

Finally, consider the term A_2. Our objective is to show that $|A_2| \leq C_2 h^{1/2}$. Therefore, define the stopping time

$$\tau_\rho := \inf\{s \geq 0 : |X(t+x) - x| > \rho\},$$

for some fixed constant $\rho > 0$. The stopping time τ_ρ is understood to be the first exit time of X from the open ball $B(x,\rho)$ with radius ρ around x. Since D is an open set under (I1)–(I3), there exist $\rho, h' > 0$ sufficiently small such that

$$[t, t+h'] \times \overline{B(x,\rho)} \subset D.$$

Here, we have $0 \le h < h'$, where h' is fixed. As a result, one has

$$\{\tau_D \le h\} \subseteq \{\tau_\rho \le h\} \subseteq \left\{ \sup_{0\le s\le h} |X(t+s\wedge\tau_\rho) - x| \ge \rho \right\}. \tag{C.3.6}$$

The first inclusion in (C.3.6) is because whenever X exits D before time h, then X must have exited the ball $B(x,\rho)$ either prior to τ_D or at τ_ρ. However, the converse is not necessarily true since, if X exits $B(x,\rho)$ at τ_ρ, it has not necessarily exited D. The second inclusion is due to the inclusion of the boundary at ρ.

Then, with (C.3.6), we get that

$$\begin{aligned}\mathbb{P}(\tau_D \le h) &\le \mathbb{P}\left(\sup_{0\le s\le h} |X(t+s\wedge\tau_\rho) - x| \ge \rho \right) \\ &\le \frac{1}{\rho^2}\mathbb{E}^{(t,x)}\left[\sup_{0\le s\le h} |X(t+s\wedge\tau_\rho) - x|^2 \right],\end{aligned} \tag{C.3.7}$$

where the last inequality is due to Markov's inequality (see Appendix A.3). To show the boundedness of the expectation in (C.3.7), recall that X is a jump-diffusion satisfying

$$\begin{aligned}X(t+s) - x = &\int_t^{t+s} \mu(u, X(u^-))\,\mathrm{d}u + \int_t^{t+s} \sigma(u, X(u^-))\,\mathrm{d}W(u) \\ &+ \int_t^{t+s} \gamma(u, X(u^-), z)\tilde{N}(\,\mathrm{d}u,\,\mathrm{d}z),\end{aligned} \tag{C.3.8}$$

for $s > 0$. Furthermore, consider the inequality

$$(a+b+c)^2 \le 3(a^2+b^2+c^2). \tag{C.3.9}$$

Combining (C.3.7) and (C.3.8), and applying (C.3.9) gives

$$\begin{aligned}\mathbb{P}(\tau_D \le h) \le &\frac{3}{\rho^2}\left\{ \mathbb{E}^{(t,x)}\left[\sup_{0\le s\le h} \left| \int_t^{t+s\wedge\tau_\rho} \mu(u, X(u^-))\,\mathrm{d}u \right|^2 \right] \right. \\ &+ \mathbb{E}^{(t,x)}\left[\sup_{0\le s\le h} \left| \int_t^{t+s\wedge\tau_\rho} \sigma(u, X(u^-))\,\mathrm{d}W(u) \right|^2 \right] \\ &\left. + \mathbb{E}^{(t,x)}\left[\sup_{0\le s\le h} \left| \int_t^{t+s\wedge\tau_\rho} \gamma(u, X(u^-), z)\tilde{N}(\,\mathrm{d}u,\,\mathrm{d}z) \right|^2 \right] \right\}.\end{aligned} \tag{C.3.10}$$

As we are working with the compact set $[t, t+h'] \times \overline{B(x,\rho)} \subset D$, the functions μ, σ, and γ are bounded in that compact set because they are continuous by (S1). Therefore, there exists $C_\mu > 0$ such that

$$\mathbb{E}^{(t,x)}\left[\sup_{0\le s\le h} \left| \int_t^{t+s\wedge\tau_\rho} \mu(u, X(u^-))\,\mathrm{d}u \right|^2 \right]$$

$$
\begin{aligned}
&\leq \mathbb{E}^{(t,x)}\left[\sup_{0\leq s\leq h}(s\wedge\tau_\rho)\int_t^{t+\tau_\rho\wedge s}\left|\mu(u,X(u^-))\right|^2\mathrm{d}u\right]\\
&\leq h\mathbb{E}^{(t,x)}\left[\int_t^{t+h\wedge\tau_\rho}\left|\mu(u,X(u^-))\right|^2\mathrm{d}u\right]\\
&\leq C_\mu h^2. \qquad\qquad (C.3.11)
\end{aligned}
$$

For the Brownian part in (C.3.10), first applying Doob's L^2-maximal inequality [64, Theorem 3.8 (iv)] followed by the Itô isometry [84, Theorem 1.17], we obtain

$$
\mathbb{E}^{(t,x)}\left[\sup_{0\leq s\leq h}\left|\int_t^{t+s\wedge\tau_\rho}\sigma(u,X(u^-))\,\mathrm{d}W(u)\right|^2\right] \qquad (C.3.12)
$$

$$
\begin{aligned}
&\leq 4\mathbb{E}^{(t,x)}\left[\left|\int_t^{t+\tau_\rho\wedge h}\sigma(u,X(u^-))\,\mathrm{d}W(u)\right|^2\right]\\
&\leq 4\mathbb{E}^{(t,x)}\left[\int_t^{t+\tau_\rho\wedge h}\left|\sigma(u,X(u^-))\right|^2\,\mathrm{d}u\right]\\
&\leq C_\sigma h. \qquad\qquad (C.3.13)
\end{aligned}
$$

Similarly for the jump part,

$$
\begin{aligned}
&\mathbb{E}^{(t,x)}\left[\sup_{0\leq s\leq h}\left|\int_t^{t+s\wedge\tau_\rho}\int_{\mathbb{R}^k}\gamma(u,X(u^-),z))\tilde{N}(\,\mathrm{d}u,\,\mathrm{d}z)\right|^2\right]\\
&\quad\leq 4\mathbb{E}^{(t,x)}\left[\int_t^{t+s\wedge\tau_\rho}\int_{\mathbb{R}^k}\left|\gamma(u,X(u^-),z))\right|^2\nu(\,\mathrm{d}z)\,\mathrm{d}u\right]\\
&\quad\leq C_\nu h, \qquad\qquad (C.3.14)
\end{aligned}
$$

where the last inequality was obtained by applying (S3), that is,

$$
\begin{aligned}
&\int_{\mathbb{R}^k}\left|\gamma(s,X(s^-),z)\right|^2\nu(\,\mathrm{d}z)\\
&\quad\leq 2\int_{\mathbb{R}^k}\left|\gamma(s,X(s^-),z)-\gamma(s,x,z)\right|^2\nu(\,\mathrm{d}z)+2\int_{\mathbb{R}^k}\left|\gamma(s,x,z)\right|^2\nu(\,\mathrm{d}z)\\
&\quad\leq C|X(s^-)-x|^2+C'|x|^2\\
&\quad\leq C''\rho^2.
\end{aligned}
$$

Combining (C.3.10)-(C.3.14), we obtain

$$
A_2=\mathbb{P}(\tau_D\leq h)^{1/2}\leq C_1h^{1/2}. \qquad (C.3.15)
$$

Hence, with (C.3.5) and (C.3.15), the residual term (C.3.4) is bounded by

$$
|R|\leq Ch^{3/2},
$$

for $C>0$. We therefore conclude that for h sufficiently small,

$$
v(t,x)=\mathbb{E}^{(t,x)}\left[\int_0^h f(t+s,X(t+s))\,\mathrm{d}s+v(t+h,X(t+h))\right]+\mathcal{O}(h^{3/2}). \qquad \square
$$

APPENDIX

D

ECONOMIC SCENARIO GENERATOR

The method and algorithm presented in this appendix are used for the pathwise simulation of stock prices in Chapter 9. We closely follow the approach presented in [72, 71].

D.1 Variance-Gamma processes and subordination

Let $S_i = \{S_i(t)\}_{t\geq 0}$ denote a one-dimensional price process of stock i, for $i = 1, \ldots, d$, on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}\}_{t\geq 0}, \mathbb{P})$. Suppose that the initial condition $S_i(0^-) = s_i$ is known and that S_i follows a pure-jump SDE

$$\mathrm{d}S_i(t) = S_i(t^-)\left(\mu_i\,\mathrm{d}t + \int_{\mathbb{R}} (e^z - 1)\,\tilde{N}_i(\,\mathrm{d}t,\,\mathrm{d}z)\right), \tag{D.1.1}$$

where μ_i and $\tilde{N}_i$ are the drift coefficient and the compensated Poisson random measure of stock i, respectively. $\tilde{N}$ is given by

$$\tilde{N}(\,\mathrm{d}t,\,\mathrm{d}z) = \begin{cases} N(\,\mathrm{d}t,\,\mathrm{d}z) - \nu(\,\mathrm{d}z)\,\mathrm{d}t, & \text{for } |z| < 1 \\ N(\,\mathrm{d}t,\,\mathrm{d}z), & \text{for } |z| < 1. \end{cases} \tag{D.1.2}$$

In the stock price model (D.1.1), the mark z is understood to be the log-jump size. ν in (D.1.2) denotes the Lévy measure. For a Variance-Gamma (VG) process, it takes the form

$$\nu(\,\mathrm{d}z) = \frac{C}{|z|}\left(e^{Gz}1_{\{z<0\}} + e^{-Mz}1_{\{z>0\}}\right)\,\mathrm{d}z,$$

where $C, G, M > 0$ are constant parameters. We assume that every stock in the portfolio is a VG process with correlated parameters $C_i, G_i, M_i > 0$.

For a VG process, the SDE (D.1.1) has the solution

$$S_i(t) = s_i e^{(\mu_i+\omega_i)t+L(t;\sigma_i,\theta_i,\nu_i)}, \qquad S_i(0^-) = s_i, \tag{D.1.3}$$

where

$$\omega_i = \frac{1}{\nu_i} \log\left(1 - \theta_i \nu_i - \frac{1}{2}\sigma_i^2 \nu_i\right), \tag{D.1.4}$$

and the log-return process $L = \{L(t)\}_{t \geq 0}$ is given by

$$L(t; \sigma_i, \theta_i, \nu_i) = \sigma_i W_i(G_i(t)) + \theta_i G_i(t), \qquad L(0^-) = 0. \tag{D.1.5}$$

Here, W_i denotes the Brownian motion of stock i and G_i is the corresponding subordinator. In particular, G_i follows a Gamma distribution with shape $1/\nu_i$ and rate ν_i, i.e.

$$G_i(t) \sim \text{Gamma}\left(\frac{t}{\nu_i}, \frac{1}{\nu_i}\right).$$

The subordinator G_i can be interpreted as a stochastic "business clock", enabling the stock price model to capture volatility clustering and non-normal returns. The clock accelerates during periods of high market activity, such as earning announcements or switches in market regime, and slows down during quieter periods. Moreover, the weighted jump term $\theta_i G_i$ in (D.1.5) represents the arrival of information or large trades which lead to discontinuous jumps in the asset prices.

The marginal parameters $\sigma_i, \nu_i > 0$ and $\theta_i \in \mathbb{R}$ are related to C_i, G_i, and M_i in the sense that

$$\begin{aligned} C_i &= \frac{1}{\nu_i} > 0, \\ G_i &= \frac{1}{\sqrt{\frac{1}{4}\theta_i^2\nu_i^2 + \frac{1}{2}\sigma_i^2\nu_i} - \frac{1}{2}\theta_i\nu_i} > 0, \\ M_i &= \frac{1}{\sqrt{\frac{1}{4}\theta_i^2\nu_i^2 + \frac{1}{2}\sigma_i^2\nu_i} + \frac{1}{2}\theta_i\nu_i} > 0, \end{aligned}$$

see [88, Chapter 5.3.7]. In practice, it is often preferable to estimate σ_i, θ_i, and ν_i directly from market data, such as European call options, rather than working with C_i, G_i, and M_i, see Appendix E.

D.2 Simulation of correlated stocks via multivariate subordination

D.2.1 Multivariate subordination

Let $\mathbf{S} = \{(S_1(t), \ldots, S_d(t))\}_{t \geq 0}$ denote a multidimensional stock price process with $d \in [1, \infty)$, where each stock S_i follows the VG process (D.1.3) with parameters σ_i, θ_i, and ν_i for $i = 1, \ldots, d$. Furthermore, define the multidimensional log-return process $\mathbf{L} = \{(L_1(t), \ldots, L_d(t))\}_{t \geq 0}$, where each L_i follows (D.1.5) with respective parameters for $i = 1, \ldots, d$.

Suppose that $Y_i = \{Y_i(t)\}_{t \geq 0}$, $i = 1, \ldots, d$, and $Z = \{Z(t)\}_{t \geq 0}$ are independent subordinators, with

$$Y_i(t) \sim \text{Gamma}\left(\frac{t}{\nu_i} - a, \frac{1}{\nu_i}\right) \quad \text{and} \quad Z(t) \sim \text{Gamma}\left(a, 1\right). \tag{D.2.1}$$

The parameter $a > 0$ in (D.2.1) is a real number. In particular, we choose

$$a = \lambda \min \left\{ \frac{1}{\nu_1}, \dots, \frac{1}{\nu_d} \right\},$$

with $\lambda \in (0,1)$. Because a is present in the shape of both Y_i and Z, it is understood to be the common activity parameter. In the "business clock" interpretation of the subordinator, Y_i is interpreted as the idiosyncratic clock of stock i, while Z is the shared market clock. Thus, both the Gamma clocks and the jump components of the stocks in the portfolio are correlated through the parameter a. The choice of λ depends on the data, but should not be too big as it would restrict the jumps of the risky asset with the smallest jump intensity $1/\nu$. In our case, we have chosen $\lambda = 0.4$ arbitrarily.

With the subordination processes Y_i and Z in mind, the Lévy process $\mathbf{G} = \{\mathbf{G}(t)\}_{t \geq 0}$ given by

$$\mathbf{G}(t) = \big(Y_1(t) + \nu_1 Z(t), \dots, Y_d(t) + \nu_d Z(t)\big)$$

is a multivariate subordinator, see [72].

Finally, let $W_i = \{W_i(t)\}_{t \geq 0}$, $i = 1, \dots, d$, a standard Brownian motion. We collect all Brownian motions into $\mathbf{W}(t) = (W_1(t), \dots, W_d(t))$. Assuming that the stocks' Brownian motions are correlated through the matrix

$$\mathbf{C} = \begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1d} \\ \rho_{12} & 1 & \cdots & \rho_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1d} & \rho_{2d} & \cdots & 1 \end{pmatrix}, \tag{D.2.2}$$

where $\rho_{ij} \in (-1,1)$ for all $i, j = 1, \dots, d$ and $i \neq j$, we use Cholesky's decomposition theorem and write $\mathbf{C} = \mathbf{\Lambda}\mathbf{\Lambda}^\top$. The diffusion matrix $\mathbf{A}$ is defined as

$$\mathbf{A} = \mathbf{D}\mathbf{\Lambda},$$

where $\mathbf{D} = \text{diag}\{\sigma_1, \dots, \sigma_d\}$. Then, the log return process $\mathbf{L}$ follows the subordinated process

$$\mathbf{L}(t) = \mathbf{A}\mathbf{W}(\mathbf{G}(t)) + \theta \mathbf{G}(t), \qquad \mathbf{L}(0^-) = 0, \tag{D.2.3}$$

where $\theta = (\theta_1, \dots, \theta_d)$. The stock price process $\mathbf{S}$ is therefore

$$\mathbf{S}(t) = \mathbf{s}e^{(\mu+\omega)t + \mathbf{L}(t)}, \qquad \mathbf{S}(0^-) = \mathbf{s} \tag{D.2.4}$$

where $\mathbf{s} = (s_1, \dots, s_d)$ is the initial vector, $\mu = (\mu_1, \dots, \mu_d)$ is the drift vector, and $\omega = (\omega_1, \dots, \omega_d)$.

D.2.2 Estimation of the Brownian correlation matrix

The correlation matrix $\mathbf{C}$ cannot be directly observed from stock market data alone and must be estimated. In view of this, let $\mathbf{R}$ denote the correlation matrix of log-returns, which can be estimated using historical stock data. Following [72, 71], suppose $\mathbf{R}$ takes the form

$$\mathbf{R} = \begin{pmatrix} 1 & \rho_{\mathbf{Y}}(1,2) & \cdots & \rho_{\mathbf{Y}}(1,d) \\ \rho_{\mathbf{Y}}(1,2) & 1 & \cdots & \rho_{\mathbf{Y}}(2,d) \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{\mathbf{Y}}(1,d) & \rho_{\mathbf{Y}}(2,d) & \cdots & 1 \end{pmatrix}, \tag{D.2.5}$$

where, for a Brownian motion correlation ρ_{ij} in $\mathbf{C}$, $\rho_{\mathbf{Y}}(i,j)$ is given by

$$\rho_{\mathbf{Y}}(i,j) = \frac{a\left(\rho_{ij}\sigma_i\sigma_j\sqrt{\nu_i\nu_j} + \theta_i\theta_j\nu_i\nu_j\right)}{\sqrt{(\sigma_i^2 + \theta_i^2\nu_i)(\sigma_j^2 + \theta_j^2\nu_j)}}. \tag{D.2.6}$$

Then, if $\rho_{\mathbf{Y}}^{\text{Market}}(i,j)$ denotes the observed log-return correlation between assets i,j and $\rho_{\mathbf{Y}}^{\text{Model}}(i,j)$ is given by (D.2.6), we estimate ρ_{ij} via constrained minimisation. Specifically, we solve

$$\min_{\substack{\hat{\rho}_{ij} \\ i,j=1,\ldots,d}} \sum_{i<j} \left(\rho_{\mathbf{Y}}^{\text{Market}}(i,j) - \rho_{\mathbf{Y}}^{\text{Model}}(i,j)\right)^2 \quad \text{s.t } \hat{\rho}_{ij} \in (-1,1) \text{ for all } i,j = 1,\ldots,d,\ i \neq j.$$

The correlation matrix $\hat{\mathbf{C}}$ is then constructed using the estimated correlation coefficients $\hat{\rho}_{ij}$.

D.2.3 Numerical scheme

With (D.2.3)-(D.2.6), the simulation of d-assets present in the portfolio is straightforward. Assume that we work on a uniform time grid $0 = t_0 < t_1 < \cdots < t_{N_t} = T$ with $\Delta t = T/N_t$ and $N_t > 2$. Furthermore, assume that the parameters a, $\mu = (\mu_1,\ldots,\mu_d)$, $\sigma = (\sigma_1,\ldots,\sigma_d)$, $\theta = (\theta_1,\ldots,\theta_d)$, $\mathbf{A}$, and $\hat{\mathbf{C}}$ are given.

At each time step t_n, we draw N_{MC} (Monte Carlo samples) of $\mathbf{W}(t_n)$, $Y_i(t_n)$, $i = 1,\ldots,d$, and $Z(t_n)$ according their respective distributions. Then all d stock prices are computed simultaneously with

$$\mathbf{S}(t_n) = \mathbf{S}(t_{n-1})e^{(\mu+\omega)\Delta t + \Delta\mathbf{L}(t_n)}, \tag{D.2.7}$$

where

$$\Delta\mathbf{L}(t_n) = \mathbf{L}(t_n) - \mathbf{L}(t_{n-1}).$$

The process is repeated for $n = 1,\ldots,N_t$. Algorithm 3 summarises this simulation procedure.

Algorithm 3 Simulation procedure of d-correlated stocks via multivariate subordination

Input: Parameters a, μ, σ, ν, θ, $\mathbf{A}$, $\mathbf{C}$

1: Compute ω_i using (D.1.4) for $i = 1, \dots, d$

2: Initialise time grid $0 = t_0 < t_1 < \cdots < t_{N_t}$, with $\Delta t = T/N_t$ and $t_n - t_{n-1} = \Delta t$ for every $n = 1, \dots, N_t$

3: Initialise $\mathbf{S}_0 = \mathbf{s}$

4: **for** $n = 1$ to N_t **do**

5: Sample N_{MC}

$$\Delta Y_{i,n} \sim \text{Gamma}\left(\frac{\Delta t}{\nu_i}, \frac{1}{\nu_i}\right), \quad \Delta Z_n \sim \text{Gamma}\left(a\Delta t, 1\right), \quad \varepsilon_n \sim \mathcal{N}(0, \mathbf{I}_d)$$

6: Construct $\Delta \mathbf{G}_n = (\Delta Y_{1,n} + \Delta Z_n, \dots, \Delta Y_{d,n} + \Delta Z_n)$

7: Compute $\Delta \mathbf{L}_n = \mathbf{A}\sqrt{\Delta \mathbf{G}_n} \odot \varepsilon_n \Delta t + \theta \odot \Delta \mathbf{G}_n$

8: Compute $\mathbf{S}_n = \mathbf{S}_{n-1} e^{\Delta \mathbf{L}_n}$

9: **end for**

APPENDIX

E

ASSET PRICE CALIBRATION USING EUROPEAN CALL OPTIONS

This appendix describes the calibration of the Variance-Gamma stock price model to observed European option prices. The objective is to choose model parameters σ, θ, and ν such that the resulting options prices are close to those quoted in the market. In this sense, calibration uses market option prices to determine suitable parameter values for the underlying stock price model.

To make the procedure computationally efficient, option prices are computed using the Fast Fourier Transform (FFT) approach introduced by Carr and Madan [27]. The calibration problem is then formulated as a numerical minimisation problem, where the distance between model prices and market prices is minimised over the admissible parameter space. The focus of this appendix is on the practical implementation of the calibration procedure. Detailed derivation of the underlying pricing theory is therefore omitted. For further details, see [27, 61, 73, 88].

E.1 Risk-neutral option pricing

A *European option* is a financial contract that gives its holder the right, but not the obligation, to buy or sell an underlying asset at a predetermined price on a fixed maturity date. A *call option* gives the right to buy the asset, while a *put option* gives the right to sell it. The predetermined price is called the strike price, denoted by K, and the fixed maturity date is denoted by T. European options can only be exercised at maturity.

Since European call and put prices are linked through the put-call parity, see [61, p.85], it is sufficient for our purposes to consider European call options, which can be priced using the FFT approach of Carr and Madan [27]. In the following, we refer to European call options simply as call options. The underlying asset is assumed to be a company's stock.

The standard approach to option pricing is based on risk-neutral pricing. Under this approach, the derivative price is obtained by discounting its expected payoff under a risk-neutral pricing measure. For a call option with strike K, maturity T, constant risk-free rate δ, and underlying stock price $S(T)$ at maturity, the time-t price is given by

$$C_T(s,K) = e^{-\delta T}\mathbb{E}_{\mathbb{Q}}[(S(T)-K)_+ \mid S(0)=s],$$

where $(x)_+ := \max(x,0)$ is the hockey stick function, and $\mathbb{Q}$ denotes the risk-neutral probability measure. For simplification, we set $t=0$. Then, the call option is priced via

$$C_T(s,K) = e^{-\delta T}\mathbb{E}_{\mathbb{Q}}[(S(T)-K)_+|S(0)=s]. \tag{E.1.1}$$

In the calibration procedure below, stocks are modeled under the risk-neutral probability measure.

E.2 Option pricing via Fourier Transforms

Let k denote the logarithm of the strike price K and $C_T(s,k)$ the price of a European call option with maturity T. Let the risk-neutral density of the logarithm price s_T be $q_T(s)$. Then the call option pricing formula (E.1.1) reads as

$$\begin{aligned} C_T(s,K) &= e^{-\delta T}\mathbb{E}_{\mathbb{Q}}[(S(T)-K)_+|S(0)=s] \\ &= e^{-\delta T}\int_k^\infty (e^s - e^k) q_T(s)\,\mathrm{d}s. \end{aligned} \tag{E.2.1}$$

Note that the function C_T is not square-integrable because $C_T \to 0$ as $K \to -\infty$. Moreover, let u be the Fourier frequency of k. Then, it can be shown that the Fourier transform of C_T is singular at $u=0$. Therefore, Carr and Madan [27] introduce the damping term $e^{\alpha k}$, where $\alpha > 0$ is a real number. Specifically, define

$$c_T(s,k) = e^{\alpha k} C_T(s,k). \tag{E.2.2}$$

Let $\hat{c}_T$ denote the Fourier transform of c_T. The Fourier transform is given by

$$\hat{c}_T(s,u) = \int_{-\infty}^{\infty} e^{iuk} c_T(s,k)\,\mathrm{d}k,$$

with $i^2 = -1$ an imaginary number. Furthermore, define the characteristic function of q_T by

$$\phi_T(u) = \int_{-\infty}^{\infty} e^{ius} q_T(s)\,\mathrm{d}s. \tag{E.2.3}$$

Then, applying Fourier transforms on c_T and q_T, it can shown that (E.2.1) can be written as

$$C_T(s,k) = \frac{1}{\pi} e^{-\alpha k} \int_0^\infty e^{-iuk} \hat{c}_T(s,u)\,\mathrm{d}u, \tag{E.2.4}$$

where

$$\hat{c}_T(s,u) = \int_{-\infty}^{\infty} e^{iuk} \int_k^\infty e^{\alpha k}(e^s - e^k) q_T(s)\,\mathrm{d}s\,\mathrm{d}k = \frac{e^{-\delta T}\phi_T(u-(\alpha+1)i)}{\alpha^2+\alpha-u^2+i(2\alpha+1)u}. \tag{E.2.5}$$

Rigorous details on the derivation of (E.2.4) and (E.2.5) can be found in [27].

E.3 From Fourier transforms to the Fast Fourier Transform

The Fast Fourier Transform (FFT) is an efficient algorithm to compute the sum

$$w(n) = \sum_{j=1}^{N} e^{-i\frac{2\pi}{N}(j-1)(n-1)} f(j), \quad \text{for } n = 0, 1, \dots, N, \tag{E.3.1}$$

where N is typically a power of 2. The algorithm was first introduced by Cooley and Tuckey in 1965, see [28]. Since the FFT algorithm has a runtime complexity of magnitude $\mathcal{O}(N \log N)$, it is more efficient than a naive discrete Fourier transform, which typically has a runtime complexity $\mathcal{O}(N^2)$. In light of this, we now aim to apply the FFT algorithm to compute a discrete version of (E.2.4).

Using the trapezoid rule, (E.2.4) can be written as

$$C_T(s,k) \approx \frac{1}{\pi} e^{-\alpha k} \sum_{j=1}^{N} e^{-iu_j n} \hat{c}_T(s,u_j)\eta, \tag{E.3.2}$$

where $u_j = \eta(j-1)$. The upper limit of the integration is now

$$a = N\eta.$$

Choosing a regular spacing λ and setting a bound on the logarithm of the strike k to range between $-$ and b, we obtain the grid

$$k_n = \varepsilon + \lambda(n-1), \quad \text{for } n = 1, \dots, N,$$

where

$$\varepsilon = \frac{1}{2} N\lambda.$$

The formula (E.3.2) can now be written as

$$C_T(s,k_n) \approx \frac{1}{\pi} e^{-\alpha k_n} \sum_{j=1}^{N} e^{-i\lambda\eta(j-1)(n-1)} e^{i\varepsilon u_j} \hat{c}_T(s,u_j)\eta. \tag{E.3.3}$$

To obtain a formula similar to (E.3.1) (on which we can apply the FFT), we set $\lambda\eta = \frac{2\pi}{N}$. Furthermore, to obtain a more accurate integration for large η, Carr and Madan [27] suggest applying Simpson's weighting rule. Applying these changes to (E.3.3), we obtain

$$C_T(s,k_n) \approx \frac{1}{\pi} e^{-\alpha k_n} \sum_{j=1}^{N} e^{-i\frac{2\pi}{N}(j-1)(n-1)} e^{i\varepsilon u_j} \hat{c}_T(s,u_j) \frac{\eta}{3} \left[3 + (-1)^j - \delta_{j-1}\right], \tag{E.3.4}$$

where δ_{j-1} is the Kronecker delta function that is unit for $j-1 = 0$ and zero otherwise. The function $\hat{c}_T$ remains unchanged and is given by (E.2.5).

E.4 Model calibration

With an efficient way to price call options, we now turn our attention to the estimation of parameters σ, θ, and ν in the Variance-Gamma (VG) model. The VG option pricing model is described in detail in [73]. As such, derivations and proofs are omitted in this appendix.

Let $L(t;\sigma,\theta,\nu)$ denote a VG-process with parameters σ, θ, and ν. Moreover, assume that the stock pays dividends. We denote by q the annualised log-dividend, estimated using historic dividends. Under the risk-neutral measure $\mathbb{Q}$, the stock price is given by

$$S(t) = se^{(\delta+\omega-q)t+L(t;\sigma,\theta,\nu)}, \quad t \geq 0, \tag{E.4.1}$$

where $S(0) = s$ is known and $\omega = (1/\nu)\log(1-\theta\nu-\frac{1}{2}\sigma^2\nu)$. The characteristic function of the logarithm of $S(T)$ is then

$$\phi_T(u) = \exp\left(iu\left(\log s + (\delta+\omega-q)T\right)\right)\left(1 - i\theta\nu u + \frac{1}{2}\sigma^2\nu u^2\right)^{-T/\nu}.$$

The option prices for N strike prices are obtained by inserting $\phi_T(u)$ into the expression of $\hat{c}_T$ (E.2.5) and then using the FFT algorithm on (E.3.4). The calibration then essentially becomes the work of "reverse engineering" the parameters σ, θ, and ν using market data. Specifically, for options data with M maturities and N strikes, we minimise the squared relative error

$$\Theta^* = \arg\min_{\Theta} \sum_{m=1}^{M}\sum_{n=1}^{N}\left(\frac{C_{T_m}^{\text{model}}(s,K_n) - C_{m,n}^{\text{market}}}{C_{m,n}^{\text{market}}}\right)^2, \tag{E.4.2}$$

where $\Theta := (\sigma,\theta,\nu)$ and $C_{m,n}^{\text{market}}$ denotes the market price of a call option with maturity T_m and strike K_n. The optimisation constraints are

$$0 < 1 - \theta\nu - \frac{1}{2}\sigma^2\nu \quad \text{and} \quad \alpha < \sqrt{\frac{\theta^2}{\sigma^4} + \frac{2}{\sigma^2\nu}} - \frac{\theta}{\sigma^2} - 1.$$

The first constraint is to keep the logarithm in ω well-defined. The second constraint is to keep the quantity

$$\phi_T(-(\alpha+1)i) = e^{\log s + (\delta+\omega)T}\left(1 - \theta\nu(\alpha+1) - \frac{1}{2}\sigma^2(\alpha+1)^2\nu\right)^{-T/\nu}$$

finite, see [27, Section 5].

The minimisation problem (E.4.2) is possibly ill-posed and the gradient of C_T^{model} is unknown. In light of this, we use the *Differential Evolution* (DE) algorithm, which is a population-based stochastic optimization algorithm that does not require gradients. The DE algorithm was first introduced by Storn and Price [93]. It is implemented using `scipy.optimize.differential_evolution` in Python.

E.5 Estimating the drift

The drift term μ is computed by replacing $\delta - q$ in (E.4.1) with μ and then estimating the resulting mean log-return from historical stock prices. The historical stock prices are sampled over a year on a daily basis. Under this specification, the stock price dynamics can be written as

$$S(t) = se^{(\mu+\omega)t+L(t;\sigma,\theta,\nu)}.$$

Let the observed stock prices be $s, S_{t_1}, \dots, S_{t_{N_t}}$ with equidistant time step Δt. The log-return over the interval $[j-1, j]$ is

$$R_j := \log\left(\frac{S_{t_j}}{S_{t_{j-1}}}\right) = (\mu+\omega)\Delta t + \Delta L_j, \quad \text{for } j = 1, \dots, N_t,$$

where $\Delta L_j := L(t_j;\sigma,\theta,\nu) - L(t_{j-1};\sigma,\theta,\nu)$. Since the VG process has stationary independent increments and

$$\mathbb{E}\left[\Delta L_j\right] = \theta\Delta t,$$

it follows that

$$\mathbb{E}[R_j] = (\mu+\omega+\theta)\Delta t.$$

Consequently, if

$$\bar{R} := \frac{1}{n}\sum_{j=1}^{N_t} R_j$$

denotes the empirical mean log-return, we estimate the drift parameter by

$$\hat{\mu} = \frac{\bar{R}}{\Delta t} - \omega - \theta.$$

APPENDIX

F

PORTFOLIO PERFORMANCE METRICS

In Chapter 9, the portfolios are evaluated based on their returns, their risks, and their overall performance. Based on [98, 76], we define the metrics used as follows:

1. **Cumulative Return:** Measures total portfolio growth over time and is defined as

$$R_{\text{cumul}} = \left(\prod_{n=1}^{N_t} (1 + R_n) \right) - 1,$$

where R_n is the portfolio return at time step n, and N_t is the time grid size. Higher values indicate effective trading strategies.

2. **Annualised Return:** Measures standardised growth in portfolio value over time, making it comparable across different durations. It is defined as:

$$R_{\text{annual}} = \left(\frac{\Pi(T)}{\pi_0} \right)^{\frac{252}{N_t}} - 1,$$

where $\Pi(0^-) = \pi_0$ is the initial portfolio value and $\Pi(T)$ is the portfolio value on the investment horizon $T < \infty$.

3. **Sharpe ratio:** Evaluates risk-adjusted returns by dividing the mean annualised return by the annualised return volatility, i.e.

$$\text{Sharpe Ratio} = \frac{\bar{R}_{\text{annual}} - \delta}{\sigma_r},$$

where δ is the risk-free rate and σ_r is the return standard deviation.

4. **Maximum Drawdown:** Measures the largest peak-to-through decline in portfolio value:

$$\text{MDD} = \max_n \left(\frac{\text{Peak}_n - \Pi(t_n)}{\text{Peak}_n} \right),$$

where $\Pi(t_n)$ denotes the portfolio value at time step n and $\text{Peak}_n = \max_{j \leq n} \Pi(t_j)$ is the running maximum up to time t_n.

5. **Calmar Ratio**: Assesses return relative to maximum drawdown, i.e.

$$\text{Calmar Ratio} = \frac{R_{\text{annual}}}{\text{MDD}}.$$

6. **Empirical Value-at-Risk:** Let $\{L_i\}_{i=1,\dots,N_{\text{MC}}}$ be a sequence of N_{MC} Monte Carlo simulated portfolio losses at time T. Here, a loss is defined as $L_i = \Pi^{(i)} - \pi_0$. Then, an empirical estimator of the Value-at-risk at the significance level α is given by

$$\widehat{\text{VaR}}_\alpha \approx L_{\lfloor N_{\text{MC}}(1-\alpha)\rfloor+1,N_{\text{MC}}},$$

where $L_{1,n} \geq \cdots \geq L_{n,n}$ are the upper-order statistics of $L_1, \dots, L_n$ and $\lfloor\cdot\rfloor$ is the ceiling operator.

7. **Empirical Expected Shortfall:** Let $\{L_i\}_{i=1,\dots,N_{\text{MC}}}$ be a sequence of N_{MC} Monte Carlo simulated portfolio losses at time T. Here, a loss is defined as $L_i = \Pi^{(i)} - \pi_0$. Then, an empirical estimator of the expected shortfall at significance level α is given by

$$\widehat{\text{ES}}_\alpha \approx \frac{1}{N_{\text{MC}}(1-\alpha)} \sum_{i=1}^{\lfloor N_{\text{MC}}(1-\alpha)\rfloor} L_{i,N_{\text{MC}}},$$

where $L_{1,N_{\text{MC}}} \geq \cdots \geq L_{N_{\text{MC}},N_{\text{MC}}}$ are the upper-order statistics of $L_1, \dots, L_{N_{\text{MC}}}$ and $\lfloor\cdot\rfloor$ is the ceiling operator.

APPENDIX

G

DATA AND CALIBRATION RESULTS

G.1 Data

The data used for calibration are obtained from the LSEG Workspace terminal. As such, the data set is assumed to be clean, and no additional data preprocessing is performed. Stock price data are collected over the period from 2024-05-14 to 2026-05-13 at a daily frequency at market close, resulting in a total of 498 observations per stock. Option prices are collected on 2026-05-13 at market close, with times to maturity of at most one year. An overview of the collected data is provided in Tables 13 and 14.

ABBN		
Valuation date		2026-05-14
Spot price (CHF)		82.86
Total samples		228
Expiry date	Days to maturity	Samples
2026-06-19	36	62
2026-07-17	64	37
2026-09-18	127	43
2026-12-18	218	54
2027-03-19	309	32

ALCC		
Valuation date		2026-05-14
Spot price (CHF)		49.61
Total samples		111
Expiry date	Days to maturity	Samples
2026-06-19	36	32
2026-07-17	64	32
2026-09-18	127	15
2026-12-18	218	18
2027-03-19	309	14

CFR		
Valuation date		2026-05-14
Spot price (CHF)		156.55
Total samples		269
Expiry date	Days to maturity	Samples
2026-06-19	36	59
2026-07-17	64	34
2026-09-18	127	70
2026-12-18	218	76
2027-03-19	309	30

GALD		
Valuation date		2026-05-14
Spot price (CHF)		158.35
Total samples		119
Expiry date	Days to maturity	Samples
2026-06-19	36	36
2026-07-17	64	29
2026-09-18	127	17
2026-12-18	218	25
2027-03-19	309	12

Continued on next page

GEBN

Valuation date	2026-05-14
Spot price (CHF)	503.40
Total samples	120

Expiry date	Days to maturity	Samples
2026-06-19	36	36
2026-07-17	64	30
2026-09-18	127	17
2026-12-18	218	21
2027-03-19	309	16

GIVN

Valuation date	2026-05-14
Spot price (CHF)	2683.00
Total samples	182

Expiry date	Days to maturity	Samples
2026-06-19	36	53
2026-07-17	64	31
2026-09-18	127	34
2026-12-18	218	35
2027-03-19	309	29

HOLN

Valuation date	2026-05-14
Spot price (CHF)	76.26
Total samples	177

Expiry date	Days to maturity	Samples
2026-06-19	36	50
2026-07-17	64	31
2026-09-18	127	32
2026-12-18	218	34
2027-03-19	309	30

LONN

Valuation date	2026-05-14
Spot price (CHF)	474.20
Total samples	177

Expiry date	Days to maturity	Samples
2026-06-19	36	48
2026-07-17	64	34
2026-09-18	127	31
2026-12-18	218	35
2027-03-19	309	29

NESN

Valuation date	2026-05-14
Spot price (CHF)	76.88
Total samples	188

Expiry date	Days to maturity	Samples
2026-06-19	36	49
2026-07-17	64	33
2026-09-18	127	35
2026-12-18	218	42
2027-03-19	309	29

NOVN

Valuation date	2026-05-14
Spot price (CHF)	116.80
Total samples	182

Expiry date	Days to maturity	Samples
2026-06-19	36	44
2026-07-17	64	29
2026-09-18	127	37
2026-12-18	218	44
2027-03-19	309	28

PGHN

Valuation date	2026-05-14
Spot price (CHF)	883.60
Total samples	173

Expiry date	Days to maturity	Samples
2026-06-19	36	47
2026-07-17	64	28
2026-09-18	127	30
2026-12-18	218	41
2027-03-19	309	27

ROPC

Valuation date	2026-05-14
Spot price (CHF)	320.00
Total samples	205

Expiry date	Days to maturity	Samples
2026-06-19	36	48
2026-07-17	64	29
2026-09-18	127	48
2026-12-18	218	53
2027-03-19	309	28

SDZ

Valuation date	2026-05-14
Spot price (CHF)	67.76
Total samples	151

Expiry date	Days to maturity	Samples
2026-06-19	36	35
2026-07-17	64	26
2026-09-18	127	32
2026-12-18	218	33
2027-03-19	309	25

SGSN

Valuation date	2026-05-14
Spot price (CHF)	84.68
Total samples	153

Expiry date	Days to maturity	Samples
2026-06-19	36	38
2026-07-17	64	29
2026-09-18	127	29
2026-12-18	218	29
2027-03-19	309	28

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SIKA		
Valuation date		2026-05-14
Spot price (CHF)		140.95
Total samples		127
Expiry date	Days to maturity	Samples
2026-06-19	36	36
2026-07-17	64	20
2026-09-18	127	23
2026-12-18	218	28
2027-03-19	309	20

SLHN		
Valuation date		2026-05-14
Spot price (CHF)		837.60
Total samples		201
Expiry date	Days to maturity	Samples
2026-06-19	36	52
2026-07-17	64	36
2026-09-18	127	40
2026-12-18	218	43
2027-03-19	309	30

SCMN		
Valuation date		2026-05-14
Spot price (CHF)		678.00
Total samples		175
Expiry date	Days to maturity	Samples
2026-06-19	36	48
2026-07-17	64	29
2026-09-18	127	32
2026-12-18	218	37
2027-03-19	309	29

SRENH		
Valuation date		2026-05-14
Spot price (CHF)		119.45
Total samples		174
Expiry date	Days to maturity	Samples
2026-06-19	36	51
2026-07-17	64	34
2026-09-18	127	29
2026-12-18	218	32
2027-03-19	309	28

UBSG		
Valuation date		2026-05-14
Spot price (CHF)		36.22
Total samples		173
Expiry date	Days to maturity	Samples
2026-06-19	36	50
2026-07-17	64	33
2026-09-18	127	30
2026-12-18	218	30
2027-03-19	309	30

ZURN		
Valuation date		2026-05-14
Spot price (CHF)		563.00
Total samples		218
Expiry date	Days to maturity	Samples
2026-06-19	36	52
2026-07-17	64	30
2026-09-18	127	62
2026-12-18	218	46
2027-03-19	309	28

Table 14: Option sample overviews for all underlyings. Prices are quoted in the Swiss Franc (CHF).

G.2 Calibration results

G.2.1 Estimated parameters

To estimate the parameters μ, σ, θ, and ν, we use the calibration procedure proposed in Appendix E. The FFT parameters were $N_{\mathrm{FFT}} = 1024$, $\eta = 0.25$, $\alpha = 1.5$, which are the same parameters Carr and Madan [27] use. When estimating the parameters using the DE algorithm, we asserted the bounds: $\sigma \in (0.05, 0.6)$, $\theta \in (-2.0, -0.0001)$, and $\nu \in (0.001, 0.45)$ to keep the estimated parameters economically interpretable. When estimating μ, we cap empirical log-returns to the 95-th quantile in order to make a more conservative estimation. The risk-free rate is the Swiss Average Rate Overnight (SARON) and is $\delta \approx 0\%$ per May 14th, 2026. The resulting estimations can be found in Table 15. The log-return correlation matrix can be found in Figure 17.

Asset	Ticker	Initial Price	Final Price
ABB Ltd	ABBN	46.88	82.86
Alcon AG	ALCC	73.06	49.61
Compagnie Financière Richemont SA	CFR	155.75	156.55
Galderma Group AG	GALD	97.95	158.35
Geberit AG	GEBN	597.20	503.40
Givaudan SA	GIVN	3997.00	2683.00
Holcim AG	HOLN	49.50	76.26
Lonza Group AG	LONN	572.20	474.20
Nestlé SA	NESN	85.21	76.88
Novartis AG	NOVN	90.52	116.80
Partners Group Holding AG	PGHN	1187.50	883.60
Roche Holding AG	ROPC	259.30	320.00
Sandoz Group AG	SDZ	38.15	67.76
SGS SA	SGSN	85.70	84.68
Sika AG	SIKA	217.80	140.95
Swiss Life Holding AG	SLHN	826.20	837.60
Swisscom AG	SCMN	538.00	678.00
Swiss Re AG	SRENH	147.25	119.45
UBS Group AG	UBSG	27.23	36.22
Zurich Insurance Group AG	ZURN	573.00	563.00

Table 13: Summary of stock price data from 2024-05-14 through 2026-05-13. Prices are quoted in the Swiss Franc (CHF).

G.2.2 Error metrics and plots

The error metrics used to evaluate the estimated parameters across all strikes and maturities are the Mean Absolute Error (MAE), the Root Mean-Squared Error (RMSE), and the Mean Absolute Percentage Error (MAPE), given by (G.2.1), (G.2.2), and (G.2.3), respectively.

$$\mathrm{MAE} = \frac{1}{MN}\sum_{m=1}^{M}\sum_{n=1}^{N}\left|C_{T_m}^{\mathrm{Model}}(s,K_n) - C_{m,n}^{\mathrm{Market}}\right| \tag{G.2.1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{MN}\sum_{m=1}^{M}\sum_{n=1}^{N}\left(C_{T_m}^{\mathrm{Model}}(s,K_n) - C_{m,n}^{\mathrm{Market}}\right)^2} \tag{G.2.2}$$

$$\mathrm{MAPE} = \frac{100}{MN}\sum_{m=1}^{M}\sum_{n=1}^{N}\left|\frac{C_{T_m}^{\mathrm{Model}}(s,K_n) - C_{m,n}^{\mathrm{Market}}}{C_{m,n}^{\mathrm{Market}}}\right|. \tag{G.2.3}$$

The error metrics above have different interpretations. MAE captures the average magnitude of pricing deviations without considering the direction of the deviation. It provides a measure for overall model accuracy. RMSE emphasizes larger errors, making it particularly sensitive to large mispricings. Finally, MAPE expresses errors relative to market prices, making it particularly useful for assessing whether low-priced instruments are accurately matched. Pricing errors for each underlying can be

TKR	μ (%)	σ (%)	θ (%)	ν (%)
ABB	12.71	29.42	-36.63	8.31
ALCC	-8.35	27.31	-28.01	8.87
CFR	0.03	30.20	-0.01	0.94
GALD	18.74	27.61	-84.67	3.64
GEBN	-0.70	21.32	-15.62	1.13
GIVN	-5.89	25.68	-33.67	38.18
HOLN	13.36	25.21	-29.84	4.99
LONN	1.13	27.89	-28.42	5.73
NESN	-3.45	18.48	-31.11	8.21
NOVN	6.16	20.60	-25.11	1.83
PGHN	-6.21	23.76	-79.28	1.19
ROPC	6.76	21.61	-16.70	1.88
SDZ	18.30	28.44	-62.67	3.45
SGSN	3.86	19.74	-42.65	10.52
SIKA	-12.97	30.29	-22.27	9.68
SLHN	6.08	18.93	-0.01	0.66
SCMN	7.67	19.56	-22.00	45.00
SRENH	2.24	22.56	-31.83	15.14
UBSG	8.61	25.65	-44.70	5.18
ZURN	3.86	17.65	-25.12	2.29

Table 15: Estimated model parameters.

found in Table 16. Note that pricing errors are relatively small across all underlying assets except for Parnters Group Holding (PGHN).

ABBN	
Metric	Value
MAE	0.116918
RMSE	0.161007
MAPE (%)	3.747744

ALCC	
Metric	Value
MAE	0.065518
RMSE	0.092814
MAPE (%)	8.608676

CFR	
Metric	Value
MAE	0.686083
RMSE	0.908786
MAPE (%)	7.469491

GALD	
Metric	Value
MAE	0.265190
RMSE	0.356102
MAPE (%)	4.030166

GEBN	
Metric	Value
MAE	1.010899
RMSE	1.690254
MAPE (%)	6.993848

GIVN	
Metric	Value
MAE	7.633179
RMSE	13.332721
MAPE (%)	15.814692

HOLN	
Metric	Value
MAE	0.860288
RMSE	1.024028
MAPE (%)	11.666620

LONN	
Metric	Value
MAE	0.538848
RMSE	0.869958
MAPE (%)	12.044501

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NESN	
Metric	Value
MAE	0.100446
RMSE	0.131476
MAPE (%)	7.265444

NOVN	
Metric	Value
MAE	0.336908
RMSE	0.699478
MAPE (%)	3.111208

PGHN	
Metric	Value
MAE	17.017618
RMSE	22.402186
MAPE (%)	22.797090

ROPC	
Metric	Value
MAE	0.874842
RMSE	1.675453
MAPE (%)	3.485799

SDZ	
Metric	Value
MAE	0.106628
RMSE	0.143644
MAPE (%)	4.105436

SGSN	
Metric	Value
MAE	0.116536
RMSE	0.152647
MAPE (%)	5.263105

SIKA	
Metric	Value
MAE	0.175852
RMSE	0.280050
MAPE (%)	9.908205

SLHN	
Metric	Value
MAE	2.232442
RMSE	3.297510
MAPE (%)	8.818372

SCMN	
Metric	Value
MAE	1.142384
RMSE	2.044142
MAPE (%)	11.760861

SRENH	
Metric	Value
MAE	0.149317
RMSE	0.223762
MAPE (%)	11.477619

UBSG	
Metric	Value
MAE	0.081952
RMSE	0.117784
MAPE (%)	2.462055

ZURN	
Metric	Value
MAE	1.287465
RMSE	1.791158
MAPE (%)	6.312769

Table 16: Error metrics for all underlyings.

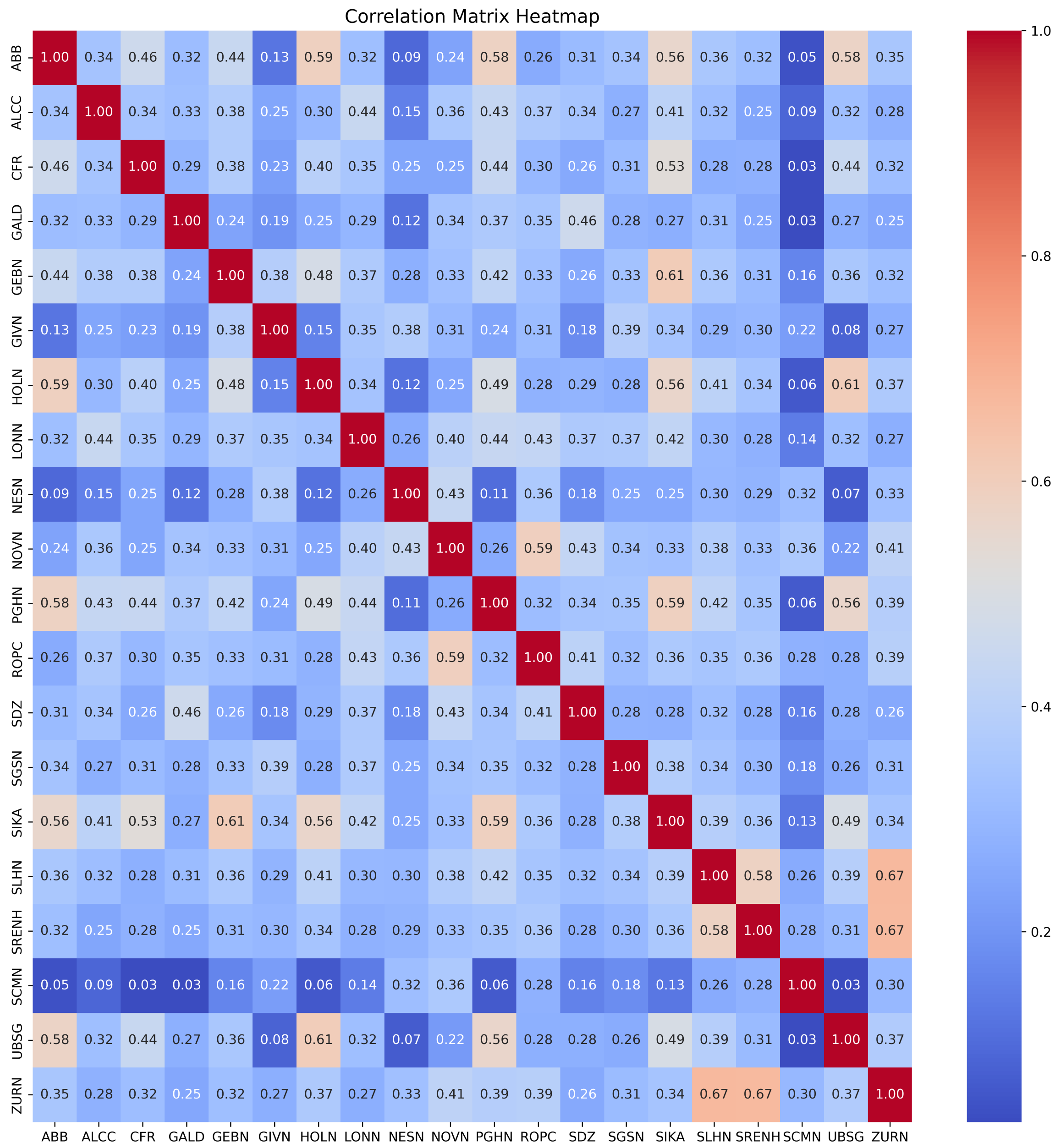

Figure 17: Log-return correlation matrix